Contents

Chapter 1

SYNTHETIC FIBRES AND PLASTICS

INTRODUCTION

You have read in your previous classes that natural fibres like cotton, wool, silk, etc., are obtained from plants or animals. The synthetic fibres, on the other hand, are made by human beings. That is why these are called synthetic or man-made fibres. Man-made fibres are spun and woven into a huge number of consumer and industrial products, including garments such as shirts, scarves, and hosiery; home furnishings such as upholstery, carpets, and drapes; and industrial parts such as tire cord, flame-proof linings, and drive belts. The chemical compounds from which man-made fibres are produced are known as polymers. Many of the polymers that constitute man-made fibres are the same as or similar to compounds that make up plastics, rubbers, adhesives, and surface coatings. Indeed, polymers such as regenerated cellulose, polycaprolactam, and polyethylene terephthalate, which have become familiar household materials under the trade names rayon, nylon, and Dacron (trademark), respectively.

MAN MADE MATERIALS

Synthetic fibres and plastics consists of polymers. Normally the things you observe around you in your routine life are made of polymers, like shirts, jeans and clothing are made up of synthetic fabrics and kids playing toys, cable wires etc are made up of plastics.

A **polymer** is a large molecule of high molecular mass formed by the repetitive bonding of many small molecules called monomers. The process by which the monomers are transformed into polymers is called **polymerisation**. As polymers are single big size molecules, they are also called **macromolecules.**

Classification of polymers on the basis of source:

(*a*) **Natural polymers:** Proteins, cellulose, starch, resins and rubber.

(*b*) **Semi–synthetic polymers:** Cellulose derivatives as cellulose acetate (rayon) and cellulose nitrate, etc.

(*c*) **Synthetic polymers:** Plastic (polythene), synthetic fibres (nylon 6, 6) and synthetic rubbers (Buna – S).

As you have already studied about natural fibres (cotton, silk, wool, etc.) and their processing in earlier classes. Now we will discuss about synthetic fibres in detail.

SYNTHETIC FIBRES

Fibres which are man made and manufactured artificially from chemicals are called synthetic fibres.

A few commonly used synthetic fibres are rayon, nylon and polyester (terylene).

Characteristics of Synthetic Fibres

Clothes made from synthetic fibres are stronger and do not wrinkle easily. They also dry easily. But they have less air spaces in them than natural fibres and do not 'breathe' so well. They also cannot absorb sweat. These properties make them unsuitable for hot and humid weather.

Advantages of Synthetic Fibres:

(i) They are tough and durable.

(ii) They are water-resistant and easy to dry.

(iii) They do not bleed colour when washed.

(iv) Unlike natural fibres, they do not shrink on washing.

(v) They do not wrinkle as they retain crease longer.

(vi) They trap the heat in our body.

(vii) They make us sweatend.

(viii) They are not very comfortable.

Disadvantages of Synthetic Fibres

A big disadvantage of synthetic fibres is that they melt on heating. If the cloths catch fire, it can be disastrous. They contain chemicals which may harm our skin. Another disadvantage is that the clothes made of synthetic fibres are not suitable to wear during hot summer weather.

RAYON

Rayon is first man-made fibre which was developed in France in 1884 to offer a cheaper alternative to silk textiles. Rayon resembles silk in appearance. Hence it is also known as artificial silk. It is produced from natural cellulose by its chemical treatment. The cellulose required for making rayon is obtained from wood pulp.

Wood pulp is first dissolved in an alkaline solution. The thick liquid produced is then passed through tiny holes of a metal cylinder which is known as spinneret to make fibres. The fibres are hardened by passing them into a bath of sulphuric acid. The fibres are then spun into yarn and woven into cloth.

Properties of Rayon

(i) Low warmth, good to drape, not durable, creases easily and needs ironing.

(ii) Rayon fibre has the same comfort property as natural fibres.

(iii) Rayon can easily be dyed in variety of colours.

(iv) Usual rayon fibres **recommended care** for dry cleaning purpose only.

(v) Rayon fabrics are soft, smooth, cool, comfortable, and highly absorbent, but they do not insulate body heat, making them ideal for use in hot and humid climates.

Uses of Rayon

(i) It is the most commonly used semi-synthetic fibre and is used in:
Textile industry for making textiles (fabrics)
- Manufacture of tyres
- Medical field for making *bandages* and *surgical dressing*.

(ii) Rayon's thin fibres have ability to allow clothing to breathe more than other fabrics, which makes it a good choice for the design of sports wear and sundresses that are expected to provide coolness and comfort.

(iii) Rayon when mixed with wool is used to make carpets.

Do You Know!!

Rayon was originally called artificial silk but now the name rayon is given to all fibres obtained by chemical treatment of cellulose. Thus, artificial silk is a polysaccharide i.e., cellulose derivative.

NYLON

It is the first man made completely synthetic fibre. First introduced by Wallace Carotters on 28th February 1935. The chemical compounds used in making nylon are now obtained from petroleum products called petrochemicals. Actually, nylon is made up of repeating units of a chemical called an amide. So, nylon is a polyamide. Nylon is a thermoplastic polymer. It was made simultaneously in New York (NY) and London (LON) thus it has derived the name NYLON.

Properties of Nylon

(i) It possesses good strength. Nylon is a very strong but light weight material. Its strength does not lost with age.
(ii) It possesses good elasticity. It stretches but also regains its original shape easily.
(iii) It is moth and alkali resistant
(iv) It is inert to many chemicals and solvents
(v) It is wrinkle resistant
(vi) Nylon fibrics absorb very little water. So nylon clothes are very easy to wash and dry.
(vii) Nylon fabrics has low resistance to sunlight.

Uses of Nylon

It is used for making
(i) fishing nets, car seat belts, strings for sports rackets etc
(ii) ropes used for mountaineering parachute fibres, tyre cords etc.
(iii) machine parts in form of plastics.
(iv) material in dress socks, swim wear, shorts, track pants, active wear, wind breakers, draperies and bedspreads.
(v) carpets, textile fibres, bristles for brushes, track suits, combat uniforms, swim wears and women stockings.
(vi) a substitute for metals in ball bearings and gears.

Do You Know!!

High temperatures are able to melt nylon. Because of this fact, when ironing items made of nylon you should set your iron at a low setting and turn the item over onto the wrong side.

CHECK Point

Nylon is natural or synthetic fibre ?

SOLUTION

Nylon is a polyamide which are natural but nylon is a man made fibre.

POLYESTERS

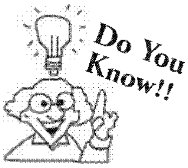
It is a general name for synthetic fibres that contain many ester groups. Alcohols and organic acids react together to make compounds called esters which polymerises to form polyesters.
Terylene or Dacron is the best example. The chemical compounds used in making polyester fibres are made from petroleum products called petrochemicals. Like nylon polyster is also a thermoplastic polymer.
Polyster blended with cotton and wool are called polycot and polywool respectively. Polycot contain 55% polyesters and 45% wool.

Properties of Polyester

(i) Polyester fabrics and fibres are extremely strong.
(ii) Polyester is very durable; resistant to most chemicals, wrinkle resistant, mildew and abrasion resistant.
(iii) Polyester is hydrophobic in nature and quick drying. It can be used for insulation by manufacturing hollow fibres.
(iv) Polyester retains its shape and hence is good for making outdoor **clothing** for harsh climates.

Uses of Polyester

(i) It is used for making shirts, trousers, sarees, and other dress materials.
(ii) It is used for making sails for sailing boats.
(iii) It is used for making fire-hoses and conveyer belts.
(iv) Due to its strength and tenacity polyester was also used to make ropes in industries
(v) Now a days PET bottles are one of the most popular uses of polyester
(vi) Polyester is also used for making nets, thread, raincoats, jackets, clothing and medical textiles.



Among synthetic fibres the polyesters, are most widely used. They account for about one-half of all synthetic fibres.

ACRYLIC

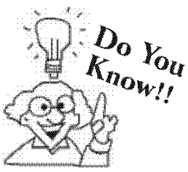
Acrylic is a synthetic fibre. This is made from a chemical called 'acrylonitrile' through Polymerisation. **Acrylic** is a commonly used clothing material. It is light weight, soft and warm, has a wool-like feel, can be easily dyed, does not shrink or wrinkle and also insect resistant.
It is easy to maintain and is often used as an alternative to expensive woolen fabrics like cashmere. It tends to form fuzz and is not as pure wool.

Properties of Acrylic

(i) It is more flexible and shock resistant than glass.
(ii) It is abrasion resistant and resistant to UV and chemical damage.
(iii) Acrylic can transmit or filter ultraviolet light and is easily cleaned.
(iv) It is easily cut, corrosion resistant and a good insulator.

Uses of Acrylic

The application of acrylic include; windows, aircraft canopies, automobile tail lights, hobby crafts, sunscreens, lighting fixtures, **furniture**; table tops, sign boards, decorating panels, windshields, camera lenses, aquariums, toys, incubators, appliances and security shields.



Acryline is extremely strong. If it breaks or gets damaged, it will not shatter.

PLASTICS

A **plastic** is a synthetic material, which can be moulded or set into any desired shape when soft, and then harden to produce articles. Some plastic have linear and some have cross links. They are usually synthetic, most commonly derived from petrochemicals, but many are partially natural.

The first man-made plastic was created by Alexander Parkes who publicly demonstrated it at the 1862 in Great International Exhibition in London.

Plastic is an organic material of high molecular weight with specific properties. Most of the plastics are polymers having long chains.

Classification of Plastics

Plastics can be broadly classified as :

(i) Thermoplastic and (ii) Thermosetting Plastics.

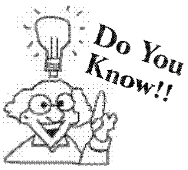
(i) Thermoplastic material : A plastic material that can be melted repeatedly by heating and can be moulded again and again into different shapes is called **thermoplastic material .**

A **thermoplastic material** becomes soft on heating and hard on cooling. These plastics can be moulded into toys, beads, buckets, telephone and television cases. Examples of thermoplastics are polythene, polyvinyl chloride (PVC) and polystyrene. Shower curtains, and pipes are most often made of PVC.

(ii) Thermosetting Plastics : A plastic substance, which once set, does not soften on heating is known as thermosetting plastic. A **thermosetting plastic** can be used only once. They are used in a situation where resistance to heat is important. Examples of thermosetting plastics are bakelite, melamine and urea formaldehyde resin. They are used in a situation where resistance to heat is important.

Differences between Thermoplastics and Thermosetting Plastics

	Property	Thermoplastics	Thermosetting Plastics
1.	Effect of heat	Soften at 70°C to 90°C	Do not soften easily
2.	Strength	Comparatively weaker	Comparatively stronger
3.	Reusability	Can be reused after melting	Can not be remelted or reused.
4.	Flexibility	They are flexible	They are hard and rigid.



Plasticizers are non volatile liquids which are added to a polymer such as PVC to make it soft and readily workable on heating.

Knowledge ENHANCER

As polymeric chains in the thermosetting plastic form cross links, this in turn prevents the plastic from melting once heated after initial forming and cooling. If you again heat it a lot it would burn, not melt. So using it in a kettle is good because it won't rust, it is electrically safe and it won't go soft and floppy when hot .Whereas thermoplastics possesses intermolecular forces this property allows thermoplastics to be remolded because the intermolecular interactions spontaneously reform upon cooling.

Applications of Some Routine Thermoplastic and Thermosetting Materials

	Name of plastic	Type of plastic	Uses of plastic
1.	(a) LDPE (Low density polyethene)	Thermoplastic	It is used in insulation of electricity carrying wires and manufacture of squeeze bottles, toys, flexible pipes (it is chemically inert, tough, flexible and poor conductor of electricity)
	(b) HDPE (High density polyethene)	Thermoplastic	It is used for making buckets, dustbins, bottles, pipes, etc. (it is comparatively more chemically inert, tougher and harder)
2.	Polytetrafluoro ethene (Teflon)	Thermoplastic	It is used in making oil seals and gaskets and also used for non-stick surface coated utensils as it is chemically inert and resistant to attack by corrosive reagents.
3.	Polyacrylonitrile (PAN)	Thermoplastic	It is used as a substitute for wool in making commercial fibres as orlon or acrilan.
4.	Polyproprene	Thermoplastic	It is used in manufacture of ropes, toys, pipes, fibres, etc.
5.	Polystyrene	Thermoplastic	It is used as an insulator, wrapping material in manufacture of toys, radio and television cabinets.
6.	Polyvinyl chloride (PVC)	Thermoplastic	It is used in manufacture of rain coats, hand bags, vinyl flooring, water pipes, etc.
7.	Bakelite	Thermosetting	It is used in making combs, phonograph records, electrical switches and handles of various utensils, etc.
8.	Melamine	Thermosetting	It is used in manufacture of unbreakable crockery

idea box

Visit your nearby market with your friends. Go to various shops which have plastic or plastic containing objects. Make a list of those objects and observe them carefully. Now discuss with your friends and make a list of objects which can be remoulded or reused.

CHECK Point

Why thermosetting plastics can not be remelted or reused ?

SOLUTION

It is because once it is melted it forms strong cross linked bonds on cooling which posses great strength.

PLASTICS – A NEW AGE MATERIAL

Plastic materials display properties that are unique when compared to other materials and have contributed greatly to quality of our everyday life. Thus plastic can easily replace other important products including wood, metals, glass etc. Consumers like it because of its versatility, durability, light weight and other qualities.

Demerits associated with other materials makes plastic such important material .For example, if plastic bottles would have to be replaced by glass. Glass is much heavier and can break. Broken glass can be dangerous when stepped on or touched. Plastics have also replaced wood items. We use plastics because they are more affordable and can be long lasting when compared with wood.

General Properties of Synthetic Plastics :

(i) Most plastics are waterproof and are unaffected by weather. They are non-biodegradable.

(ii) They are not affected by water, strong acids and alkalies. Some are however soluble in certain organic solvents.

(iii) Plastic is tough and can withstand high pressure. It is stretchable and its tensile strength is many times than natural fibre. Even a fine synthetic fibre is not easily breakable.

(iv) Plastic is easily moulded to any shape or even passed through a fine pore forming a fibre.

(v) Plastics may be transparent, translucent or opaque.

(vi) Plastics are non-conductors of heat and electricity and hence used as insulators.

(vii) Most plastics catch fire easily. They are inflammable.

Uses of Plastics:

(i) For making water bottles and waterproof containers.

(ii) For making plastic wrap materials, which is used to keep food fresh in refrigerators.

(iii) As packing materials.

(iv) For making table cloths, bathroom curtains, raincoats, waterproof shoes and fashionable footwear.

(v) Glass-reinforced plastic (GRP) or Glass fibre-reinforced plastic (GFRP) is a fibre reinforced polymer made of a plastic matrix reinforced by fine fibres made of glass. Simply known as fibre glass.

Disadvantages of Plastic

(i) A large part of our household wastes are made of plastics. We throw away plastic bags, in which we buy foodstuffs, vegetables, clothes and other materials. The plastic bags used for packing is not biodegradable and takes thousands of years for a plastic bag to disintegrate.

(ii) When burnt, plastics give off poisonous fumes. PVC on burning gives fumes of hydrogen chloride.

(iii) Recycling plastic is expensive.

(iv) They are non-renewable resources.

(v) The poisonous gaseous product produced by the decomposition of plastic can cause cancer.

(vi) They harm the environment by choking the drains.

PLASTICS AND THE ENVIRONMENT

The extensive and wide use of plastic has now become a big threat for our environment. The next time you go for shopping and carry home the things in a cute, comfy plastic carry-bag, think you are contributing your share to a deadly pollution whose ill-effects are irreversible and capable of reaching out to numerous generations to come. Plastic is one of the major toxic pollutants. Being a non-biodegradable substance, composed of toxic chemicals, plastic pollutes earth, air and water.

Plastic waste is a major environmental issue and cause of public health problems. Plastic shopping or carrier bags are one of the main sources of plastic waste in our country. Besides this visual pollution, plastic bag wastes contribute to blockage of drains and gutters, plastic bags are a threat to aquatic life when they find their way to water bodies and are harmful for aquatic animals in many ways. Furthermore, when filled with rainwater, plastic bags become breeding grounds for mosquitoes, which cause malaria. Plastic bags along with food waste can be eaten by animals and as they don't break down this can cause digestive problems in them.

When plastics are burnt, they emit toxic gases and smoke full of carbon particles. Plastics that get burried under the soil prevent the percolation of rainwater and the run away water into the soil.

Indiscriminate dumping of plastic waste on land makes the land infertile.

Like all solid waste, the primary strategy for effectively managing plastics are reduce, reuse, recycle and recover.

Reduction of plastic waste can be done by using

(i) products with less plastic packaging jute

(ii) carrybags made up of cloth, jute or paper.

Recycling of plastic waste should be emphasized. House hold plastic waste should be collected and forwarded for recycling. One should support and promote plastic recycling schemes in their local area.

Reuse the plastic materials. Used plastic bags and bottles should not be thrown away they should be reused if possible.

The fourth R is to recover material which cannot be reused or recycled should be recovered.

CONNECTING TOPIC

In addition to fibre and plastic there are many other important man-made materials and substances which we use in our routine life. Medicines for health care, kitchen crockery made up of glass. Rubber used in tyres and detergents for washing and cleaning purposes. These materials are disscussed one by one in brief.

SOAPS AND DETERGENTS

Soap : Fatty acid salts of sodium and potassium are known as soaps. These are prepared by the action of fatty acids with sodium hydroxide or potassium hydroxide.

$$\text{Fatty acid + sodium hydroxide} \rightarrow \text{Soap + glycerol.}$$

Detergents are sodium salt of long chain sulphonic acids or alkyl hydrogen sulphate.

Advantages of detergents over soaps

(i) Detergents can be used for laundering even with hard water as they are soluble even in hard water.

(ii) Detergents possess better cleansing properties than soaps.

Disadvantages of detergents over soap.

Detergents are prepared from hydrocarbons, while soaps are prepared from edible fatty oils. Thus they are non biodegradable.

Saponification

It is the process of making of soap by the hydrolysis of fats and oils with alkalies. Both soaps and detergents are soluble in water and act as **surfactants** which reduce the surface tension of water to a great extent. This increases the water - fabric interaction as a consequence of which dirt particles, grease spots etc are washed away effectively. In other words soaps and detergents enhance the cleansing action of water.

RUBBERS

Natural rubber : It is naturally occurring polymer of high molecular weight obtained from rubber trees. It possesses more than 300 percent elasticity. Natural rubber contains deniable bonds. It is a polymer of monomer called isoprene.

Natural rubber is a poly cis-isoprene. It is prepared from latex (obtained from rubber tree by coagulation with acetic acid). It is a soft and tacky material. Gutta percha on the other hand is poly-isoprene. It is a hard and horny material.

Synthetic rubber : It is identical to natural rubber but possesses better properties. The two important forms of artificial rubber are. (i) Thiokol and (ii) Neoprene

Thiokol is produced by chemical reaction of dichloroethane with sodium polysulphide.

Neoprene is produced by polymerisation of 2-Chloro-butadiene.

Vulcanization of rubber : The process of heating natural rubber with sulphur is known as vulcanization of rubber. The cross linking of rubber molecules, improves strength and elasticity of rubber considerably.

Natural rubber is soft, sticky and possesses low strength where as vulcanized rubber is comparatively hard, non-sticky and shows better strength. It possesses good elasticity and is abrasion resistant.

The tensile strength of natural rubber is low but of vulcanized rubber is high.

Natural rubber can be used in the temperature range of 10-60°C whereas vulcanised rubber can be used in the range of 45°C to 110°C.

The elasticity of vulcanised rubber is high whereas that of natural rubber is low.

Natural rubber is soluble in organic solvents like ether, petrol and carbon disulphide whereas vulcanised rubber is practically insoluble in organic solvents.

GLASS

It consists of a mixture of two or more silicates.

Preparation of Glass :

Common glass (or soft glass) : It is used to make bottles, glass wares etc. and is obtained by heating together silica (in the form of sand), sodium carbonate or sodium sulphate and chalk or lime stone (calcium carbonate). Some broken glass and a little coke are usually added. The glass so prepared consists of silicates of sodium and calcium.

$$Na_2CO_3(s) + \underset{\text{(silica)}}{SiO_2(s)} \rightarrow Na_2SiO_3\,(s) + CO_2\,(g)$$

$$Na_2SO_4(s) + SiO_2(s) \rightarrow Na_2SiO_3\,(s) + SO_3\,(g)$$

$$CaCO_3(s) + SiO_2(s) \rightarrow CaSiO_3\,(s) + CO_2\,(g)$$

Hard glass : For preparation of hard glass K_2CO_3 is used in place of Na_2CO_3. It consists of a mixture of calcium and potassium silicates.

Physical Properties of Glass

Hard, rigid, high viscosity, bad conductor of heat and electricity, Brittle etc.

Chemical Properties of Glass

(i) It is resistant to action of air and acids except hydroflouric acid.

(ii) It is alkaline in nature.

(iii) It slowly reacts with water to form alkaline solution.

Types of Glass :

(i) Silica glass : For this type of glass the raw material used is 100% pure form of *quartz*. It is quite *expensive*. It is used in the manufacture of laboratory apparatus. It has low thermal expansion. Its softening point is very high and it is resistant to a wide variety of chemicals.

(ii) Alkali silicate glass : For it the raw materials used are sand and soda. It is also called **water glass** because it is soluble in water and used only as a solution. It is generally used to make gums and adhesives.

(iii) Lead glass : For this type of glass lead oxide is added to ordinary glass. The addition of lead oxide increases the density and also the refractive index. This type of glass is used for the manufacture of ornamental glass ware, decorative articles etc.

(iv) Optical glass : This type of glass is used in the manufacture of optical instruments like binoculars, spectacles, lenses, prisms, telescopes, microscopes etc. It is transparent and can be grounded into the required shape. It generally contains phosphorus, and lead silicates with little cerium oxide which absorbs UV radiations.

(v) Processed glass : The properties and applications of glass also depend upon the processing of glass.
Some types of processed glass and their applications are given here :

Processed glass	Applications
(a) Laminated glass	Used for doors and windows of automobiles. (It has high strength)
(b) Fibre glass	Used for reinforcing purpose (It has enough tensile strength)
(c) Foam glass	Used for civil construction and insulation purposes (it is light weight).
(d) Opaque glass	In it non transparent glass filters the light entering into it. Thus provides an aesthetic look

(vi) Borosilicate glass : It contains silica and boron oxide and small amount of oxides of sodium and aluminium. It is resistant to a wide variety of chemicals due to this property it is used in the manufacture of laboratory ware.

Etching of Glass :

HF dissolves glass due to formation of soluble fluorosilicates. The glass surface is coated with wax and the places where design is to be made, are exposed to aqueous HF by removing the wax surface. Aqueous HF is poured on the exposed part. After some time it is poured out and wax is removed from the surface. The exposed part of glass is engraved.

MEDICINES

Drug

Chemical substances used for curing, diagnosing and treatment of diseases are called drugs. The branch of chemistry which deals with the treatment of diseases using suitable chemicals is known as chemotherapy.

Requirements of a Substances to act like a Drug

(i) Its target should be localised at the site of action.

(ii) It should not make host tissue resistant against the effect of drug after use for some time.

(iii) It should not produce any side effects of long term use.

(iv) It should be effective within therapeutic range.

Types of Drugs

(i) Antipyretics :
Those drugs which are used to bring down body temperature during fever. e.g., Paracetamol, Aspirin etc.

(ii) Antacids :
Those substances which neutralize the excess acid and raise the pH to an appropriate level in stomach are called antacids. e.g., magnesium hydroxide, aluminium hydroxide gel etc.

(iii) Antihistamines :
The drugs which interfere with the natural action of histamine. These are widely used for treatment of hay fever, conjuctivitis, snoozing, itching of eyes, nose and throat etc. e.g., diphenylhydramine, promethazine etc.

(iv) Tranquillizers :

Drugs which are used for the treatment of stress, fatigue, mild and severe mental diseases are called tranquillizers. e.g., equanil, valium, serotonin, barbituric acid, luminal, seconal etc.

(v) Analgesics :

Drugs which reduce pain without reducing consciousness, mental confusion, paralysis or some other disturbance of the nervous system are called analgesics. e.g., aspirin, paracetamol, morphine diacetate etc.

(vi) Antimicrobials :

Drugs used to cure diseases caused by microorganisms such as bacteria, viruses, fungi etc., are called antimicrobials. e.g., sulphanilamide

(vii) Antibiotics :

These are chemical substances which are produced by some microorganism and can be used to either inhibit the growth or even kill other microorganism which cause infections. e.g., penicillin, chloramphenicol, erythromycin etc.

(viii) Antiseptics :

Those chemical substances which prevent the growth of micro-organism or may even kill them. They are safe to be applied on living tissues. e.g., furacin, soframycin, boric acid, iodine, bithional etc.

(ix) Disinfectants :

Those chemical substances which kill micro-organisms but are not safe to be applied on the living tissues. These are generally used to kill the micro-organism present in the drains, toilets, floors etc. e.g., 0.2% solution of phenol acts as an antiseptic whereas 1% solution acts as a disinfectant.

(x) Antifertility drugs :

Chemical substances used to check pregnancy in women are known as anti-fertility drugs or oral contraceptives. e.g., Norethindrone, Mifepristone etc.

SUMMARY

- There are two types of fibre - One which are obtained from natural sources and other which are man-made.
- Fibres which are obtained from natural sources are called natural fibres. E.g. cotton, silk, wool, etc.
- Fibres which are manufactured artificially are called synthetic fibres e.g. rayon, nylon, polyester etc.

  Rayon : It is a semi-synthetic fibre. It is prepared from natural cellulose. It is also known as artificial silk.
- Types of rayon

  (i) Viscose rayon (ii) Cuprammonium rayon (iii) Acetate rayon
- **Nylon :** It is a polyamide. It is manufactured by using adipic acid and hexamethylenediamine.
  Since both compounds used contain six carbon atoms in the molecule so it is referred as **Nylon 6,6.**
  It is a condensation polymer.
- Nylon is used in making of different types of clothes, ropes, socks, curtains, sleeping bags, parachutes, etc. The fibre of nylon is stronger than a steel wire of same thickness.
- **Polyester :** It is the general name for synthetic fibre that contains many ester groups. $\left[\begin{array}{c} \mathrm{O} \\ \| \\ -\mathrm{C}-\mathrm{O}- \end{array}\right]$
- Terylene is one of the most famous types of polyester.
- Fabrics made from polyester fibre are almost wrinkle-free, easy to wash and have shiny appearance.
- PET (Polyethylene terepthalate) is very famous term for polyester.
- Acrylic:Acrylic is man-made fibre. Since, acrylic resembles wool so it is also known as artificial wool or synthetic wool.
- Acrylic is cheaper than natural wool and can be dyed in various colour.
- Acrylic is used in making sweaters, blanket, and other many clothes.

- **Plastics :** A **plastic** is a synthetic material, which can be moulded or set into any desired shape when soft and then hardened to produce articles.
- **Classification of plastics**
  (i) Thermoplastic materials
  (ii) Thermosetting materials
- Plastics which can be easily bent or deform on heating are known as thermoplastic. PVC and Polythene are the examples of thermoplastics.
- Plastics which do not get deformed or softened on heating when mold once, are called thermosetting plastics. Bakelite and melamine are the examples of thermosetting plastics.
- Substances which get decomposed through the natural processes, such as action of bacteria, etc. are called biodegradable substances. For example; potato peels, peels of other vegetable, food stuffs, fruit, paper, cotton cloths, wood, etc.
- Substances which either do not decompose or take many years to get decomposed through the natural process, are called non-biodegradable substance, e.g. tin, aluminium, plastics, etc.
- Plastic is a non-biodegradable substance. If it is left or thrown, it takes many years to get decomposed or either does not get decomposed.
- Burning the plastic can result in release of many harmful gases in the atmosphere. This can lead to air pollution.
- For dealing with the menace of plastic waste, we need to follow the three Rs, i.e. Reduce, Reuse and Recycle.

1 EXERCISE

Fill in the Blanks :

DIRECTIONS : *Complete the following statements with an appropriate word / term to be filled in the blank space(s).*

1. .... is called artificial silk.

2. A thermosetting plastic can be used ....

3. Base raw material for rayon is ....

4. Tensile strength of nylon is .... than that of cotton.

5. Melamine is a thermo .... plastic.

6. .... is the synthetic fibre used as wool.

7. The synthetic fibres are also known as .... fibres

8. The first fully synthetic fibre was ....

9. A plastic used for making crockery is ....

True / False :

DIRECTIONS : *Read the following statements and write your answer as true or false.*

1. Rayon is a semi-synthetic fibre

2. Nylon is corrosion resistant

3. PVC is a thermo-setting plastic.

4. Bakelite is thermosetting plastic.

5. Polyester is a thermoplastic.

6. Common name of PETE is terylene

7. Polystyrene foam is used as an insulating and packaging material

8. Plastics do not corrode or rust.

9. Most plastics are biodegradable

10. Plastic melamine is fire resistant.

11. Bakelite is a soft and flexible plastic.

12. Polysters are purely synthetic fibres.

13. Rayons are derived from cellulose.

14. Acrylics are crimpy and are a substitute for natural wool.

15. Synthetic fibres are generally hydrophilic.

16. We can mould plastics into various shapes because of the property of elasticity in them.

Match the Following :

DIRECTIONS : *Following question contains statements given in two columns which have to be matched. Statements (A, B, C and D) in column I have to be matched with statements (p, q, r and s) in column II.*

1. Match Column-I with Column-II and select the correct answer using the code given below in the columns.

Column-I	Column-II
(A) Rayon	(p) Polymer of diacid and diamine
(B) Nylon	(q) Terylene
(C) Polyester	(r) Regenerated cellulose
(D) Polystyrene	(s) Light foam used for insulating and packing.

(a) A – (r); B – (p); C – (q); D – (s)
(b) A – (s); B – (p); C – (q); D – (r)
(c) A – (r); B – (q); C – (p); D – (s)
(d) A – (r); B – (s); C – (q); D – (p)

2. Match Column-I with Column-II and select the correct answer using the code given below the columns.

Column-I	Column-II
(A) Thermoplastics	(p) Endoscopes
(B) Glass fibre	(q) Bakelite
(C) Thermosetting plastics	(r) polypropylene
(D) Optical fibre	(s) Used in making space suits

(a) A – (p); B – (s); C – (q); D – (r)
(b) A – (r); B – (s); C – (q); D – (p)
(c) A – (s); B – (r); C – (q); D – (p)
(d) A – (r); B – (s); C – (p); D – (q)

3. Match Column-I with Column-II and select the correct answer using the code given below the columns.

Column-I	Column-II
(A) Natural fibre	(p) Terylene
(B) Synthetic fibre	(q) Rain coats
(C) Semisynthetic fibre	(r) Wool
(D) Rubber	(s) Terry wool

(a) A – (r); B – (p); C – (s); D – (q)
(b) A – (q); B – (p); C – (s); D – (r)
(c) A – (r); B – (p); C – (q); D – (s)
(d) A – (s); B – (p); C – (r); D – (q)

4. Which of the following represents the correct match for items in Column I with those in Column II

	Column -I		**Column-II**
(A)	Nylon	(p)	Thermoplastic
(B)	PVC	(q)	Thermosetting plastic
(C)	Bakelite	(r)	Fibre

(a) A – (q), B – (r), C – (p)
(b) A – (q), B – (p), C – (r)
(c) A – (r), B – (p), C – (q)
(d) A – (r), B – (q), C – (p)

Very Short Answer Questions:

DIRECTIONS : *Give answer in one word or one sentence.*

1. What is a monomer?
2. What is a polymer?
3. Define polymerisation.
4. Define tensile strength of a material.
5. Name any one synthetic polyamide fibre.
6. Give two examples of thermosetting plastics.
7. Give an important use of nylon.
8. Name any two commonly used plastics.
9. What is the common name of rayon ?
10. Name a man-made fibre which was made without using natural raw materials.
11. Name two articles made by rayon.
12. How many types of arrangement of units in plastics ? Name them.
13. Why plastics are not ecofriendly ?
14. What are 4Rs?

Short Answer Questions :

DIRECTIONS : *Give answer in 2-3 sentences.*

1. What is plastic? Name different types of plastics.
2. Give the uses of thermosetting plastics.
3. Why have synthetic fibres became so popular?
4. How is rayon prepared?
5. What is rayon ? Why is it called an artificial silk?
6. Why is Polyster quite suitable for making dress materials ?
7. Why should we not wear synthetic clothes while working in kitchen ?
8. Why is it convenient to store plastic containers than metals?
9. Write the uses of plastics in health care industry.
10. Say no for polythene bags and say yes for paper bags. Comment on this slogan.

Long Answer Questions :

DIRECTIONS : *Answer the following questions in detail.*

1. Explain thermoplastics and thermosetting plastics with examples.
2. Write down the name of any 5 fibres/plastics and also mention one use of each.

S.No.	Fibres/Plastics	Uses
1		
2		
3		
4		
5		

2 EXERCISE

Text-Book Exercise :

1. Explain why some fibres are called synthetic.
2. Mark (✓) the correct answer.
   Rayon is different from synthetic fibres because
   (a) it has a silk like appearance
   (b) it is obtained from wood pulp
   (c) its fibres can also be woven like those of natural fibres
3. Fill in the blanks with appropriate words:
   (a) Synthetic fibres are also called ______ or _______ fibres.
   (b) Synthetic fibres are synthesised from raw material called ________.
   (c) Like synthetic fibres, plastic is also a __________ .
4. Give examples which indicate that nylon fibres are very strong.
5. Explain why plastic containers are favoured for storing food.
6. Explain the difference between the thermoplastic and thermosetting plastics.
7. Explain why the following are made of thermosetting plastics.
   (a) Saucepan handles
   (b) Electric plugs/switches/plug boards
8. Categorise the materials of the following products into 'can be recycled' and 'cannot be recycled'.
   Telephone instruments, plastic toys, cooker handles, carry bags, ball point pens, plastic bowls, plastic covering on electrical wires, plastic chairs, electrical switches.
9. Rana wants to buy shirts for summer. Should he buy cotton shirts or shirts made from synthetic material? Advise Rana, giving your reason.
10. Give examples to show that plastics are noncorrosive in nature.
11. Should the handle and bristles of a tooth brush be made of the same material? Explain your answer.
12. 'Avoid plastics as far as possible'. Comment on this advice.
13. Match the terms of column I correctly with the phrases given in column II.

	Column-I		Column-II
(i)	Polyester	(a)	Prepared by using wood pulp
(ii)	Teflon	(b)	Used for making parachutes and stockings
(iii)	Rayon	(c)	Used to make non-stick cookwares
(iv)	Nylon	(d)	Fabrics do not wrinkle easily

14. 'Manufacturing synthetic fibres is actually helping for conservation of forests'.
    Comment.
15. Describe an activity to show that thermoplastic is a poor conductor of electricity.

Exemplar Questions :

1. Plastic is used for making a large variety of articles of daily use and these articles are very attractive. But it is advised to avoid the use of plastic as far as possible. Why ?
2. A lady went to the market in buy a blanket. The shopkeeper showed her blankets made of acrylic fibres as well as made of wool. She preferred to buy an acrylic blanket. Can you guess why ?
3. Terrycot is made by mixing two types of fibres. Write the names of the fibres.
4. Rohit took with him some nylon ropes, when he was going for rock climbing. Can you tell why he selected nylon ropes instead of ropes made of cotton or jute ?
5. Despite being very useful it is advised to restrict the use of plastic. Why is it so ? Can you suggest some methods to limit its consumption ?

HOTS Questions :

1. Why is rayon called a regenerated fibre?
2. Why is nylon called a polyamide?
3. What is the application of nylon in medical field ?
4. Give the names of two articles made by nylon.
5. Why is the name nylon given to the first fully synthetic fibre ?
6. Why is thermosettings plastic stronger than thermoplastic?

3 EXERCISE

Multiple Choice Questions :

DIRECTIONS : *This section contains 23 multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.*

1. Which of the following statements is true for thermoplastic and thermosetting plastic?
(a) Thermosettings are permanent setting resins but thermoplastics are not, since they can be softened on cooling.
(b) Thermoplastics are less brittle as compared to thermosettings because of the absence of long chain polymers.
(c) Thermosettings are more brittle as compared to thermoplastics because of the presence of strong bonds.
(d) The chemical nature of thermoplastics can be changed by repeated heating and cooling.

2. Select the one that is a natural polymer.
(a) Nylon 6,6 (b) Polyester
(c) Cellulose (d) none of these

3. Select the synthetic polymer whose filament was first obtained that could be streched several times its original length.
(a) Nylon (b) polyester
(c) Acrylic (d) none of these.

4. The hard plastic covers of telephones are made of polymers of
(a) styrene (b) acrylonitrile
(c) ethene (d) none of these

5. Plastics are
(a) salts (b) acids
(c) polymers (d) solvents

6. For which of the following is polythene not used?
(a) construction materials of plastics
(b) Plastic pipes
(c) Bread - wrappers
(d) candles

7. Which of the following are synthetic fibres:
(a) Orlon, polypropylene and nylon
(b) Jute, ramie and hemp
(c) Wool, silk and cotton
(d) All of these

8. A synthetic fibre obtained from renewable resources is
(a) Nylon. (b) Polyester.
(c) Rayon. (d) Acrylic.

9. The major raw materials for rayon is obtained from
(a) wood. (b) coal.
(c) petroleum. (d) both coal and petroleum.

10. Which of the following are advantages of synthetic fibres:
(a) They are tough and durable.
(b) They are water resistant and easy to dry.
(c) They do not wrinkle.
(d) All of these.

11. Terrywool is obtained by mixing
(a) polyester and wool.
(b) cotton and wool.
(c) polyester and cotton.
(d) cotton and silk.

12. Plastics
(a) are light.
(b) can be spun into fibres to make cloth and carpets.
(c) can be coloured and moulded into any desired shape.
(d) (a), (b) and (c).

13. Synthetic fibres are obtained from
(a) plants (b) animals
(c) petroleum (d) All of these

14. Which of these fabrics will you prefer on a hot and humid day?
(a) Nylon (b) Silk
(c) Cotton (d) Wool

15. Natural fibres are
(a) Nylon and Rayon (b) Nylon only
(c) Rayon silk (d) None of these

16. Synthetic fibres are
(a) Rayon and Cotton
(b) Nylon, Polyester and Jute
(c) Rayon, Nylon, and Polyester
(d) None of these

17. Correct full form of abbreviation PETE is
(a) Polyethylene terephthalate
(b) Polyethane terephthalate
(c) Polyethene terephthalate
(d) None of these

18. Which of the following synthetic fibre is good absorbent?
(a) Nylon (b) Rayon
(c) Polyester (d) Acrylic

19. Which of the following is thermosetting plastics?
(a) Polyvinyl chloride (b) Polyester
(c) Polypropylene (d) Bakelite

20. Polycot is obtained by mixing
(a) polyester and cotton
(b) polyester and wool
(c) cotton and wool
(d) None of these

21. The first man made fibre is
(a) Nylon (b) Polyester
(c) Rayon (d) Cotton

22. The plastics which do not remould again on heating are called
(a) Thermosetting plastics
(b) Thermoplast plastics
(c) Both of these
(d) None of these

23. Which of the following groups contain all synthetic substances ?
(a) Nylon, Terylene, Wool
(b) PVC, Polythene, Bakelite
(c) Cotton, Polycot, Rayon
(d) Acrylic, Silk, Wool

More than One Option Correct :

DIRECTIONS : *This section contains 5 Multiple Choice Questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE OR MORE may be correct.*

1. Plastics
(a) are organic material of high molecular weight
(b) are polymers having long chains
(c) are made by the process of polymerisation
(d) only (a) and (b) are correct

2. Pick out natural fibres out of the following ?
(a) Cotton (b) Jute
(c) Nylon (d) Wool

3. Which of the following are thermo plastic polymer?
(a) Melamine (b) PVC
(c) Polythene (d) Nylon

4. Which of the following is/are common properties of plastics ?
(a) Non-reactive (b) Durable
(c) Light in weight (d) Good conductor of electricity

5. Which of the following is/are synthetic fibres
(a) Angora (b) Rayon
(c) Nylon (d) Polyester

Passage Based Questions :

DIRECTIONS : *Study the given paragraph(s) and answer the following questions.*

Passage-1

Synthetic fibres are the result of extensive research by scientists to improve on naturally occuring animal and plant fibres. Before synthetic fibres were developed, artificially manufactured fibres were made from cellulose, which comes from plants. These fibres are called cellulose fibres. Synthetic fibres account for about half of all fibre usage, with applications in every field of fibre and textile technology. Although many classes of fibre based on synthetic polymers have been evaluated as potentially valuable commercial products, four of them - Nylon, polyester, acrylic and polyolefin - dominate the market. These four account for approximately 98 percent by volume of synthetic fibre production, with polyester alone accounting for around 60 percent.

1. Synthetic fibres are result of extensive research on
(a) Natural fibres
(b) Glass fibres
(c) Semi-synthetic fibres
(d) Optical fibres

2. Which of the following is a semi-synthetic fibre?
(a) Nylon (b) Polyester
(c) Cellulose acetate (d) Acrylic

3. Which of the following synthetic fibre has most wide applications?
(a) Nylon (b) Polyester
(c) Acrylic (d) Polyolefin

Passage-2

Rahul one day thought to clean his messy room while cleaning he found items like old T-shirts, cable wire, cycle tyres, bottle of shaving foam, table cloth, broken toys, pain killer strips, stop cork of glass bottle and tube of face wash.

4. Which of the following is a synthetic fibre ?
(a) table cloth (b) broken toys
(c) pain killer strips (d) stop cock of glass bottle

5. Which of of the following comes under category of plastic?
(a) pain killer strips (b) cycle tyre
(c) broken toys (d) old T-shirts

6. Which of the following is/are polyester(s) ?
(a) T-shirts (b) pain killer strip
(c) cycle tyre (d) broken toys

Assertion & Reason :

DIRECTIONS : *Each of these questions contains an assertion followed by reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.*

(a) If both **Assertion** and **Reason** are **correct** and Reason is the **correct explanation** of Assertion.

(b) If both **Assertion** and **Reason** are correct, but Reason is **not the correct explanation** of Assertion.

(c) If **Assertion** is **correct** but **Reason** is **incorrect**.

(d) If **Assertion** is **incorrect** but **Reason** is **correct**.

1. **Assertion :** Clothes made up of synthetic fibre are unsuitable for hot and humid weather.
Reason : Synthetic clothes not breathe so well

2. **Assertion :** Bakelite is used for making electric switches and plugs.
Reason : It is a thermosetting polymer and an insulator.

3. **Assertion :** Acrylic fabrics are used in making socks and shawls.
Reason : Acrylic fabrics are replacement of wool.

4. **Assertion :** Plastics are non-biodegradable.
Reason : They cannot be decomposed by microorganisms.

5. **Assertion :** Synthetic fibres are polymers that are synthesised by human beings by chemical process.
Reason : Polymers occur in nature also.

6. **Assertion :** Nylon is used for making rock climbing ropes, parachutes, car seat belts and fishing nets.
Reason : Nylon is a very strong fibre.

7. **Assertion :** Handles of utensils are made up of thermoplastics.
Reason : Thermoplastics get soften on heating.

8. **Assertion :** Plastic being light in weight and easy to handle is used as a substitute for glass and metal.
Reason : We should avoid plastics as far as possible.

Multiple Matching Questions :

DIRECTIONS : *Following question has four statements (A, B, C and D) given in Column I and four statements (p, q, r and s) in Column II. Any given statement in Column I can have correct matching with one or more statement(s) given in Column II. Match the entries in column I with entries in column II.*

1.

Column-I	Column-II
(A) Rayon	(p) Polymer
(B) Nylon	(q) Semi-synthetic fibre
(C) Artificial silk	(r) Synthetic fibre
(D) Polyester	(s) Regenerated fibre.

Integer Type Questions :

DIRECTIONS : *Following are integer based questions. Each question, when worked out will result in one integer from 0 to 9 (both inclusive).*

1. Number of synthetic fibres among the following
Nylon, Jute, Wool, Cotton.

2. Among the following number of materials that cannot be recycled.
Cooker handles, Electrical switches, Plastic chairs, Plastic toys, Plastic bowls.

4 ADVANCED EXERCISE
BASED ON CONNECTING TOPICS

DIRECTIONS (Qs. 1-15) : *This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.*

1. Which of the following statements, is not correct, about glass.
(a) Because of its high viscosity glass exists in solid state
(b) There is no definite melting point for glass.
(c) The silicate units in glass are arranged in a way that is quite similar to the arrangement found in liquids.
(d) Glass is a solid because it has a regular crystalline arrangement.

2. One of the properties of glass is its transparency. This property of glass is due to
(a) its high viscosity
(b) regular arrangement of silicate units in glass.
(c) irregular arrangement of silicate units in glass.
(d) its high coefficient of thermal expansion.

3. Washing soaps are potassium and sodium salts of
(a) Dicarboxylic acids (b) fatty acids
(c) mineral acids (d) none of these

4. Natural rubber is soft but vulcanized rubber is hard. Vulcanised rubber also resists.
(a) drops of acid rain (b) cold temperatures
(c) jerking movements (d) wear and tear due to friction

5. When glass is heated, it
(a) does not melt at a fixed temperature
(b) vapourises
(c) melts only above 1000°C
(d) None of these

6. Glass is a transparent substance obtained by heating silica with oxides or carbonates of metals. Glass is a mixture of
(a) phosphates (b) sulphates
(c) oxides (d) silicates

7. Which of the following is tranquilizer ?
(a) aspirin (b) paracetamol
(c) equanil (d) diphenylhydramine

8. Which of the following drug will behave like analgesics as well as antipyretic ?
(a) Boric acid (b) Aspirin
(c) Sulphanilamide (d) Penicillin

9. Which substance can acts like antiseptic as well as disinfectant ?
(a) Furacin
(b) Phenol
(c) Aluminium hydroxide gel
(d) Penicillin

10. Soft soaps are
(a) sodium and potassium salt
(b) sodium salt of fatty acids
(c) potassium salt of fatty acids
(d) potassium salt of sulphonic acids

11. An antibiotic with a broad spectrum
(a) Kills the antibodies
(b) Acts on a specific antigen
(c) Acts on different antigens
(d) Acts on both the antigens and antibodies

12. Washing soap can be prepared by saponification with alkali of which of the following oil
(a) Rose oil (b) Paraffin oil
(c) Groundnut oil (d) Kerosene oil

13. Which of the statements about glass are correct?
1. Glass is a super-cooled liquid having infinite viscosity.
2. Violet coloured glass is obtained by adding MnO_2.
3. Glass is a man-made silicate.
4. Glass is a crystalline substance.

Select the correct answer using the codes given below.
(a) 1, 2 and 4 (b) 2, 3 and 4
(c) 1, 2 and 3 (d) 1 and 3

14. The major component used in preparation of different types of glasses is
(a) silica (b) sodium borate
(c) calcium silicate (d) sodium silicate

15. Consider the following statements
1. Soap cannot be used in acidic water.
2 Ionic part of a soap is $—COO^{-} \cdot Na^{+}$.
3. Soap dissolves in water faster then detergent.

Which of the statements given above is/are correct?
(a) 1 and 2 (b) 2 and 3
(c) 3 only (d) 1 only

DIRECTIONS (Qs. 16-21) : *This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE OR MORE may be correct.*

16. Alkali silicate glass
(a) is also called silicate glass
(b) is also called water glass
(c) is generally used to make gums and adhesives.
(d) None of the above is correct.

17. Which statement(s) is/are correct for silica glass?
(a) For it the raw material used is 100% quartz.
(b) It is quite cheap
(c) Its softening point is very high
(d) It has a very high thermal expansion

18. In the manufacture of lead glass
(a) Oxide of lead is added to ordinary glass
(b) Oxide of lead increases the density of glass
(c) Oxide of lead increases the refractive index of glass
(d) None of the above is correct.

19. Vulcanisation of rubber
(a) involves heating of natural rubber with sulphur.
(b) improves strength and elasticity of natural rubber to a great extent
(c) increases the tensile strength of natural rubber.
(d) results in cross - linking of rubber molecules.

20. Aspirin is an example of
(a) analgesic (b) antifertility drug
(c) antipyretics drug (d) tranquilizers

21. Vulcanised rubber
(a) is natural rubber heated with sulphur
(b) can be used in the range of 45° to 110°C
(c) can be used in the range of 10-60°C
(d) natural rubber is heated with silicon.

DIRECTIONS (Qs. 22-24) : *Study the given paragraph(s) and answer the following questions.*

Passage

Like other living beings, plants require food for their growth. A wide variety of elements are required for many different purposes for the nourishment of plants. Among these carbon, oxygen and hydrogen are principal nutrients. Nitrogen, phosphorus, potassium, magnesium, sulphur, calcium and iron play significant role for the growth and development of plants. These elements are called major plant nutrients. Some other elements like manganese, molybdenum, copper, zinc, iron, chlorine are used by the field crops in very small quantities. These are called minor plant nutrients.

22. Which of the following is/are a minor plant nutrient.
(a) manganese (b) magnesium
(c) molybdenum (d) calcium

23. Select the principal nutrient(s) out of the following
(a) Hydrogen (b) Copper
(c) Nitrogen (d) None of these

24. Select the major plant nutrient(s) out of the following
(a) nitrogen (b) phosphorus
(c) potassium (d) zinc

DIRECTIONS (Qs. 25-26) : *Each of these questions contains an Assertion followed by reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.*

(a) If both **Assertion** and **Reason** are **correct** and Reason is the **correct explanation** of Assertion.
(b) If both **Assertion** and **Reason** are correct, but Reason is **not the correct explanation** of Assertion.
(c) If **Assertion** is **correct** but **Reason** is **incorrect**.
(d) If **Assertion** is **incorrect** but **Reason** is **correct**.

25. **Assertion :** Glass is not a solid.
Reason : Its molecules are randomly arranged

26. **Assertion :** For silica glass the raw material used is 100% pure form of quartz.
Reason : Silica glass is very expensive

DIRECTIONS (Q. 27) : *Each question contains statements given in two columns which have to be matched. Statements (A, B, C, D) in column I have to be matched with statements (p, q, r, s) in column II.*

27.

Column-I	**Column-II**
A. Phenol solution	(p) Antiseptic
B. Aspirin	(q) Medicine
C. Bithronal	(r) Antipyretic
D. Aluminium hydroxide gel	(s) Disinfectants
	(t) Analgesic

DIRECTIONS (Q. 28) : *Following are integer based questions. Each question, when worked out will result in one integer from 0 to 9 (both inclusive).*

28. How many of them are processed glass?
Lead glass, laminated glass, opaque glass, silica glass

SOLUTIONS

Brief Explanations of Selected Questions

1 EXERCISE

Fill in the Blanks :

1.	Rayon.	**2.**	Only once.	**3.**	cotton
4.	more	**5.**	set	**6.**	acrylic
7.	man-made	**8.**	nylon	**9.**	melamine

True / False :

1.	True	**2.**	True	**3.**	False	**4.**	True
5.	True	**6.**	True	**7.**	True	**8.**	True
9.	False	**10.**	True	**11.**	False	**12.**	True
13.	True	**14.**	True	**15.**	False	**16.**	False

Match the Following :

1. (a) A – (r); B – (p); C – (q); D – (s)
2. (b) A – (r); B – (s); C – (q); D – (p)
3. (a) A – (r); B – (p); C – (s); D – (q)
4. (c) A – (r); B – (p); C – (q)

Very Short Answer Questions :

1. A monomer is a molecule that forms the basic unit for Polymer.
2. Polymers are very large molecules that are made up of thousands of monomer unit that are bonded together in a repeating pattern.
3. A chemical process that combines several monomers to form a polymer or polymeric compound.
4. The tensile strength of a material is the maximum amount of tensile stress that it can take before failure.
5. Nylon 6,6.
6. Bakelite and melamine.
7. It is used for making fabrics in textile industry.
8. Polyethylene (PE) and Polystyrene (PS)
9. Rayon is known as artificial silk.
10. Nylon
11. Socks and tents
12. There are two types of arrangement of units in plastics Linear and cross-linked.
13. Plastics take several years to decomposes, so are not ecofriendly.
14. 4Rs stands for Reduce, Reuse, Recycle and Recovers.

Short Answer Questions :

1. Plastic is a material consisting of a wide range of synthetic or semi-synthetic organic compounds that are malleable and can be molded into solid objects.
Types of plastics : Polythylene (PE), Polystyrene(PS), Polyvinyl Chloride(PVC), PETE etc.
2. In making telephone receivers, radio, television and photographic goods, ice-cube trays etc.
3. Because synthetic fibres are lighter, cheaper, easily washable, fast dryable etc.
4. Rayon is a natural-based material that is made from the cellulose of wood pulp or cotton.
5. Rayon is synthetic fibre having properties similar to that of silk. So it is called artificial silk. It is obtained by chemical treatment of wood pulp.
6. Polyester is a synthetic fibre which is quite suitable to make dresses and other clothes due to its properties. It does not wrinkle easily. It remains crisp and is easy to wash. It is light and durable. It takes very less time to dry.
7. Synthetic fibres melt very soon on heating. If the clothes catch fire, it can be disastrous. The fabrics melt and stick to the body of person wearing it. We should therefore not wear synthetic clothes while working in the kitchen.
8. Plastic containers seem most convenient than metal containers. This is because of their light weight, lower price, good strength and easy handling. Being lighter as compared to metals, plastics are used in cars, aircrafts and spacecrafts also.
9. Plastics are used as drug delivery devices, capsules, blood bags, catheters, as lenses, and also in knee replacement therapy.
10. Plastic is a non degradable substance. It remains in environment for thousands of year where as paper is a degradable substance which degrades. Plastic has a very bad effect on our environment as it is a hazardous substance. So we must use paper bag instead of plastic bag.

Long Answer Questions :

1. There are following two type of plastics :
(i) Thermoplastics (ii) Thermosetting plastics.
(i) **Thermoplastics :** The plastics which get deformed easily by heating and can be bent easily are known as thermoplastics. Polythene and PVC are the examples of some thermoplastics.
(ii) **Thermosetting plastics :** The plastics which when moulded once cannot be softened again by heating are called thermosetting plastics. Bakelite and Melamine are two most common examples of thermosetting plastics.

2.

S.No.	Fibres/plastics	Uses
1	Terylene	Shirts and other dresses
2	Polyethylene Terephthalate (PET)	Bottles and Utensils
3	Teflon	Non stick cook wares
4	Bakelite	Electrical switches and handles of utensils
5	Melamine	Floor tiles and Kitchen wares

2 EXERCISE

Text-Book Exercise :

1. Some fibres are not obtained from plants or animals, instead of that they are made by human beings by chemical processes. That is why these kind of fibres are called synthetic fibres. These are made of small units that join together to form long chains called polymers. Polyester, rayon, nylon, acrylic etc., are some examples of synthetic fibres.

2. (b) Rayon is obtained from wood pulp.

3. (a) artificial, man-made
(b) petrochemicals
(c) polymer

4. Nylon fibres are very strong fibres that is why nylon is used for making rock climbing ropes, parachutes, car seat belts etc. A nylon thread is stronger than steel wire of same thickness or diameter.

5. Plastic containers are favoured for storing food because of their :
(i) light weight
(ii) lower price
(iii) good strength
(iv) easy handling

6.

Property	Thermoplastics	Thermosetting
1. Effect of heat	Soften at 70°C to 90°C	Do not soften easily comparatively stronger.
2. Strength	Comparatively weaker	Comparatively stronger
3. Reusability	Can be reused after melting	Can not be remelted or reused.
4. Flexibility	They are flexible	They are hard and rigid.

7. (a) Thermosetting plastics are used for making saucepan handles because these plastics are poor conductors of heat and do not get softened on heating.
(b) Electric plugs/switches/plug boards are made up of thermosetting plastics because these plastics are poor conductors of heat and electricity. Therefore prevent the flow of current through it.

8.

Can be recycled	Cannot be recycled
1. Plastic toys	1. Cooker handles
2. Carry bags	2. Electrical switches
3. Ball point pens	3. Telephone instruments
4. Plastic covering on electrical wires	
5. Plastic bowls	

9. Rana should buy cotton shirts for summer because shirts made of synthetic material do not absorb sweat easily and uncomfortable to wear in summer whereas cotton being a good absorber of water can soak the sweat easily and pores in it allow evaporation and keeps our body cool.

10. Plastics are noncorrosive in nature that is why they are used to make various containers, utensils and laboratory equipments to store food, chemicals etc.

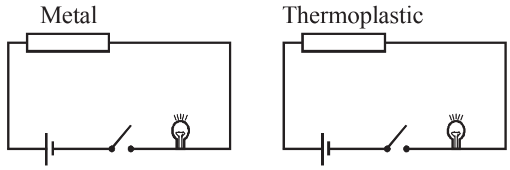
11. Handle and bristles of a toothbrush should be made of different materials. The handle of toothbrush should be hard and strong so that it can give firm grip whereas bristles should be made of soft and flexible material.

12. Plastics are not environment friendly. They take several years to decompose. They do not burnt completely. During burning of plastics lots of poisonous fumes are released into the atmosphere causing air pollution.
Polythene bags, carelessly thrown in garbage can be eaten by cow and other animals may choke their respiratory system and even cause death.
The polythene bags carelessly thrown here and there are responsible for clogging the drains.
The only solution to reduce these problems is to reduce the use of plastics. Therefore, we should avoid plastics as far as possible.

13. (i) - (d), (ii) - (c), (iii) - (a), (iv) - (b)

14. Synthetic fibres are man-made fibres. They are obtained by chemical processing of petrochemicals and do not require cutting of trees or hunting of animals. Therefore manufacturing synthetic fibres is actually helping for conservation of forests.

15. To show that thermoplastic is a poor conductor of electricity design a circuit by taking a small piece of thermoplastic material, a bulb, wires, a piece of metal and a battery. Set up the circuit (as shown below) first with the metal and then with the thermoplastic material.



On passing current, bulb in case of metal glows, but not in case of thermoplastic. This proves that thermoplastic is a poor conductor of electricity.

Exemplar Questions :

1. Due to its non-biodegradable nature it causes environmental pollution.

2. Acrylic blankets are cheap, light in weight, more durable and are available in variety of colours and designs. They can be easily washed at home.

3. Terylene and cotton.

4. Nylon ropes are strong elastic and lighter as compared to cotton or jute ropes.

5. Plastic is a non-biodegradable material and as such it causes land pollution. At the same time burning such materials in the form of garbage causes serious air pollution.
Some ways to limit its consumption are :
1. Reducing the use of platics whenever possible use paper bag.
2. Reusing it for some other purpose thereby decreasing its consumption.
3. Recycling of plastic. It requires the palstic to be collected, sorted, chopped, melted and remoulded.

HOTS Questions :

1. Rayon is called regenerated fibre, because starting material is a natural product i.e. wood or paper pulp.

2. Since nylon contains many amide groups in each molecule so it is called a polyamide.

3. It is used for making bandages and surgical dressings.

4. Parachutes, ropes for rock climbing
5. Since the fibre was synthesized simulatneously in New York and London, it was called nylon based on the initial letters of the names of cities.
6. Due to presence of cross linked arrangement of units it is difficult to break apart the units in thermosetting plastic, while due to lack of cross linkages thermoplastics are soft and weaker.

3 EXERCISE

Multiple Choice Questions :

1. (c) Thermoset plastics are usually harder, stronger and more brittle than thermoplastics.
2. (c) Cellulose, proteins, starch etc., are natural polymer.
3. (a) Nylon is more stretchy.
4. (d) These are made of phenol formaldehyde resins.
5. (c) Plastics are polymer
6. (d) Candles are hydrocarbon of higher group
7. (a) Orlon, polypropylene and nylon are prepared synthetically by chemical methods thus they are synthetic fibres.
8. (c) Rayon as it is obained from cellulose
9. (a) Wood is the major raw material of rayon
10. (d)
11. (a) Terrywool is obtained by mixing polyester (synthetic fibre) and wool (natural fibre)
12. (d) All are correct options
13. (c) Nylon and polyester are synthetic fibres obtained from plastic material we get mainly from petroleum.
14. (c) Wool retains air between the fibres. That is why it is fluffy. This enables it to retain body heat. Woolen clothes are therefore, good for wearing in cold weather.
15. (d) None of the above is natural, all are synthetic fibres.
16. (c) Rayon, Nylon and Polyster are synthetic fibres.
17. (a) Polyethylene terephthalate
18. (b) Rayon have appreciable absorbent properties.
19. (d) Bakelite is a thermosetting plastics
20. (a) Polyester + Cotton = Polycot
21. (c) **22.** (a) **23.** (b)

More Than One Option Correct :

1. (a, b, c) **2.** (a, b, d) **3.** (b, c, d,)
4. (a, b, c)
5. (b, c, d)
Angora wools is obtained from angora goats, which are found in hilly areas like Jammu and Kashmir. It is a natural fibre.

Passage Based Questions :

1. (a) **2.** (c)
3. (b) polyester alone accounts for around 60% use of synthetic fibre
4. (a) **5.** (c) **6.** (a, b)

Assertion & Reason :

1. (a) Clothes made up of synthetic fibres have less air spaces in them than the natural fibres and do not 'breathe' so well.
2. (a) Bakelite is used for making electrical switches and plugs because it is an insulator.
3. (a) Acrylic fabric is synthetic wool which is also called orlon.
4. (a)
5. (b) Synthetic fibres are man-made polymers. Polymers also occur in nature for e.g., cotton is a polymer called cellulose which is made up of a large number of glucose units.
6. (a) Being a strong fibre nylon has strong load bearing capacity that is why nylon is used to make rock climbing ropes, parachutes, car seat belts and fishing nets.
7. (d) Handles of utensils are made up of thermosetting plastics because they do not soften on heating.
8. (b) Plastics are non-biodegradable and take several years to decompose. They adversely affect our ecosystem. Thus we should avoid plastics as far as possible.

Multiple Matching Questions :

1. $A \rightarrow (p, q, s)$, $B \rightarrow (p, r)$, $C \rightarrow (p, q, s)$, $D \rightarrow (p, r)$

Integer Type Questions :

1. 1
Nylon
2. 2
Cooker handles and Electrical switches.

4 EXERCISE

1. (d) Glasses are amorphous.
2. (c) **3.** (b)
4. (d) Due to this it is used in tyres.
5. (a) **6.** (d)
7. (c) Equanil is a tranquilizer
8. (b) **9.** (b) **10.** (c)
11. (c) Broad spectrum antibiotics act on different antigens.
12. (c) Any oils which are good for eating or cooking, can be used in making soap. One of the best is said to be Coconut oil. Groundnut, Shea butter, Cocoa butter, Sun flower and many other vegetable oils are also used.
13. (d) Glass is an amorphous substance.
14. (a) The major component used in the preparation of different types of glasses is silica.
15. (a) Detergent dissolves in water faster than soap.
16. (a, b, c) **17.** (a, c) **18.** (a, b, c)
19. (a, b, c, d) **20.** (a, c) **21.** (a, b)
22. (a, c) **23.** (a) **24.** (a,b, c)
25. (a) **26.** (b)
27. $A \rightarrow (p, q, s)$, $B \rightarrow (q, r, t)$, $C \rightarrow (p, q)$, $D \rightarrow (q)$
28. 2
Laminated glass and opaque glass are type of processed glass.

Chapter 2

MATERIALS : METALS AND NON-METALS

INTRODUCTION

If you think about world around us you can think of many materials like, coins, cars and buses in which we ride, airplanes, utensils at your home, electric wires, electric appliances and various other things. These all are made up of metals. Without metals it is difficult to imagine our life. They all possess almost similar properties like hardness, solid and good conductor of electricity. Cars and buses you ride in are made of steel, which is mostly iron. Airplanes are made of aluminium. Many coins are combinations of zinc with copper, nickel etc. It is not necessary that all hard, solid materials are metals.

They could be non-metals also like coke or charcoal. Now think of any five objects that do not contain or that are not made up of metal. Some of them might be soft and smooth, such as an animal's fur, soft cotton etc. but some may be harder such as your study table or your plastic toys.

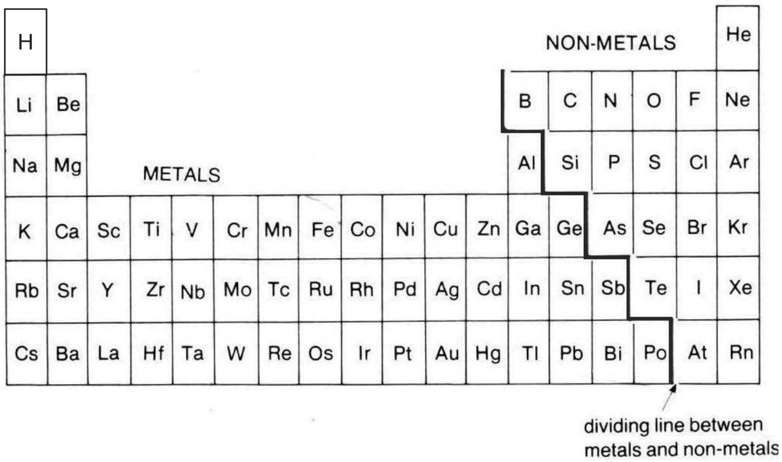
Actually our world is full of materials that contain little or no metal. These materials have a wide variety of properties ranging from soft to hard, from flexible to breakable, and from solid to gaseous. Almost all things which are present in our surrounding are made up of metals and non-metals. They play significant role in our life.

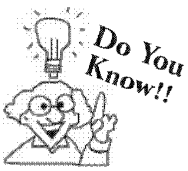
We have already learnt that a pure substance which can neither be broken down into simpler substance nor formed from two or more simpler substances by any known physical or chemical means is called an **element**.

Elements can be roughly divided into **metals, non-metals** and **metalloids.**

METALS AND NON-METALS :

At present the number of elements known to us is approximately 117. Out of these about 24 are non-metals and rest are metals. There are about 83 naturally occurring elements. For convenience in study all elements are placed in a systematic manner in a chart which is known as periodic table.





Some metal like steel is so strong. They can support great weights. The wires made up of steel are able to hold up a bridge. Where as some metals such as sodium and potassium, are so soft they can be cut with a knife.

METALS

A category of elements that usually have a shiny surface, are generally good conductors of heat and electricity, and can be melted or fused, hammered into thin sheets, or drawn into wires.

Physical Properties of Metals

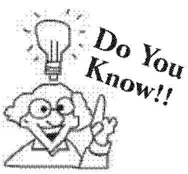
(i) They have a metallic *lustre* (shine).
(ii) They *conduct heat and electricity*.
(iii) They are *ductile* (*i.e.* can be drawn into wires).
(iv) They are *malleable* (*i.e.* can be hammered into thin sheets).
(v) They are *sonorous* (*i.e.* make a tinkling sound when hit). The sound produced is called **metallic sound** or **metallic clink.**
(vi) They are hard and have a *high density*.
(vii) They have a *high melting point* and are usually *solid* at room temperature.
(viii) They have *high tensile strength* (*i.e.* they are strong enough to bear heavy loads).
(ix) They can form alloys.

CHECK Point

1. **Why are pots and pans used for cooking are made of metal?**
2. **Why cooking utensils are not provided with metal handle.**

SOLUTION

1. It is because of high melting point and hard character of metals.
2. It is because metals are good conductors of heat and electricity. That's why most of pots and pans have wooden or plastic handles.

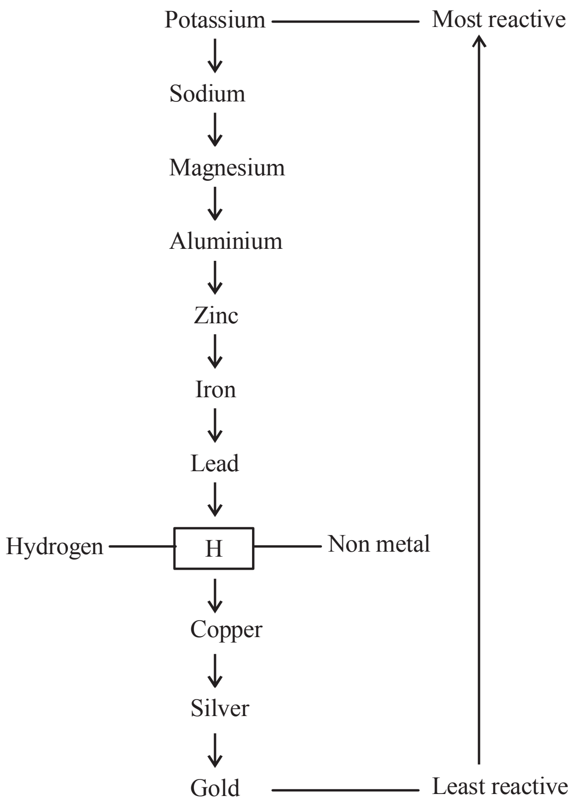


Mercury (Hg) is the only metal which is found in liquid state at room temperature whereas in case of some metals you have to raise the temperature upto 3,400°C to melt them.

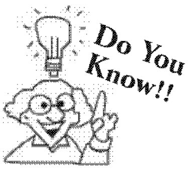
Metals can be divided broadly into two groups *i.e.* **reactive metals** and **every day metals.** Reactive metals are too reactive to be used to make articles of daily use. We use articles made of certain metals such as copper, iron, aluminium etc. in our every day life. Such metals are called **every day metals.** Since every day metals are not very reactive so they do not react quickly with water vapour or oxygen in the air. However there is a great variation between these two groups of metals. For example: Gold does not tarnish in air but iron rusts slowly in air.

Chemical Behavior of Metals

Chemical nature of metals can be explained on the basis of a reactivity series.



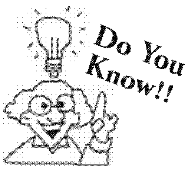
Reactivity Series of the Metals



Sodium and Potassium are stored in Kerosene oil due to its high reactivity. Owing to their high reactivity the surface of alkali metals tarnish very quickly.

Reactivity series :

Metals along with hydrogen (a non-metal) are arranged in order of their reactivity in a series, called the **activity series**. This series helps us in understanding the reactions of metals. In the reactivity series the reactive metals are at the top and less reactive are at the bottom.



Metals undergo similar kind of reactions. However the "vigour" with which they react is not the same. Some metals are more reactive than others.

Chemical Properties of Metals

1. Reaction with oxygen :

Metals can donate electrons to oxygen molecule, therefore metal form oxides by combining with **oxygen**. Oxides of metals are basic in nature.

$$2Mg + O_2 \longrightarrow 2\,MgO$$
$$4Na + O_2 \longrightarrow 2\,Na_2O$$
$$2Ca + O_2 \longrightarrow 2\,CaO$$

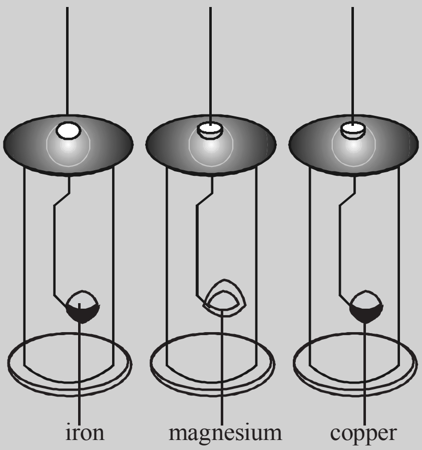
Metals like Na and K react immediately.
Mg like metal have to be heated.
Cu and Fe reacts very slowly only upon heating upto high temperature.
Copper is least reactive of these.

Do You Know!!

With oxygen, chromium and manganese form acidic oxides such as chromium trioxide and manganese (VII) oxide, in addition to basic oxides. Aluminium, zinc and tin form amphoteric oxides (i.e. oxides having both acidic and basic properties).

IDEA BOX idea box

In your school laboratory heat, iron, magnesium and copper then put the metals into gas jars filled with oxygen. The diagram shows the reaction.



(a) Which metal burns brightest?
(b) Which metal burns dullest?

2. Reaction with water :

Different metals react differently with water. On reaction with cold water Na and K reacts violently to form their hydroxides with evolution of hydrogen gas. For example

$$2Na + 2H_2O \longrightarrow 2NaOH + H_2\uparrow$$

Calcium starts floating because the bubbles of hydrogen gas formed stick to the surface of the metal.
Magnesium does not react with cold water. It reacts with hot water to form magnesium hydroxide and hydrogen. It also starts floating due to the bubbles of hydrogen gas sticking to its surface.

$$Mg(s) + H_2O\,(l) \longrightarrow MgO\,(s) + H_2(g)$$

Do You Know!!

Metals like aluminium, iron and zinc do not react either with cold or hot water. But they react with steam to form the metal oxide and hydrogen.

$$2Al(s) + 3H_2O(g) \longrightarrow Al_2O_3(s) + 3H_2(g)$$

$$3\,Fe(s) + 4H_2O(g) \longrightarrow Fe_3O_4(s) + 4H_2O(g)$$

3. Reaction with acids :

Metals react with dilute acids to form their respective salts with the evolution of hydrogen gas. Generally metals react with dilute hydrochloric acid (dil. HCl) to form corresponding chlorides with evolution of hydrogen gas. For example:

$$Mg + 2\,HCl \longrightarrow MgCl_2 + H_2\uparrow$$

$$Zn + 2HCl \longrightarrow ZnCl_2 + H_2\uparrow$$

Metals reacts with sulphuric acid to yield their corresponding sulphates with evolution of hydrogen gas.

$$Mg + H_2SO_4 \longrightarrow MgSO_4 + H_2\uparrow$$

$$Zn + H_2SO_4 \longrightarrow ZnSO_4 + H_2\uparrow$$

Do You Know!!

The more reactive metals, sodium and potassium react violently with dilute acids. The reaction is slower for less reactive metals. Copper, silver and gold do not react with acids. Magnesium reacts fastest and lead reacts slowest with dilute acids.

CHECK Point

Why we can not store lemon pickle in an aluminium utensil?

SOLUTION

We cannot store the lemon pickle in aluminium utensil because aluminium is a metal and lemon is acidic. The acid react with metal to give hydrogen which would spoil the food and make it inedible.

Knowledge ENHANCER

When a metal reacts with dilute nitric acid, then hydrogen gas is not evolved. Because nitric acid is a strong oxidising agent. So, as soon as hydrogen gas is formed in the reaction between a metal and dilute nitric acid, the nitric acid, oxidises this hydrogen to water. Very dilute nitric acid, however reacts with magnesium and manganese to evolve hydrogen gas.

IDEA BOX

Find a penny from 1983 or later. Rub one edge on sandpaper to scrap away the copper. Place the penny in a foam cup. Add vinegar to a depth of 1-2 cm. Then wait for 24 hours. Then describe changes you found. What property of metals have you demonstrated?

4. Reaction with bases :

Only some metals such as aluminium, zinc and lead react with strong bases such as sodium hydroxide to release hydrogen gas.

5. Reaction of metal with solution of other metal salt :

Different metals possessed different reactivity some are more and some are less reactive as discussed before in reactivity series. A more active metal displaces less active metal from its salt solution. For example aluminium and iron displace copper from solution of copper sulphate.

$$2Al(s) + 3CuSO_4(aq) \longrightarrow Al_2(SO_4)_3 + 3Cu$$

$$Fe(s) + CuSO_4(aq) \longrightarrow Cu(s) + FeSO_4(aq)$$

Magnesium and zinc also displace copper from copper sulphate solution.
Displacement reaction does not occur in following cases.

$$Cu(s) + FeSO_4(aq) \longrightarrow \text{No reaction}$$
$$Ag(s) + Al_2(SO_4)_3\,(aq) \longrightarrow \text{No reaction}$$

A few more displacement reactions are:

$$Cu(s) + AgNO_3(aq) \longrightarrow CuNO_3(aq) + Ag(s)$$
$$Zn(s) + Pb(NO_3)_2(aq) \longrightarrow Zn(NO_3)_2(aq) + Pb(s)$$

IDEA BOX — idea box

Take equal amount of iron sulphate solution in your school lab in a three glass beakers A, B and C simultaneously. Now put coins made of copper, silver and aluminium in beakers A, B and C respectively for two minutes. Then take out each coin and notice is there any change in any coin?

Do You Know!!

Aluminium corrodes less than iron inspite of being more reactive.

Summary the Chemical Reactions for Metals :

		Reaction with O_2	Reaction with H_2O	Reaction with mineral acid
Potassium	K	Burn vigorously to form oxides	With cold water forms hydrogen gas and alkaline hydroxide solution. React with decreasing vigour down the series	Violent reaction to give hydrogen gas and salt solution
Sodium	Na			
Calcium	Ca	Burn with decreasing vigour down the series		React to form hydrogen gas and salt solution with decreasing vigour down the series
Magnesium	Mg			
Aluminium	Al		No reaction with cold water. With steam forms hydrogen gas and oxide	
Zinc	Zn			
Iron	Fe			

CORROSION

Corrosion is the process of slowly eating away of metal due to attack of atmospheric gases and water on the surface of metal. The most common example of corrosion is the **rusting of iron**.
Rusting is the corrosion of iron on exposure to atmosphere

$$2Fe\,(s) + \frac{3}{2}O_2\,(g) + xH_2O\,(l) \longrightarrow Fe_2O_3.xH_2O\,(s)$$

i.e., rust is hydrated iron (III) oxide
When a copper vessel is exposed to moist air for long, it acquires a dull green coating. The green material is a mixture of copper hydroxide ($Cu(OH)_2$) and copper carbonate ($CuCO_3$). The reaction involved is

$$2Cu + \underbrace{H_2O + CO_2 + O_2}_{\text{moist air}} \longrightarrow Cu(OH)_2 + CuCO_3$$

The major problem of corrosion occurs with iron (or steel) as it is used as a structural material in industries like construction, infrastructure, bridges, rail transport power transmission, ship building, automobiles, heavy industries etc.

Methods of Preventions

Rusting of iron can be prevented by avoiding direct contact with air and moisture by using following methods:

(*i*) Applying grease and oil on the exposed parts of iron articles.

(*ii*) Painting surfaces of iron articles.

(*iii*) Galvanising the surface of iron articles i.e. by coating the surface of iron with zinc. During galvanisation, iron articles are dipped in molten zinc and then, after being taken out of molten zinc the articles are coated.

(*iv*) Rusting of iron can also be prevented by **alloying**.

CHECK Point

Explain an activity to show rust is basic in nature.

SOLUTION

Collect some amount of rust after a reaction between iron, oxygen and water. Dissolve it in a very little amount of water. Shake well the mixture of rust and water. Test the solution with the red and blue litmus papers. We observe that the red litmus paper turns into blue. It shows that the nature of rust is basic.

CONNECTING TOPIC

EXTRACTION OF METALS

Occurrence of Metals

Metals occur in the nature both in *free state* and in *combined state*.

Free state: The major source of metals is earth's crust. Some of the metals found in free state are *platinum, gold, silver and copper.* These elements can occur in free state because these are *less reactive*.

Combined state: Most of the metals occur in combined state. Naturally - occurring metallic compounds are called **minerals.** In the combined state, metal occur in the form of their compounds such as **oxides**, **sulphides**, **sulphates**, **carbonates**, **nitrates**, **chloride** etc. Those minerals from which an element can be extracted economically is called an **ore**. Examples of some ores of important metals are bauxite ($Al_2O_3.2H_2O$) for aluminium, magnetite (Fe_3O_4) for iron, zinc blende (ZnS) for zinc, cuprite (Cu_2O) for copper etc.

Metallurgy

As you see that most of metals naturally occur in combined state. Thus they need to be separated from rest of impurities for various applications. This process of extraction of metal from ore for various purposes is called metallurgy. The complete process of extraction of metals from their ores and then refining them for use is called **metallurgy.**

Detail study of metallurgical operations are studied in higher classes. Here we are giving you brief introduction about various stages involved in metallurgy.

The metallurgy of a metal involves three main operations

1. Concentration or dressing of ore. 2. Reduction 3. Purification or refining.

1. Concentration

The removal of impurities like soil, sand, stones and useless silicates (Gangue) from the ores is known as concentration. It is done by physical as well as chemical method.

Physical methods

(i) **Hydraulic washings (gravity separation)**

This process of concentration of ores is based on the fact that *ore* and *gangue* particles are having *different densities. Heavy ore* are separated from *lighter gangue particles* by washing with a stream (flow) of water.

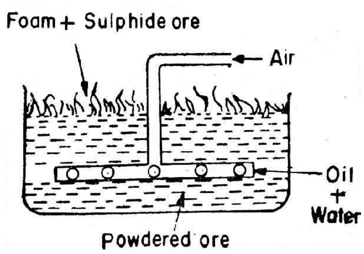
Ores of tin and lead are concentrated by this process.

(ii) **Froth flotation process**

This method is exclusively used for the concentration of sulphide ores like galena (PbS), zinc blende (ZnS) etc. It is based on the *different wetting characteristics of the ore and gangue particles with water and oil.*

The gangue particles are preferentially wetted by water while the ore particles by oil. The crushed ore is treated with water and pine oil in a tank. Pine oil acts as **frothing agent** and produces froth. Air is bubbled through the mixture. Ore particles are preferentially wetted by the oil and floats on water surface in the form of froth. The impurities settle at the bottom.

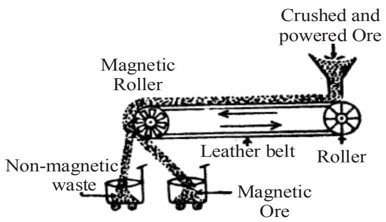
The concentrated ore is skimmed off from the top.



Froth flotation process

(iii) **Magnetic separation**

This process is used for separating magnetic materials from non-magnetic materials. As shown in figure a magnetic separator consists of a leather belt moving on two rollers. One end consists of a magnetic roller, when the ore reaches the magnetic end, the non-magnetic material falls down vertically while the magnetic material is attracted by the magnet. This process is used for separating ores like magnetite (Fe_3O_4), chromite [$Fe(CrO_2)_2$] etc.



Magnetic separation

Chemical methods

(i) **Calcination :** Heating the carbonate ore to a high temperature in the absence of air is known as **calcination.** Organic matter and volatile impurities are eliminated and **metal oxide** is formed.

Example:

$$\underset{\text{(Zinc Carbonate)}}{ZnCO_3} \xrightarrow{\text{heat}} \underset{\text{(Zinc Oxide)}}{ZnO} + CO_2$$

(ii) **Roasting :** This process is usually used in case of sulphide ores. The ore is heated in the presence of air at a temperature below its melting point and result in the formation of metal oxide or metal sulphate.

Example:

$$\underset{\text{(Zinc blende)}}{2\,ZnS} + \underset{\text{(air)}}{3O_2} \xrightarrow{\text{heat}} 2\,ZnO + 2SO_2$$

(iii) **Leaching :** It is a chemical method used for the concentration of ore in which crude powdered ore is allowed to react with suitable reagent. Ore reacts with added reagent to form soluble complex while impurities remain undissolved. Insoluble impurities are separated by filtration and ore is reprecipitated from soluble complex.

e.g. $\rightarrow$ This process is used for concentration of Al, Ag, Au ores etc.

2. Reduction to free metal

(i) **Smelting** : In this process the metallic oxides are reduced to metals by suitable reducing agents. The commonly used reducing agents are carbon, and carbon monoxide.

Examples:

$$\underset{\text{(zinc oxide)}}{ZnO} + C \longrightarrow Zn + \underset{\text{(carbon monoxide)}}{CO}$$

$$Fe_2O_3 + 3C \longrightarrow 2\,Fe + 3\,CO$$

$$Fe_2O_3 + 3CO \longrightarrow 2\,Fe + 3\,CO_2$$

(ii) **Self reduction process :** This process is used for reducing metals having *low reactivity.* Mercuric oxide (HgO) can be reduced to mercury (Hg) by heating above 300 ºC.

$$\underset{\text{(mercuric oxide)}}{2\,HgO} \xrightarrow[300°C]{\text{heat}} 2Hg + O_2$$

Copper can be also obtained by self-reduction process.

(iii) **Reduction by aluminium (Gold-schmidt aluminothermic process) :** In this process the metal oxides are thoroughly mixed with Al powder and a little BaO_2 in a crucible. The reaction is brought about by introducing a lighted Mg ribbon. Since the reaction is exothermic the temperature reached to about 3000°C. Metals like Cr and Mn can be extracted by this method.

$$Cr_2O_3 + 2Al \longrightarrow Al_2O_3 + 2Cr$$

$$3\,MnO_2 + 4Al \longrightarrow 2Al_2O_3 + 3Mn$$

(iv) **Electrolytic reduction:** Highly electropositive metals (e.g. sodium. potassium, calcium etc.) can not be reduced by chemical reducing agents. These are obtained by electrolytic reduction of their fused chlorides.

Examples : Sodium and magnesium are obtained by this method. These metals are deposited at the cathode during electrolysis.

$$\underset{\substack{\text{(Sodium Chloride)}\\\text{(fused or molten)}}}{2NaCl} \xrightarrow{\text{Electrolysis}} \underset{\substack{\text{(Sodium metal)}\\\text{(at cathode)}}}{2Na} + \underset{\substack{\text{(Chlorine)}\\\text{(at anode)}}}{Cl_2}$$

$$MgCl_2 \xrightarrow{\text{Electrolysis}} Mg + Cl_2$$

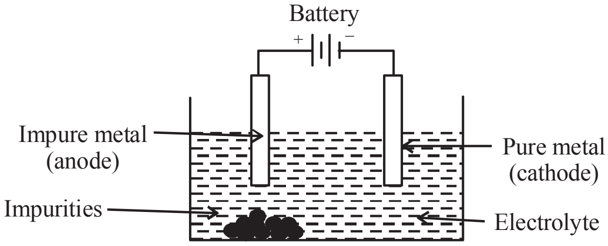
Metals extracted from their ores are usually contaminated with impurities. These are purified by following methods.

3. Refining

(i) **Liquation :** It is used for metals having low melting point
This method is used to purify readily fusible metal (e.g., Sn, Pb and Bi etc.) from infusible impurities. When crude metal is heated, the metal having low melting point melts and flows down on inclined surface leaving behind impurities. e.g. Purification of Sn containing impurities of high melting metals like Fe, Cu, W etc.

(ii) **Distillation :** Low boiling point metals like zinc, mercury etc. can be purified by this method. The pure metal distills over while the impurities are left behind.

(iii) **Refining by Electrolysis or electrolytic refining :** Large number of metals like copper, silver, gold and zinc are purified by this method. The metal of highest purity are obtained by this method.
Various steps involved in this method are :
(*a*) The block of *impure metal* is made **anode** by connecting to the positive terminal of the battery.
(*b*) A thin strip of *pure metal* is made **cathode** by connecting to the negative terminal of the battery.
(*c*) A *water soluble salt of the metal* acts as an **electrolyte**.



Electrolytic refining

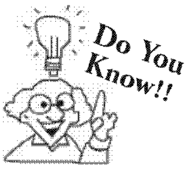
The arrangement is shown in figure above. When the battery is switched on, current flows in the circuit. Impure metal from the anode dissolved by the electrolyte and pure metal gets deposited on the cathode.

(iv) **Thermal decomposition method :**
This method may be of different type and in this method, the metal is converted into suitable volatile compound which is decomposed at high temperature to give pure metal.

(v) **Zone-refining or fractional crystallization :**
This is another method used to obtain metals of very high purity. Metals like Si, Ge, Ga etc. used in semiconducter devices are purified by this method. This method is based on difference in solubilities of impurities in molten and solid state.

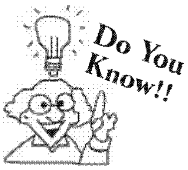
Uses of Metals

Taking advantage of the strength, hardness and rigidity of metals, they are used in making machinery, automobiles, aeroplanes, trains, satellites and various industrial gadgets.



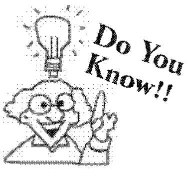
An alloy of iron and niobium is exceptionally strong and is used in the construction of nuclear reactors.

Uses of iron : One of the most important use of iron ore is in the manufacture of steel. Iron is combined with various other combinations of elements including carbon, chromium, nickel, silicon, molybdenum etc. to make different varieties of steel which is an excellent material for various forms of construction work. As steel can withstand high pressure as well as temperature. It is widely used for making outdoor household things like decorative wrought iron fencing, wrought iron trollis, wrought iron furniture and so on. Metallic iron possesses natural magnetic properties and hence it is used for making permanent magnets as well as electromagnets.



Iron is an important part of a large molecule called haemoglobin, which carries oxygen in our bloodstream. Haemoglobin also gives blood its bright red color.

Uses of aluminium: It is used to make house-hold article and for making aeroplanes (because of its lightness). It is also used in making electrical cables (because it is quite cheaper than copper). Aluminium foils (used for wrapping food stuffs, medicines, chocolates, cigarettes etc.)



When red hot iron is hammered, it gives off lots of sparks. These sparks are iron burning in oxygen from the air.

Knowledge ENHANCER

Aluminium : *This is the most abundant metal on planet Earth (and the third most abundant of all chemical elements). It forms about one-twelfth by weight of the Earth's outer layer, the crust. Pure aluminium is very light but not very strong. However, when combined with other elements such as copper, magnesium or silicon, it forms extremely strong alloys. Also it does not rust, unlike steel.*

Uses of copper : It is used in electrical gadgets (because it is a very good conductor of electricity). It is also used for making radiators of automobiles.

Uses of magnesium : It is used in flares and flash bulb in photography (because it burns in air with dazzling white flame). Alloys of magnesium being light and strong are used in manufacture of air-crafts and other vehicles. Magnesium, being resistant to corrosion, is used to line water pipes and tanks.

Uses of zinc : A coating of zinc is provided on iron and steel to prevent **corrosion** of iron. The coated metal is called **galvanised iron.**

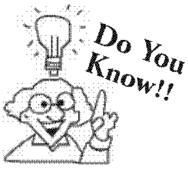
Noble metals and their uses : Gold, silver and platinum are known as noble metals. These metals are not affected by air, water or acids.

Uses of silver :

(i) Used in jewellery (because it is too soft).
(ii) Used to make coins and medals.
(iii) Used to make contacts in some kind of electrical equipments.
(iv) Used in electroplating. A thin layer of silver is deposited on another metal by electrolysis.
(v) Compounds of silver are used in photography. (These compounds are sensitive to light). For example silver iodide, silver bromide, silver chloride etc.
(vi) Tooth cavities are filled with dental amalgams which are alloys of silver, tin etc. with mercury.

Uses of gold :

(i) Used in jewellery and ornaments.
(ii) Used to make coins and medals.
(iii) Used as a standard of currency for trade.
(iv) Used in dentistry. Artificial teeth were earlier usually made of silver or gold.
(v) Used for gold lettering.



Gold is a famous symbol of wealth. But rarer metals such as platinum and palladium command higher prices for specialized engineering and electronic uses.

Uses of platinum :

(i) Used to make surgical instruments and chemical equipments.
(ii) Alloys of silver and platinum are used for making electrical parts and bearings.
(iii) Used to make fine jewellery.
(iv) Used as a catalyst in many reactions.

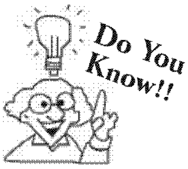
ALLOYS

The capacity of metals to combine with other metals is called **alloy formation.** (i.e., homogenous mixture) Examples of alloys are brass, bronze and steel. Useful alloys combine the best properties of two or more metals into a single substance.

For example, copper is a fairly soft and malleable metal. But when mixed with tin, it forms bronze, which can be cast into statues that last hundreds of years. Brass is an alloy of copper and zinc. Pure iron rusts very easily, but when mixed with carbon, chromium, and vanadium, iron forms stainless steel. Knives and forks made of stainless steel can be washed over and over again without rusting.

Some of the alloys along with their composition and uses are listed in table given below.

	Alloy	Composition	Uses
1.	Brass	Cu = 80%, Zn = 20%	For making utensils and cartridges.
2.	Bronze	Cu = 90%, Sn = 10%	For making statues, medals, ships, coins and machines
3.	Solder	Sn = 50%, Pb = 50%	For joining metals, solding wire and electronic components etc.
4.	Duralumin	Al = 95.5%, Cu = 3%, Mn = 1.0%, Mg = 0.5%	Used in bodies of aircrafts, kitchen ware and automobile parts etc.
5.	German Silver	Cu = 60%, Zn = 20%, Ni = 20%	For making utensils and ornaments
6.	Gun metal	Cu = 90%, Sn = 10%	For Gears and castings etc.
7.	Bell metal	Cu = 80%, Sn = 20%	For bells, gangs etc.
8.	Magnalium	Al = 90%, Mg = 10%	For balance beams, light instruments.
9.	Type metal	Pb = 82%, Sb = 15%, Sn = 3%	For casting type
10.	Stainless steel	Fe, Ni, Cr, C	For utensils, cutlery etc.



Solder is an alloy made up of lead and tin. It melts at a low temperature. It is used for fastening other pieces of metal together.

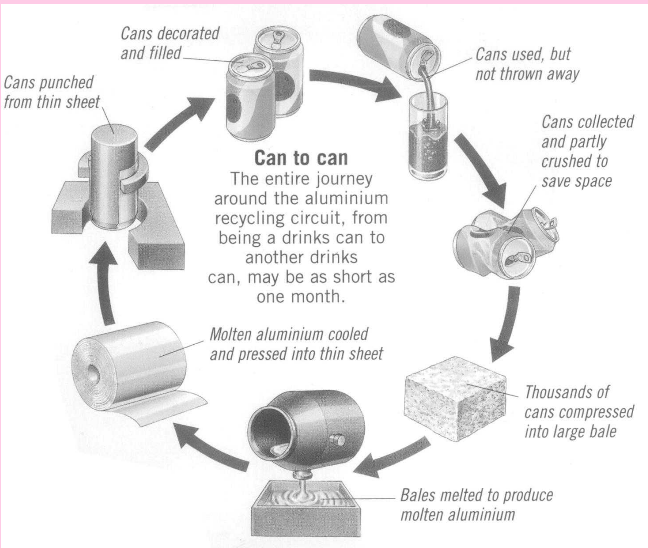
Knowledge ENHANCER

We should conserve and recycle metals

Because our planet is chock-full of metal-containing compounds, it is difficult to imagine how we could ever encounter a shortage of metals. Experts suggest that however, if we continue with our present rate of consumption, such shortages will occur within the next two centuries. The problem is not a shortage of metal-containing compounds but rather a shortage of ores from which these compounds can be extracted at a reasonable cost.

Recycling metals

Most metals occur combined with minerals and other substances, spread through rocks known as ores. It takes huge amounts of time and energy to dig or mine the ores, and then extract and purify the metals from them. Recycling helps to reduce these problems. For example, recycled aluminium uses only one-twentieth of the energy which is needed to produce aluminium from its ore. Many other metals can also be recycled, including irons and steel, and even the silver and gold used in electrical circuits and false teeth!



NON-METALS

A **non-metal** is an element that can easily form negative ion **(anions)** by gaining electrons, *e.g.*

$$\underset{\text{(Chlorine)}}{Cl} + e^- \longrightarrow \underset{\text{(Chloride ion)}}{Cl^-}$$

$$\underset{\text{(Oxygen)}}{O} + 2e^- \longrightarrow \underset{\text{(Oxide ion)}}{O^{2-}}$$

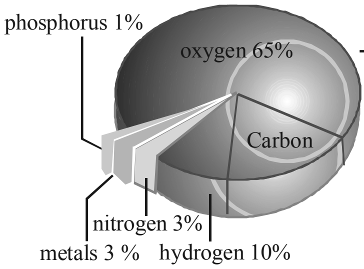


Chart shows that our body contains 3% metals the rest is made up of non metals.

Do You Know!!

The most abundant non-metal in the earth's crust is oxygen, which constitutes about 50% of the earth's crust. The next abundant non-metal is silicon which constitutes about 26% of the earth's crust.

Physical Properties of Non-metals:

(i) At room temperature non-metals are either solids or gases, except bromine (Br) which is a liquid.

(ii) They are lustreless (*i.e.* dull). Except graphite and Iodine.

(iii) They are generally brittle.

(iv) They are bad conductors of heat and electricity (except graphite).

(v) They are non-sonorous (*i.e.* they do not produce a metallic **clink).**

CHECK Point

Name a non-metal that is a liquid at room temperature ?

SOLUTION

Bromine.

Do You Know!!

As phosphorus is naturally not stable as an element. So, phosphorus in nature is always found in compounds.

Chemical Properties of Non-metals

Non-metals are more reactive with metals than with other non-metals. Generally non-metal react with each other at a high temperature.

1. Action of Air : Non-metals do not react with air at room temperature. Except white phosphorus which burns in air, at room temperature to form its oxide. Thus to avoid contact with air, phosphorus is stored under water.

Non-metals reacts with oxygen to form either acidic or neutral oxides. Acidic oxides dissolve in water to form acids.

Hydrogen is being burnt in air produces a **"pop" sound** to form water

$$2H_2 + O_2 \xrightarrow{\text{Ignition}} 2H_2O$$

Do You Know!!

Hydrides like H_2S, H_2Se and H_2Te burns in the atmosphere of oxygen with blue flame forming dioxides.

$$2H_2S + 3O_2 \longrightarrow 2H_2O + SO_2$$

Carbon reacts with oxygen to form carbon dioxide, which dissolve in water to form carbonic acid. Carbon reacts with insufficient amount of oxygen to form carbon monoxide, which is a neutral oxide and does not dissolve in water.

$$C(s) + O_2(g) \longrightarrow CO_2(g)$$
$$CO_2(g) + H_2O(l) \longrightarrow H_2CO_3(aq)$$
$$CO(g) + H_2O(l) \longrightarrow \text{No reaction}$$

- Sulphur reacts with oxygen to form sulphur dioxide, an acidic oxide, which dissolves in water to form sulphurous acids.

$$S(s) + O_2(g) \longrightarrow SO_2(g)$$
$$SO_2(g) + H_2O(l) \longrightarrow H_2SO_3(aq)$$

- Phosphorus reacts with oxygen to form phosphorus pentaoxide, an acidic oxide, which dissolves in water to from phosphoric acid.

$$2P(s) + 5/2\, O_2(g) \longrightarrow P_2O_5(s)$$
$$P_2O_5(s) + 3\,H_2O(l) \longrightarrow 2H_3PO_4(aq)$$

2. Action of water: Generally non-metals do not react with water. However chlorine dissolves in water and form an acidic solution.

$$2Cl_2 + 2H_2O \xrightarrow[\text{or heat}]{\text{Sunlight}} 2HCl + \underset{\text{(Hypochlorous acid)}}{2HOCl}$$

On passing *steam* over red hot coke (carbon), we get a gaseous mixture of carbon monoxide and hydrogen which is known as water gas.

$$\underset{\text{(red hot)}}{C} + \underset{\text{(steam)}}{H_2O} \longrightarrow \underset{\text{(water gas)}}{CO + H_2}$$

3. Action of acids : Non-metals do not react with dilute acids.

idea box

Instruction : Do this experiment with your teacher as it can be dangerous.
In your school's chemistry lab heat iron, aluminium and copper. Then put the metal in to gas jars filled with chlorine which is very reactive and dangerous gas.
(a) Which metal burn brightest in chlorine ?
(b) Give the descending order of 3 metals with respect to their tendency to burn in presence of chlorine.

COMPARISION BETWEEN METALS & NON-METALS

METALS	NON-METALS
Metals are generally solids. (exception : mercury, gallium)	Non-metals are found in all three states.
Metals are heavy, (exception : sodium, potassium, magnesium)	Non-metals are generally light in weight.
They are hard and nonbrittle. (exception : sodium, potassium and lead which are soft)	Solid non-metals are hard but brittle.
They are good conductors of heat and electricity. (exception : lead)	They are bad conductors of heat and electricity, (exception : graphite)
They are ductile and malleable.	They are neither ductile nor malleable.
Their melting point and boiling point are generally high.	Their melting point and boiling point are generally low.
They generally produce ringing sound on collision.	They do not produce ringing sound.
They are generally lustrous and can be polished.	They are generally non-lustrous and cannot be polished.

Uses of Non-metals

Non-metals find a large number of uses in our life.

Uses of hydrogen :

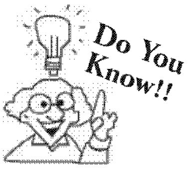
(i) Used to convert vegetables oil into vegetable ghee. (The process is called **hydrogenation** of oils).
(ii) Used in preparation of ammonia (NH_3) gas which is used in fertilizers.
(iii) H_2 with metals like nickel, palladium etc is used as common reducing agent in organic chemistry.

Do You Know!!

Although hydrogen makes up more than 90 percent of the atoms in the universe, it makes up only 1.0 percent of the mass of Earth's crust, oceans, and atmosphere. Hydrogen is rarely found on Earth as a pure element. Most of it is combined with oxygen as water.

Uses of oxygen:

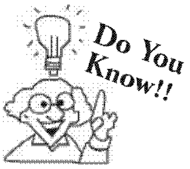
(i) It is essential for survival of life.
(ii) It is essential for combustion process.
(iii) Used in oxy-acetylene torch which is used to cut and weld metals.
(iv) It is used in manufacture of acids (*e.g.* H_2SO_4, HNO_3 etc).
(v) Liquid oxygen is used to burn rocket fuel.



Ozone (O_3) is an allotrope of oxygen. There is a ozone layer near the top of the atmosphere. This ozone layer is very valuable as it stop harmful rays from the sun. These are rays that cause cancers and other problems.

Uses of nitrogen:

(i) Used in manufacture of ammonia (NH_3), nitric acid (HNO_3) and fertilizers.
(ii) Used to preserve food (because nitrogen is inert *i.e.* less reactive).
(iii) Used in explosives *e.g.* ammonium nitrate (NH_4NO_3).
(iv) Used by plants to manufacture proteins.
(v) Liquid nitrogen is used to preserve blood, corneas and other donated organs.



Sulfur and nitrogen are found in coal and oil. When coal is burned, the sulfur and nitrogen form sulfur dioxide and an oxide of nitrogen. These compounds then combine with moisture in the air and form sulfuric acid and nitric acid. These are a few of the dangerous pollutants in the air.

Uses of carbon :

(i) Diamond (a crystalline form of carbon) is used as a gem.
(ii) Diamond is used for cutting rocks or glass.
(iii) Graphite (a crystalline form of carbon) is used as an electrode (because graphite is a good conductor of electricity).
(iv) Graphite is used as a lubricant (solid lubricant) in machines (becuause of high m.p.)
(v) Graphite is used in manufacture of lead pencils.

CHECK Point

What happens when you hit
(a) the lead strip
(b) the carbon rod.

SOLUTION

The lead strip get bent, carbon rod is broken in to two pieces.

Uses of sulphur :

(i) Used in manufacture of dyes, matches, fire work and gun powder.
(ii) Used in many medicines (*e.g.* skin ointments)
(iii) Used as *fungicide* to kill harmful fungi.
(iv) Used in *vulcanization* of rubber.
(v) Used in manufacture of sulphuric acid (H_2SO_4).

Uses of chlorine :

(i) Used in manufacture of PVC (poly-vinyl chloride) which is a plastic.
(ii) Used as *water disinfectant* to kill germs.
(iii) Used as a bleaching agent in paper and textile industry.
(iv) Used in manufacture of hydrochloric acid (HCl).
(v) Used in manufacture of pesticides like **Gammexene.**

Uses of iodine :

(i) Used as an *antiseptic*.
(ii) Used in preparation of "tincture of Iodine".
(iii) Used in medicines.
(iv) Iodized salt contains a compound of iodine which helps to prevent diseases like goitre.

Uses of noble gases :

The gases Helium (He), Neon (Ne), Argon (Ar) etc. are known as noble gases because of their non-reactive nature.

(i) Helium is used to fill balloons.
(ii) A mixture of helium and oxygen is used by divers for respiration.

(iii) Neon is used for illumination. It glows orange red in advertising signs.
(iv) Argon is used to fill electric bulbs.
(v) A mixture of argon and mercury vapours is used in advertising. This mixture glows green in advertisement signs.

Do You Know!!

Few elements have both metallic and non-metallic properties. And a few elements, such as the noble gases, have neither kind of properties.

Uses of phosphorus

(i) Used in manufacture of phosphoric acid and super phosphate fertilizer.
(ii) Used in preparation of fire works, smoke screens and in match stick industry.
(iii) Used in making phosphorus bronze alloy which is corrosion resistant.

Do You Know!!

Soft drinks are mainly carbonated water but also contains phosphoric acid; this is made of hydrogen, oxygen and phosphorus

Uses of silicon

(i) Used for making *silicon steel* alloy and also in preparation of a polymer called **silicone.**
(ii) Used in making semi-conductor devices such as transistors.

Knowledge ENHANCER

Metalloids : *There are six elements called metalloids (Boron, silicon, germanium, arsenic, antimony and tellurium. The metalloids have some of the characteristics of metals and some of non-metals. The most common metalloid is silicon (Si). It combines with oxygen to form a number of familiars substances, including sand, glass and cement. The most useful property of metalloids is their varying ability to conduct electricity. As the amount of electricity conducted by a metalloid depends upon temperature, exposure to light, or the presence of small amount of impurities. For this reason, metalloids such as silicon and germanium (Ge) are used to make semiconductors.*

idea box

Observe 10 hard, brittle substances from your home and surroundings which you use in your routine life like plastic toys, kitchen dishes, nails, table, chair, fan, motor pump etc. Classify these substances into category of metals and non-metals. Tell your friends to do the same and compare the results.

SUMMARY

Elements are broadly classified as metals and non-metals.

Metal is an element which can form positive ion (cation) by losing electrons. Metals are also called **electropositive elements.**

Metals can occur both in free and combined state.

Most of the metals occurring in combined state are obtained from their ores.

When metals occur in free state, then it is in uncombined state Au, Ag, Pt occur in free state.

Reactive metals such as Mg, Ca, Al, Fe, etc. occur in combined state in the form of their oxides, sulphides, sulphates, carbonates, nitrates, chlorides etc.

Elements which show characteristics of both metals and non-metals are called metalloids *e.g.* Si, Ge etc.

Various steps which are necessary to obtain metals from their ores are *concentration of ores, reduction, refining.*

Metals are generally hard, strong solids, They have a metallic lustre. They are sonorous, malleable, ductile, good conductors of heat and electricity.

The oxygen of the air acts on metals to form their oxides.

Metals on combustion produce basic or amphoteric oxides.

Metals along with hydrogen, are arranged in decreasing order of reactivity in the **activity series.** This series helps us to understand the chemical behaviour of metals.

The **metal activity** series is K > Na > Ca > Mg > Al > Zn > Fe > Sn > Pb [H] Cu > Hg > Ag

Metals more reactive than hydrogen ⟷ (K … Pb); Metals less reactive than hydrogen ⟷ (Cu … Ag)

A more active metal displaces a less active metal from the solution of its compound soluble in water.

Metals higher than hydrogen in the activity series displace hydrogen from water and acids.

Gold, silver and platinum are **noble metals.** They are not affected by water or acids.

Naturally occurring metallic compounds are called minerals.

The minerals from which metal can be extracted profitably are called **ores.**

The impurities of sand and rocky materials that are present in ores are called **gangue** or **matrix.**

Calcination : It is the process of heating of ore in absence of air.

Roasting : It is the process of heating of ore in presence of air.

Smelting : It involves the heating of oxide ore with carbon so as to reduce the metal oxide to metal.

Alloy : A uniform mixture of two or more metals or metal and non-metal in known as alloy, *e.g.* steel is an alloy of iron and carbon.

Refining : The process of purification of metal is called refining. It is carried out to obtain metal of maximum purity.

Corrosion : The wastage of metal layer by layer due to formation of metal compounds on the surface of metal is called corrosion.

Rusting : The corrosion of iron is called rusting. In moist air a piece of iron rusts.

Amalgam : An amalgam is an alloy of mercury and one or more metals.

Metals are used in making machinery, automobiles, aeroplanes, trains, satellites, industrial gadgets etc.

A **non-metal** is an element which can form negative ions (**anion**) **by gain of electrons.**

Non-metals are lustreless (except graphite and iodine), brittle (if solid), non-sonorous, bad conductors of heat and electricity (except graphite).

Except *white phosphorus,* non-metals do not react with air at ordinary temperature.

In general, non-metals are non-reactive towards water.

Some reactive non-metals are kept in water to protect them from the influence of air, *e.g.* if white phosphorus is kept in air it catches fire so it is kept in water.

Majority of non-metals do not react with acids.

Non-metals on combustion produce **acidic oxides.**

1 EXERCISE

Fill in the Blanks :

DIRECTIONS : *Complete the following statements with an appropriate word / term to be filled in the blank space(s).*

1. Metals combine with oxygen to form .... oxides.

2. Metals above hydrogen in the activity series can displace .... from dilute acids.

3. .... is a non-metal that is a good conductor of electricity.

4. .... is a non-metal that is lustrous.

5. Iodine is used as an antiseptic in the form of .... solution.

6. The chief ore of iron is ....

7. The black part of a pencil is made of ....

8. .... .is a metal that exist as a liquid at room temperature.

9. Liquid .... .is used to preserve donated organs.

10. Copper corrodes in the presence of ________ and water.

True / False :

DIRECTIONS : *Read the following statements and write your answer as true or false.*

1. Metals are electropositive elements.

2. Different metals have same reactivities with water and dilute acids.

3. Naturally occurring metallic compounds are called minerals.

4. Bauxite is an ore of zinc.

5. Bromine is a gaseous non-metal.

6. In the activity series silver is above copper

7. Iodine is lustrous and is a metal

8. Non-metals can easily lose electrons to form cations.

9. Non-metals are less dense and have low m.p. and b.p. as compared to metals.

10. Sodium and potassium occur in nature in free state.

11. Dental amalgams contain gold and mercury.

12. Magnesium burns in air with dazzling white light.

Match the Following :

DIRECTIONS : *Following question contains statements given in two columns which have to be matched. Statements (A, B, C and D) in column I have to be matched with statements (p, q, r and s) in column II.*

1. **Match the Column**

Column I	Column II
(A) Brass	(p) Homogeneous mixture of copper and zinc
(B) Bronze	(q) Homogeneous mixture of iron and carbon
(C) Steel	(r) Homogeneous mixture of copper and tin
(D) Solder	(s) Homogeneous mixture of tin and lead

Very Short Answer Questions:

DIRECTIONS : *Give answer in one word or one sentence.*

1. Why does gold and silver occur in native state?

2. Why is white phosphorous kept under water?

3. Name one metallic element that is more reactive than hydrogen.

4. Give one example, with equation, of displacement of hydrogen by a metal from acid.

5. What happens when iron nails are put into copper sulphate solution.

6. Which metal can be extracted from bauxite?

7. Why iron sheets are coated with zinc?

8. What is galvanising?

9. Can copper displace iron from iron sulphate solution? Give reason.

10. Which non-metal is used in the manufacture of fertilizers?

11. Name the metal which is converted into its foils which are used for packing of medicine tablets.

12. Coal and pencil lead does not show the property of malleability. Can we call them metal?

13. What happens when a non-metal reacts with acid?

14. Which gas is always produced when a metal reacts with a dilute acid?

Short Answer Questions :

DIRECTIONS : *Give answer in 2-3 sentences.*

1. Give an example of a metal which
   (a) is liquid at room temperature
   (b) is best conductor of heat
   (c) can be easily cut with knife

2. What are the two necessary conditions for the rusting of iron?

3. Differentiate between metals and non-metals.

4. Write equations for the reaction of
   (a) iron with steam
   (b) calcium with water
   (c) potassium with water

5. What is reactivity series? How does it help us in predicting reactivities of various metals?

6. What are uses of sulphur.

7. What type of oxides are formed when non-metals combine with oxygen?

8. Name the following :
   (a) A metal used in hot water system
   (b) A metal used in long distance cable wires
   (c) A metal added to gold to harden it

9. What happens when a copper vessel is exposed in moist air?

Long Answer Questions :

DIRECTIONS : *Answer the following questions in detail.*

1. Observe the following figure, complete the reaction(s) and tell in which beaker/beakers displacement reactions take place.

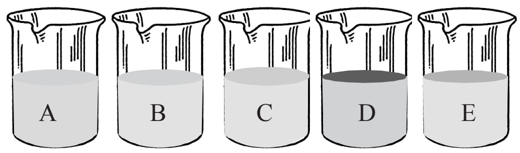
Beaker A : Copper Sulphate ($CuSO_4$) + Zinc granule (Zn),

Beaker B : Copper Sulphate ($CuSO_4$) + Iron nail (Fe)

Beaker C : Zinc Sulphate ($ZnSO_4$) + Copper turnings (Cu),

Beaker D : Ferrous sulphate ($FeSO_4$) + Copper turnings (Cu).

Beaker E : Zinc Sulphate ($ZnSO_4$) + Iron nail (Fe),



2. (A) Explain the physical properties of non-metals.
   (B) Give an example of a non-metal which :
      (a) is a solid at room temperature
      (b) possesses lustre
      (c) is a good conductor of electricity
      (d) is a good conductor of heat

3. Define the following terms :
   (a) Minerals (b) Ores
   (c) Metallurgy (d) Malleability
   (e) Ductility

2 EXERCISE

Text-Book Exercise :

1. Which of the following can be beaten into thin sheets?
   (a) Zinc (b) Phosphorus
   (c) Sulphur (d) Oxygen

2. Which of the following statements is correct?
   (a) All metals are ductile.
   (b) All non-metals are ductile.
   (c) Generally, metals are ductile.
   (d) Some non-metals are ductile.

3. Fill in the blanks :
   (a) Phosphorus is very ____________ non-metal.
   (b) Metals are ____________ conductors of heat and ____________ .
   (c) Iron is ____________ reactive than copper.
   (d) Metals react with acids to produce ____________ gas.

4. Mark 'T' if the statement is true and 'F' if it is false.
   (a) Generally, non-metals react with acids. ( )
   (b) Sodium is a very reactive metal. ( )
   (c) Copper displaces zinc from zinc sulphate solution. ( )
   (d) Coal can be drawn into wires. ( )

5. Some properties are listed in the following Table. Distinguish between metals and non-metals on the basis of these properties.

Properties	Metals	Non-metals
1. Appearance		
2. Hardness		
3. Malleability		
4. Ductility		
5. Heat Conduction		
6. Conduction of Electricity		

6. Give reasons for the following :
   (a) Aluminium foils are used to wrap food items.
   (b) Immersion rods for heating liquids are made up of metallic substances.
   (c) Copper cannot displace zinc from its salt solution.
   (d) Sodium and potassium are stored in kerosene.

7. Can you store lemon pickle in an aluminium utensil? Explain.

8. Match the substances given in Column A with their uses given in Column B.

Column A	Column B
(i) Gold	(a) Thermometers
(ii) Iron	(b) Electric wire
(iii) Aluminium	(c) Wrapping food
(iv) Carbon	(d) Jewellery
(v) Copper	(e) Machinery
(vi) Mercury	(f) Fuel

9. What happens when
   (a) Dilute sulphuric acid is poured on a copper plate?
   (b) Iron nails are placed in copper sulphate solution?
   Write word equations of the reactions involved.

10. Saloni took a piece of burning charcoal and collected the gas evolved in a test tube.
    (a) How will she find the nature of the gas ?
    (b) Write down word equations of all the reactions taking place in this process.

11. One day Reeta went to a jeweller's shop with her mother. Her mother gave old gold jewellery to the goldsmith to polish. Next day when they brought the jewellery back, they found that there was a slight loss in its weight.
    Can you suggest a reason for the loss in weight?

Exemplar Questions :

1. A purple coloured non-metal forms a brown solution in alcohol which is applied on wounds as an antiseptic. Name the non-metal.

2. Zinc sulphate forms a colourless solution in water. Will you observe any colour on adding copper turning in it?

3. Why are bells made of metals?

4. Paheli bought a statue made of copper. To her surprise it acquired a dull green coating after a couple of months. Explain the reason.

5. Some of the following statements are incorrect. Find the incorrect statements and correct them.
   (a) The property of metals by virtue of which they can be drawn into wires is called ductility.
   (b) Metals are good conductor of electricity but poor conductor of heat.
   (c) Articles made of metals produce ringing sound when struck hard.

(d) Oxides of non-metals and metals are acidic in nature.

(e) A less reactive metal replaces a more reactive metal from its salt solution in water.

HOTS Questions :

1. Explain why, aluminium is more reactive than iron yet there is less corrosion of aluminium when both are exposed to air?
2. In a solution of silver nitrate, a copper plate was dipped. After some time, silver from the solution was deposited on the copper plate. Which metal is more reactive - copper or silver? Why?
3. Can aluminium vessels be employed for storage of copper sulphate solution? Explain your answer.
4. It is advised that food stuffs with acid components should not be stored in the utensils made of iron, aluminium or copper, why?
5. Of the two metals **A** and **B, A** is found in native state and **B** occurs in combined state in nature. Which of the two will be present first in the activity series of metals?
6. The tarnished copper vessels are cleaned with lemon juice? Explain why these sour substances are effective in cleaning the vessels.
7. Give reasons
   (a) Platinum, gold and silver are used to make jewellery.
   (b) Sodium, potassium and lithium are stored under oil.
   (c) Aluminium is a highly reactive metal, yet it is used to make utensils for cooking.

3 EXERCISE

Multiple Choice Questions :

DIRECTIONS : *This section contains **21** multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.*

1. Select the one that could displace copper from a solution of copper sulphate.
   (a) Silver (b) Mercury
   (c) Tin (d) Gold
2. Which of the following is a liquid metal?
   (a) Mercury (b) Bromine
   (c) Aluminium (d) Sodium
3. The property of metals to be hammered into their sheets is called
   (a) Malleability (b) Ductility
   (c) Tensile strength (d) Sonorous nature
4. Select the metal that is soft
   (a) Aluminium (b) Copper
   (c) Sodium (d) Lead
5. Metals combine with oxygen to form an oxide. The nature of oxide formed is ....
   (a) Acidic (b) Basic
   (c) Neutral (d) Amphoteric
6. The process of eating away of metals layer by layer due to formation of metal compound on surface is called
   (a) Galvanisation (b) Amalgam formation
   (c) Corrosion (d) Vulcanization
7. Which of the following metals has no reaction even with steam.
   (a) Sodium (b) Calcium
   (c) Iron (d) Silver
8. The most characteristic property of metals is their tendency to
   (a) Form basic oxides (b) Form hydrides
   (c) Lose electrons (d) Gain electrons
9. The process of protecting iron, from rusting, by coating with zinc is called
   (a) Rusting (b) Roasting
   (c) Smelting (d) Galvanizing
10. The correct order of metals in the activity series is
    (a) $Cu > Zn > Mg > Ca$ (b) $Ca > Zn > Cu > Mg$
    (c) $Zn > Mg > Cu > Ca$ (d) $Ca > Mg > Zn > Cu$.
11. Sodium reacts with cold water to form
    (a) sodium hydroxide and hydrogen
    (b) sodium hydroxide and oxygen
    (c) sodium hydride and oxygen
    (d) None of these
12. The reactivities of iron, magnesium, sodium and zinc towards water are in the order
    (a) $Fe > Mg > Na > Zn$ (b) $Zn > Na > Mg > Fe$
    (c) $Na > Mg > Zn > Fe$ (d) $Mg > Na > Fe > Zn$.
13. Select the one that occurs in native form.
    (a) Calcium (b) Aluminium
    (c) Sodium (d) Gold
14. Antimony and arsenic can be classified as
    (a) Metals (b) Non metals
    (c) Metalloids (d) Any one of these

15. Select the property that is associated with non-metals.
(a) Low density
(b) Low melting point
(c) Poor conductor of electricity
(d) All of the above

16. For vulcanization of rubber we use which one of the following non-metal?
(a) Sulphur (b) Phosphorus
(c) Chloride (d) Carbon

17. Non-metals are majorly present as
(a) Solids (b) Liquids
(c) Gases (d) Both (a) and (c)

18. When MgO is dissolved in water, $Mg(OH)_2$ is obtained. A red litmus paper dipped in this solution turns blue; this shows that the solution is _________ in nature.
(a) acidic (c) neutral
(c) alkaline (d) reactive

19. Which of the following methods is suitable for preventing an iron frying pan from rusting?
(a) Applying grease
(b) Applying paint
(c) Applying a coating of zinc
(d) All of the above.

20. Which of the following statement is false?
(a) Metals are good conductors of heat and electricity.
(b) Gold, Silver and Zinc are most malleable metals.
(c) Mercury is the only liquid metal.
(d) Bromine is the only liquid non-metal.

21. Which of the following statement regarding non-metals is true?
(a) Non-metals are of two types only solids and gases.
(b) Non-metals reacts with oxygen to form basic oxides generally.
(c) Non-metals are non-lustrous with dull apppearence. Graphite, an allotrope of Carbon and iodine have shining lustrous surfaces.
(d) Non-metals replace hydrogen from acids.

More than One Option Correct :

DIRECTIONS : *This section contains 7 Multiple Choice Questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE OR MORE may be correct.*

1. Which of the following will produce an acidic oxide on reaction with oxygen?
(a) Calcium (b) Carbon
(c) Sulphur (d) Nitrogen

2. Which element(s) of the following will produce basic oxides on reaction with oxygen?
(a) Sodium (b) Magnesium
(c) Aluminium (d) Carbon

3. Which of the following exists in liquid state?
(a) Mercury (b) Gallium
(c) Bromine (d) Helium

4. Which of the following show lustre?
(a) Sodium (b) Graphite
(c) Iodine (d) Neon

5. Which of the following metal(s) is/are classified as every day metal(s)?
(a) Sodium (b) Aluminium
(c) Iron (d) Copper

6. Which of the following lie above hydrogen in the activity series?
(a) Aluminium (b) Silver
(c) Iron (d) Lead

7. Which of the following will displace copper from copper sulphate solution?
(a) Nickel (b) Tin
(c) Lead (d) Platinum

Passage Based Questions :

DIRECTIONS : *Study the given paragraph(s) and answer the following questions.*

Passage - 1

Though most metals undergo similar kind of reactions. The "vigour" with which they react is not the same. Some are more reactive than the others. Metals along with hydrogen (a non-metal) are arranged in order of their reactivity in a series called **activity series**

Activity decrease ↓

K
Ca
Na
Mg
Al
Zn
Fe
H
Cu
Hg
Ag
Au

1. Which of the following is most reactive?
(a) Sodium (b) Potassium
(c) Calcium (d) Magnesium

2. Which of the following can displace Iron from $FeSO_4$ Solution
   (a) Zinc (b) Copper
   (c) Silver (d) None of these
3. Which of the following metals cannot displace copper when dropped in copper sulphate solution?
   (a) Magnesium (b) Iron
   (c) Silver (d) None of these

Passage - 2

Reactive metals like sodium and potassium react with water with evolution of hydrogen gas and form corresponding hydroxides whereas the less reactive metals like magnesium, zinc, aluminium and iron are too low in reactivity to react with cold water. They react with steam when red hot forming corresponding oxides.

4. Which of the following metals forms corresponding hydroxide on reaction with water?
   (a) Mg (b) Na
   (c) Fe (d) Zn
5. Which of the following metals do not react with cold water?
   (a) Na (b) K
   (c) Mg (d) Ca
6. Which of the following metals react with steam only
   (a) Na (b) Mg
   (c) Al (d) Pt

Assertion & Reason :

DIRECTIONS : *Each of these questions contains an Assertion followed by reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.*

(a) If both **Assertion** and **Reason** are **correct** and Reason is the **correct explanation** of Assertion.
(b) If both **Assertion** and **Reason** are correct, but Reason is **not the correct explanation** of Assertion.
(c) If **Assertion** is **correct** but **Reason** is **incorrect**.
(d) If **Assertion** is **incorrect** but **Reason** is **correct**.

1. **Assertion :** Silver and gold are found in free state in nature.
   Reason : Silver and gold are noble metals.
2. **Assertion :** The elements sodium, potassium, aluminium are classified as metals because of their tendency to form cations.
   Reason : Metals are malleable, ductile and good conductors of heat and electricity.
3. **Assertion :** Zinc is used in galvanization of iron
   Reason : Galvanization increases the life of iron articles by protecting them from rusting.
4. **Assertion :** When a piece of aluminium is put in a solution of copper sulphate we get a colourless solution.
   Reason : Aluminium lies above copper in the activity series.

Multiple Matching Questions :

DIRECTIONS : *Each question has four statements (A, B, C and D) given in Column I and four statements (p, q, r and s) in Column II. Any given statement in Column I can have correct matching with one or more statement(s) given in Column II. Match the entries in column I with entries in column II.*

1.

Column I	Column II
A. Graphite	(p) Non-metal
B. Platinum	(q) Lustrous
C. Mercury	(r) highly unreactive
D. Bromine	(s) Metal

Integer Type Questions :

DIRECTIONS : *Following are integer based questions. Each question, when worked out will result in one integer from 0 to 9 (both inclusive).*

1. Metal lies above hydrogen in the activity series.
   Aluminium, Zinc, Sodium, Gold, Silver
2. How many among the following exists in liquid state?
   Murcury, Bromine, Sodium, Helium

4 ADVANCED EXERCISE
BASED ON CONNECTING TOPICS

DIRECTIONS (Qs. 1-16) : *This section contains 16 multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.*

1. The alloy brass consists of
(a) Copper and Tin
(b) Copper and Zinc
(c) Copper and Aluminium
(d) Tin and Lead

2. Duralumin is an alloy containing
(a) Magnesium and aluminium
(b) Magnesium and copper
(c) Magnesium,copper and aluminium
(d) Magnesium,copper, manganese and aluminium.

3. Sulphide ores are usually concentrated by
(a) Gravity separation (b) Froth flotation process
(c) Calcination (d) Hydraulic washing

4. The process to remove unwanted impurities from the ore is called
(a) purification (b) calcination
(c) bassimerisation (d) concentration

5. Heating of concentrated ore in absence of air for conversion into oxide ore is known as
(a) roasting (b) calcination
(c) reduction (d) none of these

6. The reducing agent in thermite process is
(a) Al (b) Mg
(c) BaO_2 (d) MnO_2

7. The method of zone refining of metals is based on the principle of
(a) Greater solubility of the impurity in the molten state than in the solid
(b) Greater mobility of the pure metal than that of the impurity
(c) Higher melting point of the impurity than that of the pure metal
(d) Greater noble character of the solid metal than that of the impurity

8. Calcination is the process of heating the ore
(a) In a blast furnace (b) In absence of air
(c) In presence of air (d) None of these

9. The chief ore of aluminium is
(a) Cryolite (b) Feldspar
(c) Kaolin (d) Bauxite

10. Electrolytic reduction method is used in the extraction of
(a) Highly electronegative elements
(b) Highly electropositive elements
(c) Transition metals
(d) Noble metals

11. Bronze is a mixture of
(a) Pb + Sn (b) Cu + Sn
(c) Cu + Zn (d) Pb + Zn

12. Chemical method used in concentration of ore is known as
(a) bleaching (b) leaching
(c) roasting (d) calcination

13. The process of extraction of metal from its ores, is known as
(a) concentration
(b) calcination
(c) purification
(d) metallurgy

14. Roasting is generally done in case of the following
(a) Oxide ores (b) Silicate ores
(c) Sulphide ores (d) Carbonate ores

15. The rocky material present in ore is known as
(a) Gangue (b) Flux
(c) Slag (d) None of these

16. Select the correct statement.
(a) A mineral can not be an ore
(b) All ore can not be a mineral
(c) All ores are minerals
(d) All minerals are ores

DIRECTIONS (Qs. 17-20) : *This section contains 4 multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE OR MORE may be correct.*

17. Which of the following alloys contain tin as one of the constituents?
(a) Bronze (b) Solder
(c) Gun metal (d) Monel

18. The process of zone refining is used in the purification of
(a) Si (b) Al
(c) Ag (d) Cu

19. The role of calcination in metallurgical operations is
(a) to remove moisture
(b) to decompose carbonate
(c) to make oxporous
(d) to achieve all the above

20. Which of the following metals is extracted by the electrometallurgical method ?
(a) Cu (b) Fe
(c) Na (d) Ag

DIRECTIONS (21-23) : *Study the given paragraph(s) and answer the following questions.*

Passage

The ores obtained from the earth contain large quantities of foreign matter. These unwanted impurities, e.g. earth particles, rocky matter, sand limestone etc. present in an ore are called gangue or matrix. Prior to the extraction of the metal from the ore, it is necessary to separate, the ore from the gangue. This separation can often be achieved by physical means since mineral and gangue generally occur as separate solid phases. The process of removal of gangue from the ore is technically known as concentration or ore dressing and the purified ore is known as concentrate.

21. Which of the following is a chemical method of concentration of ores?
(a) Froth floatation process
(b) Magnetic separation
(c) Hydraulic washing
(d) Leaching

22. The ores that are concentrated by floatation method are
(a) Carbonates (b) Sulphides
(c) Oxides (d) Phosphates

23. Magnetic separation is used to increase concentration of
(a) Horn Silver (b) Calcite
(c) Haematite (d) Magnesite

DIRECTIONS (Qs. 24-26) : *Each of these questions contains an Assertion followed by reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.*

(a) If both **Assertion** and **Reason** are **correct** and Reason is the **correct explanation** of Assertion.
(b) If both **Assertion** and **Reason** are correct, but Reason is **not the correct explanation** of Assertion.
(c) If **Assertion** is **correct** but **Reason** is **incorrect**.
(d) If **Assertion** is **incorrect** but **Reason** is **correct**.

24. Assertion : Leaching is a process of reduction.
Reason : Leaching involves treatment of the ore with a suitable reagent so as to make it soluble while impurities remains insoluble.

25. Assertion : Copper obtained after bassemerization is known as blister copper.
Reason : Blisters are produced on the surface of the metal due to escaping of dissolved SO_2.

26. Assertion : Lead, Tin and bismuth are purified by liquation method.
Reason : Lead, Tin and bismuth have low m.p. as compared to impurities.

DIRECTIONS (Qs. 27) : *Each question contains statements given in two columns which have to be matched. Statements (A, B, C, D) in column I have to be matched with statements (p, q, r, s) in column II.*

27.

	Column I		Column II
(A)	Iron	(p)	Refining by liquation
(B)	Aluminium	(q)	Electrolyte refining
(C)	Tin	(r)	Reduction by smelting
(D)	Silver	(s)	Leaching

	A	B	C	D
(a)	r	s	p	q,s
(b)	p	q	s	r
(c)	q	s	p	r
(d)	r	q	s	p

DIRECTIONS (Qs. 28-29) : *Following are integer based questions. Each question, when worked out will result in one integer from 0 to 9 (both inclusive).*

28. How many of the given elements are extracted by electrolytic refining?
Pb, Hg, Ag, Zn, Zr and Ge

29. How many of the following metals are extractexd by self reduction process?
Zn, Fe, Al, Hg, Cu, Na and Mg

SOLUTIONS

Brief Explanations of Selected Questions

1 EXERCISE

Fill in the Blanks :

1. Basic 2. Hydrogen
3. Graphite (a form of carbon)
4. Graphite (a form of carbon) or Iodine.
5. Alcoholic 6. Haematite
7. Graphite 8. Mercury
9. Nitrogen 10. Carbondioxide, Oxygen

True / False :

1.	True	**2.**	False	**3.**	True	**4.**	False
5.	False	**6.**	False	**7.**	False	**8.**	False
9.	True	**10.**	False	**11.**	False	**12.**	True

Match the Following :

1. A – (p); B – (r); C – (q); D – (s)

Very Short Answer Questions :

1. Gold and silver possess negligible or very less reactivity. Due to this they occur in native state in nature.
2. It burns easily in air because of its *low ignition temperature* so it is kept under water.
3. Sodium. [Any metal above hydrogen in activity series can be named]
4. $Mg + 2\,HCl \rightarrow MgCl_2 + H_2$
5. It results in change of colour of solution from blue to green. The colour change occurs due to displacement of Cu by Fe from $CuSO_4$ solution as shown by following equation
$$\underset{\text{(blue)}}{Fe + CuSO_4} \rightarrow \underset{\text{(green)}}{FeSO_4} + Cu$$
6. Aluminium (Al)
7. This is done to prevent rusting of iron.
8. It is the process of depositing a thin layer of zinc on the surface of metal.
9. No, copper is less reactive than iron.
10. Nitrogen.
11. Aluminium (Al)
12. No, they are not metals.
13. Non-metal do not react with acids.
14. Hydrogen

Short Answer Questions :

1. (a) Mercury or gallium
   (b) Silver
   (c) Sodium
2. (i) Air or oxygen and
   (ii) Water [i.e. wet air]
3. Refer text
4. (a) $3Fe(s) + 4H_2O(g) \longrightarrow Fe_3O_4(s) + 4H_2(g)$
   (b) $2Ca + 2H_2O \longrightarrow Ca(OH)_2 + H_2\uparrow$
   (c) $2K + 2H_2O \longrightarrow 2KOH + H_2\uparrow$
5. Refer text
6. Uses of sulphur :
   (i) Used in manufacture of dyes, matches, fire work and gun powder.
   (ii) Used in many medicines (*e.g.* skin ointments)
   (iii) Used as *fungicide* to kill harmful fungi.
7. Acidic oxides.
8. (a) Copper, because copper is a good conductor of heat and electricity.
   (b) Aluminium, because aluminium is light metal and is a good conductor of electricity.
   (c) Copper
9. When a copper vessel is exposed in moist air for a long time, it aquires a dull green coating. The green substance is the mixture of copper hydroxide [$Cu(OH)_2$] and copper carbonate ($CuCO_3$)
$$2Cu + \underset{\text{(moist air)}}{H_2O + CO_2 + O_2} \longrightarrow \underset{\text{(Green coating)}}{Cu(OH)_2 + CuCO_3}$$

Long Answer Questions :

1. Beaker A :
   $CuSO_4 + Zn \longrightarrow ZnSO_4 + Cu$
   Beaker B :
   $CuSO_4 + Fe \longrightarrow FeSO_4 + Cu$
   Beaker C :
   $ZnSO_4 + Cu \longrightarrow$ No reaction
   Beaker D :
   $FeSO_4 + Cu \longrightarrow$ No reaction.
   Beaker E :
   $ZnSO_4 + Fe \longrightarrow$ No reaction.
   In beaker A and B displacement reaction takes place.
2. **(A)** Physical properties of non-metals :
   (i) At room temperature non-metals are either solids or gases, except bromine (Br) which is a liquid.
   (ii) They are lustreless (*i.e.* dull). Except graphite and Iodine.
   (iii) They are generally brittle.
   (iv) They are bad conductors of heat and electricity (except graphite).
   (v) They are non-sonorous (*i.e.* they do not produce a metallic clink).
   (B) (a) Sulphur

(b) Iodine
(c) Graphite (a form of carbon)
(d) Graphite (a form of carbon)

3. Refer to the text

2 EXERCISE

Text-Book Exercise :

1. (a) Zinc being a metal can be beaten into thin sheets. This property of metals is called malleability. Phosphorus, sulphur and oxygen are non-metals therefore do not show this property.

2. (c) Mercury (metal), which is liquid at room temperature cannot be drawn into wires and is not ductile.

3. (a) reactive (b) good, electricity
(c) more (d) hydrogen

4. (a) (F) Non-metals generally do not react with acids.
(b) (T) Sodium reacts vigorously with oxygen and water.
(c) (F) Zinc is more reactive than copper, therefore copper cannot displace zinc from zinc sulphate solution.
(d) (F) Coal (carbon) being a non-metal cannot be drawn into wires.

5.

Properties	Metals	Non-metals
1. Appearnace	Lustrous	Dull in appearance
2. Hardness	Generally hard	Generally soft
3. Malleability	Generally malleable	Non-malleable
4. Ductility	Generally Ductile	Non-ductile
5. Heat conduction	Good Conductors	Poor Conductors
6. Conduction of electricity	Good Conductors	Poor Conductors

6. (a) Aluminium foils are used to wrap food items because aluminium is highly malleable i.e. it can be easily beaten into thin sheets and it does not react with food items.
(b) Metals are good conductor of heat. That is why immersion rods for heating liquids are made up of metallic substances.
(c) A less reactive metal cannot displace a more reactive metal from its salt solution. Copper being less reactive than zinc cannot displace zinc from its salt solution.
(d) Sodium and potassium are very reactive. They can quickly react in air and water and easily catch fire when come in contact with air. So, they are stored in kerosene.

7. No, Aluminium being a metal reacts with lemon pickle which contains acid. Metals on reaction with acids release hydrogen gas. Due to which pickle can be spoiled.

8. (i) – (d), (ii) – (e), (iii) – (c), (iv) – (f), (v) – (b), (vi) – (a)

9. (a) Copper reacts with dilute sulphuric acid to produce copper sulphate and hydrogen gas is liberated.

$$\underset{\text{Copper}}{Cu} + \underset{\substack{\text{Sulphuric}\\\text{acid}}}{H_2SO_4} \longrightarrow \underset{\substack{\text{Copper}\\\text{Sulphate}}}{CuSO_4} + \underset{\text{Hydrogen}}{H_2}$$

(b) Iron being more reactive than copper, displaces copper from copper sulphate solution. Blue colour of copper sulphate solution fades due to formation of iron sulphate.

$$\underset{\text{Iron}}{Fe} + \underset{\substack{\text{Copper}\\\text{Sulphate}}}{CuSO_4} \longrightarrow \underset{\substack{\text{Iron}\\\text{Sulphate}}}{FeSO_4} + \underset{\text{Copper}}{Cu}$$

10. (a) She should add some water in the test-tube in which gas is collected. Cover the test- tube and shake it well. Test the solution with blue and red litmus. Solution will turn blue litmus red. Hence gas is acidic in nature.
(b) Charcoal on burning reacts with oxygen to form carbon dioxide gas.

$$\underset{\substack{\text{Charcoal}\\\text{(Carbon)}}}{C} + \underset{\text{Oxygen}}{O_2} \longrightarrow \underset{\substack{\text{Carbon}\\\text{dioxide}}}{CO_2}$$

Carbon dioxide on reaction with water forms carbonic acid.

$$\underset{\substack{\text{Carbon}\\\text{dioxide}}}{CO_2} + \underset{\text{Water}}{H_2O} \longrightarrow \underset{\substack{\text{Carbonic}\\\text{acid}}}{H_2CO_3}$$

Carbonic acid turns blue litmus paper red.

11. Gold jewellery is polished by dipping jewellery in a mixture of hydrochloric acid and nitric acid called aqua regia. Outer layer of gold dissolves in aqua regia and inner shiny layer appears. Because of the loss of upper layer of jewellery its weight is reduced.

Exemplar Questions :

1. iodine
2. No, because displacement reaction does not takes place.
3. Metals are sonorous.
4. The green material is a mixture of copper hydroxide and copper carbonate formed due to reaction of copper with moist air (water, oxygen and carbon dioxide).
5. Statement (b), (d) and (e) are not correct.
(b) Metals are good conductor of electicity and also good conductor of heat.
(d) Oxides of non-metals are acidic in nature and oxides of metals are basic in nature.
(e) A more reactive metal replaces a less reactive metal from its salt solution in water.

HOTS Questions :

1. Because of its higher reactivity aluminium is expected to get corroded easily on exposure to moist air, however it is not the case. When aluminium metal is exposed to atmosphere (moist air), it combines with oxygen to form a thin but strong film of aluminium oxide Al_2O_3 on its surface. This film acts as a protective film and reduces the chemical reactivity of aluminium and protects it from corrosion. In case of iron the oxide film formed is weak and brittle that's why it appears as a rust and results into corrosion.

2. Since silver gets deposited on copper plate which was dipped in a solution of silver nitrate hence it is clear that copper metal have displaced the silver from silver nitrate solution. This shows that copper is more reactive than silver.
3. No. Aluminium is more reactive than copper so it will displace copper from copper sulphate solution. It will cause holes in the vessel and thus we can not store copper sulphate solution in an aluminium vessel.
4. When food stuffs with acid components are stored in vessels made of copper, iron or aluminum these metals may react with acidic components to produce certain poisonous materials. Due to this it is advised not to store food stuffs with acidic components in such vessels.
5. **B** [**A** is found in native state so it is less reactive and **B** is more reactive]
6. The basic copper carbonate that is formed on the surface of copper vessel reacts with acid present in lemon juice and gets dissolved. Thus it gets readily removed.
7. (a) As platinum, gold and silver are least reactive metals thus does not affected by air, water and other chemicals hence found to be most suitable for making jewellery.
   (b) Sodium, Potassium and lithium are stored under oil because, these are highly reactive metals and as they come in contact with air and moisture, get covered with a carbonate layer.

$$4Na + O_2 \longrightarrow 2Na_2O \xrightarrow{H_2O} 2NaOH \xrightarrow{CO_2} Na_2CO_3$$

$$2Na + 2H_2O \longrightarrow 2NaOH + H_2.$$

   (c) Aluminium is a good conductor of heat, therefore it is used for making utensils.

3 EXERCISE

Multiple Choice Questions :

1. (c) Tin (Sn); [It lies above copper in Activity series]
2. (a) Mercury
3. (a) **4.** (c)
5. (b) Metal oxides are basic in nature.
6. (c) **7.** (d) **8.** (c) **9.** (d)
10. (d) **11.** (a) **12.** (c)
13. (d) Gold occurs in native form, others occur in combined form.
14. (c) **15.** (d) **16.** (a)
17. (d) Non-metals are majorly available in the form of solids and gases there is only one non-metal bromine which is liquid at room temperature.
18. (c) The solution of magnesium oxide is alkaline in nature and that is why it changes red litmus paper to blue when dipped in it.
19. (c)
20. (b) Gold and Silver are most malleable metals whereas zinc metal is non-malleable and brittle.
21. (c)

More Than One Option Correct :

1. (b, c, d) Non-metals produce acidic oxides.
2. (a, b). **3.** (a, b, c) **4.** (a, b, c)
5. (b, c, d) **6.** (a, c, d)
7. (a, b, c) All these lie above copper in activity series.

Passage Based Questions :

1. (b) Potassium
2. (a) Zinc
3. (c) Silver being less reactive than copper cannot displace copper from copper sulphate solution
4. (b) Sodium reacts with water there by forming sodium hydroxide

$$2Na(s) + 2H_2O(l) \longrightarrow 2NaOH(aq) + H_2(g)\uparrow$$

5. (c) Magnesium do not react with cold water. It reacts with hot water by forming its oxide.
6. (c) Aluminium have low reactivity toward cold water. It reacts with steam only according to following equation.

$$2Al(s) + 3H_2O(g) \longrightarrow Al_2O_3(s) + 3H_2(g)\uparrow$$

Assertion & Reason :

1. (a) Noble metals are highly non-reactive metals. Noble metals can occur in free state.
2. (b)
3. (a)
4. (a) Aluminium displaces copper from copper sulphate solution to form aluminium sulphate which is a colourless solution.

Multiple Matching Questions :

1. A – q ; B – r ; C – s ; D – p

Integer Type Questions :

1. **3**
   Aluminium, Zinc and Sodium
2. **2**
   Murcury and Bromine

4 EXERCISE

1. (b) Brass is 80% copper + 20% zinc.
2. (d) It is Al = 95.5%, Cu = 3%; Mn = 1.0%, Mg = 0.5%.
3. (b)
4. (d)
5. (b) Calcination involves heating of the ore below its fusion temperature in absence of air.
6. (a) Aluminium works as reducing agent in thermite process because it displaces Fe from Fe_2O_3. Hence, option (a) is correct.

7. (a) Zone refining is based on the difference in solubility of impurities in molten and solid state of the metal. This method is used for obtaining metals of very high purity.

8. (b) **Calcination** is the heating of the ore in a suitable furnace in absence of air much below its melting point to cause decomposition and elimination of volatile products.

$$PbCO_3 \rightarrow PbO + CO_2$$

The process is generally applied to hydrated oxide or carbonate ores.

9. (d) Bauxite may be written as $Al_2O_3.2H_2O$ Cryolite is $Na_3[AlF_6]$. Feldspar are mock forming minerals. Kaolin is a Clay (China Clay).

10. (b) Highly electropositive element Na is extracted by electrolytic reduction.

11. (b) Bronze is a mixture of Cu + Sn
Option (b) is correct.

12. (b) Leaching is a chemical method of concentration of ores. Leaching involves the treatment of the ore with a suitable reagent as to make it soluble while impurities remain insoluble. The ore is recovered from the solution by suitable chemical method.

13. (d)

14. (c) In this process sulphides ores are converted into oxide ores $2ZnS + 3O_2 \rightarrow 2ZnO + 2SO_2\uparrow$

15. (a) **16.** (c)

17. (a, b, c)
Bronze - 88% Cu + 12% Sn
Solder - 11% Sn + 37% Pb + 42% Bi
Gun Metal - 86% Cu + 9.5% Sn + 2.5% Pb + 2% Zn
Monel - 60% Ni + 33% Cu + 7% Fe

18. (a) Zone refining is used to purify materials, mainly semiconductors like Si.

19. (d)

20. (c) Na is obtained by elctrolytic reduction being electropositive in nature.

21. (d)

22. (b)

23. (d)

24. (d) Assertion is false but reason is true. Leaching is a process of concentration.

25. (a) Both assertion and reason are correct and reason is the correct explanation of assertion.

26. (a)

27. (a) A – r ; B – s; C– p; D – q, s

28. **2**
Silver and Zinc are refined by electrolytic refining.

29. **2**
Hg and Cu can be reduced by self-reduction process.

Chapter 3

COAL AND PETROLEUM

INTRODUCTION

The world depends on fossil fuels for its energy. **Crude oil**, **coal** and **natural gas** formed from the prehistoric matter of plants, animals, zooplankton and other life that was buried sometimes miles deep inside the Earth and subjected to high temperatures and high pressure over millions of years. These three so-called fossil fuels include a wide assortment of carbon-based substances.

Humans have been using coal as a fuel for hundreds or even thousands of years, and coal fueled steam engines in trains long before the liquid fuel-powered engines of today's cars were in use.

Humans have known about **petroleum**, or **crude oil**, for centuries, but the substance wasn't considered terribly interesting until the mid 1800s, when it was distilled into kerosene and found to be a good, cheap alternative . Americans and others worldwide quickly adopted petroleum and learned to make an unending stream of useful products from it. Simultaneously, a worldwide obsession with striking oil was born.

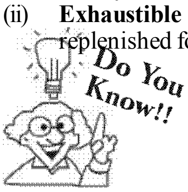
COAL AND PETROLEUM

Mankind in order to meet its basic (food, clothes and shelter) and advance needs utilizing available resources at rapid rate. Thus use of resources by human being is much faster in comparison to rate at which these resources can be replenished by nature. Some of the resources used by humans are natural like the air we breathe, water we drink, sunlight, soil, forests, fossil fuels, etc., and others are man-made resources like plastic, paper, metal alloys, glass, fabric etc.
Plants and animals are living natural resources. Air water, soil and minerals are non-living natural resources.

Types of Natural Resources

(i) **Inexhaustible/renewable resources :** Those natural resources which can be replenished or reproduced easily. For example, sunlight is a resource which will never run out as the sun is expected to last for another 5 billion years. Oxygen is renewable because it is replaced in the atmosphere as plants release oxygen during photosynthesis.
Many of the inexhaustible resources gets replenished with time quickly. But some of the inexhaustible resources also depleting as they take longer time to replenish like ground water, etc.

(ii) **Exhaustible resources :** Those natural resources which take very long time to replenish. Thus once used they cannot be replenished for mankind. For example sources like fossil fuels, top fertile layer of soil, minerals, forests, etc.



Some non-renewable resources such as metal can be used again and again. These non-renewable resources can be recycled.

Coal, Petroleum and Natural Gas are very important natural resources, and play a vital role in modern society. They are found in the earth's crust. Their easy availability and specific characteristics make them very important in the growth of industry. At present they are the chief sources of energy worldwide.

COAL

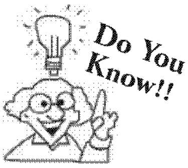
Coal, a fossil fuel, is black in colour and hard as stone. Earlier it was used to cook food and to run railway engines.
It is the largest source of energy for the generation of electricity worldwide, as well as one of the largest worldwide anthropogenic sources of carbon dioxide emissions. Gross carbon dioxide emissions from coal usage are slightly more than those from petroleum and about double the amount from natural gas. Coal is our most abundant fossil fuel resource.

Composition of Coal :

Coal is a complex mixture of organic chemical substances containing carbon, hydrogen and oxygen in chemical combination, together with smaller amounts of nitrogen and sulphur.

Deposits :

The distribution of coal deposits is not uniform in the earth's crust. To the total coal reserves in the world, Asia contributes about one third whereas North and South America contribute more than half. India has large deposits of coal. It is estimated that India has about 80 billion tones of proven coal deposits. The coal deposits are spread over in the states of Jharkhand, Madhya Pradesh and west Bengal.



The electricity formed by using coal, is known as thermal electricity and about 50% of the electricity in the United States is generated from coal.

Knowledge ENHANCER

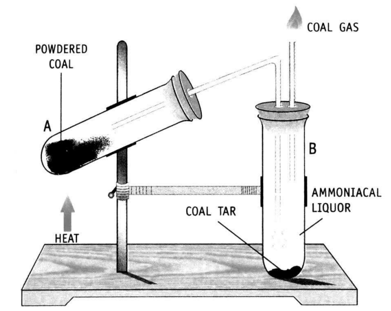
Different types and their uses

Different varieties of coal are formed as a result of carbonisation known as peat, lignite or brown coal, sub-bituminous coal, bituminous coal, and anthracite. The carbon content, also called the rank of the coal, increases progressively from lignite, a low rank coal to anthracite, a high rank coal. Different amounts of heat and pressure during the geochemical stage of coal development cause these differences in rank. It is not due to the kind of plants the coal is formed from.
Out of these four, anthracite coal is the hardest coal which has the highest content 94-98% of carbon. Anthracite and bituminous coal are the most important and are mainly used in industries. Bituminous coal is softer than anthracite and contains about 82% carbon. Lignite is youngest variety of coal and is brown in colour. It contain 70% carbon. Since coal is of low cost and reliable source of energy, it is used for the generation of electricity.

Formation of Coal :

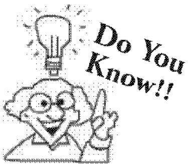
It is believed that millions of years ago the ground below the forests was split open by natural forces such as earthquakes and volcanoes. The forests were buried in the earth. Thus, the plants had no contact with oxygen. Successive layers of sediments sealed the buried plants. Over millions of years these deposits were subjected to tremendous pressure and heat which finally transformed them into coal. The chemical process involved in the transformation of plant matter into coal is called the carbonization of plant matter.

Destructive distillation of coal: Heating coal in absence of air is called destructive distillation of coal. It produces coke (non-volatile residue), coal tar (black thick viscous liquid), coal gas (mixture of combustible gases) and ammonical liquor (solution of ammonia in water). Destructive distillation of coal is carried out by heating coal strongly to 1000°C in the absence of air.



Destructive distillation of coal

Now let us discuss about various products obtained from destructive distillatioin of coal and their uses.



When coal is preserved under the right pressure and temperature for ten thousands of years you may get diamonds from these coal pieces. Diamond and coal are both made up of carbon particles. Diamond is one of the purest form of carbon.

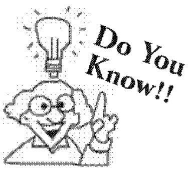
Coke

Coke is the solid carbonaceous material derived from destructive distillation of low-ash, low-sulphur bituminous coal. It is an almost pure form of carbon. It is a good fuel and burns with no smoke. The products obtained by the decomposition of coal are coal gas, carbon, coal tar, liquor ammonia and coal gas. Cokes from coal are grey, hard, and porous.

Coke is usually produced from coal; the process is called destructive distillation. Volatile constituents of the coal—including water, coal-gas, and coal-tar—are driven off by baking in an airless furnace or oven at temperatures as high as 2,000 degrees Celsius. The greater the volatile matter in coal, the more by-product can be produced, but too low or too high a level of volatile matter in the coal results in inferior coke produced in respect to coke quality properties. It is generally considered that levels of 26-29 % of volatile matter in the coal blend is good for coking purposes. Thus different types of coal are proportionally blended to reach acceptable levels of volatility before the coking process begins.

Uses : Coke is used as a fuel and as a reducing agent in smelting iron ore in a blast furnace.

Since smoke-producing constituents are driven off during the coking of coal, coke forms a desirable fuel for stoves and furnaces in which conditions are not suitable for the complete burning of bituminous coal itself.



Coke may burn with little or no smoke under combustible conditions, while bituminous coal would produce much smoke.

Coal Tar

Coal tar is a mixture of different carbon compounds. It is a thick, black liquid. The fractional distillation of coal tar gives many chemical substances which are used in the preparation of dyes, explosives, paints, synthetic fibres, drugs and pesticides. Some of these chemical substances are benzene, toluene, phenol and aniline.

On further fractionation of coal tar a large number of compounds are obtained, out of which 200 compounds have been isolated yet.

Coal Gas

Coal gas is mainly a mixture of hydrogen, methane and carbon monoxide. The gases present in coal gas are combustible, and hence it is an excellent fuel. It has a high calorific value. It was used for lighting houses, factories and streets in Mumbai (Bombay) until 1950. It was also used for cooking until recently.

Ammonical Liquor

The ammonia produced as a result of destructive distillation of coal is absorbed in water. The aqueous solution of ammonia, i.e. ammonium hydroxide solution is called ammonical liquor. It is used in the preparation of nitrogeneous fertilizers such as ammonium sulphate and ammonium superphosphate.

Knowledge ENHANCER

Coke was one of the materials used in the heat shielding on NASA's Apollo program space vehicles. This material has been used most recently as the heat shielding on the Mars Pathfinder vehicle. Although not used in modern day space shuttles, NASA had been planning to utilize coke and other materials for the heat shield for its next generation space craft, named Orion.

CONNECTING TOPIC

CHARCOAL

Charcoal is the light, black residue consisting of impure carbon and remaining ash obtained by removing water and other volatile constituents from animal and vegetation substances. Charcoal is usually produced by slow pyrolysis, the heating of substances like wood, sugar, bones or other organic matter in the absence of oxygen. The resulting soft, brittle, lightweight, black, porous material resembles coal.

Types of charcoal :

(i) wood charcoal (ii) bone charcoal (iii) sugar charcoal

(i) Wood charcoal : Wood charcoal is obtained by burning wood in limited supply of air.

Properties :

1. It is a *bad conductor* of heat and electricity.
2. It burns in air at about 400°C to form CO_2 gas

$$C + O_2 \xrightarrow{400°C} CO_2$$

3. In limited supply of air it forms carbon monoxide (CO)

$$2C + O_2 \xrightarrow{heat} 2CO$$

4. When steam is passed over red hot carbon (wood charcoal), a *mixture of* CO *and* H_2 called **water gas** is formed.

$$\underset{\text{(red hot)}}{C} + \underset{\text{(steam)}}{H_2O} \longrightarrow \underbrace{CO + H_2}_{\text{water gas}}$$

5. Wood charcoal has a property of **adsorption.**

Uses :

1. Wood charcoal is used as a *fuel* for *igniting fire*.
2. Wood charcoal is used in *gas masks to absorb poisonous gas*.
3. Wood charcoal is used as a *constituent* of *gun powder.*

(ii) Animal charcoal or Bone charcoal : It is obtained by heating bones in a retort in the absence of air.

Uses :

1. Animal charcoal is used for *decolourising brown sugar.* [Because of its property of absorbing colouring matter]
2. It is used for *purifying organic liquids*.
3. It is used as *artists colour.*

(iii) Sugar charcoal : It is obtained by dehydration of cane sugar with concentrated sulphuric acid.

$$\underset{\text{(Sucrose)}}{C_{12}H_{22}O_{11}} \xrightarrow{\text{conc. } H_2SO_4} \underset{\text{(Sugar charcoal)}}{12C(s)} + 11H_2O(l)$$

Sugar charcoal is a *very pure form of carbon.*

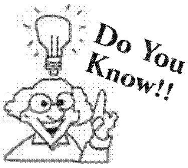
It is used in *preparation of artificial diamonds.*

PETROLEUM

Petroleum is a naturally occurring oil that consists chiefly of hydrocarbons (compounds containing carbon and hydrogen) with some other elements, such as sulphur, oxygen and nitrogen. It is now known that petroleum contains hydrocarbons of the paraffin series, with upto 100 or more carbon atoms in the chain. The unrefined form of petroleum is called crude oil.

Petroleum is also called rock oil i.e. petro = rock, oleum = oil.

It is believed that petroleum was formed from organisms living in the sea. The remains of these organisms were deposited in shallow depressions in the sea bed long, long, ago. These were covered by layers of sand and clay which compressed these remains. Over a period of millions of years, the organic matter present in the dead organisms underwent a series of processes before being finally transformed into petroleum. The petroleum migrated from the source rock to be entrapped in large underground reservoirs beneath Impermeable rocks. It often floats over a layer of water and is held in this position under pressure beneath a layer of natural gas.



Petroleum is obtained by drilling through the earth and impervious (non-porous) rock above it. The world's first oil well to yield petroleum was dug in Pennsylvania, USA, in 1859. In India, oil was first struck in Madum in Assam in 1867. The crude oil pumped out form a well is a black liquid. Because of its importance in today's world, it is referred to as ***black gold****.*

Refining

Petroleum is a mixture of several hydrocarbons. It is a foul - smelling brown black liquid. It also contains water, salt and rocky materials. It cannot be used in this crude form either as a fuel or as a basic material to produce other useful components. Before being put to use, it has to be purified or refined. The process of separating the various components of petrol from one another is known as the refining of petroleum. This is done by a process called fractional distillation which is based on the fact that the different components of petroleum have distinctly different boiling points.

They are separated in a large fractionating column. Crude oil is piped to the refinery from a well. It is washed with acid and alkali solutions to remove the basic and acidic impurities respectively. Crude oil is now heated to about 673 K and fed at the base of fractionating column. All the components except asphalt are in the vapour state. As the mixture of hot vapours rises up in the column, it begins to cool. The component with the highest boiling point condenses first and is collected. Those with low boiling points condense later. The residual gases escape uncondensed from the upper part of the column. The various components condensed at different heights of the column are collected separately. The components obtained at different heights in order from the bottom are asphalt, lubricating oil, paraffin wax, fuel oil, diesel, kerosene, petrol and petroleum gas.

Some important fractions of petroleum and their uses

Fraction	Boiling Point	Number of Carbon Atom	Products Obtained/Uses
Petroleum gas	below 40° C	1–4	LPG (Liquefied petroleum gas)– fuel for home and industry
Gasoline and naphtha	40°–160° C	4–10	Petrol–motor fuel, aviation fuel, solvent for dry-cleaning
Kerosene	150°–250° C	10–16	Kerosene–domestic fuel, jet engine fuel
Light oil	250°–350° C	16–20	Diesel–fuel for motor vehicles (especially heavy vehicles), electric generators
Heavy oil	250°–350° C	20–25	Useful organic chemicals
Residue	over 350° C	over 25	Lubricating oil–lubricating machinery; Paraffin wax–candles, vaseline, ointments etc; Asphalt (bitumen)–paints, road surfaces

CHECK Point

Why is petroleum refined?

SOLUTION

Petroleum or crude oil that is pumped out from oil wells is not pure. It is a mixture of several compounds. It is refined and converted into various components so that they can be used for various purpose.

Petrochemicals

The substances which are obtained form petroleum and natural gas are called petrochemicals. These are used in the manufacture of detergents, synthetic fibres like polyster and nylon, polythene and plastics. Petroleum is also called black gold because of its wide application as a fuel in industry and home.

CONNECTING TOPIC

Petroleum gas : It is a mixture of ethane, propane and butane. Its main constituent is butane which burns giving off a lot of heat. Butane is easily liquefied under high pressure. In the liquid form it is supplied in cylinders and is commonly known as Liquefied petroleum Gas (LPG). It is a colourless, odourless and inflammable gas. A domestic gas cylinder contains about 14.2 kg of LPG. A strong - smelling substance called ethyl mercaptan is added to the LPG to detect the leakage of gas from the cylinder. On being lighted, it burns with a blue flame. One gram of LPG produces about 50 kJ of heat.

LPG should be used with care. Any accidental leakage can cause an explosion. If there is any leakage of gas from the cylinder, the following precautions should be immediately taken -

- Any open flame in the vicinity of the gas should immediately be extinguished.
- Electric bulbs should also be switched off.
- All doors and windows of the room in which the cylinder is kept should be opened to allow the gas to escape.
- The tube and joints attached to the cylinder should be systematically checked for defects.

NATURAL GAS

Occurence :

Natural gas is naturally occuring mixture of gaseous hydrocarbons. It is found in porous sedimentary rock in the earth's crust, usually in association with petroleum deposits.

Some times natural gas is also found at places where there are no petroleum deposits. But most wells produce natural gas as well as petroleum. Infact natural gas is obtained as a co-product in petroleum mining.

Natural gas is found along with petroleum in reservoirs under the ground. It is chiefly made up of methane, though butane and propane are also present in small proportions. It can be easily transported through pipes and is a clean non-polluting fuel. It is stored under high pressure as **compressed natural gas (CNG).** CNG is used for power generation. In some parts of India, e.g. Vadodara, CNG is supplied to homes and factories through pipes and directly used as a fuel. CNG is also now being used as a non-polluting fuel for vehicles. Natural gas is used as a starting material for the manufacture of a number of chemicals and fertilizers.

Deposits of natural gas are found in India in Tripura, Rajasthan, Maharashtra and in the Krishna-Godavari delta.

Formation :

Natural gas is formed from the decomposition of organic matter buried under sea beds millions of years ago, by a process called anaerobic fermentation. Anaerobic fermentation is the process of decomposition of complex organic matter, in absence of air by means of micro-organisms.

Composition :

Natural gas consists mainly of methane (about 85%), ethane (up to about 10%), propane (about 3%) and butane. Carbon dioxide, nitrogen, oxygen, hydrogen sulphide and sometimes helium may also be present.

Uses :

(i) Natural gas burns to produce heat. It can be used in homes and factories as a fuel. In homes and factories, it is supplied through pipes. Vadodara in Gujarat has a network of such pipelines. Though the transportation of gas through pipelines is costly in the beginning, it become economical in the long run. Its transportation does not need any additional storage. Such a network of pipelines is being planned to facilitate the use of this energy resource in several parts of our country.

(ii) Compressed Natural Gas (CNG) has been found to be an alternative to petrol as automobile fuel.

(iii) Natural gas is rich source of hydrogen gas which is needed in manufacture of chemical fertilizers.

CONSERVATION OF NATURAL RESOURCES

Why do we need to conserve our natural resources ?

- These natural resources are essential to meet our routine needs.
- With the rapid increase in population these days the amount or quantity of resources required to meet our needs also increases vastly.
- Rapid industrialisation and urbanisation results into depletions of forests and other natural resources.
- Mis-management or inefficient use of natural resources as a result of unawareness of their importance.

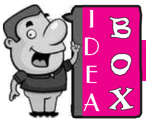
Many of the important natural resources like minerals, coal and petroleum are exhaustible and takes millions of years to replenish.

Excessive use of many natural resources like coal, petroleum, forests results into contamination of environment. Following are some examples,

(i) Burning of fossil fuels, cutting forests leads to air pollution and global warming.

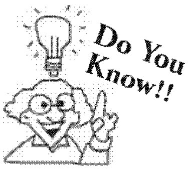
(ii) When petroleum products comes into contact with soil and water it results into their contamination.

(iii) Accidental oil spillages on sea leads to massive water pollution.



idea box

Find ways in the classroom to reduce, reuse, and recycle. Brainstorm different ideas together and create guidelines to share to your classmates and family members. You can find ways to reduce waste in your school, and reuse materials to conserve natural resources. If possible, with the help of your teacher start a recycling drive and challenge schoolmates to see how much they can recycle. Nearly everything we use can be reduced, reused, or recycled!



In India, the Petroleum Conservation Research Association (PCRA) advises people with methods of saving petrol/diesel while driving.

Some tips:

- *Drive at a constant and moderate speed as far as possible.*
- *Switch off the engine at traffic signals or at places where you have to wait.*
- *Ensure correct tyre pressure.*
- *Ensure regular maintenance of the vehicle.*

SUMMARY

There are two types of natural resources: Inexhaustible Natural Resources and Exhaustible Natural Resource

Examples of inexhaustible natural resources are air, sunlight, etc.

Example of exhaustible natural resources are coal, petroleum, forest, minerals, etc.

Coal and petroleum are known as fossil fuels since they are formed from remains of living organism.

Coal is used to produce electricity in thermal power stations.

Coal gives carbon dioxide when burnt in air.

Coal is mined from earth and processed to obtain many useful products, such as coal, coke, coal tar, etc.

Coke is almost pure form of carbon.

It is obtained from destructive distillation of a particular type of coal.

Coal tar is obtained as one of the by-products while obtaining the coke or coal gas.

Coal tar is used in the manufacturing of many products; such as synthetic dyes, drugs, explosive, perfumes, plastics, roofing material, etc. as starting material.

Petroleum is a dark oily liquid having unpleasant odour.

Petroleum is formed from the dead remains of living organisms which lived in sea and were buried millions of year ago below the sea bed.

After extraction, petroleum is refined to obtain petrol, diesel, lubricating oil, wax, paraffin, bitumen, kerosene, etc. Place where refining process of petroleum is done is called petroleum refinery.

Natural gas is naturally occurring hydrocarbon gas. Natural gas is a mixture of methane, higher alkanes, carbon dioxide, etc.

Natural gas is found in deep underground rock formation as fossil fuel. It is one of the most important fuel.

Natural gas can be transported through pipeline and can be compressed to smaller volume.

Compressed natural gas (CNG) is used for running vehicle. Natural gas can be used directly for burning as fuel. It is a clean fuel as it creates least pollution.

Natural gas is used as starting material in many industries, such as fertilizer and chemical.

In India natural gas reserves are found in Tripura, Rajasthan, Maharashtra, and Krishna and Godavari delta.

Petroleum, coal, natural gas, etc. are natural resources but available in limited amount. Thus, it is necessary to use them with care.

1 EXERCISE

Fill in the Blanks :

DIRECTIONS : *Complete the following statements with an appropriate word / term to be filled in the blank space(s).*

1. Natural gas mainly consists of ....
2. The main products obtained from the destructive distillation of coal are .... compounds .... tar and .... gas.
3. We can separate petroleum into its various fractions by fractional/distillation because the different fraction have different ....
4. The process of conversion of wood into coal is called ....
5. .... is the variety of coal with maximum carbon content.
6. Petroleum and natural gas were formed from dead ....
7. In fractional distillation, hydrocarbons with the .... (highest/lowest) boiling points condense first.
8. Petroleum or crude oil pumped out from .... is not pure.
9. Coal mining causes many .... impacts on environment.
10. These days .... is being used in vehicles in place of petrol and diesel.
11. Lignite contains about .... percent carbon.
12. Coal gas contains ....and .... gases.
13. Producer gas is a mixture of .... and ....
14. The crude oil is heated to a high temperature and passed into .... column.
15. Formation of fissil fuels is a very .... process.
16. The black thick liquid with .... smell is known as coal tar.
17. Excessive burning of fossil fuels is a major cause of ....

True / False :

DIRECTIONS : *Read the following statements and write your answer as true or false.*

1. The percentage of carbon in *anthracite* is 60%.
2. Petroleum is called black gold.
3. Bituminous coal is softer than anthracite.
4. Petroleum is a fossil fuel
5. Petroleum is formed by the decomposition of remains of marine
6. The refined form of petroleum is called crude oil
7. Petrol and diesel are obtained from a natural resources called petroleum
8. Coal is the purest form of carbon
9. The destructive distillation of coal gives coke, coal tar, coal gas, etc.
10. Peat contains 96% carbon and burns without smoke
11. The distribution of coal deposits is uniform in the earth's crust
12. Coke is 98% carbon like charcoal it is good fuel and burns without smoke
13. Coal tar is a mixture of different carbon compounds. It is a thick, black liquid
14. Ammoniacal liqour contains ammonia produced as a result of destructive distillation of coal is absorbed in water
15. Methane is released during mining of coal.
16. Petroleum is formed from the fossils remains of dead animals.
17. Inhaling coal dust is useful for humans.
18. Carbonisation is formation of coal from the carbon content of dead plants.
19. Compressed natural gas is a non polluting fuel.
20. Petroleum gets its name from petra and oleum means rock oil.
21. Natural gas can be sent to the houses through pipes.
22. Oxygen in air is an exhaustible natural resource.
23. Wildlife is an exhaustible natural resource.
24. CNG is less polluting fuel than petrol and diesel.

Match the Following :

DIRECTIONS : *Following question contains statements given in two columns which have to be matched. Statements (A, B, C and D) in column I have to be matched with statements (p, q, r and s) in column II.*

1.

	Column-I		Column-II
A.	Natural gas	(p)	It is produced in marshy areas by the action of bacteria feeding on dead vegetation.
B.	L.P.G	(q)	Contains mainly methane.
C.	Kerosene	(r)	Used as a domestic fuel.
D.	Bio-gas	(s)	Obtained as a liquid fraction of refining of petroleum

2.

	Column-I		Column-II
A.	Kerosene	(p)	Road surfacing
B.	Paraffin wax	(q)	Fuel for vehicles
C.	Bitumen	(r)	Manufacturing candles
D.	Gasoline	(s)	Jet engine fuel

Very Short Answer Questions:

DIRECTIONS : *Give answer in one word or one sentence.*

1. What is producer gas ?
2. What is water gas ?
3. What is meant by carbonization ?
4.

Fuel	In year 2011
Petroleum	23
Nuclear power	5
Solid biomass and organic waste	23
Natural gas	8
Coal	41

On the basis of the data given in the table above what is the primary source of energy in India ?

5. Give name of a non-polluting fuel which can be used in vehicles.
6. Why is petroleum called black gold ?
7. Name the gas which is produced when coal is heated in absence of air.
8. Where was the first oil well drilled ?
9. Expand PCRA.
10. Which gas is produced when coal burns in air ?
11. Is kerosene a fossil fuel ?
12. Which fuel is used in jet aeroplanes ?
13. Name the petroleum product used to manufacture candles, vaseline, grease, polish, etc.

Short Answer Questions :

DIRECTIONS : *Give answer in 2-3 sentences.*

1. Classify the following as exhaustible and inexhaustible natural resources :
   Water, Minerals, Coal, Natural Gas, Petroleum, Sunlight, Wind, Forests
2. How is petroleum formed?
3. Why are coal, petroleum and natural gas called fossil fuels?
4. List five things that you can do to help in preventing an energy crisis.
5. Why do different gases condense to liquids on different trays in the fractionating column ?
6. What do you mean by emission standard used in automobiles ?
7. What are the measures given by PCRA for saving petrol and diesel ?
8. Why is depletion of coal, petroleum and forest a matter of concern ?
9. (a) What is coal?
   (b) Where is coal found?

Long Answer Questions :

DIRECTIONS : *Answer the following questions in detail.*

1. Discuss some important methods to conserve our natural resources.
2. Sunlight and water are inexhaustible natural resources. How can they help in saving exhaustible resources ?
3. Write down two uses of each of the natural resources given in table:

S.No.	Natural Resources	Uses
1	Coal	
2	Coke	
3	Coal tar	
4	Coal Gas	

2 EXERCISE

Text-Book Exercise :

1. What are the advantages of using CNG and LPG as fuels?
2. Name the petroleum product used for surfacing of roads.
3. Describe how coal is formed from dead vegetation. What is this process called?
4. Fill in the blanks :
   (a) Fossil fuels are __________, ___________ and ____________ .
   (b) Process of separation of different constituents from petroleum is called ___________.
   (c) Least polluting fuel for vehicle is _________.
5. Tick True/False against the following statements :
   (a) Fossil fuels can be made in the laboratory. (T/F)
   (b) CNG is more polluting fuel than petrol. (T/F)
   (c) Coke is almost pure form of carbon. (T/F)
   (d) Coal tar is a mixture of various substances. (T/F)
   (e) Kerosene is not a fossil fuel. (T/F)
6. Explain why fossil fuels are exhaustible natural resources.
7. Describe characteristics and uses of coke.
8. Explain the process of formation of petroleum.
9. The following Table shows the total power shortage in India from 1991– 1997. Show the data in the form of a graph. Plot shortage percentage for the years on the Y-axis and the year on the X-axis.

S. No.	Year	Shortage (%)
1	1991	7.9
2	1992	7.8
3	1993	8.3
4	1994	7.4
5	1995	7.1
6	1996	9.2
7	1997	11.5

Exemplar Questions :

1. What does CNG stand for and why is it considered to be a better fuel than petrol ?
2. Write two important uses of coke.
3. Name the products obtained and their uses when coal is processed in industry.
4. Write some important uses of the various constituents of petroleum.
5. Name the petroleum product used as the fuel for stoves, lamps and jet aircrafts.

HOTS Questions :

1. What is meant by refining of petroleum ? What is a refinery?
2. How does coal mining affect the environment ?
3. Justify that water is a limitless resource.
4. How many substances are found in coal tar ?
5. Name some non-conventional or alternative sources of energy.
6. All renewable resources are inexhaustible. Do you agree ?
7. In a petroleum well, crude oil is found above water. Which two peoperties must petroleum have to form a layer above water ?

3 EXERCISE

Multiple Choice Questions :

DIRECTIONS : *This section contains 33 multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.*

1. Select the one that is **not** derived from fossil fuel.
(a) LPG (b) Kerosene
(c) Diesel (d) Biogas

2. The conversion of wood into coal occurs by a biochemical process that takes over millions of years and is known as ....
(a) catenation (b) carbonisation
(c) pyrolysis (d) destructive distillation

3. Water gas is
(a) $CO + CO_2$ (b) $CO + N_2$
(c) $CO + H_2$ (d) $CO + N_2 + H_2$

4. Main constituent of marsh gas is
(a) ethane (b) acetylene
(c) ethyne (d) methane

5. Petroleum refining is –
(a) distillation of petroleum to get different fractions of it.
(b) obtaining aromatic compounds from aliphatic compounds present in petroleum.
(c) cracking of petroleum to get gaseous hydrocarbons.
(d) purification of petroleum.

6. The natural petroleum contains
(a) saturated hydrocarbons.
(b) cyclic saturated hydrocarbons.
(c) compounds of sulphur.
(d) All of these.

7. Kerosene is used as fuel because it is
(a) less volatile (b) more volatile
(c) cheap (d) abundantly available

8. The origin of petroleum is indicated by the fact that
(a) Its constituents can be separated by fractional distillation.
(b) Petroleum was formed from living sea organisms which after death get deposited in sea bed for long period of time.
(c) Petroleum contains traces of chlorophyll.
(d) Oil fields are located with the help of seismograph.

9. Gasoline is obtained from crude petroleum oil by its
(a) fractional distillation
(b) vacuum distillation
(c) steam distillation
(d) pyrolysis

10. Natural gas is a very important fossil fuel because
(a) it is easy to transport through pipes and emits no pollutants.
(b) it is a mixture of various constituents
(c) it is used as a fuel in many industries
(d) it is a tough porous and black substance

11. Natural gas mainly contains
(a) ethane (b) butane
(c) propane (d) methane

12. Kerosene is used as fuel
(a) in home and industry
(b) for heavy motor vehicles
(c) for stove, lamps and jet aircraft
(d) for electric generator

13. We can use coke
(a) as an oxidising agent (b) as a reducing agent
(c) in printers ink (d) as electrode.

14. Coal tar is a
(a) black, thick liquid.
(b) dark, oily liquid.
(c) tough, porous and black substance.
(d) gas.

15. Now-a-days coal gas is used as a source of
(a) light (b) heat
(c) electricity (d) steam

16. Which type of coal has highest percentage of carbon?

(a) Anthracite (b) Bituminous

(c) Peat (d) Lignite

17. Coal gas is a mixture of –

(a) $CH_4 + H_2 + CO$ (b) $C_4H_{10} + H_2$

(c) $C_4H_{10} + H_2O$ (d) $C_2H_6 + H_2 + O_2$

18. Gasoline is the name of

(a) Crude oil

(b) The fraction of petroleum condenses with in the range of 343–393 K and have composition of $C_7 - C_9$.

(c) The mixture of uncondensed gases produced in the distillation of crude oil

(d) The mixture of residue and gas oil obtained in the distillation of crude oil

19. The order of appearance of the following with rising temperature during the refining of crude oil is

(a) kerosene oil, gasoline, diesel

(b) diesel, gasoline, kerosene oil

(c) gasoline, diesel, kerosene oil

(d) gasoline, kerosone oil, diesel

20. Petrol and diesel are obtained from a natural resources called

(a) sun (b) petroleum

(c) coal tar (d) coal gas

21. The process of heating coal in the absence of air is called

(a) Fractional distillation

(b) Distillation

(c) Destructive distillation

(d) None of these

22. Which is the best variety of coal?

(a) Peat (b) Lignite

(c) Anthracite (d) Bituminous

23. Natural gas mainly contains

(a) ethane (b) butane

(c) propane (d) None of these

24. The non-combustible element of producer gas is

(a) CO (b) N_2

(c) H_2 (d) H_2O

25. Destructive distillation of coal produces

(a) coal gas (b) natural gas

(c) biogas (d) None of these

26. Lignite coal contains how many percentage of carbon?

(a) 60% (b) 67%

(c) 64% (d) 88%

27. What is the actual composition of natural gas.

(a) CH_4(85%), C_2H_6(10%), C_3H_8 (3%) and small amount of C_4H_{10}

(b) CH_4 (10%), C_2H_6 (85%), C_3H_8 (3%) and small amount of C_4H_{10}

(c) CH_4 (88%), C_2H_6 (10%) and small amount of C_3H_8 and C_4H_{10}

(d) CH_4 (65%), C_2H_6 (20%), C_3H_8 (10%) and 5% of C_4H_{10}

28. Which of the following is boiling range of diesel oil

(a) 573 K to 623 K (b) 423 K to 573 K

(c) 622 K to 673 K (d) 373 K to 623 K

29. Which of the following sources of energy can be a good alternative to coal in a power station ?

(i) Geothermal energy

(ii) Energy from water

(iii) Energy from petrol

(iv) Energy from plants

(a) (i) and (ii) only (b) (iv) only

(c) (i), (ii) and (iii) only (d) All of the above

30. Read the following statements and mark the correct ones from the given options.

(i) Coal, petroleum and natural gas are called fossil fuels.

(ii) Coal and natural gas are two exhaustible substances.

(iii) Coke is used in manufacture of steel

(iv) Fossil fuels are present in limited quantities

(a) (i) and (ii) (b) (i) and (iv)

(c) (i), (ii) and (iii) (d) (i), (ii), (iii) and (iv)

31. Besides the risk of pollution, fossil fuels also pose a risk of

(a) global warming (b) water pollution

(c) leakage (d) explosion

32. Consider the following statements.

1. Natural gas can be supplied to homes and factories through pipes.
2. Natural gas is obtained by fractional distillation of crude oil.
3. Natural gas is a cleaner fuel because on burning only water is produced.
4. Natural gas is an exhaustible source of energy like fossil fuels.

Which alternative has the correct statements ?

(a) 1 and 3 (b) 2 and 3

(c) 1 and 4 (d) 1, 3 and 4

33. Coal is a fossil fuel and it cannot be prepared in a laboratory or industry because the formation of coal

1. is a very slow process
2. need very low pressure and low temperature
3. need very high pressure and high temperature
4. causes air pollution

Select the correct alternative

(a) 1 and 2 (b) 2 and 4

(c) 1 and 3 (d) 4 and 3

More than One Option Correct :

DIRECTIONS : *This section contains 7 Multiple Choice Questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE OR MORE may be correct.*

1. Select the correct statements about coke

(a) It is a crystalline form of carbon

(b) It contains about 90% carbon

(c) It is used as reducing agent

(d) It is a black porous substance

2. Choose the products obtained from fossil fuels.

(a) Coal (b) Coke

(c) Coal tar (d) Oxygen

3. Which of the following is not obtained on fractional distillation of petroleum?

(a) Gasoline (b) Light oil

(c) Coal gas (d) Kerosene

4. Which of the following are produced from coal tar?

(a) Synthetic dyes (b) Drugs

(c) Perfumes (d) Soap and detergent

5. Which of these is/are not fossil fuels ?

(a) Coal (b) LPG

(c) Biogas (d) Natural gas

6. Which of the following statements is/are true ?

(a) Petroleum is a fossil fuel.

(b) Petroleum is formed by the decomposition of remains of marine.

(c) The refined form of petroleum is called crude oil.

(d) Petrol and diesel are obtained from a natural resource called petroleum.

7. Mark the incorrect statements.

(a) Fossil fuels can be made in laboratory

(b) CNG is more polluting than petrol

(c) Petroleum is a mixture of various oxides of carbon

(d) Coal tar is a mixture of various substances

Passage Based Questions :

DIRECTIONS : *Study the given paragraph(s) and answer the following questions.*

Passage

When petroleum is heated in a fractionating column, various fractions are obtained at various heights of the column. More volatile liquid goes upto the top and least volatile liquid remains at the bottom. Various gaseous fractions are condensed according to their boiling points.

1. Which will condense near the top, petrol, diesel oil, fuel oil or kerosene ?

(a) Petrol (b) Diesel oil

(c) Fuel oil (d) Kerosene

2. What can you say about the boiling points of liquids that collect at the bottom as residue ?

(a) Their boiling points must be very low.

(b) Their boiling points must be much higher

(c) Their temperature is equal to the temperature of column

(d) None of the above

3. As the gas reaches at the height where temperature is equal to or just below its boiling point it will

(a) condense to form a liquid

(b) remain in gaseous state

(c) condense to form a solid

(d) escape from the column

Assertion & Reason :

DIRECTIONS : *Each of these questions contains an Assertion followed by reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.*

(a) If both **Assertion** and **Reason** are **correct** and Reason is the **correct explanation** of Assertion.

(b) If both **Assertion** and **Reason** are correct, but Reason is **not the correct explanation** of Assertion.

(c) If **Assertion** is **correct** but **Reason** is **incorrect**.

(d) If **Assertion** is **incorrect** but **Reason** is **correct**.

1. **Assertion :** Coal gas is a mixture of methane, hydrogen and carbon monoxide.

Reason : It is obtained when coal is burnt in excess of air.

2. **Assertion :** Anthracite is the purest form of coal.

Reason : It contains about 50% carbon.

3. **Assertion :** CNG and LPG are clean fuels.

Reason : They do not leave any residue on burning.

4. **Assertion :** Petroleum or crude oil pumped out of oil well is not pure.

Reason : Petroleum is refined to get various fraction which can be used for a specific purpose.

5. **Assertion :** Petrol is more volatile than diesel oil.
Reason : Petrol condenses near the top of the column than diesel oil.

6. **Assertion :** Air and sunlight are inexhaustible natural resources.
Reason : Air and sunlight are present in unlimited quantity in nature.

Multiple Matching Questions :

DIRECTIONS : *Each question has four statements (A, B, C and D) given in Column I and four statements (p, q, r and s) in Column II. Any given statement in Column I can have correct matching with one or more statement(s) given in Column II. Match the entries in column I with entries in column II.*

1.

	Column-I		Column-II
A.	Peat	(p)	98% carbon
B.	Lignite	(q)	90% carbon
C.	Anthracite	(r)	70% carbon
D.	Lampblack	(s)	60% carbon

Integer Type Questions :

DIRECTIONS : *Following are integer based questions. Each question, when worked out will result in one integer from 0 to 9 (both inclusive).*

1. Number of exhaustable resources among the following is
Air ,Forests , coal , sunlight, petroleum

2. How many products are obtained from destructive distillation of coal

3. How many among the following are fossil fuels
Coal, petroleum,Charcoal,Wood.

4. The given reaction represents formation of producer gas:
$2CO + O_2 + 4N_2 \rightarrow xCO + 4N_2$
Here x is

4 ADVANCED EXERCISE
BASED ON CONNECTING TOPICS

DIRECTIONS (Qs. 1-8) : *This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.*

1. Activated charcoal is used in gas masks because
(a) it is a good adsorbent
(b) it is a good reducing agent
(c) it burns without smoke
(d) it is highly active

2. LPG is a mixture of
(a) $C_6H_{12} + C_6H_6$ (b) $C_4H_{10} + C_3H_8$
(c) $C_2H_2 + C_2H_4$ (d) $CH_4 + C_2H_4$

3. On the basis of following features identify the correct process
I. This process can be carried out with or without catalyst
II. This process is carried out to meet the increasing demands of gasoline
III. In this process higher hydrocarbons breakdown to give smaller hydrocarbons
(a) Refining (b) Destructive distillation
(c) Cracking (d) Rerforming

4. On the basis of following features identify the correct gas
I. This gas is commonly known LPG.
II. This gas is commonly used as domestic fuel
III. A strong smelling substance called ethyl mercaptan is added to detect the leakage of this gas.
(a) Ethane (b) Butane
(c) Propane (d) Butene

5. $$C_{12}H_{22}O_{11} \xrightarrow[H_2SO_4]{Conc.} 12C + 11H_2O$$
Which of the following is obtained in the above reaction ?
(a) Animal charcoal (b) Sugar charcoal
(c) Coke (d) Wood charcoal

6. Ethyl mercaptan is added to LPG
(a) to give colour to it (b) to give volume to it
(c) to give smell to it (d) to make it liquid

7. Mark the incorrect statement.
(a) Burning of coal in a sufficient amount of oxygen produces carbon dioxide.
(b) When coal burns in insufficient amount of oxygen, carbon monoxide is formed.
(c) Charcoal is a better fuel than kerosene to be used as fuel for cooking at home.
(d) LPG is considered to be a good fuel for domestic use.

8. Butane gas is used for fillling cylinders to be used as LPG because
(a) it is easily available
(b) it is easily compressed into a liquid and stored in cylinders
(c) it is stored in gaseous state only in the cylinder
(d) it is the cheapest ga available

DIRECTIONS (Qs. 9-11) : *This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE OR MORE may be correct.*

9. Choose the correct statements :
(a) Animal charcoal is obtained by heating bones in a retort in absence of air.
(b) Animal charcoal is used for decolourising brown sugar.
(c) Sugar charcoal is used in preparation of synthetic diamonds
(d) Adsorption is a surface phenomenon.

10. On dehydration of cane sugar with concentrated sulphuric acid, we get
(a) Sugar charcoal
(b) an allotrope of carbon that is amorphous
(c) an allotrope of carbon that is a very pure form of carbon.
(d) wood charcoal.

11. Which of the following precaution (s) should be taken if in case there is any leakage of gas.
(a) All doors and windows of the room in which cylinder is kept should be opened.
(b) All lights of the room should be turned on.
(c) Any open flame nearby should be immediately extinguished.
(d) All of the above are incorrect.

DIRECTIONS (Qs. 12-14) : *Study the given paragraph(s) and answer the following questions.*

Passage

Liquefied petroleum gases are a feedstock for the petrochemical industry. LPGs undergo a very complex process called pyrolysis, taking place at very high temperatures. The resulting products are called olefins - ethylene and propylene. These are subsequently polymerized and transformed into polymers or plastic, such as polyethylene, polypropylene, or other products. Thus, the polyethylene bags, disposable dishware, food packing and wrapping materials we use in everyday life are produced from liquefied gases.

12. Liquefied Petroleum Gas (LPG) consists of mainly ?
(a) Methane Ethane and Hexane
(b) Ethane Hexane and Nonane
(c) Methane Butane and Propane
(d) Ethane Nonane and Methane

13. Products obtain during pyrolysis of Liquefied Petroleum Gas are
(a) Ethylene and propylene
(b) Methylene and Ethylene
(c) Butylene and Propylene
(d) Ethane and butane

14. After polymerization Liquefied Petroleum Gas (LPG) is converted to get
(a) Camphor
(b) Polymers
(c) Coal
(d) None of these

DIRECTIONS (Qs. 15-19) : *Each of these questions contains an Assertion followed by reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.*

(a) If both **Assertion** and **Reason** are **correct** and Reason is the **correct explanation** of Assertion.
(b) If both **Assertion** and **Reason** are correct, but Reason is **not the correct explanation** of Assertion.
(c) If **Assertion** is **correct** but **Reason** is **incorrect**.
(d) If **Assertion** is **incorrect** but **Reason** is **correct**.

15. **Assertion :** The main constituent of LPG is butane.
Reason : A small quantity of mercaptan is added to cooking gas cylinders supplied for domestic use.

16. **Assertion :** Sugar charcoal is prepared by dehydration of sugar.
Reason : Sulphuric acid is a dehydrating agent.

17. **Assertion :** Charcoal does not produce any smoke.
Reason : Charcoal contains only carbon.

18. **Assertion :** Bone charcoal cannot act as a reducing agent
Reason : Since bone charcoal contains maximum percentage of calcium phosphate so it can not act as a reducing agent.

19. **Assertion :** Wood charcoal is used as one of the constituents of gun powder.
Reason : Word charcoal is used as a deodorant and disinfectant.

DIRECTIONS (Qs. 20-21) : *Each question contains statements given in two columns which have to be matched. Statements (A, B, C, D) in column I have to be matched with statements (p, q, r, s) in column II.*

20.

	Column-I		Column-II
A.	Produce less pollutants on burning	(p)	LPG
B.	Ethyl marcaptan is added to check its leakage	(q)	Wood charcoal
C.	Used as a fuel in automobiles	(r)	CNG
D.	Fuel obtained by burning wood in limited supply of air	(s)	Sugar charcoal

21.

	Column-I		Column-II
A.	Crude oil refining	(p)	Petrol
B.	Domestic and jet engine fuel	(q)	Coal gas
C.	Coal processing	(r)	Kerosene
D.	Mixture of H_2, CH_4 and CO	(s)	Coal Tar

DIRECTIONS (Qs. 22-23) : *Following are integer based questions. Each question, when worked out will result in one integer from 0 to 9 (both inclusive).*

22. How many of them are the main components of LPG.
C_3H_8, C_4H_{10}, C_2H_6, CH_4, C_2H_2, C_2H_4, C_5H_{12}

23. How many types of charcoal are known?

SOLUTIONS

Brief Explanations of Selected Questions

1 EXERCISE

Fill in the Blanks :

1. Methane
2. Ammonium, Coal, Coal
3. Boiling point
4. Carbonisation
5. Anthracite
6. Organism
7. highest
8. oil wells
9. adverse (bad)
10. CNG
11. 38
12. hydrogen, methane, carbon monoxide
13. carbon monoxide and nitrogen
14. fractionating
15. slow
16. unpleasant
17. air pollution

True / False :

1. False
2. True
3. True
4. True
5. True
6. False : The unrefined form of petroleum is called crude oil
7. True
8. False.
9. True
10. False.
11. False.
12. True
13. True
14. True
15. True
16. True
17. False : Inhaling coal dust is harmful for humans.
18. True
19. True : CNG is a cleaner and non-polluting fuel.
20. True : Petra = rock, oleum = oil.
21. True
22. False
23. True
24. True

Match the Following :

1. $A \rightarrow (q)$, $B \rightarrow (r)$, $C \rightarrow (s)$, $D \rightarrow (p)$
2. $A \rightarrow (s)$, $B \rightarrow (r)$, $C \rightarrow (p)$, $D \rightarrow (q)$

Very Short Answer Questions :

1. Producer gas is a mixture of flammable gases (CO & H_2) and nonflammable gases (mainly N_2 and CO_2).
2. Water gas is a mixture of carbon mono oxide and H_2.
3. It refers to heating of a substance in absence of air.
4. Coal is the primary source of energy in India.
5. Compressed natural gas or CNG is a non-polluting fuel.
6. Petroleum is called black gold because of its important uses in all the industries and daily life.
7. Coal gas.
8. The first oil well was drilled at Pennsylvania, USA in 1859.
9. Petroleum Conservation Research Association.
10. Carbon dioxide is produced when coal burns in air.
    $C + O_2 \rightarrow CO_2$
11. Yes, It is obtained from fractional distillation of petroleum.
12. The fuel used in Jet aeroplanes is unleaded Kerosene (Jet A) or a neptha-kerosene blend (Jet B).
13. Praffin wax

Short Answer Questions :

1. **Exhaustible natural resources :**
   Minerals, Coal, Natural Gas, Petroleum, Forests
   Inexhaustible natural resources :
   Water, Sunlight, Wind
2. Petroleum is formed from organisms living in the sea. As these organisms die, their bodies settle down at the bottom of the sea and get covered with layers of sand and clay. Over millions of years, absence of air, high temperature and pressure transform the dead organisms into petroleum and natural gas.
3. Coal,Petroleum and Natural gas are called fossil fuels because they are formed from dead plants and animals deep buried under the rocks millions of year ago.
4. Five things that can help in preventing an energy crisis are:
   (i) Hydroelectric power stations, though they produce only 12% of the total power in India, should be encouraged further.
   (ii) Switch to renewable energy.
   (iii) Install energy-efficient appliances in the home.
   (iv) Save water and do not waste food.
   (v) Rely less on fossil fuels.
5. When the gases are heated in a fractionating column they start rising up. As each gas reaches a height where the

temperature of the furnace is just equal to or below the boiling point of the gas it condenses to form a liquid, for example, when steam rises in a column and the moment it touches a surface cooler than 100°C. it condenses to form water.

6. Euro I, Euro II, etc., are vehicular emission standards aimed at reducing atmospheric pollution. They ensure better and more efficient running of the vehicles without any wastage of the fuel. Indirectly they help in saving the fossil fuels.

7. The measures given by PCRA are
- (i) Drive at a constant and moderate speed as far as possible.
- (ii) Switch off the engine at traffic lights or at a place where you have to wait.
- (iii) Ensure correct tyre pressure.
- (iv) Ensure regular maintenance of the vehicle.

If we follow the tips given by PCRA then we can save more petrol or diesel.

8. To meet the fuel requirement of our ever increasing population, we are overusing coal and petroleum on a large scale. It is apprehended that they may get exhausted in the near future. Similarly, trees in forests are being cut down recklessly for human settlement and establishment of industries. Though the forests are renewable natural resources, their regeneration requires a lot of time. The depletion of forests also results in the disappearance of wildlife.

9. (a) Coal is a complex mixture of compounds of carbon, hydrogen and oxygen. Small amount of nitrogen and sulphur compounds are also present in coal.

(b) Coal is usually found in coal mines deep under the surface of the earth.

Long Answer Questions :

1. Few measures to manage and conserve our natural resources are :
- (i) Reduce : Resources should be used sparingly and when absolutely necessary for e.g., save electricity by not wasting it.
- (ii) Recycle : Recycling saves the natural resources so that their sustainability is maintained.
- (iii) Control pollution : Some of the important steps to check environmental pollution are
  - (a) Identification of the sources and causes of pollution
  - (b) Estimation of capacity of environment to tolerate pollution.
- (iv) Biodegradable waste should be converted to non-toxic and more efficient fuels which will decrease the pressure on the fossil fuels.
- (v) Alternative sources of energy should be utilized.

2. Sunlight, wind and water are inexhaustible natural resources. They can be utilized as alternative sources of energy. They will reduce our dependence on coal and petroleum. Use of solar heaters and wind mills should be done. Sunlight can be used to produce electricity. Though water is inexhaustible source of energy the abnormal increase in human population has resulted in water crisis. Water should be conserved and various methods like water harvesting should be done for sustainability of water as a resource.

3.

S.No.	Natural Resources	Uses
1	Coal	1. Cooking fuel 2. Used as a fuel in railway engines to produce steam
2	Coke	1. Manufacture of steel 2. Extraction of metals
3	Coal tar	1. Synthetic dyes and drugs 2. Sealing agent against leakage
4	Coal Gas	1. Street light 2. Industrial fuel

2 EXERCISE

Text-Book Exercise :

1. Advantages of using CNG and LPG as fuels are as follows
- (i) They can be easily transported in cylinders through pipelines.
- (ii) They are clean fuels and gives no ash particles after burning.
- (iii) They can be used directly for burning in homes and factories.
- (iv) They can give lot of heat energy when burnt.

2. Bitumen, a petroleum product is used for surfacing of roads.

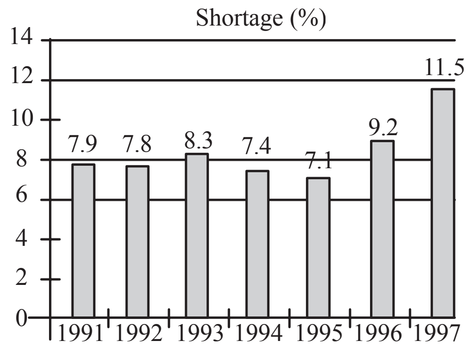
3. About 300 million years ago the earth had dense forest in low lying wetland areas. Due to natural processes, like flooding, these forests got buried under the soil. As more soil deposited over them, they were compressed. Their temperature also rose as they sank deeper and under high pressure and temperature these dead plants got slowly converted into coal.

The process of conversion of dead vegetation into coal is called carbonisation.

4. (a) coal, petroleum, natural gas
(b) refining
(c) CNG

5. (a) (F) Fossil fuels are formed by natural process from the dead remains of living organisms.
(b) (F) CNG is a cleaner fuel than petrol.
(c) (T) Coke is an almost pure form of carbon.
(d) (T) Coaltar is a mixture of about 200 substances.
(e) (F) Kerosene is a constituents of petroleum hence it is a fossil fuel.

6. Fossil fuels require millions of year to form from the dead remains of living organisms at high temperature and pressure. They cannot be formed in laboratory. They are used at much faster rate than rate of their formation. Due to which fossil fuels are exhaustible in nature.
7. **Characteristics of Coke:**
   (i) It is tough, porous and black substance.
   (ii) It is an almost pure form of carbon.
   (iii) It burns much cleaner than coal.
   Uses:
   (i) It is used in manufacture of steel.
   (ii) Used in extraction of many metals.
8. Petroleum was formed from organisms living in the sea. These dead organisms got settled at the bottom of the sea and got covered with layers of sand and clay. Over millions of years, absence of air, high temperature and high pressure transformed the dead organisms into petroleum.
9.



Exemplar Questions :

1. CNG stands for Compressed Natural Gas. It is considered to be a better fuel because it is less polluting.
2. It is used for extraction of many metals and also for the manufacture of steel.
3. Coal when processed in industry gives coke, coal tar and coal gas.
   Coke is used in the manufacture of steel and in extraction of many metals.
   Coal tar is used as starting material for manufacturing various sustances such as synthetic dyes, drugs, explosives, perfumes, paints, etc.
   Coal gas is used as fuel.
4. • Petroleum gas in liquid form (LPG) – used as fuel for home and industry.
   • Petrol – used as fuel for automobile and aviation.
   • Kerosene – used as fuel for stoves, lamps and for jet aircrafts.
   • Diesel – used as fuel for heavy motor vehicles, electric generators.
   • Lubricating oil – used for lubrication.
   • Praffin wax – used in ointments, candles, vaseline etc.
   • Bitumen – used in paints and road surfacing.
5. Kerosene is used as fuel for stoves, lamps and jet aircrafts.

HOTS Questions :

1. The process of separation of various fractions of petroleum by the process of fractional destillation is called refining of petroleum. The plant (place) where crude oil is precessed and refined into more useful petroleum products, is called a refinery.
2. Coal mining, causes many adverse environmental impacts. During mining of coal, a lot of methane, which is a dangerous 'greenhouse gas,' is released. It also interferes with groundwater and water table levels. Inhaling coal dust is harmful for humans. Burning of coal, in addition greatly contributes to 'greenhouse gas' emissions, climate change and global warming. Strip mining (removed of soil and rock) severely alters the landscape, which damages the environmental value of the surrounding land. Mountain top removal to extract coal is taken to be a large negative change to the environment.
3. Water is a limitless resource but by unwise use and cutting of trees water resources can be depleted.
4. Approximately 200 substances are present in coal tar.
5. Solar energy, wind energy, tidal energy, ocean thermal energy and geothermal energy are some of the non-conventional sources of energy.
6. Yes, all renewable resources are inexhaustible as they are present in unlimited quantity in nature and are not likely to be exhausted by human activities , the resources can be used again and again. for example air sunlight etc.
7. In a petroleum well crude oil is found above water. The two properties due to which it form layer above water are:
   (i) It is lighter than water.
   (ii) it is insoluble in water.

3 EXERCISE

Multiple Choice Questions :

1. (d) All others are obtained by refining of petroleum which is a fossil fuel.
2. (b)
3. (c) water gas is a mixture of CO and H_2
4. (d)
5. (a)
6. (d) Petroleum is a naturally occuring crude oil containing saturated, cyclic, unsaturated hydrocarbons with some other elements, such as sulphur, oxygen and nitrogen.
7. (b)
8. (b)
9. (a) Fractional distillation is used because the difference between the boiling point of different components is less.
10. (a) Natural gas is a very important fossil fuel because it is easy to transport through pipes.
11. (d)

12. (c) Kerosene is used as fuel in stoves, lamps and jet aircrafts.
13. (b) Coke can be used as a reducing agent in metallurgy.
14. (a) Coal tar is a black, thick liquid.
15. (b) Now-a-days, coal gas is used as a source of heat.
16. (a) It is hard coal and contains about 96% carbon whereas peat, lignite and bituminous contians 60, 67 and 88% of carbon respectively.
17. (a) **18.** (b) **19.** (d)
20. (b) Petrol and diesel are obtained from a natural resource called petroleum.
21. (c) The process of heating coal to a high temperature in the absence of air is called destructive distillation of coal. It yields products such as coke, coal gas, etc.
22. (c) **23.** (d) **24.** (b) **25.** (a)
26. (b) **27.** (a) **28.** (a) **29.** (a)
30. (d) All statements are correct.
31. (a) Fossil fuels produce greenhouse gases like carbon dioxide which are responsible for global warming.
32. (d) **33.** (c)

More Than One Option Correct :

1. (a, b, c, d)
2. (b, c)
3. (b, c)
   Coal gas is obtained by destructive distillation of coal.
4. (a, b, c)
   Synthetic dyes, drugs, perfumes all are made from coal tar.
5. (a, c, d)
   LPG is not a fossil fuel as it is obtained from distillation of petroleum. The rest three are fossil fuels as they are formed under earth's crust.
6. (a, b, d)
   The unrefined form of petroleum is called crude oil.
7. (a, b, d)

Passage Based Questions :

1. (a) Petrol condenses near the top of the column since it is more volatile than other three.
2. (b) Their boiling points must be much higher than the temperature at the bottom of fractionating column hence they do not boil.
3. (a) As the gas reaches a height where temperature is equal to or just below its boiling point, it will condense to form a liquid.

Assertion & Reason :

1. (c) Coal gas is obtained by destructive distillation of coal.
2. (c) Anthracite contains more than 90% carbon.
3. (a)
4. (b) Petroleum is refined by fractional distillation to get various useful fractions.
5. (b)
6. (a) Air and sunlight are not likely to be exhausted by human activities hence they are called inexhaustible natural resources.

Multiple Matching Questions :

1. A $\rightarrow$ (s), B $\rightarrow$ (r), C $\rightarrow$ (q), D $\rightarrow$ (p)

Integer Type Questions :

1. **3**
   Petroleum ,coal and forest are exhaustable.
2. **4**
   Coke , coal tar, coal gas and ammonium compounds.
3. **2**
   Petroleum and Coal.
4. **2**

4 EXERCISE

1. (a)
2. (b) LPG is a mixture of C_4H_{10} (Butane), C_3H_8 (propane) and C_2H_6 (ethane). Main constituent is butane (C_4H_{10}).
3. (c) **4.** (b)
5. (b) Conc. H_2SO_4 dehydrates sugar to give sugar charcoal.
6. (c) Since constiuent hydrocarbons of LPG do not have any smell, ethyl mercaptan, a strong smelling agent is added to it to detect the leakage.
7. (c)
8. (b) Butane can be easily compressed and liquefied to be filled into cylinders.
9. (a, b, c, d)
10. (a, b, c)
11. (a, c)
12. (c) **13.** (a) **14.** (b)
15. (b)
16. (b) Sulphuric acid removes water from sugar and converts it to sugar charcoal.
17. (a) Charcoal is a processed fuel hence burns completely without any smoke.
18. (a)
19. (b)
20. A - (p, r), B - (p), C - (r), D - (q)
21. A - (p, r), B - (r), C - (q, s), D - (q)
22. **2**
    The main components of LPG are hydrocarbon containing three or four carbon atoms. The main components of LPG thus are propane (C_3H_8) and Butane (C_4H_{10}).
23. **3**
    There are three types of charcoal
    (i) Wood charcoal
    (ii) Boan charcoal
    (iii) Sugar charcoal

Chapter 4

COMBUSTION AND FLAME

INTRODUCTION

In last chapter you have studied about variety of fuels and their importance to meet our routine energy needs. You are now well aware of fuels used at domestic scale like candle to produce light, wood or coal to cook food in villages, LPG to cook food in towns, petrol and diesel to run motor vehicles etc. Some uses of fuels at massive scale are use of coal to produce electricity in coal fired power stations, natural gas is used for heating and lightning purposes through network of pipelines. Therefore energy is produced when fuels are burnt but following are some questions which may arise in the mind of reader.

- Do all materials burn and can be used as a fuel ?
- Different fuels burns in a similar manner or not ?
- Why different fuels are needed for different purposes ?

In this chapter we will find out the answers to the above questions and will study the chemical process of burning and the types of flame produced during the process.

COMBUSTION

Combustion is a chemical process in which a substance (fuel) burns in the presence of air (oxygen) with the release of heat and light, the flame. On combustion the burning fuel gets oxidised (combines with oxygen) to form new substances. Sometimes, combustion is accompanied with the production of sound as well. The sound produced may be a hissing sound, crackling sound or an explosion.

$$C + O_2 \rightarrow CO_2 + \text{Heat}$$
$$CH_4 + O_2 \rightarrow CO_2 + H_2O + \text{Energy}$$

Combustibility is a measure of how easily a substance will catch fire. On the basis of combustibility substances are broadly divided into two categories.

A substance that burns in air or oxygen to produce heat and light is called a combustible substace, e.g., paper, wood, kerosene, LPG.

A substance that does not burn in air or oxygen is called a non-combustible substance, e.g., stone, metal.

You have seen that a combustible substance that can be used to produce heat at a reasonable cost is called a fuel.

Thus every combustible substance which burns cannot be used as a fuel.

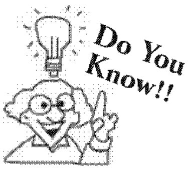
Conditions Necessary for Combustion :

(i) **Combustible substance:** Firstly substance to be burnt must be combustible (the substance must catch fire easily). Some of the combustible substances are fossil fuels and their processed products, wood, paper, alcohols, ether etc.

(ii) **Supporter of combustion:** The presence of a supporter of combustion is another condition for combustion to take place. They are nothing but the substances which help in combustion of a combustible substance. Oxygen and air are the supporter of combustion. Air contains 21% of oxygen (a supporter of combustion) by volume and 78% of nitrogen by volume which is neither combustible nor a supporter of combustion. That is why combustible substances burn at moderate rate in air but burn at a very fast rate in oxygen.

You might have observed or aware of the fact that air is blown into the traditional chullahs to start fire. This shows that air or oxygen is a supporter of combustion.

Another example to support the above fact is that it is recommended to cover a person with blanket whose clothes catches fire to cut off air supply.



Sun also produces heat and light but it is not due to combustion as combustion cannot take place there due to absence of combustion supporter oxygen. The heat and light produced in the sun is a result of nuclear reactions occurring there which you will study in higher classes.

(iii) **Ignition temperature:** Another condition for combustion to take place is that the combustible substance should be heated so that its temperature reaches to its ignition temperature. No substance can burn below its ignition temperature. The lowest temperature at which a substance catches fire is called ignition temperature.

Have you ever throught why a matchstick does not burn at its own ? A matchstick burns only when it is rubbed on the side of a matchbox.

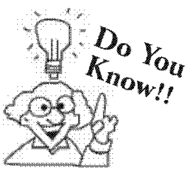
Incident of forest fires due to ignition of dry grass occurs in extreme hot conditions only.

A paper can burn by bringing burning matchstick near it but a piece of wood or coal does not.

Cooking oil catches fire when kept in open pan for prolong heating.

A kerosene oil or petrol ignites with little spark but this is not the case with wood or coal.

All the above experiences or facts indicates that different substances catch fire or ignites at different temperatures. Thus every substance has a particular ignition temperature at which it ignites.



An empty paper cup with air in it catches fire immediately by bringing burning matchstick near it whereas paper cup filled with water or wetted with water does not as heat supplied by matchstick gets transferred to water by conduction thereby not allowing paper to reach its ignition temperature.

Inflammable Substances :

Flammable and inflammable are two words that are commonly confusing. Flammable and inflammable means exactly the same thing that burns easily. Those substances which has low ignition temperature and burns easily are called inflammable substances. Examples or CNG, LPG, Petrol, Kerosene, alcohol etc. Special care should be taken while handling and storing such substances.

Contents of Matchsticks :

A matchstick is a small stick of wood with a solidified mixture of flammable chemical deposited on one end, when that end is struck on a rough surface, the friction generates enough heat to ignite the chemicals and produce a small flame.

Woods used to make matchsticks must be porous enough to absorb various chemicals, and rigid enough to withstand the bending forces encountered when the match is struck. They should also be straight-grained and easy to work, so that they may be readily cut into sticks. White pine and aspen are two common woods used for this purpose.

In earlier days a mixture of antimony trisulphide, potassium chlorate and white phosphorous with some glue and starch was applied on head of a match. However white phosphorous proved to be dangerous. These days the head of the safety match contain antimony trisulphide and potassium chlorate. The rubbing surface has powdered glass and little red phosphorous which is much less dangerous.

idea box

Collect around ten substances namely wood, plastic, cloth, coal, Aluminium foil, Tin can, Rubber sheet, Dry leaves, Kerosene oil, cow dung cake.

Now burn these materials one by one and observe carefully.

(1) Which substances, readily catches fire on burning producing less smoke and

(2) Which catches fire on prolonged burning producing a lot of smoke.

Based on the observation classify the substance as combustible and non-combustible material.

CONTROL OF FIRE

- As we all know that fire is must to fulfill our routine needs but if we do not take necessary measures it can goes out of control and can be very disasterous, it can ruin forests, huge buildings and even towns.

  Any fire needs three things to be present: Fuel, Air (oxygen) and Heal. Thus in case of fire accident, fire can be extinguished (put off) by eliminating any of the three requisites of combustion.

  (a) Removal of combustible material, the fuel.

  (b) Cutting off the supply of air (oxygen)

  (c) Lowering the temperature of the fuel to below its ignition temperature.

- The process of extinguishing a fire is called fire fighting. In case of fire accident, fire-fighters rush to the accident site with a tanker full of water, a pumping device and other fire-fighting equipment like axe, rope and ladders.

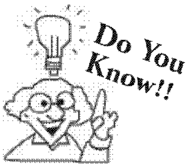
Methods of Controlling Fire:

(i) **By using water :** The first important method is to spray the burning material with water. This helps in two ways: (a) lowering the temperature of the burning materials, and (b) putting off flames by cutting off the supply of oxygen to the flames.

Thus the fire is extinguished.

Water is a good extinguisher in case of paper and wood fire. But in case of electrical equipments water cannot be used because water conducts electricity and can harm those trying to put out the fire.

Moreover in the case of oil fires, to extinguish the fire, the burning material is not sprayed with water. Water is heavier than oil. On spraying water it sinks and oil floats, and the fire continues.



It is not necessary that heat and light are always produced during oxidation.

(ii) By using carbondioxide: For fire involving electrical equipments and inflammable materials such as petrol, carbon dioxide (CO_2) is the best extinguisher. Carbon dioxide (CO_2), being heavier than oxygen, covers the fire like blanket. As the contact between the fuel and oxygen is cut off, the fire is controlled. CO_2 does not harm electrical equipments.

CO_2 can be stored and transported at a high pressure in cylinders like LPG. When knob of cylinder is opened CO_2 expands enormously and cools down. Thus it not only forms a blanket around the fire, it also brings down the temperature of the fuel.

CHECK Point

Water should not be used to put off electrical fires or burning oil and pertrol. Instead sand or soil should be used. Why?

SOLUTION

Water should not be used for oil or pertrol fires because oil and pertrol are lighter than water hence they float on water and keep burning on the surface. Water may conduct electricity and chances of short circuiting will be more.

CONNECTING TOPIC

FIRE EXTINGUISHER

A fire extinghuisher is a device used for putting of fires.

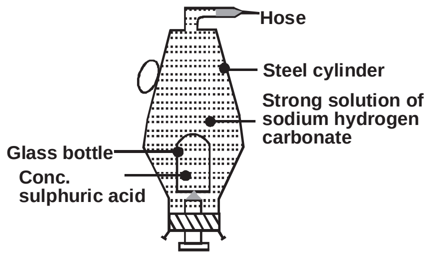
Different types of CO_2 fire extinguisher used for putting off fires are following :

Types of fire extinguishers

(i) "**Dry powder' fire extinguisher :** It contains a mixture of sand and baking soda (*i.e.* sodium bicarbonate). When thrown over the burning material, baking soda is decomposed by the heat of fire to produce *carbon dioxide*, which being heavier than air cuts off the contact between air and fire. (*i.e.* the CO_2 layer is formed between the flame and air)

$$2NaHCO_3 \xrightarrow{heat} Na_2CO_3 + H_2O + CO_2 \text{ (Carbon dioxide)}$$

(ii) "**Soda - acid" fire extinguisher :** It consists of a metallic cylinder filled with a solution of sodium bicarbonate ($NaHCO_3$). At the bottom of the cylinder, a thin sealed glass tube containing concentrated sulphuric acid (H_2SO_4) is placed. This tube is surrounded by a fixed wire gauze. A plunger with a sharp end is placed at the bottom of cylinder in such a way that the sharp end is placed at the sealed thin glass tube. On the top of the cylinder, a nozzle which is sealed with wax is provided.



Soda-acid fire extinguisher

When the plunger is hit against the floor, the sharp tip of the plunger breaks the glass test-tube. The acid in the test-tube reacts with sodium bicarbonate to produce carbon dioxide gas which is forced through the nozzle. The wax seal is broken off, thus forcing CO_2 gas out through the nozzle. The fire is put off when the gas is directed against the fire.

$$2NaHCO_3 + H_2SO_4 \longrightarrow Na_2SO_4 + 2H_2O + 2CO_2$$

(iii) **"Foam-Type" fire extinguisher :** It is similar to **soda-acid** fire extinguisher. However sodium bicarbonate solution also contains a substance **Saporin** or **turkey red oil** which produces a foam with the gas and the solution issuing from the nozzle. The foam being light will float on the surface of oil fires, thus cutting off the air supply.

TYPES OF COMBUSTION

Explosive Combustion

A combustion in which a substance burns suddenly and produce heat, light and sound with the help of heat or pressure is called explosive combustion or explosion. When a cracker is ignited it explodes with a big-bang producing heat and light (sparkles). Same is true with the firing of a gun. Big sound is produced alongwith combustion. Firewood burns with a crackling sound.

Spontaneous Combustion

In this type of combustion a material suddenly bursts into flames without the application of any apparent cause.eg. Spontaneous forest fires are sometimes due to the heat of the sun or lighting strike and sometimes due to the carelessness of human beings spontaneous combustion takes place at room temperature. The heat required for spontaneous combustion is produced inside, the substance by its slow oxidation. Spontaneous combustion is usually undergone by those substances which have quite low ignition temperature. For example white phosphorus burns in air at room temperature.

Rapid Combustion

This is a combustion in which a substance burns rapidly and produces heat and light with the help of external heat for example cooking gas (in our homes), spirit, petrol and camphor go to flames even with a spark from a gas lighter. These materials are highly inflammable. You might have observed "Highly Inflammable" written on petrol tankers. This is a warning to keep a flame away from these tankers;

Slow Combustion

This type of combustion occurs at moderate speed. In this combustion, fuel is not burnt completely. This results in the production of smoke due to unburnt carbon particles. Burning of wood, coal, wax candle, pieces of paper or such other materials is a slow combustion.

FLAME

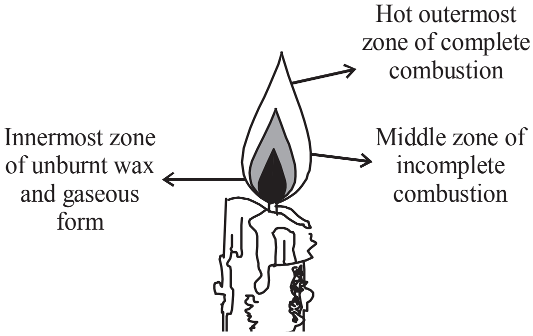
Do you know that everything that burns does not produce flame. A flame is produced as a result of rapid combustion when combustible substance is in the vapour state and burns strongly in the presence of air. For example, kerosene oil, molten wax and mustard oil rise through the wick and are vaporised during burning and form flames.

Wick is a loosely twisted thread made from absorbent fibre, which is generally cotton or it is a spongy material. One end of the wick in a lamp is kept dipped in the liquid fuel, the other end is kept free in the air. The free end of the wick is also soaked in oil or fuel and is lighted. The wick catches fire and bears the flame.

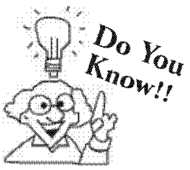
A candle mainly has three zones.

(i) **Inner most zone or Dark zone :**

It is the least hot or the coolest part of the flame and it is dark.This zone is surrounding the wick of the candle. It consists of unburnt wax vapours, which have carbon particles hence the black colour. No combustion takes place in this zone.



The Candle Flame



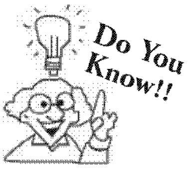
(i) The blowpipe is used by gold smiths to blow air into the outermost zone of flame so as to make it hotter. This hot flame is used to melt silver or gold.

(ii) The greater the number of carbon atoms in a molecule of a fuel, the greater is the capacity to produce heat.

(ii) **Middle or luminous zone :** It is the largest zone of candle flame and is moderating hot. It is bright and luminous.
The wax vapours do not burn completely in the middle zone as the supply of oxygen is inadequate. It leaves a black soot and other residues on an object placed here. The carbon particles glow emitting a yellow light.

(iii) **Outermost non-luminous zone :** This zone of the flame is thin and blue in colour. This is the hottest zone of the flame. Because of adequate supply of oxygen, complete combustion occurs. The temperature of this zone is maximum around 1800°C.

- There is a small region near the base of the flame. Vaporised wax gets oxidised to carbon monoxide and carbon monoxide burns completely with a blue flame in this zone.

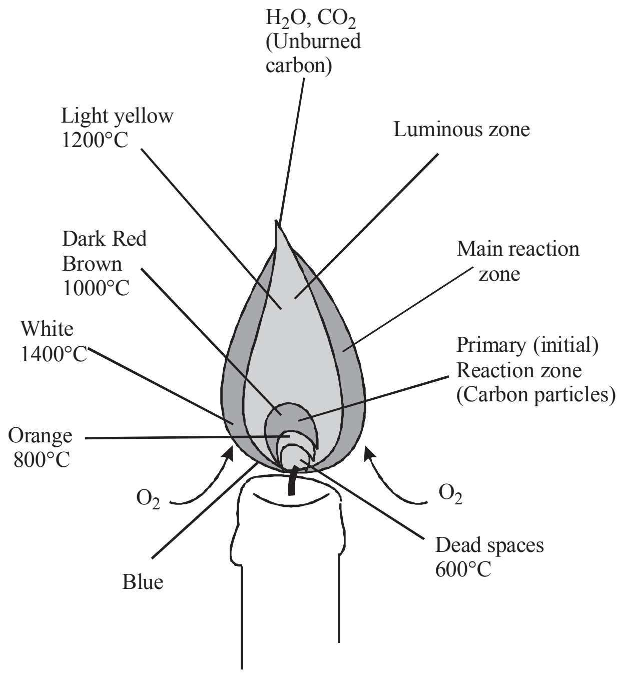


Luminous zone of a flame is mainly due to incomplete burning of carbon. In the case of fuel gas (LPG or CNG) there is no soot formation. The gases burn with a blue flame oxidising hydrocarbons completely.

CONNECTING TOPIC

Chemical Description of a Burning Candle :

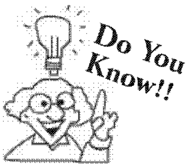
Twenty five percent of the energy created by combustion is given off as heat. A small portion of the radiant heat from the candle flame feeds back into the candle and melts the wax to fuel the flame. The candle flame surface itself is the place where fuel (wax vapor) and oxygen mix and burn at high temperatures, radiating heat and light. Heat from the flame is conducted down the wick and melts the wax at the wick base. The liquid wax rises up the wick because of capillary action. As the liquid wax nears the flame, the flame's heat causes it to vaporize and combine with oxygen. The vapors are drawn into the flame where they ignite. The heat produced melts more wax, and so on. Carbon dioxide and water is produced and forms and reforms many carbon rich particles (soot). The soot is drawn up into the top part of the flame where increased temperature causes the carbon to luminesce and burn. Fresh oxygen from the surrounding air is drawn into the flame primarily because of convection currents that are created by the released heat. Hot gases produced during burning are less dense than the cooler surrounding air. They rise upward, and, in doing so, draw the surrounding air, which contains fresh oxygen, into the flame. Solid particles of soot that form in the region between the wick and flame are also carried upward by the convection currents. They ignite and form the bright yellow tip of the flame. The upward flow of hot gases causes the flame to stretch out in a teardrop shape.
The wax is actually a complex mix of various long molecules that all have a chain of carbon atoms as a "backbone." Many hydrogen atoms are attached to the carbon atoms.



Candle Flame reaction zones, emissions, and Temperature

FUELS

The materials which are burnt to produce heat energy are known as fuels. Wood, coal, domestic gas (LPG), kerosene oil, diesel, petrol *etc*. are all fuels.



Glowing charcoal does not produce a flame.
Wood at first gives flame but later glows like charcoal.
When fuels burn completely they produce a flame without smoke.
When a fuel is not completely burnt, it produces a smoky flame.

CHECK Point

Gaseous fuels are better fuels than solid and liquid fuels. Explain.

SOLUTION

Gaseous fuels leave no residue after burning also they have high calorific value because their combustion is complete.

Characteristics of a Good Fuel:

While selecting an ideal fuel for domestic or industrial purpose we should keep in mind that the fuel selected must possess the following characteristic properties.

1. It should possess high calorific value.
2. It should have proper ignition temperature. The ignition temperature of the fuel should neither be too low nor too high.
3. It should not produce poisonous products during combustion. In other words, it should not cause pollution on combustion.
4. It should have moderate rate of combustion.
5. Combustion should be easily controllable i.e., combustion of fuel should be easy to start or stop as and when required.
6. It should not leave behind much ash on combustion.
7. It should be easily available in plenty.
8. It should have low moisture content.
9. It should be cheap.
10. It should be easy to handle and transport.

Calorific Values of Different Fuels

Fuel	Calorific Value (kJ/kg)
Cow dung cake	6000-8000
Wood	17000-22000
Coal	25000-33000
Petrol	45000
Kerosene	45000
Diesel	45000
Methane	50000
CNG	50000
LPG	55000
Biogas	35000-40000
Hydrogen	150000

Uses of Fules

Fuels are very important for our life and without them our life is impossible. Following are some uses of common fuels which make them important for us.

(i) Gasoline is the most obvious fuel used in daily life. It runs our cars, school buses and other vehicles every day.

(ii) Natural gas can be used to power the heating systems, stove tops, water heaters and dryers in our homes.

(iii) Oil and natural gas play important role in our daily lives, and in some ways you might not expect: Hydrocarbons are the building blocks of vital products such as plastics, pharmaceuticals and more.

(iv) Many electrical plants burn coal as the primary fossil fuel for powering the electrical supply for homes across the country.

Types of Fuels :

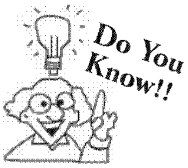
(a) **Solid fuels :** Dung cakes, firewood, straw and other agricultural wastes, charcoal, coal (mined fossil fuel) like steam coke and soft coke, paraffin wax, tallow (animal fat) and may be camphor.

(b) **Liquid fuels:** Fossil fuel (petroleum) or crude oil from which petrol (also called gasoline), kerosene, diesel and other fuel products are obtained.

(c) **Gaseous fuels :** LPG (Liquid petroleum gas) being used as cooking gas in our households, CNG (Compressed natural gas) obtained from natural oil wells. Natural gas is methane which is given out from putrefying organic matter, biogas (prepared in our villages from animals dung and farm waste) and also being collected and supplied from sewage plants, Coal gas (gas obtained from heating hard coke and converting it into coke), water gas (prepared by passing steam over red hot coke), butane gas obtained from natural gas, and acetylene, a gas made by adding calcium carbide to water. Acetylene produced is with a little unpleasant smell and was at one time primarily used for welding purposes. Flame from acetylene is hot and is also used for cutting metals.

Calorific Value of Fuel or Fuel Efficiency :

- Calorific value of a fuel is the amount of heat energy given out on burning one unit of fuel in the presence of fully required amount of oxygen for its burning.
- Calorific value is used to compare the efficiency of two fuel.
- Hydrogen has a highest calorific value of 150 kJ/gm.
- Any fuel containing oxygen in it has less calorific value because oxygen supports combustion but does not burn.
- Fuel having higher mass percentage of H have high calorific value eg. Calorific value of $CH_4 > C_2H_6$.



Scientists are working on using hydrogen as a fuel, as it has the highest calorific value and creates no pollution after burning. Some success has been achieved and hydrogen is expected to the fuel of the future.

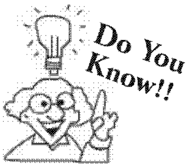
Impacts of Burning Fuels on Environment

Increase in consumption of fuel lead to many harmful effects on environment. Following are some points which clearly explain these effects.

1. Burning of carbon containing fuels like wood, coal, coke, petroleum etc., results into discharge of ash and unburnt carbon particles in the air. These fine particles are called SPM (Suspended Particulate Matter). SPM leads to respiratory diseases like asthma, bronchitis etc.
2. Incomplete combustion of coal, gasoline and other fossil fuels liberates carbon monoxide and unburnt hydrocarbons. Carbon monoxide (CO) combines with haemoglobin in the blood to form carboxy-haemoglobin and makes it incapable of transporting oxygen. Many hydrocarbons are carcinogenic and hence, are serious health hazards.
3. Combustion of most fuels results into discharge of carbon dioxide in the atmosphere. Moreover deforestation also reduces number of trees absorbing carbon dioxide. Thus these two factors results into elevated levels of carbon dioxide in atmosphere. Carbon dioxide has ability to absorb infrared radiation emitted by the earth and re-radiate it back. This effect is known as 'greenhouse effect'. It is speculated that if the concentration of CO_2 in atmosphere increases significantly this could have the effect of increasing the average temperature of the earth. This phenomena is called global warming which leads to rise in the sea level, causing floods in the coastal areas.

4. Bruning of coal and diesel results into discharge of sulphur dioxide into atmosphere. It irritates the nose, throat, and airways to cause coughing, wheeing, shortness of breath, or a tight feeling around the chest. Similarly automobile exhaust contains oxides of nitrogen which leads to respiratory problems. Collectively these oxides of nitrogen and sulphur dissolves in rain water and forms corresponding nitric and sulphuric acid thereby making rain water acidic leading to acid rain.

Acid rain causes acidification of lakes and streams and contributes to the damage of trees at high elevations (for example, red spruce trees above 2,000 feet) and many sensitive forest soils. In addition, acid rain accelerates the decay of building materials and paints, including irreplaceable buildings, statues, and sculptures that are part of our nation's cultural heritage.



Due to heavy pollution caused by traditional fossil fuels petrol and diesel automobiles are being replaced by CNG these days as it causes less pollution problems.

Measures to Conserve Fuels

As you all know how important fuels are for us and we have limited stock of fuels thus conservation of fuels is very important . Following are some measures we can take at individual level to conserve fuel.

- Next time when your mother is going to cook, suggest her that all material needed for cooking should be available before she switch on the gas.
- Suggest your father that he should regularly check tyre pressure of his car.
- When you go out for short distance prefer walking rather than using car/motorbike.
- Prefer public transportation for travelling rather than using private vehicles.

1 EXERCISE

Fill in the Blanks :

DIRECTIONS : *Complete the following statements with an appropriate word / term to be filled in the blank space(s).*

1. .... is the substance which reacts with oxygen and turns to release energy.

2. During an explosion a large amount of.............. is given out.

3. The .... zone in a candle flame is the zone of no combustion.

4. The amount of heat liberated when one kilogram of a fuel is completely burnt in sufficient supply of oxygen is called.............. of the fuel.

5. .... is a gaseous fuel which can be stored in cylinders.

6. On burning, solid fuels leave residue and..............

7. .... has corroded the marble structure of Taj Mahal.

8. .... is used to control fires involving electrical equipments.

9. Ignition temperature of a log of wood is.............. than a splinter.

10. Respiration is a .... combustion.

11. Combustion of white phosphorus in air is an example of .... combustion.

12. When the clothes of a person catch ...., the person is covered with a ....

13. The substances which have very .... ignition temperature and can easily catch fire with a flame are called .... substances.

True / False :

DIRECTIONS : *Read the following statements and write your answer as true or false.*

1. To extinguish petrol fire we can use water.

2. Carbon dioxide is effective for fire fighting.

3. Carbon monoxide is neither combustible nor a supporter of combustion.

4. In luminous zone of candle a incomplete combustion of wax vapours takes place with liberation of maximum heat energy.

5. Wood is a better fuel than LPG.

6. It is difficult to burn dry leaves but green leaves catch fire easily.

7. Colour of outer zone of a candle flame is blue while middle zone is yellow in colour.

8. Incomplete combustion of a fuel gives carbon dioxide.

9. Charcoal does not vaporise, so it does not produce a flame.

10. Oxides of sulphur and nitrogen are produced by burning of coal and diesel which cause acid rain.

11. Carbon dioxide, water vapours, heat and light are the products of combustion.

12. At airport and petrol pump soda acid fire extinguishers are not used.

13. Magnesium is a non-combustible metal.

Match the Following :

DIRECTIONS : *Following question contains statements given in two columns which have to be matched. Statements (A, B, C and D) in column I have to be matched with statements (p, q, r and s) in column II.*

1.

	Column-I		Column-II
A.	Materials which burn readily	(p)	Oxygen
B.	Heat produced by burning unit mass of fuel	(q)	Ignition temperature
C.	Point of catching fire	(r)	Calorific value
D.	Supporter of combustion	(s)	Combustible

Very Short Answer Questions:

DIRECTIONS : *Give answer in one word or one sentence.*

1. What is a flame ?

2. What is the cause of smoky flame ?

3. What is a fire extinguisher ?

4. Name any two types of fire extinguishers commonly used.

5. How does water extinguish fire ?

6. What is the colour of LPG flame ?

7. Why does charcoal not produce flame ?
8. Name three substances used to extinguish fire.
9. Coal is used in power generation. What is the power generated in such a way known as ?
10. Why is sand used for fire extinguished to be efficient ?
11. On what parameters is a fuel considered to be efficient ?
12. What are the fine particles released on burning carbon-containing fuels called ?
13. What is the full form of CNG and LPG?
14. What is SI unit of calorific vlaue?
15. Is rusting of iron a combustion process.

Short Answer Questions :

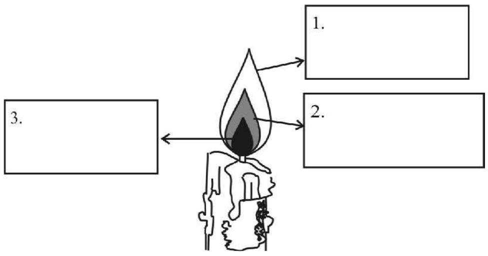
DIRECTIONS : *Give answer in 2-3 sentences.*

1. Why does a loud noise produced when a fire cracker is burst?
2. Give reasons for the following :
   (a) The use of diesel and petrol as fuel in automobiles is being replaced by CNG.
   (b) A matchstick starts burning by rubbing it on the side of the match box.
3. What are the harmful effects of increasing consumption of fuels ?
4. Write the colours of the flame produced by
   (a) Kerosene lamp
   (b) Candle
   (c) Bunsen burner
5. Give any three examples of solid, liquid and gaseous fuels.
6. What are the disadvantages of using wood as a fuel ?

Long Answer Questions :

DIRECTIONS : *Answer the following questions in detail.*

1. What are the various problems caused by combustion of fuels ?
2. Explain the phenomenon of explosion with suitable examples.
3. (a) Label different zones of candle flame in the following figure.



   (b) Explain the three different zones of candle flame. What kind of combustion occurs in each?
4. Write down the examples of each types of fuel in the given table.

S.No.	Solid fuel	Liquid fuel	Gaseous fuel
1			
2			
3			
4			

5. The calorific values of petrol and CNG are 45000 kJ/kg and 50,000 kJ/kg, respectively. If you have vehicle which can run on petrol as well as CNG, which fuel will you prefer and why ?
6. Although wood has a very high calorific value, we still discourage its use as a fuel. Explain.

2 EXERCISE

Text-Book Exercise :

1. List conditions under which combustion can take place.
2. Fill in the blanks:
   (a) Burning of wood and coal causes ________ of air.
   (b) A liquid fuel, used in homes is ________.
   (c) Fuel must be heated to its ________ before it starts burning.
   (d) Fire produced by oil cannot be controlled by ________.
3. Explain how the use of CNG in automobiles has reduced pollution in our cities.
4. Compare LPG and wood as fuels.
5. Give reasons:
   (a) Water is not used to control fires involving electrical equipment.
   (b) LPG is a better domestic fuel than wood.
   (c) Paper by itself catches fire easily whereas a piece of paper wrapped around an aluminium pipe does not.
6. Make a labelled diagram of a candle flame.
7. Name the unit in which the calorific value of a fuel is expressed.
8. Explain how CO_2 is able to control fires.
9. It is difficult to burn a heap of green leaves but dry leaves catch fire easily. Explain.
10. Which zone of a flame does a goldsmith use for melting gold and silver and why?
11. In an experiment 4.5 kg of a fuel was completely burnt. The heat produced was measured to be 180,000 kJ. Calculate the calorific value of the fuel.
12. Can the process of rusting be called combustion? Discuss.
13. Abida and Ramesh were doing an experiment in which water was to be heated in a beaker. Abida kept the beaker near the wick in the yellow part of the candle flame. Ramesh kept the beaker in the outermost part of the flame. Whose water will get heated in a shorter time?

Exemplar Questions :

1. Anu wants to boil water quickly in a test tube. On observing the different zones of the flame, she is not able to decide which zone of the flame will be best for boiling water quickly. Help her in this activity.
2. Boojho wants to separate the following materials as combustible and non-combustible. Can you help him ?
3. You are provided with three watch glasses containing milk, petrol and mustard oil, respectively. Suppose you bring a burning candle near these materials one by one, which material(s) will catch fire instantly and why ?
4. Manu was heating oil to fry potato chips. The cooking oil all of a sudden caught fire; he poured water to extinguish the fire. Do you think this action was suitable. If yes, why ? If not, why not ? In such a condition what should Manu have done ?
5. Two glass jars A and B are filled with carbon dioxide and oxygen gases, respectively. In each jar a lighted candle is placed simultaneously. In which jar will the candle remain lighted for a longer time and why?
6. People usually keep Angethi/burning coal in their closed rooms during winter season. Why is it advised to keep the door open ?
7. Cracker on ignition produces sound. Why ?

HOTS Questions :

1. CO_2 is chief gas responsible for global warming (Green house effect), why ?
2. Why does the flame of candle rise up to good height on burning ?
3. Hydrogen has the highest calorific value, but it is not used as a fuel. Why ?
4. A flame always points upwards. Why do you think this is so?
   (**Hint :** Gases produced in a flame are hot, and hence lighter)
5. Only gases burn with a flame. When you burn wood it initially burns with a flame. Later it only glows without a flame. What do you think the answer is?
6. When a fuel is burnt, carbon dioxide (or carbon monoxide) and water vapour are given out. Can you name one fuel which burns without giving off water vapour?
7. Your LPG gas stove at home is giving a yellow flame. What can this mean?
8. Why do forest fires occur during hot summers?

3 EXERCISE

Multiple Choice Questions :

DIRECTIONS : *This section contains 31 multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.*

1. Water is used to extinguish fire because water
(a) raises the ignition temperature of the burning substance.
(b) lowers the ignition temperature of the burning substance.
(c) cools the burning substance to a temperature below its ignition temperature.
(d) conducts the heat.

2. Which of the following has highest calorific value?
(a) Petrol (b) Coke
(c) Natural gas (d) Kerosene

3. Combustion is a .... reaction accompanied by heat and light –
(a) reduction (b) redox
(c) substitution (d) oxidation

4. Outermost zone of flame is called
(a) Luminous zone (b) Non-luminious zone
(c) Dark zone (d) None of these

5. The fuel having lowest calorific value is
(a) H_2 (b) $C_6H_{12}O_6$
(c) kerosene (d) CH_4

6. Incomplete combustion of fuel produces
(a) CO_2 (b) CO
(c) H_2O (d) H_2

7. Combustion is the process of
(a) vaporisation of fuel
(b) oxidation of fuel
(c) burning of fuel with the evolution of heat and light
(d) production of explosive sound

8. Vapourised wax gets oxidised in
(a) blue zone (b) dark zone
(c) luminous zone (d) non-luminous zone

9. Hottest part of the flame?
(a) blue zone (b) dark zone
(c) luminous zone (d) non-luminous zone

10. Select the quality that a good fuel should possess.
(a) High calorific value
(b) A low calorific value
(c) It should burn too fast
(d) It should burn too slow.

11. Select the fuel with highest calorific value ?
(a) LPG (b) Kerosene oil
(c) Wood (d) Hydrogen gas

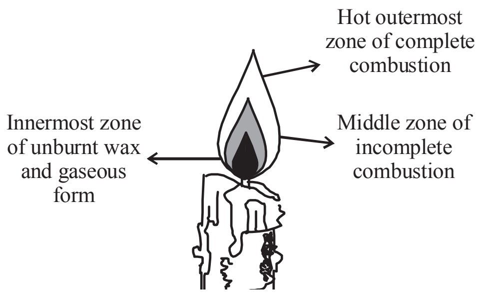
12. Which one is the best fire extinguisher ?
(a) SO_2 (b) CO_2
(c) NO_2 (d) O_2

13. The colour of luminous zone of a candle flame is
(a) black (b) blue
(c) yellow (d) None of these

14. Which of the following is example of spontaneous combustion?
(a) Burning of matchstick
(b) Combustion of fuel by motor vehicles.
(c) Lightning of kerosene lamp.
(d) Combustion of coal dust in coal mines.

15. Molten wax on heating releases
(a) water and carbon dioxide
(b) carbon and hydrogen
(c) carbon dioxide and hydrogen
(d) water and carbon

16. Observe the following figure carefully and answer the following question.



The Candle

Which zone of candle flame in above figure is the hottest zone?
(a) Innermost zone
(b) Middle zone
(c) Outermost zone
(d) All flames are equally hot

17. Which of the following is correct definition of ignition temperature?
(a) The lowest temperature at which a substance catches fire spontaneously.
(b) The lowest temperature at which a substance catches fire.
(c) The lowest temperature at which a substance changes its physical state.
(d) The lowest temperature required for a chemical reaction to take place.

18. Which of the following is correct definition of calorific value of fuel?
(a) It is the amount of heat required for burning one unit of fuel in presence of oxygen.
(b) It is the amount of heat energy given out on burning one unit of fuel.
(c) It is the amount of heat energy given out on burning one unit of solid fuel.
(d) It is the amount of heat energy given out on burning one unit of gaseous fuel.

19. Which of the following is best explanation of statement: Metabolism of glucose is considered as combustion
(a) It involves reaction of glucose molecule with oxygen
(b) Products obtained are CO_2 and H_2O therefore it is a combustion reaction occurs at slow and steady rate.
(c) This reaction occur as rapid as combustion reaction.
(d) This reaction release enormous amount of energy

20. Which of the following is best explanation of statement: Water should never be used to extinguish oil fires.
(a) Water is heavier than oil, on spraying water it sinks and oil floats, and the fire continues.
(b) Water is lighter than oil, on spraying water it sinks and oil floats and the fire continues.
(c) Water reacts with oil and form a new flammable compound.
(d) Oil fire is very intense it convert water in to steam and evaporate it off.

21. LPG burns with a
(a) non-luminous flame
(b) luminous flame
(c) yellow flame
(d) both (a) and (b)

22. In improved chulhas, smoke is removed by
(a) pipes (b) windows
(c) doors (d) chimney

23. Digestion of food is in an example of
(a) incomplete combustion
(b) complete combustion
(c) slow combustion
(d) rapid combustion

24. In presence of water, ignition temperature of paper
(a) decreases (b) increases
(c) remains constant (d) can decrease or increase

25. The head of the safety match contains a mixture of
(a) antimony trisulphide and potassium chlorate.
(b) potassium chlorate and red phosphorus
(c) antimony trisulphide and white phosphorus
(d) glass powder and potassium chlorate

26. Place a piece of burning charcoal an iron plate and cover it with a plastic jar. The charcoal stops burning because
(a) its ignition temperature is lowered
(b) supply of oxygen is out off
(c) it becomes cold after some time
(d) None of the above

27. Why do fire brigades pour water on the fire ?
(a) Water cools the combustible material so that its temperature is brought below its ignition temperature.
(b) This prevents the fire from spreading.
(c) Water vapours help in cutting off the supply of air.
(d) All of the above

28. The substance that does not burn with flame is
(a) LPG (b) dry grass
(c) camphor (d) charcoal

29. Choose the correct statement about inflammable substances from the following. They have
(a) low ignition temperature and cannot catch fire easily
(b) high ignition temperature and can catch fire easily
(c) low ignition temperature and can catch fire easily
(d) high ignition temperature and cannot catch fire easily

30. Choose the incorrect statement from the following
Forest fires are usually due to :
(a) carelessness of humans
(b) cutting of trees
(c) heat of sun
(d) lightning strike

31. Shyam was cooking potato curry on a chulha. To his surprise he observed that the copper vessel was getting blackened from outside. It may be due to :
(a) is readily available.
(b) producers a large amount of heat.
(c) leaves behind many undersirable substances.
(d) burns easily to air at a moderate rate.

More than One Option Correct :

DIRECTIONS : *This section contains* **6** *Multiple Choice Questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE OR MORE may be correct.*

1. Water can be used as fire extinguisher to put out –
(a) burning wood (b) burning oil
(c) burning cloth (d) burning charcoal

2. Which is/are not fuel(s) ?
(a) Wood (b) Carbon monoxide
(c) Nitrogen (d) Oxygen

3. Which of the following gases cause acid rain ?
(a) Ammonia (b) Hydrogen
(c) Oxides of Nitrogen (d) Sulphur dioxide

4. Burning is accompanied with
(a) evolution of heat (b) evolution of energy
(c) evolution of light (d) evolution of electricity

5. We can prevent energy crisis if we
(a) use non-renewable sources of energy
(b) slow down the use of fuels like coal and petroleum
(c) start using sources like wind, sunlight, hydro energy, etc.
(d) use wood & coke instead of coal & petroleum

6. Fire extinguishers extinguish the fire by
(a) cutting off the supply of air
(b) bringing down the temperature of fuel
(c) supplying air to the fuel
(d) bringing down the temperature of fire

Passage Based Questions :

DIRECTIONS : *Study the given paragraph(s) and answer the following questions.*

Passage-1

Any substance that can be burnt or otherwise consumed to produce heat energy is called a fuel. Wood, natural gas, petrol, kerosene, diesel, coal, and LPG are commonly used as fuels. Fuel serves as a major part of our energy requirement. Petroleum, a major fuel is widely used in our everyday lives and also used to power automobiles. Usually fuels are compounds of carbon and hydrogen, thus they combine with oxygen on burning and liberate carbon dioxide and water.

1. Which fuels are used for running automobiles?
(a) Wood (b) Coal
(c) Diesel (d) Charcoal

2. Which of the following is/are carbon fuel
(a) Wood (b) Coal
(c) Petroleum (d) All of these

3. Fuel may be
(a) Solid (b) Liquid
(c) Gas (d) All of these

Passage-2

Carbon fuels such as wood, coal, petroleum release unburnt carbon particles in the environment. These particles are very dangerous pollutants and cause respiratory diseases for example asthma.When fuels are incompletely burnt, they release carbon monoxide gas in the atmosphere. This gas is very dangerous as it is poisonous in nature. The combustion of fossil fuels also releases a large amount of carbon dioxide in the atmosphere. Carbon dioxide is a green house gas which is responsible for global warming. Burning of coal and diesel releases sulphur dioxide gas. This gas is extremely corrosive and suffocating in nature. Petrol gives off oxides of nitrogen. The oxides of sulphur and nitrogen get dissolved in rain water and form acids.

4. Unburnt carbon particles causes
(a) Stomach infections (b) Respiratory problems
(c) Brain infections (d) Throat problems

5. Global warming is caused due to____ concentration of CO_2 in air
(a) Decreased (b) increased
(c) Both (a) & (b) (d) None of these

6. Burning of coal and diesel releases
(a) NO_2 (b) CO_2
(c) SO_2 (d) CO

Assertion & Reason :

DIRECTIONS : *Each of these questions contains an assertion followed by reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.*

(a) If both **Assertion** and **Reason** are **correct** and Reason is the **correct explanation** of Assertion.
(b) If both **Assertion** and **Reason** are correct, but Reason is **not the correct explanation** of Assertion.
(c) If **Assertion** is **correct** but **Reason** is **incorrect**.
(d) If **Assertion** is **incorrect** but **Reason** is **correct**.

1. **Assertion :** Innermost zone of the flame is cooler than the outer zones and is dark.
Reason : The middle zone is largest zone of flame and this zone gives soot and smoke.

2. **Assertion :** Luminous zone of a flame is mainly due to incomplete burning of carbon
Reason : In luminous zone hydrogen burns with a brilliant yellow luminous flame.

3. **Assertion :** Calorific value of methane is greater than ethane
Reason : Fuels having higher percentage of hydrogen have higher calorific value.

4. **Assertion :** Fire can be extinguished by cutting off the supply of oxygen.
Reason : Oxygen is a supporter of combustion

5. **Assertion :** Candle burns with a flame whereas coal does not.
Reason : Coal does not vapourise on combustion.

6. **Assertion :** Diesel and petrol can not be used as household fuels for cooking.
Reason : They are highly non-flammable substances.

7. **Assertion :** Coal fire can be started by using a lighted matchstick
Reason : Coal has a high ingnition temperature.

Multiple Matching Questions :

DIRECTIONS : *Following question has four statements (A, B, C and D) given in Column I and four statements (p, q, r and s) in Column II. Any given statement in Column I can have correct matching with one or more statement(s) given in Column II. Match the entries in column I with entries in column II.*

1.

	Column-I		Column-II
A.	Oxides of sulphur and nitrogen	(p)	Respiratory irritants
B.	Burning of gasoline in automobiles	(q)	Rapid combustion
C.	Fuel burnt in vapour phase	(r)	Acid rain
D.	Use of LPG as domestic fuel	(s)	Flame

Integer Type Questions :

DIRECTIONS : *Following are integer based questions. Each question, when worked out will result in one integer from 0 to 9 (both inclusive).*

1. How many among the following are solid fuel ?
Petrol, LPG, Coal, Wood, Cow dung cake, Charcoal , CNG, Biogas.

2. How many among the following are\ produces flame
Candle, Magnesium, Camphor, Coal

4 ADVANCED EXERCISE
BASED ON CONNECTING TOPICS

DIRECTIONS (Qs. 1-8) : *This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.*

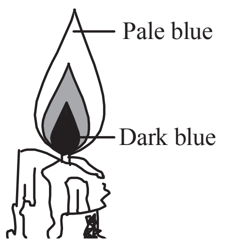
1. Which of the following substances is used in place of H_2SO_4 in foam type of fire extinguisers ?
(a) Saponin (b) Aluminium sulphate
(c) Sodium carbonate (d) Alcohol

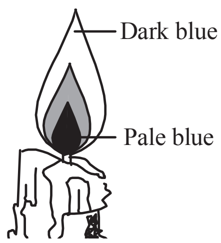
2. Hydrocarbon burns to produce
(a) CO_2 + water vapour + Heat + light
(b) $CO_2 + O_2$
(c) $CO_2 + H_2$ + energy
(d) $CO + H_2$

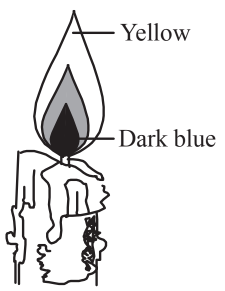
3. Daily use candles (paraffin wax) contain
(a) higher saturated hydrocarbon
(b) lower saturated hydrocarbon
(c) higher unsaturated hydrocarbon
(d) lower unsaturated hydrocarbon

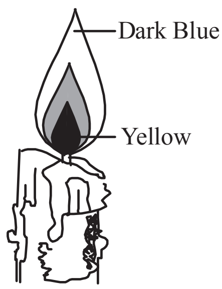
4. The salt used in soda acid type extinguisher is
(a) potassium chlorate
(b) sodium nitrate
(c) sodium bicarbonate
(d) potassium sulphate

5. Which of the following diagram shows a non-luminous flame?

(a) 

(b) 

(c) 

(d) 

6. CO_2 is stored as a liquid in cylinder at
(a) high pressure (b) low pressure
(c) high temperature (d) low temperature

7. CO_2 is given off by chemicals like
(a) $NaHCO_3$ (b) $CaSO_4$
(c) Na_2SO_4 (d) H_2SO_4

8. Which among the following statements is *false*?
(a) Oil slick in sea water increases D.O. value.
(b) The main reason for river water pollution is industrial and domestic sewage discharge.
(c) Surface water contains a lot of organic matter mineral nutrients and radioactive materials.
(d) Oil spill in sea water causes heavy damage to fishery.

DIRECTIONS (Qs. 9-13) : *This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE OR MORE may be correct.*

9. Which of the following is/are advantage(s) of solid fuels ?
(a) Their ash content is high
(b) They are easy to transport
(c) Cost of production is low
(d) Easy to store

10. Which of the following is/are advantage(s) of liquid fuels?
(a) Require less furnace space for combustion
(b) Clean in use
(c) Can be stored indefinitely
(d) Leaves less ash content

11. Which of the following is/are advantage(s) of gaseous fuels?
(a) They have high heat content
(b) They are clean in use
(c) They can be lighted at moment's notice
(d) They can be transported through network of pipelines

12. Which of the following is/are disadvantage(s) of gaseous fuels ?
(a) very large storage tanks are needed
(b) higher possibilities of fire hazards
(c) they are costly
(d) they leaves no ash on burning

13. The zone that does not produce light in the candle flame is
(a) non-luminous zone
(b) luminous zone
(c) invisible zone
(d) All zones give light

DIRECTIONS (Qs. 14-16) : *Study the given paragraph(s) and answer the following questions.*

Passage

Fire extinguishers are more popular, compared to the remaining Fire protection systems. There are a variety of fires. Each fire type , cannot treated the same way that, is why we have to use different fire extinguishers for different fires. For example, a water type fire extinguisher cannot be used for an electrical fire. Fire extinguishers work on the principle that by isolating oxygen from, the fuel and ignition; the Fire triangle is effectively broken. Hence, the fire stops immediately. However, Fire extinguishers are supposed to be used for small fire or fire in an infant stage.

14. Which of these fire extinguisher is most suitable for electrical fires?
(a) CO_2 (b) water
(c) wet chemical (d) foam

15. Which of these materials would you use a water fire extinguisher on?
(a) paper (b) wood
(c) plastic (d) all of these

16. Besides a CO_2 fire extinguisher, which type of fire extinguisher can you use on electrical fires?
(a) Wet chemical fire extinguisher
(b) Foam fire extinguisher
(c) Dry powder fire extinguisher
(d) Water fire extinguisher

DIRECTIONS (Qs. 17-19) : *Each of these questions contains an Assertion followed by reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.*

(a) If both **Assertion** and **Reason** are **correct** and Reason is the **correct explanation** of Assertion.
(b) If both **Assertion** and **Reason** are correct, but Reason is **not the correct explanation** of Assertion.
(c) If **Assertion** is **correct** but **Reason** is **incorrect**.
(d) If **Assertion** is **incorrect** but **Reason** is **correct**.

17. **Assertion :** Water fire extinguishers can be used for solids only, such as wood, paper, plastics and fabrics
Reason : water is a bad conductor of electricity

18. **Assertion:** the substances classified as fuel must necessarily contain one or several of the combustible elements
Reason: carbon, hydrogen, sulphur are combustible substances.

19. **Assertion:** The heat of the flame vaporizes the liquid wax .
Reason: These vaporized molecules are drawn up into the flame

DIRECTIONS (Qs. 20) : *Each question contains statements given in two columns which have to be matched. Statements (A, B, C, D) in column I have to be matched with statements (p, q, r, s) in column II.*

20.

	Column-I		Column-II
A.	Petrol	(p)	Water
B.	electrical wires	(q)	Foam
C.	textile	(r)	Dry powder
D.	flammable gases	(s)	CO_2 gas

(a) A-q, r, s ; B- r, s ; C- p, q, r ; D- r
(b) A-s ; B- r, s ; C- q, r ; D- r
(c) A-r, s ; B- s ; C- p, r ; D- r
(d) A-q, r ; B- r ; C- p, q, r ; D- s

DIRECTIONS (Qs. 21-22) : *Following are integer based questions. Each question, when worked out will result in one integer from 0 to 9 (both inclusive).*

21. Candle flame contain how many flames ?

22. How many of the following are liquid fuels ?
Paraffin wax, kerosene, diesel, CNG, LPG, charcoal and acetylene.

SOLUTIONS

Brief Explanations of Selected Questions

1 EXERCISE

Fill in the Blanks :

1. Fuel.
2. heat
3. innermost
4. calorific
5. CNG
6. smoke
7. Acid rain
8. Carbon dioxide
9. higher
10. slow
11. spontaneous
12. fire, blanket
13. low, inflammable

True / False :

1. False
2. True
3. False [**Note :** Carbon monoxide is a combustible gas and burns in air with a pale blue flame]
4. False : Luminous zone is moderately hot part of the flame.
5. False : LPG does not leave any residue and has high calorific value.
6. False : Ignition temperature of green leaves is higher due to presence of water. Hence, do not burn easily.
7. True : Outer zone of a flame is blue while middle zone is yellow in colour.
8. False : Incomplete combustion of a fuel gives carbon monoxide.
9. True
10. True
11. True
12. True
13. False : Magnesium is a combustible metal.

Match the Following :

1. A $\rightarrow$ (s), B $\rightarrow$ (r), C $\rightarrow$ (q), D $\rightarrow$ (p)

Very Short Answer Questions :

1. Flame is a zone of combustion of gaseous substances accompanied by evolution of heat and light.
2. Incomplete combustion
3. It is a device used for putting off fires.
4. Soda-acid type, foam type.
5. Water cools down the fuel below its ignition temperature hence fire gets extinguished.
6. LPG burns with a blue flame.
7. Charcoal does not vaporise, so it does not produce a flame.
8. Water, sand and carbon dioxide.
9. The power generated by the help of coal is known as thermal power.
10. Putting sand on fire, cuts off the supply of oxygen to the combustible substance, hence fire is put off.
11. The efficiency of fuel is determined by its calorific value.
12. The fine particles released on burning carbon containing fuels are called suspended particulate matter (SPM) and are harmful especially to lungs.
13. CNG - Compressed Natural Gas
    LPG - Liquified Petroleum Gas
14. The SI unit of calorific value is J/kg
15. No, this process does not produce heat or light.

Short Answer Questions :

1. The loud noise produced when a fire cracker is burst is due to the excessive amount of energy being instantaneously released into the air, making it expand faster than the speed of sound. This rapid release produces a shock wave, and we hear a loud boom.
2. (a) CNG produces very small amount of harmful products. It is a clean fuel. Hence, diesel and petrol as fuel are being replaced by CNG in automobiles.

   (b) The head of the safety match contains antimony trisulphide and potassium chlorate. The rubbing surface has powdered glass and a little red phosphorus. When the match is struck against the rubbing surface, some red phosphorus gets converted into white phosphorus. This immediately reacts with potassium chlorate in the matchstick head to produce enough heat to ignite antimony trisulphide and starts the combustion.
3. Harmful effects of increasing consumption of fuels are

   (i) Burning of carbon containing fuels like wood, coal and petroleum release unburnt carbon particles which create respiratory problems like asthma.

   (ii) Incomplete combustion of coal, gasoline and other fossil fuels releases gases like carbon monoxide.

(iii) Oxides of sulphur and nitrogen released by burning of coal and diesel cause acid rain causing damage to buildings and soil.
(iv) Excessive use of fuel cause global warming.

4. (a) Kerosene lamp-yellow flame.
(b) Candle-yellow flame.
(c) Bunsen burner flame-blue flame.

5. (i) Solid fuels – Coal, wood, dung cakes
(ii) Liquid fuels – Petrol, kerosene, diesel
(iii) Gaseous fuels – CNG, LPG, gobar gas

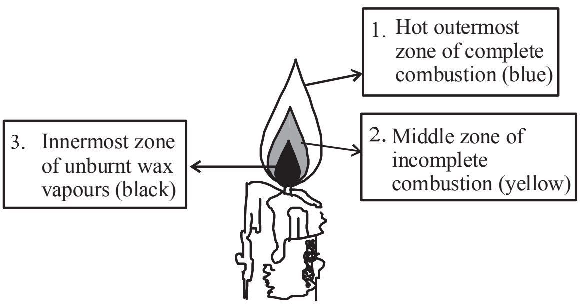
6. Burning of wood gives a lot of smoke which is harmful for human beings. Also, trees provide us with useful substances which are lost when wood is used as a fuel. Moreover cutting of trees leads to deforestation which is harmful to the environment.

Long Answer Questions :

1. Following are the major problems caused due to combustion of fuels
(i) Fuels, such as wood, coal, petrol, diesel, kerosene, LPG, etc., when burnt, produce gases like carbon monoxide, carbondioxide, oxides of nitrogen and sulphur. In addition to these gases, generally in most solid fuels, smoke is also produced. These gaseous products and unburnt carbon particles cause air pollution.
(ii) The solid fuels, such as wood and coal, after burning produce a large amount of residue (soot, ash) after combustion. This ash if not disposed off properly, may cause air and water pollution.
(iii) Pollutants in the air cause many diseases such as asthma and other respiratory problems in human beings.
(iv) Carbon dioxide produced by burning fuels causes rise in temperature of thermosphere. This is called global warming.
(v) Oxides of sulphur and nitrogen dissolve in rain water and cause acid rain which is harmful for buildings, crops and soil.

2. Fire crackers are made of chemicals that are highly combustible. Thus, when they burn, they explode i.e., they release a lot of energy in an instant, in the form of heat, light and sound. When a combustible substance burns very rapidly with the evolution of tremendous heat, light and sound energy, it is called an explosion.

3. (a)



(b) The candle flame has three zones–
(i) **Non-luminous zone** is the outermost zone of the flame where complete combustion takes place.
(ii) **Luminous zone** is the middle zone where incomplete combustion occurs.
(iii) **Dark zone** is the innermost zone near the wick where no combustion occurs.

4.

S.No.	Solid fuel	Liquid fuel	Gaseous fuel
1	Coal	Diesel	CNG
2	Wood	Kerosene	Biogas
3	Cow dung cake	Petrol	LPG
4	Charcoal	Alcohol	Hydrogen

5. **Hint :** CNG, because the calorific value of CNG is higher than that of petrol. Therefore CNG will be more economical. At the same time it produces the least air pollutants.

6. **Hint :**
(i) Wood produces lot of air pollution.
(ii) Use of wood as fuel encourages cutting of trees leading to deforestation.

2 EXERCISE

Text-Book Exercise :

1. Conditions for combustion are as following:
(i) Presence of air (oxygen)
(ii) Presence of inflammable substance
(iii) Ignition temperature

2. (a) Pollution
(b) Liquefied petroleum gas (LPG)
(c) Ignition temperature
(d) Water

3. CNG is a cleaner fuel as compared to petrol and diesel. Combustion of these fuels results in formation of unburnt carbon particles and harmful gases. CNG produces less harmful gases on burning, due to this CNG is now being used in automobiles in many cities, which has helped in reducing pollution in our cities.

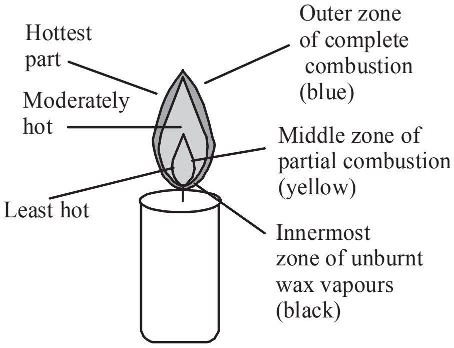
4. LPG as a fuel is better than wood. LPG burns without giving smoke while burning wood produces many air pollutants. Wood is obtained from trees; therefore using wood as a fuel causes deforestation. LPG has more fuel efficiency than wood. Hence LPG is favoured over wood.

5. (a) Water is a good conductor of electricity. Due to which the person trying to control fire involving electrical equipment with water might get electric shock.
(b) LPG is a better domestic fuel than wood because LPG burns without smoke and its storage and transportation is also easier than wood. Using wood as a fuel results in blackening of pots and pan due to

release of carbon particles during burning of wood. Carbon particles and gases released during burning of wood may also cause respiratory disorder.

(c) Aluminium is a metal therefore it is a good conductor of heat. When piece of paper wrapped around aluminium is heated, heat is transferred from the paper to the metal and the paper does not attain its ignition temperature. Hence paper does not burn.

6.



7. The calorific value of a fuel is expressed in kilo joule per kg (kJ/kg).

8. CO_2 controls fire in following ways:

(i) CO_2 being heavier than oxygen covers the fire like a blanket. This cut off the contact between the fuel and oxygen and the fire is controlled.

(ii) CO_2 when released from cylinder, expands enormously in volume and cools down. In this way, it also bring down the temperature of the fuel and control the fire.

9. Ignition temperature of green leaves are higher than dry leaves due to presence of moisture in green leaves. This is why it is difficult to burn heap of green leaves.

10. Goldsmith uses outermost zone of a flame for melting gold and silver because it is the hottest zone of a flame.

11. Calorific value is amount of energy produced on combustion of 1 kg of a fuel.

4.5 kg of fuel produced heat = 180,000 kJ

$$1 \text{ kg of fule produced heat} = \frac{180{,}000 \text{ kJ}}{4.5 \text{ kg}}$$

$$= 40{,}000 \text{ kJ/kg}$$

Hence, calorific value of fuel = 40,000 kJ/kg

12. Both rusting and combustion involves oxidation process but rusting cannot be termed as combustion because combustion is an exothermic process while heat is not produced during rusting.

13. The water in Ramesh's beaker will heat up in shorter time. This is because the outermost zone of a flame is the hottest zone. Abida had kept the beaker in yellow zone which is less hot so, it will take more time to heat up the water.

Exemplar Questions :

1. Anu should keep her test tube in the outermost zone which is the hottest part of the flame.

2. (a) Combustible – charcoal, straw, card board, paper, candle wood.

(b) Non-combustible – chalk, stone, iron rod, copper coin, glass.

3. Petrol will catch fire instantly because it is highly inflammable.

4.
- Water is not suitable for fires involving oil.
- Manu should have switched off the flame of the burner and put a lid on the frying pan. By doing this the contact between fuel and oxygen is cut off and the flame will go off.

5. In jar B, because oxygen is a supporter of combustion.

6. Due to insufficient availability of oxygen in the closed room carbon monoxide gas is produced which can kill persons sleeping in that room.

7. Sudden formation of large amount of gas due to chemical reactions.

HOTS Questions :

1. CO_2 absorbs strongly in IR region and its increased concentration in atmosphere decreases the loss of heat from earth by radiation.

2. The surrounding air pushes the flame up because hot, burning gases comprising the flame expand and thus become less dense than the surrounding air, hence the flame moves up.

3. Hydrogen has the highest calorific value but it is dangerous to handle, because it burns with explosion and is highly inflammable.

4. Hot air is less dense and lighter as compared to cold air.Flame heats the air around it .hot air rises up and cold air tries to take space from bottom area of the flame.this makes a circulation effect thus making the flame points upward.

5. Wood contains many chemicals like resin, lignin. When heated, these chemicals broke up and release combustible gases. That's why wood burns initially with a flame. when these gases are exhausted, wood burns without flame.

6. Charcoal is pure form of carbon. When it burns with oxygen, it forms carbon dioxide and no water vapours are given off.

7. If fuel and air mixture is correct, gas stove gives blue flame. When air supply (oxygen) is insufficient, fuel does not burn completely thus giving a yellow flame. Gas burner requires cleaning and air shutter adjustment to provide sufficient air.

8. There are many reasons for forest catching fire during hot summers.

(i) Dry leaves and wood are flammable. Hot air and friction on dried wood is sufficient to raise temperature to its ignition level.

(ii) Long concentration of sunlight also raises the temperature causing fire.

(iii) A small fire from a left cigarette butt or camp fire can cause burning of dried leaves and wood.

3 EXERCISE

Multiple Choice Questions :

1. (c) When water is poured over the burning substance water absorbs heat from the substance. Because of this the temperature of burning substance falls below its igntion temperature and it stops burning.

2. (c) As the fuel with more percentage of hydrogen has more calorific value.

3. (d)

4. (b) Outermost zone of candle flame is non-luminous it is poorly visible and is slightly blue. It is the hottest part of the flame.

5. (b)

6. (b)

7. (c) In combustion process fuel is burnt with formation of CO_2 and H_2O as products along with evolution of heat and light.

8. (a) Vaporised wax gets oxidised to carbon monoxide and carbon monoxide burns completely with a blue flame in this zone.

9. (d) Outermost non-luminous zone is the hottest zone where complete oxidation (burning) of the fuel is taking place.

10. (a) A good fuel should have high calorific value. It should not burn too fast nor too slow.

11. (d)

12. (b) Carbon dioxide (CO_2) is the best fire extinguisher.

13. (c)

14. (d) Spontaneous combustion of coal dust has resulted in many disastrous fires in coal mine.

15. (a)

16. (c) It is the hottest zone of candle flame.

17. (b)

18. (b)

19. (b)

20. (a)

21. (a) Due to complete combustion, LPG burns with a non-luminous flame.

22. (d) Chimney helps in removing the smoke by releasing it at a height.

23. (c) All biochemical reactions are slow reactions.

24. (b) Water increases the ignition temperature of an object.

25. (a)

26. (d) Due to non-availability of oxygen, charcoal stops burning.

27. (d)

28. (d)

29. (c)

30. (c)

31. (c)

More Than One Option Correct :

1. (a, c, d)
In case of oil fires, to extinguish the fire water is not effective. As water is heavier than oil. Thus on spraying water it sinks and oil floats, and the fire continues.

2. (b, c, d)
Wood is a fuel. Fuel is the source of heat energy for domestic and industrial purposes.

3. (c, d) Oxides of nitrogen and sulphur dioxide causes acid rain.

4. (a, b, c)
Burning is accompanied by the evolution of heat and light with energy.

5. (b, c)

6. (a, b)

Passage Based Questions :

1. (c)

2. (d)

3. (d)

4. (b)

5. (b)

6. (c)

Assertion & Reason :

1. (b)

2. (b)

3. (a) Percentage of hydrogen in comparison to carbon is greater in methane than ethane.

4. (a)

5. (a) Only those substances give a flame which are converted to vapours.

6. (d) They cannot be used as household fuels since they catch fire very easily, i.e., inflammable.

7. (d) A coal fire cannot be started by using a lighted matchstick.

Multiple Matching Questions :

1. $A \rightarrow (p, r)$; $B \rightarrow (q, r)$; $C \rightarrow (s)$; $D \rightarrow (q, s)$

Integer Type Questions :

1. **4**
Wood, Cow dung cake, Charcoal

2. **3**
Candle, Magnesium,Camphor

4 EXERCISE

1. (b)
2. (a)
3. (a) Candles consists of paraffin wax which generally contains saturated hydrocarbons of length more than then C_{24}.
4. (c) Sodium bicarbonate reacts with sulphuric acid to give carbon dioxide.
5. (a)
6. (a)
7. (a)
8. (a) Oil slick in sea water decreases D.O value.
9. (b, c, d)
10. (a, b, c)
11. (a, b, c, d)
12. (a, b, c)
13. (a, c)
    Luminous zone produces yellow light in the candle flame.
14. (a)
15. (d)
16. (c)
17. **(c)**
18. **(b)**
19. **(a)**
20. (a) A - q, r, s ; B - r, s ; C - p, q, r ; D - r
21. **3**
    Candle flame contains three main zones dark inner zone, luminous and non-luminous zone.
22. **2**
    Kerosene and diesel are liquid fuels.

Chapter 5

LANGUAGE OF CHEMISTRY

INTRODUCTION

During every moment of our existence we depend upon chemistry directly or indirectly.
Chemical sciences include the study of substances or chemicals (the chemical elements), what they are made of, and how they differ from each other in their many properties and features. Another very important area of chemistry is how substances or chemicals change when they combine or react together.

We know that chemistry is a branch of science which deals with study of matter and various changes it undergoes. It deals with the preparation, properties, reactions and structures of chemical elements and compounds.

For convenience the study of chemistry is sub-divided into various branches such as:

(i) **Inorganic chemistry :** It is the branch of chemistry that deals with the study of the compounds (generally excluding carbon compounds) obtained from mineral or inanimate sources.

(ii) **Organic chemistry :** It is the branch of chemistry that deals with the compounds of carbon and hydrogen obtained from animal and plant sources. In it we study about carbohydrates, fats, proteins, vitamins etc.

(iii) **Physical chemistry :** It is the branch of chemistry that deals with the physical principles and conditions that govern the chemical processes.

(iv) **Analytical chemistry :** It is the branch of chemistry that deals with the study of the methods of detection and determination of elements and compounds.

(v) **Industrial chemistry :** It is the branch of chemistry that deals with the study of chemical processes involved in the manufacture of industrial products.

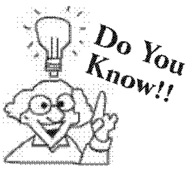
(vi) **Bio-chemistry :** It is the branch of chemistry that deals with the study of chemical processes taking place in living organism.

(vii) **Nuclear chemistry :** It is the branch of chemistry that deals with the study of the chemistry of radio-active substances and the energy changes taking place in the nucleus of the atom.

(ix) **Pharmaceutical chemistry :** Chemistry dealing with pharmaceutical preparations and drug study.

(x) **Medicinal chemistry :** Study of structure-activity relationship, pharmacological activities.

(xi) **Material chemistry :** Covers solid state chemistry, both inorganic and organic, and polymer chemistry, especially as directed to the development of materials with novel and/or useful optical, electrical, magnetic, catalytic, and mechanical properties.



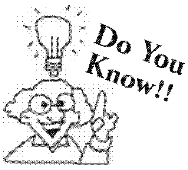
Some other branches of chemistry are instrumental Chemistry - deals with design and theory of various instruments desired to study the chemical change.
Astrochemistry - deals with the chemical aspects of space.
Geochemistry - Deals with the chemical aspects of earth crust

CHECK Point

Why chemistry is sub-divided into many branches?

SOLUTION

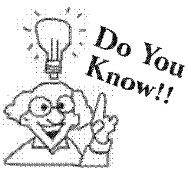
In order to make sure easy and systematic study of the subject.



The most abundant type of atom in the universe is the hydrogen atom. Nearly 74% of the atoms in the Milky Way galaxy are hydrogen atoms.

CHEMISTRY AND ITS LANGUAGE

We know that every science has its own technical terms which is frequently used and which is considered as its language. In chemistry we find that in addition to technical terms it also uses certain expression like H_2, O_2, N_2, H_2O, H_2SO_4, NaOH, NaCl, KCl etc. which in the language of chemistry stand for names of certain definite chemical compounds. To have a better understanding of the language of chemistry it is essential to know some of the chemical terms which are very frequently used. Here we give a few terms commonly used in chemistry.



To measure the masses of different atoms and to determine probable molecular formula from decomposition data, atomic mass scale was developed. Initially in the 19th century, first atomic scale was made by using hydrogen (the lightest element) as reference. H atom was arbitrarily assigned by a mass of 1 and mass of other atoms were assigned relative to it.

Knowledge ENHANCER

There are different kinds of atoms. About 92 of them occur naturally, while the remainder are made in labs. The first new atom made by man was technetium, which has 43 protons. New atoms can be made by adding more protons to an atomic nucleus. However, these new atoms (elements) are unstable and decay into smaller atoms instantaneously.

(i) ***Atom :*** *Atoms are the building blocks of all matter. It is the smallest particle of an element that maintains its chemical identity throughout all chemical and physical changes. Atoms of one element are different from those of the other elements. Today, we know that atoms are not truly indivisible they are themselves made up of particles (protons, neutrons, electrons, etc.).*

(ii) ***Molecule :*** *The smallest particle of an element or compound which can exist independently in nature is termed as molecule. Molecules are formed by the combination of two or more atoms in a constant ratio. If same type of atoms are present in molecule then it is termed as Homoatomic molecule.*
For example : N_2 (Nitrogen molecule), O_2 (Oxygen molecule), Cl_2 (Chlorine molecule)
If two or more than two types of atoms are present in any molecule then it is termed as heteroatomic molecule.
For example – H_2O (water), CO_2 (Carbon dioxide), H_2O_2 (Hydrogen peroxide) etc.

(iii) ***Atomic mass or Atomic weight :*** *Atomic mass of an element is defined as the mass of one atom of an element compared to 1/12 th of the mass of an atom of C-12.*
Thus when we say that atomic mass of oxygen is 16, it means that an atom of oxygen is 16 times heavier as compared to 1/12th of the mass of an atom of C-12.
Atomic mass is expressed in a.m.u. (atomic mass unit).
Atomic mass unit *(amu) is the 1/12th of the weight of an atom of C-12.*

CHECK Point

In the latest system of atomic scale, C-12 has been choozen as reference with atomic mass 12.000 u. Why?

SOLUTION

Because by doing so, atomic masses of maximum number of elements can be made whole number or nearly whole number.

Atomic weights of some common elements

Element	Atomic mass	Element	Atomic mass
Hydrogen	1	Fluorine	19
Lithium	7	Sodium	23
Boron	11	Magnesium	24
Carbon	12	Aluminium	27
Nitrogen	14	Chlorine	35.5
Oxygen	16	Calcium	40
Chromium	52	Bromine	80
Manganese	55	Silver	108
Iron	56	Tin	119
Copper	63.5	Iodine	127
Zinc	65	Sulphur	32
Phosphorus	31	Potassium	39
Barium	137	Platinum	197
Mercury	200	Lead	207

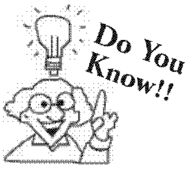
(iv) **Gram atomic mass or gram atomic weight :** The atomic mass of an element when expressed in grams is known as **gram atomic mass** or simply as **gram atom (g-atom)**. Thus one g-atom of carbon (C-12) weighs 12.0 g.

(v) **Molecular mass or molecular weight :** Molecular weight is calculated by adding the atomic weights of all the constituent atoms present in a molecule. For example.

Molecular weight of a molecule of hydrogen (H_2)

$$= 2 \times \text{atomic weight of hydrogen} = 2 \times 1 = 2 \text{ amu}$$

(vi) **Gram molecular mass:** The molecular mass of a substance when expressed in grams is known as **gram-molecular mass** or simply **gram-mole** (**g-mole**) for the sake of convenience it is expressed simply as **mole**. Thus one **mole** (or **g-mole**) of water weighs 18g (molecular weight of water = 18)



Macromolecules like polymers, proteins and carbohydrates have high molecular mass and its determination is difficult.

SYMBOLS

We use many symbols in mathematics simply to avoid writing full and lengthy terms so as to save time and to be precise. e.g. to write Angle A = Angle B, we write ∠A = ∠B.

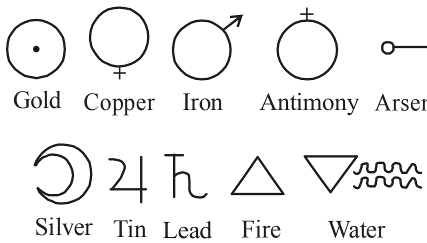
Similarly, if we have to write "triangle", "Parallel", "Since", then we denote these term by symbols, Δ, ||, ∵ respectively.

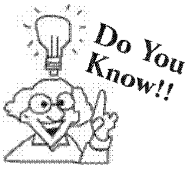
A stenographer uses shorthand to save time in taking down notes but symbols of stenographer are totally different from symbols we use in mathematics.

The need for symbols was also felt by chemists as the science of chemistry advanced. The larger number of elements and compounds needed a language which could represent the full meaning in short space and short time.

Many of the early symbols for the elements, such as those used by alchemists prior to seventeenth century, were rather puzzling. The alchemists who were interested in transforming base metals like lead into gold, used symbols that could not be easily interpreted by others. Some of the symbols used in alchemy are given below.

Symbols used in Alchemy

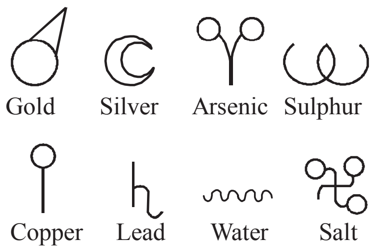




It is not possible to write all traffic rules to manage traffic run smoothly. Thus traffic signals are designed to manage traffic run smooth. Similarly symbols are designed to represent various element to study them easily.

The idea of using symbols in chemistry was originated by Greeks. The symbols for some metals used by ancient Greeks are shown below.

Ancient Greek

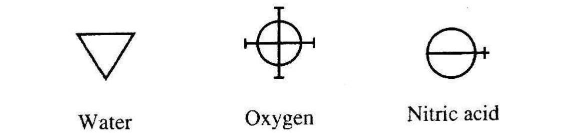


The Alchemists used strange symbols and puzzling signs to record their findings. They adopted the ancient Hindu and Greek astrologers symbols to represent some metals and other elements.

The symbols showed resemblance between metals and some heavenly bodies like Sun, Moon, Mars, Venus, Saturn etc.

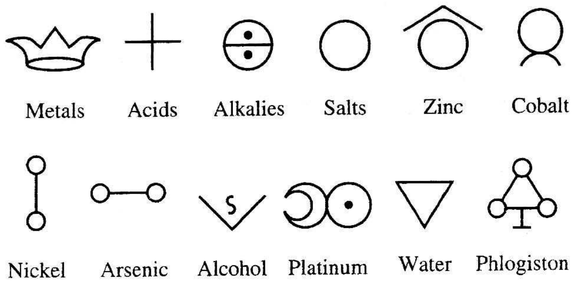
Various other systems used to write symbols are shown below :

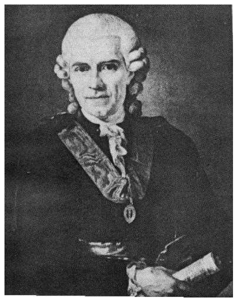
1782 Antoine Lavoisier



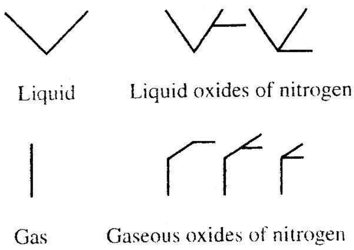

Antonine Lavoisier

1793 Torben Olof Bergman



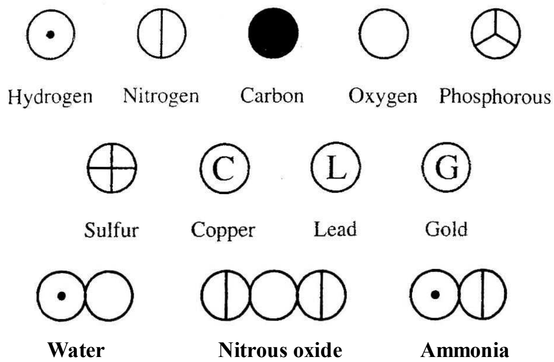

Torben Olof Bergman

1787 J. H. Hassenfratz and Adet




J. H. Hassenfratz

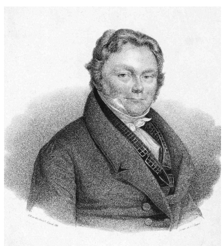
1808 John Dalton




John Dalton

1814 J. J. Berzelius

Cu + O	S + 3O
Copper Oxide	Sulfur trioxide


Berzelius

Modern

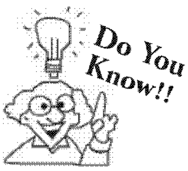
CuO	SO_3
Copper oxide	Sulfur trioxide
$CuSO_4$	K_2SO_4
Copper Sulfate	Potassium Sulfate

However it was found difficult to use these symbols developed by various scientists at different times.

The scientist who suggested a method of representing elements using the English letters (Capital as well as small) is J.J. Berzelius. The system that we use today is very close to that proposed by Berzelius. (Jons Jakob Berzelius, a Swedish scientist).

In the system used at present, the symbol for an element consists of first one or two letters of the name of the element (e.g. H for hydrogen, He for helium, Li for lithium). When names of two elements start with the same first two letters (e.g. magnesium and manganese), the symbol used first letter and a later letter (Mg for magnesium and Mn for manganese).

Berzelius said, "*I shall, therefore, take for the chemical sign the initial letter of common (or Latin/Greek) name of each chemical element*." Thus the element carbon was assigned the symbol C, Hydrogen H, Oxygen O, Nitrogen N, Phosphorus P, Sulphur S and so on...... If the first letter is common to two or more elements, we shall use both, the initial letter and the first letter they have not in common.



Sometimes when the names of two elements start with same just two letter. We use completely different letter to represent one of them like in case of potassium and polonium where potassium is represented by letter K and polonium as Po.

For example

Element	Symbol
Carbon	C
Copper	Cu (Latin, *Cuprum*)
Cobalt	Co (Latin, *Cobaltum*)
Calcium	Ca (Latin *Calx*)
Chromium	Cr (Greek, *chrom*)
Potassium	K (Greek, *Kalium*)
Krypton	Kr (Greek, *Kryptos*)
Antimony	Sb (Latin, *Stibium*)
Tin	Sn (Latin, *Stannum*)
Silicon	Si (Latin, *Silex*)

To represent a compound, Berzelius just joined the symbols of these elements contained in one molecule of that compound.
The method used at present to write symbols for elements is illustrated in the following tables:

(i) Elements with symbols based on non-english names

English Name	Symbol	Non-English Name	Language
Antimony	Sb	Stibium	Latin
Copper	Cu	Cuprum	Latin
Gold	Au	Aurum	Latin
Iron	Fe	Ferrum	Latin
Lead	Pb	Plumbum	Latin
Mercury	Hg	Hydrargyrum	Greek
Potassium	K	Kalium	Latin
Silver	Ag	Argentum	Latin
Sodium	Na	Natrium	Latin
Tin	Sn	Stannum	Latin
Tungsten	W	Wolfram	German

(ii) For some of the elements the first letter of its English name is used as symbol to represent that element in short form. Only capital letters are used.

Some examples to illustrate the point are listed in table below:

S. No.	Name	Symbol
1.	Hydrogen	H
2.	Carbon	C
3.	Nitrogen	N
4.	Oxygen	O
5.	Fluorine	F
6.	Sulphur	S
7.	Boron	B
8.	Phosphorus	P
9.	Iodine	I

(iii) When the names of the two elements start with the same letter, the second letter or a prominent letter is added to the first letter. When two letters are used the first letter is in capital and the second letter is always a small one.

Some examples to illustrate the point are listed in table below:

S. No.	Element	Symbol
1.	Carbon	C
2.	Calcium	Ca
3.	Cadmium	Cd
4.	Chlorine	Cl
5.	Phosphorus	P
6.	Platinum	Pt
7.	Palladium	Pd
8.	Boron	B
9.	Barium	Ba
10.	Bromine	Br
11.	Beryllium	Be

(iv) Some elements are named after the names of the scientists. For example.

	Element	Name of the Scientists	Symbol
1.	Curium	Madam curie	Cm
2.	Einsteinium	Albert Einstein	Es
3.	Fermium	Enrico Fermi	Fm
4.	Nobelium	Alfred nobel	No
5.	Mandelevium	Mendeleev	Md

(v) Some elements are named after the names of the countries and laboratories. For example:

	Element	Name of the Countries Laboratories	Symbol
1.	Berkelium	City of Berkley	Bk
2.	Californium	University of California	Cf
3.	Polonium	Poland	Po
4.	Americium	America	Am
5.	Ruthenium	Russia	Ru
6.	Germanium	Germany	Ge

(vi) Some elements are named after the names of the planets. For example:

	Element	Name of the planet	Symbol
1.	Uranium	Uranus	U
2.	Neptunium	Neptune	Np
3.	Plutonium	Pluto	Pu

SIGNIFICANCE OF A SYMBOL

It has both quantitative and qualitative significance.

Quantitative significance

The *symbol* of an element denotes *one atom* of the element as well as its *gram atomic weight*. e.g. The symbol N denotes

(i) 1 atom of nitrogen.

(ii) 14 parts by weight of nitrogen because 1 atom of nitrogen is 14 times heavier than $\frac{1}{12}th$ of the weight of an atom of carbon (C-12)

(iii) *One gram-atom* of nitrogen (i.e. 14 g of nitrogen)

Qualitative significance

Qualitatively the symbol represents the name of the element. e.g. **N** represents nitrogen.

(For writing the chemical formula the knowledge of valency is essential. So we will first learn about valency)

VALENCY

During the formation of molecules of compounds, atoms combine in certain fixed proportions. This is because of the fact that different atoms have different combining capacities.

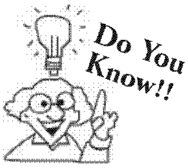
The combining capacity of an atom or radical is known as its **valency.**

The valency is measured in terms of hydrogen atoms or oxygen atoms. The valency of hydrogen is taken as one and is selected as the standard of valency.

Valency of an element can be defined as the number of hydrogen atoms which combine with an atom of element.

Following table illustrates the point:

Molecule	*Description*	*Valency of element*
Water (H_2O)	It contain two atoms of hydrogen in combination with one atom of oxygen.	Valency of oxygen is **2** (one atom of oxygen combines with the 2 atoms of hydrogen)
Methane (CH_4)	Four atoms of hydrogen combine with one atom of carbon to form CH_4	Valency of carbon is **4**
Ammonia (NH_3)	Three atoms of hydrogen with one atom of nitrogen to form NH_3	Valency of nitrogen is **3.**



The valence of an element is not always equal to the highest combining capacity : exceptions are there like ruthenium, osmium and xenon, which have valences by six but form compounds with oxygen in the +8 oxidation state, and chlorine, which has a valence of five but a valency of +7 [in perchlorates].

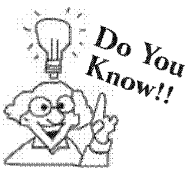
Since all atoms do not combine with hydrogen so the valency of the element is also defined in term of other elements like chlorine or oxygen.

Valency with respect to chlorine

Since the valency of chlorine is 1 (as in HCl) the number of chlorine atoms with which one atom of an element can combine is called its **valency**. e.g. In sodium chloride (NaCl), one atom of sodium (Na) combines with 1 atom of chlorine (Cl), therefore the valency of sodium is **1**.

Valency with respect to oxygen

Valency can also be defined as double the number of oxygen atoms with which one atom of an element can combine because the valency of oxygen is 2 (as evident from H_2O). e.g., In calcium oxide (CaO), one atom of calcium combines with one atom of oxygen so the valency of calcium is **2** ($2 \times 1 = 2$).



The concept of valency was first introduced in 1850. The term valency has been derived from the latin word 'valentia' which means capacity

CHECK Point

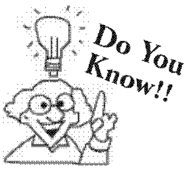
What is the valency of tin ?

SOLUTION

Since valency is defined as the number of chlorine atom with which one atom of an element can combine is called its valency. Thus tin shows two valency –2 and 4 in compounds $SnCl_2$ and $SnCl_4$ respectively.

IONS OR RADICALS

In addition to atoms and molecules, a third type of particles occurs in substances. These particles, called ions, are atoms or group of atoms that carry an electrical charge.



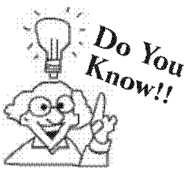
The size of cation is always smaller than its corresponding atom whereas the size of anion is always greater than its corresponding atom.

CHECK Point

Is there any differences between property of atom and ion ? Explain with example.

SOLUTION

The identity of elements is not altered when it changed in to ion, however the properties of ion differ from those of atoms e.g., Magnesium metal is appropriately reactive but its divalent ion is inert in nature because of stable electronic configuration.



A complex ion consists of an aggregate of atoms with a net charge. They consists of central metallic atom or ion to which one or more electron donating molecules is/are attached. In some complex ions, such as sulphate SO_4^{2-}, the atoms are so tightly bound together that they act as a single unit. e.g., $[Zn\ (NH_3)_4]^{2+}$

An **ion** is formed when electrons are removed from or added to an atom or group of atoms (see below).

$$\underset{\text{(Chlorine atom)}}{Cl} \xrightarrow{\text{Add 1 electron}} \underset{\text{(Chloride ion)}}{Cl^-}$$

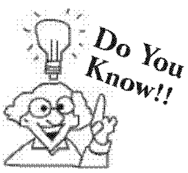
$$\underset{\text{(Chlorine atom)}}{Na} \xrightarrow{\text{Remove 1 electron}} \underset{\text{(Sodium ion)}}{Na^+}$$

$$\underset{\text{(Oxygen molecule)}}{O_2} \xrightarrow{\text{add 2 electrons}} \underset{\text{(Peroxide ion)}}{O_2^{2-}}$$

When electrons is/are removed the resulting ion is called a **cation or basic radical.** A cation is positively charged ion. (e.g. Na^+)

When electron is/are added the resulting ion is called an **anion or acidic radicals.** An anion is negatively charged ion (e.g. , Cl^-, O_2^{2-})

An ion or radical is classified as monovalent, divalent, trivalent or tetravalent when the number of charge over it is 1, 2, 3 or 4 respectively.



The identity of the element is not altered when it changes to an ion, however the properties of ion differ from those of the atoms. e.g. sodium metal is very reactive but sodium ion is unreactive.

The ionic charge represents the number of electrons lost (if positive ion) or number of electron gained (if negative ion). The charge on ion is indicated in the symbol or formula by a superscript number followed by the + or – sign. Removing one electron from sodium atoms (Na) creates sodium ion (Na^+). Sodium ion (Na^+) is an example of monoatomic ion (i.e. an ion formed from one atom).

A list of ions and their charges is given below :

Positive Radicals *(Cations or Basic Radicals)*

Radical	Nature	(Cations or Basic Radicals)
Sodium (Na^+)	Monovalent,	Monoatomic
Potassium (K^+)	Monovalent,	Monoatomic
Lithium (Li^+)	Monovalent,	Monoatomic
Ammonium (NH_4^+)	Monovalent,	Polyatomic
Barium (Ba^{2+})	Divalent,	Monoatomic
Calcium (Ca^{2+})	Divalent,	Monoatomic
Zinc (Zn^{2+})	Divalent,	Monoatomic
Magnesium (Mg^{2+})	Divalent,	Monoatomic
Nickel (Ni^{2+})	Divalent,	Monoatomic
Cobalt (Co^{2+})	Divalent,	Monoatomic
Aluminium (Al^{3+})	Trivalent,	Monoatomic
Chromium (Cr^{3+})	Trivalent,	Monoatomic

Negative Radicals (*Anions or Acid Radicals*)

Radical	Nature	(Anions or Acid Radicals)
Hydride (H^-)	Monovalent,	Monoatomic
Chloride (Cl^-)	Monovalent,	Monoatomic
Bromide (Br^-)	Monovalent,	Monoatomic
Iodide (I^-)	Monovalent,	Monoatomic
Sulphide (S^{2-})	Divalent,	Monoatomic
Oxide (O^{2-})	Divalent,	Monoatomic
Nitride (N^{3-})	Trivalent,	Monoatomic
Peroxide (O_2^{2-})	Divalent,	Diatomic
Hydroxide (OH^-)	Monovalent,	Diatomic
Nitrate (NO_3^-)	Monovalent,	Tetra-atomic
Nitrite (NO_2^-)	Monovalent,	Tri-atomic
Hypochlorite $(ClO)^-$	Monovalent,	Diatomic
Sulphite (SO_3^{2-})	Divalent,	Tetra-atomic
Sulphate (SO_4^{2-})	Divalent,	Polyatomic
Phosphate (PO_4^{3-})	Trivalent,	Polyatomic

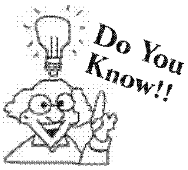
Knowledge ENHANCER

Those metallic elements which show variable valency (more than one valency states) form positive ions carrying different charges. The ion carrying lower charge is given the suffix "ous" and the ion with higher charge is given the suffix "ic" or the value of charge is shown by Roman numeral's (such as I, II, III, IV etc.) in parenthesis which is followed by the name of the metal.

Following examples illustrate the point

***Positive Radicals** (Cation or basic Radicals)*

Metallic Element	*Cation formed*	*Name of cation*
Copper (Cu)	Cu^+ *(Monovalent)*	*Cuprous or Copper (I)*
	Cu^{2+} *(Divalent)*	*Cupric or Copper (II)*
Iron (Fe)	Fe^{2+} *(Divalent)*	*Ferrous or Iron (II)*
	Fe^{3+} *(Trivalent)*	*Ferric or Iron (III)*
Lead (Pb)	Pb^{2+} *(Divalent)*	*Plumbous or Lead (II)*
	Pb^{4+} *(Tetravalent)*	*Plumbic or Lead (IV)*
Tin (Sn)	Sn^{2+} *(Divalent)*	*Stannous or stannum (II)*
	Sn^{4+} *(Tetravalent)*	*Stannic or stannum (IV)*
Mercury (Hg)	Hg^+ *(Monovalent)*	*Mercurous or Mercury (I)*
	Hg^{2+} *(Divalent)*	*Mercuric or Mercury (II)*



Both positive and negative ions occur naturally in air. However, the environment we live in today has far more sources of positive ions than in past, creating an electrical imbalance in the air and our bodies.
They can damage cells and are believed to be the reason for the deterioration of our physical and emotional well being having a role in the aging process and cancer.

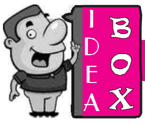
The **valencies** of some of the **elements** and **radicals** are listed in the following Tables.(Table-1 and Table-2)

Table – 1 : Cation or Basic Radicals

Name	Symbol		Valency
Aluminium	Al		3
Ammonium	NH_4^+		1
Antimony	Sb		3
Barium	Ba		2
Bismuth	Bi		3
Cadmium	Cd		2
Calcium	Ca		2
Chromium	Cr		3
Cobalt	Co		2
Copper	Cu	Cuprous or Copper (I)	1
		Cupric or Copper (II)	2
Hydrogen	H		1
Iron	Fe	Ferrous or Iron (II)	2
		Ferric or Iron (III)	3
Lead	Pb	Plumbous or Lead (II)	2
		Plumbic or Lead (IV)	4
Magnesium	Mg		2
Manganese	Mn	Managanous or Manganese (II)	2
		Managanic or Manganese (III)	3
Mercury	Hg	Mercurous or Mercury (I)	1
		Mercuric or Mercury (II)	2
Nickel	Ni		2
Potassium	K		1
Silver	Ag		1
Sodium	Na		1
Strontium	Sr		2
Tin	Sn	Stannous or Tin (II)	2
		Stannic or Tin (IV)	4
Zinc	Zn		2

Table – 2 : Anion or Acid Radicals

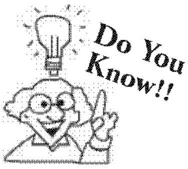
Name	Symbol	Valency
Fluoride	F	1
Chloride	Cl	1
Hypochlorite	ClO	1
Chlorate	ClO_3	1
Bromide	Br	1
Hypobromite	BrO	1
Iodide	I	1
Ferricyanide	$Fe(CN)_6$	3
Arsenite	AsO_3	3
Arsenate	AsO_4	3
Zincate	ZnO_2	2
Meta-Aluminate	AlO_2	1
Aluminate	AlO_3	3
Stannate	SnO_3	2
Silicate	SiO_3	2
Hypoiodite	IO	1
Iodate	IO_3	1
Sulphide	S	2
Sulphite	SO_3	2
Bisulphite	HSO_3	1
Sulphate	SO_4	2
Bisulphate	HSO_4	1
Thiosulphate	S_2O_3	2
Nitrite	NO_2	1
Nitrate	NO_3	1
Nitride	N	3
Hydroxide	OH	1
Oxide	O	2
Hydride	H	1
Peroxide	O_2	2
Carbonate	CO_3	2
Bicarbonate	HCO_3	1
Carbide	C	4
Phosphate	PO_4	3
Phosphite	HPO_3	2
Phosphide	P	3
Borate	BO_3	3
Acetate	CH_3COO	1
Cyanide	CN	1
Manganate	MnO_4	2
Permanganate	MnO_4	1
Chromate	CrO_4	2
Dichromate	Cr_2O_7	2
Ferrocyanide	$[Fe(CN)_6]$	4



idea box

Names of few ions are mentioned :
Sodium, Ammonium, Phosphate, Sulphate, Nitrate, Peroxide, Calcium, Aluminium, Nitride, Phosphide, Stannous, Cuprous, Carbonate, Hydride and Zinc.

1. ***Divide them into categories of anions and cations.***
2. ***Divide them into three categories of monovalent, divalent and trivalent ion.***



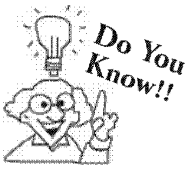
Acidic media changes litmus paper to red colour where as Alkaline media changes litmus paper strip to blue colour

ACIDS, BASES AND SALTS

As now you are familiar with terms like acidic or basic radicals. So you must aware about general concepts of acids, bases and salts.
Acids : In earlier discoveries different scientist gave different definition to explain acid but most commonly acid is a substance which donates proton (H^+) or when dissolved in water yields hydronium ions (H_3O^+) or hydrogen ions.
For example HCl is an acid which dissolves in water to give hydronium ion.

$$HCl + H_2O \longrightarrow H_3O^+ + Cl^-$$

Other examples are HNO_3, H_2SO_4, H_2CO_3 etc.



- *H_2SO_4 is used in almost all car batteries.*
- *Lactic acid is formed when milk is fermented to yoghurt.*
- *H_2CO_3 (carbonic acid) present in soft drinks decomposes to form bubbles of carbon dioxide (CO_2).*

Base : It is a substance which ionize to give OH^- ions in aqueous solution or a compound which combined with the hydronium ion (H_3O^+) of an acid to form salt and water only.

$$\underset{\text{Base}}{NaOH} \xrightarrow{\text{aqueous media}} Na^+ + OH^-$$

$$\underset{\text{Base}}{Mg(OH)_2} + \underset{\text{Acid}}{H_2SO_4} \longrightarrow \underset{\text{Salt}}{MgSO_4} + \underset{\text{Water}}{2H_2O}$$

Other examples $Al(OH)_3$, NH_4OH etc.
Salts : An ionic compound which if dissolved in water, dissociates to yield a positive ions rather than hydrogen ion (H^+ ion) and a negative ions other than hydroxyl ion (OH^- ion).

$$KCl \xrightarrow{\text{aqueous media}} K^+ + Cl^-$$

Other examples are NaCl, $MgCl_3$, $Al_2(SO_4)_3$ etc.

CHECK Point

1. Aluminium nitrate [$Al(NO_3)_3$] is a salt or acid?

2. What is aqua regia ?

SOLUTION

1. It is a salt.
2. Mixture of one part of conc. HNO_3 and three parts of con. HCl.

idea box

As we know that litmus solution can be used to identify acid or alkaline nature of substance. You can collect 5-10 things of daily household like citrus fruits, soaps, antacid, paracitamol solution in water etc, and analyse their nature.

CHEMICAL FORMULA

Formula of elements: The molecule of an element is denoted by writing the symbol of the element and, to the right and below it, a number expressing the number of atoms in the molecule. e.g.

H_2 denotes one molecule of Hydrogen containing two atoms in combination.

P_4 denotes one molecule of phosphorus containing four atoms in combination.

S_8 denotes one molecule of sulphur containing eight atoms in combination.

The formula of those elements whose atoms are capable of independent existence is the same as the symbol of the element. e.g.

Name	Symbol	Chemical Formula
Helium	He	He
Neon	Ne	Ne
Argon	Ar	Ar
Krypton	Kr	Kr
Iron	Fe	Fe
Mercury	Hg	Hg

Thus the formula is the symbolic expression for a molecule and a molecule of an element may consist of one or more atoms e.g. He, Ne, Ar... etc (one atom), O_2, H_2, N_2, Cl_2, Br_2 .... etc (two atoms), P_4, S_8 (more than 2 atoms)

Formula of compound : A molecule of a compound may be made up of atom of different elements linked up together chemically and in definite proportion by weight. e.g. Iron sulphide is made up of Iron (Fe) and sulphur(S) in the fixed ratio of 56 : 32 i.e. 56 parts of iron combine with 32 parts of sulphur by weight. Thus iron sulphide consists of one atom of iron (atomic weight of iron = 56) and one atom of sulphur (atomic weight of sulphur = 32) and its chemical formula is FeS.

In the formula both the elements are written as their symbols which always begin with a capital letter.

We have adopted the same method to represent the formula of a compound that we had adopted to represent the formula of an element.

In a chemical compound at least two symbols must appear because at least two elements must be present in a chemical compound. In the formula of a compound, the number to the right of a symbol and below it, expresses the number of atoms of the element present, the number 1 being omitted. This is quite evident from the formula of some compounds.

Formula of some common compounds are :

Compound	Formula
Hydrochloric acid	HCl
Sulphur dioxide	SO_2
Sulphur trioxide	SO_3
Sulphuric acid	H_2SO_4
Carbon dioxide	CO_2
Nitric acid	HNO_3
Sodium carbonate	Na_2CO_3
Sodium hydroxide	NaOH
Ammonia	NH_3
Potassium hydroxide	KOH
Potassium nitrate	KNO_3
Calcium carbonate	$CaCO_3$

How to write the formula of compound?

Various steps to be followed in writing the formula of a compound are as follows:

(i) Write the symbols of the two radicals (positive radical and negative radical) side by side with valencies at the top. Write the positive radical to the left and the negative radical to the right hand side.

(ii) Cancel the common factor, if any, from the valencies, to get their simple whole number ratio.

(iii) Apply criss-cross rule (i.e., shift the valencies cross-wise to lower right of the radical). Always enclose the compound radical in bracket before writing any numeral at the lower right corner.

To illustrate the above rules few examples are given below :

Example 1

Hydrogen Chloride

Radical name	Hydrogen	Chloride
Radical symbol	H	Cl
Radical nature	Basic	Acidic
Valency	+ 1	– 1

Usually subscript 1 is not written in the final formula.

H^{1+} Cl^{-1}

H_1Cl_1 or HCl

Cl_1 H_1

Example 2

Sodium Sulphate

Radical name	Sodium	Sulphate
Radical symbol	Na	(SO_4)
Radical nature	Basic	Acidic
Valency	+ 1	– 2

Na^{1+} SO_4^{-2}

$Na_2(SO_4)_1$ or Na_2SO_4

$(SO_4)_1$ Na_2

Example 3

Silver Iodide

Radical name	Silver	Iodide
Radical nature	Basic	Acidic
Valency	+ 1	– 1

Usually subscript 1 is not written in the formula.

Ag^{1+} I^{1-}

Ag_1I_1 or AgI

I_1 Ag_1

Based on those examples, complete the following :

1. **Barium Bicarbonate**

Radical name	....	....
Radical nature	....	....
Valency	....	....

Ba HCO_3

i.e., Ba $(HCO_3)_2$

$(HCO_3)_2$ Ba_1

2. **Ferric Phosphate**

Radical name	....	....
Radical nature	....	....
Valency	....	....

Fe PO_4

i.e., $Fe_3(PO_4)_3$ or $FePO_4$

$(PO_4)_3$ Fe_3

3. **Silver Sulphide**

Radical name	....	....
Radical nature	....	....
Valency	....	....

Ag S

i.e., Ag_2S

S_1 Ag_2

4. **Calcium Chloride**

Radical name	....	....
Radical nature	....	....
Valency	....	....

Ca Cl

Ca_1Cl_2 or $CaCl_2$

Cl_2 Ca_1

5. Hydrogen Sulphide

Radical name ....

Radical nature ....

Valency ....

H S $\rightarrow H_2S_1$ or H_2S

S_1 H_2

6. Ferrous Sulphate

Radical name ....

Radical nature ....

Valency ....

Fe SO_4 $\rightarrow Fe_2(SO_4)_2$ or $FeSO_4$

$(SO_4)_2$ Fe_2

7. Barium Carbonate

Radical name ....

Radical nature ....

Valency ....

Ba CO_3 $\rightarrow Ba_2(CO_3)_2$ or $BaCO_3$

$(CO_3)_2$ Ba_2

8. Phosphorus Trichloride

Radical name ....

Radical nature ....

Valency ....

P Cl $\rightarrow P_1Cl_3$ or PCl_3

Cl_3 P_1

CHECK Point

Write down the formula of compound Ammonium phosphate and Barium sulphate?

SOLUTION

1. Ammonium phosphate $[(NH_4)_3PO_4]$
2. Barium sulphate $[BaSO_4]$

The Formulae of some common compounds are given in the following table. (Table-3)

Table – 3 : Formulae of Some Common Compounds

Compound	Formula	Compound	Formula
Ammonium chloride	NH_4Cl	Nitric oxide	NO
Ammonium sulphate	$(NH_4)_2SO_4$	Dinitrogen trioxide	N_2O_3
Ammonium phosphate	$(NH_4)_3PO_4$	Nitrogen dioxide	NO_2
Aluminium bromide	$AlBr_3$	Dinitrogen pentoxide	N_2O_5
Aluminium carbonate	$Al_2(CO_3)_3$	Ammonia	NH_3
Aluminium phosphate	$AlPO_4$	Sulphur dioxide	SO_2
Barium chloride	$BaCl_2$	Hydrogen fluoride (Hydrofluoric acid)	HF
Barium sulphate	$BaSO_4$	Hydrogen chloride (Hydrochloric acid)	HCl
Calcium bromide	$CaBr_2$	Hydrogen bromide (Hydrobromic acid)	HBr
Calcium carbonate	$CaCO_3$	Hydrogen iodide (Hydoiodic acid)	HI
Carbon monoxide	CO	Hydrogen nitrate (Nitric acid)	HNO_3
Carbon dioxide	CO_2	Hydrogen nitrite (Nitrous acid)	HNO_2
Methane	CH_4	Hydrogen chlorate (Chloric acid)	$HClO_3$
Ethane	C_2H_6	Hydrogen hydroxide (Water)	HOH or H_2O
Nitrous oxide	N_2O	Hydrogen sulphite (Sulphurous acid)	H_2SO_3

Compound	Formula	Compound	Formula
Hydrogen sulphate (Sulphuric acid)	H_2SO_4	Potassium nitrate	KNO_3
Hydrogen carbonate (Carbonic acid)	H_2CO_3	Ammonium nitrate	NH_4NO_3
Hydrogen sulphide	H_2S	Potassium chlorate	$KClO_3$
Hydrogen phosphate (phosphoric acid)	H_3PO_4	Ammonium hydroxide	NH_4OH
Hydrogen borate (boric acid)	H_3BO_3	Magnesium sulphite	$MgSO_3$
Potassium fluoride	KF	Barium sulphate	$BaSO_4$
Sodium chloride	NaCl	Calcium carbonate	$CaCO_3$
Potassium bromide	KBr	Barium oxide	BaO
Silver iodide	AgI	Copper sulphide	CuS
		Iron phosphate	$FePO_4$

Significance of chemical formula : Like the symbols, a formula has also qualitative as well as quantitative significance.

Qualitative significance :

Qualitatively, the formula represents the name of the substance and the names of various elements present in the substance. e.g.

$\mathbf{H_2}$ indicates that it consists of only hydrogen.

$\mathbf{H_2O}$ indicates that it consists of hydrogen and oxygen.

$\mathbf{HNO_3}$ indicates that it consists of hydrogen, nitrogen and oxygen.

Quantitative significance

Quantitatively the chemical formula represent,

(i) One molecule of the substance (element or compound)

(ii) The actual number of atoms of each element present in one molecule of the substance (element or compound)

(iii) The number of parts by weight of the substance (molecular weight and number of parts by weight of each element).

SUMMARY

- Chemistry is a branch of science which deals with study of matter and various changes it undergoes. It deals with the preparation, properties, reactions and structures of chemical elements and compounds.
  Chemistry is sub-divided into various branches such as : Inorganic chemistry, Organic chemistry, Physical chemistry, Analytical chemistry, Industrial chemistry, Bio-chemistry, Nuclear chemistry, Pharmaceutical chemistry, Medicinal chemistry, Material chemistry etc.
- **Atomic Mass or Atomic Weight :** Atomic mass of an element is defined as the mass of one atom of an element compared to 1/12 th of the mass of an atom of C-12.
- **Gram Atomic Mass or Gram Atomic Weight :** The atomic mass of an element when expressed in grams is known as **gram atomic mass** or simply as **gram atom (g-atom)**.
- **Molecular Mass or Molecular Weight :** Molecular weight is calculated by adding the atomic weights of all the constituent atoms present in a molecule.
- **Gram Molecular Mass :** The molecular mass of a substance when expressed in grams is known as **gram-molecular mass** or simply **gram-mole** (**g-mole**) for the sake of convenience it is expressed simply as **mole**.
- A **symbol** is defined as an abbrevation or short hand sign for full name of an element.
- Some elements are named after the names of scientists, countries and laboratories and after the names of planet.
- The combining capacity of an atom or radical is known as its **valency**.
- When electrons is/are removed from an element the resulting ion is called a **cation or basic radical.** A cation is positively charged ion. (e.g. Na^+).
- When electron is/are added to an element the resulting ion is called an **anion or acidic radicals.** An anion is negatively charged ion (e.g. , Cl^-, O_2^{2-}) .
- The molecule of an element is denoted by writing the symbol of the element and, to the right and below it, a number expressing the number of atoms in the molecule.
- Chemical formula has both qualitative and quantitative significance.

1 EXERCISE

Multiple Choice Questions :

DIRECTIONS : *This section contains **18** multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.*

1. The symbols of tin and mercury are respectively
(a) Ti and Me (b) me and Ti
(c) Sn and Hg (d) Me and Sn

2. Which one of the following information is conveyed by the symbol of an element?
(a) The name of the element
(b) The atomic mass of the element
(c) The atomic number of the element
(d) All the above

3. $Na_2S_2O_3$ represent the compound
(a) sodium sulphate (b) sodium sulphite
(c) sodium thiosulphate (d) None of these

4. Which one is a bivalent ion?
(a) sodium (b) calcium
(c) sulphide (d) both (b) and (c)

5. The chromate and dichromate ions are respectively
(a) CrO_4^{2-} and $Cr_2O_7^{2-}$ (b) $Cr_2O_7^{2-}$ and CrO_4^-
(c) CrO_4^- and CrO_5^- (d) CrO_4^{2-} and $Cr_2O_5^{2-}$

6. The formula of sodium pyrophosphate is
(a) $Na_2P_2O_7$ (b) Na_3PO_4
(c) $Na_4P_2O_7$ (d) Na_3PO_3

7. The branch of chemistry which deals with study of physical properties and conditions is
(a) physical chemistry
(b) Analytical chemistry
(c) Nuclear chemistry
(d) Pharmaceutical chemistry

8. The branch of chemistry which deals with study of the methods of detection and determination of elements and compounds is
(a) Physical chemistry (b) Nuclear chemistry
(c) Analytical chemistry (d) Bio chemistry

9. What is the valency of sulphur in sulphur dioxide (SO_2)?
(a) 3 (b) 2
(c) 6 (d) 4

10. Molecular weight of water is
(a) 16 amu (b) 12 amu
(c) 10 amu (d) 18 amu

11. Which of the following is a symbol of palladium ?
(a) B (b) Pd
(c) Be (d) Ag

12. Hg is a symbol of
(a) Lead (b) Tin
(c) Antimony (d) Mercury

13. Valency of an atom or radicals is
(a) ionisation energy (b) electron affinity of atom
(c) its combining capacity (d) size of atom

14. When electrons are added the resulting ion is called
(a) basic radical (b) acidic radicals
(c) neutral radical (d) None of these

15. Acid turns litmus paper
(a) Blue (b) Yellow
(c) Red (d) None of these

16. Amphoprotic substances are those
(a) which can donate a proton
(b) which can accept a proton
(c) which can accept and donate proton
(d) which can donate hydroxyl ion

17. One molecule of sulphur contains
(a) Two atoms of sulphur (b) Eight atoms of sulphur
(c) Four atoms of sulphur (d) One atom of sulphur

18. Chemical formula of Aluminium sulphate is
(a) $Al_2(SO_4)_3$ (b) $AlSO_4$
(c) $Al_3(SO_4)_2$ (d) None of the above

More than One Option Correct :

DIRECTIONS : *This section contains 7 Multiple Choice Questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE OR MORE may be correct.*

1. Which of the following is / are bivalent radicals?
(a) Sulphate (b) Bisulphate
(c) Sulphite (d) Sulphide

2. Which of the following are triatomic?
(a) H_2O (b) CO_2
(c) CO_3^{2-} (d) NaCl

3. In organic chemistry we study about
(a) Carbohydrates (b) Fertilizers
(c) Proteins (d) Ceramics

4. Which of the following are acidic radicals ?
(a) PO_4^{3-} (b) Mg^{2+}
(c) CO_3^{2-} (d) NO_3^-

5. Acids are those substances which
   (a) Furnish H_3O^+ in aqueous solution
   (b) Lowers the pH of solution
   (c) Furnish OH^- in aqueous solution
   (d) Increase the pH of solution
6. Chemical formula gives information about
   (a) which elements are present in compound
   (b) physical properties of compound
   (c) nature of compound
   (d) total number of atoms of each element present in compounds
7. Which of the following ion is divalent ?
   (a) SO_4^{2-} (b) PO_4^{3-}
   (c) Cu^{2+} (d) Sn^{2+}

Passage Based Questions :

DIRECTIONS : *Study the given paragraph(s) and answer the following questions.*

Passage

The formula of a binary **compound**, i.e., a compound formed by two elements only, is obtained by transposing their valencies. Suppose an element. A has a valency y and element B has a valency x. Then the compound formed between A and B usually has the formula A_xB_y. The subscripts should be divided by a common factor, if any.

$$\overset{y}{A}\ \ \overset{x}{B} \Rightarrow A_xB_y$$

$$\overset{4}{A}\ \ \overset{2}{B} \Rightarrow A_2B_4 \Rightarrow AB_2$$

1. In which of the following compounds phosphorus (P) has valency 3?
   (a) PCl_5 (b) PCl_3
   (c) PCl_2 (d) None of these
2. What is the valency of the underlined elements in following compounds respectively.
   $\underline{Ag}I$, $\underline{Mg}_3N_2$, $\underline{Na}Cl$, $\underline{Ca}Cl_2$
   (a) 1, 2, 2, 2 (b) 1, 2, 1, 1
   (c) 1, 2, 1, 2 (d) 2, 1, 1, 2
3. How many times greater is the valency of N in NH_3 than that of Cl in HCl?
   (a) 3 (b) 2
   (c) 4 (d) 5

Assertion & Reason :

DIRECTIONS : *Each of these questions contains an assertion followed by reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.*

(a) If both **Assertion** and **Reason** are **correct** and Reason is the **correct explanation** of Assertion.
(b) If both **Assertion** and **Reason** are correct, but Reason is **not the correct explanation** of Assertion.
(c) If **Assertion** is **correct** but **Reason** is **incorrect**.
(d) If **Assertion** is **incorrect** but **Reason** is **correct**.

1. **Assertion :** The combining capacity of an atom or radical is known as its valency.
   Reason : The valency of an element is always a whole number.
2. **Assertion :** A cation is formed after the removal of electrons.
   Reason : An element always looses one electron to form cation.
3. **Assertion :** The molecular mass of NaOH is 40.
   Reason : The molecular mass of a compound is defined as the sum of the atomic weights of all the constituent atom present in a molecule.
4. **Assertion :** One mole of any substance is equals to its molecular weight.
   Reason : Number of moles $\rightarrow \dfrac{\text{Mass of substance in grams}}{\text{Molecular weight}}$

Multiple Matching Questions :

DIRECTIONS : *Following question has four statements (A, B, C and D) given in Column I and four statements (p, q, r and s) in Column II. Any given statement in Column I can have correct matching with one or more statement(s) given in Column II. Match the entries in column I with entries in column II.*

1.

	Column I		Column II
A.	SO_4^{2-}	(p)	Bivalent
B.	Fe^{2+}	(q)	Monoatomic
C.	N^{3-}	(r)	Trivalent
D.	PO_4^{3-}	(s)	Polyatomic

Integer Type Questions :

DIRECTIONS : *Following are integer based questions. Each question, when worked out will result in one integer from 0 to 9 (both inclusive).*

1. Find the value of x in $NaxBO_3$.
2. What is the valency of Carbon in CH_4.
3. Calculate the sum of the valencies of Helium, Phosphorus and neon.

SOLUTIONS

Brief Explanations of Selected Questions

1 EXERCISE

Multiple Choice Questions :

1. (c) **2.** (a) **3.** (c) **4.** (d) **5.** (a)
6. (c) **7.** (a) **8.** (c)
9. (a) Sulphur has valency 4 in sulphur dioxide (SO_2).
10. (d)
11. (b) **12.** (d) **13.** (c) **14.** (b)
15. (c) **16.** (c) **17.** (b) **18.** (a)

More Than One Option Correct :

1. (a, c, d) **2.** (a, b) **3.** (a, c)
4. (a, c, d) **5.** (a, b) **6.** (a, d)
7. (a, c, d)

Passage Based Questions :

1. (b) In PCl_3, phosphorus has valency 3.
2. (c) $\overset{1}{Ag}\ \overset{1}{I} \Rightarrow AgI$

$\overset{2}{Mg}\ \overset{3}{N} \Rightarrow Mg_3N_2$

$\overset{1}{Na}\ \overset{1}{Cl} \Rightarrow NaCl$

$\overset{2}{Ca}\ \overset{1}{Cl} \Rightarrow CaCl_2$

3. (a) N in NH_3 has 3 valency and Cl in HCl has 1 valency. Thus valency of N in NH_3 is three time greater than valency of Cl in HCl.

Assertion & Reason :

1. (b)
2. (c) An element can loose one to four electron to form mono, di, tri and tetra valent cation respectively.
3. (a) **4.** (a)

Multiple Matching Questions :

1. $A \to (p, s)$, $B \to (p, q)$, $C \to (r, q)$, $D \to (r, s)$

Integer Type Questions :

1. The formula of sodium borate is Na_3BO_3.
$\therefore x = 3$
2. 4
3. $P(15) = 2, 8, 5,$
Valency $= 5$ $Ne(10) = 2, 8$
$\therefore$ Valency $= 0$,
$\therefore$ Sum of valency $= 5$
$He(2) = 2$, $\therefore$ Valency $= 0$

Chapter 6

CLASSIFICATION OF MATTER ON THE BASIS OF CHEMICAL COMPOSITION (ELEMENT, COMPOUND, MIXTURE)

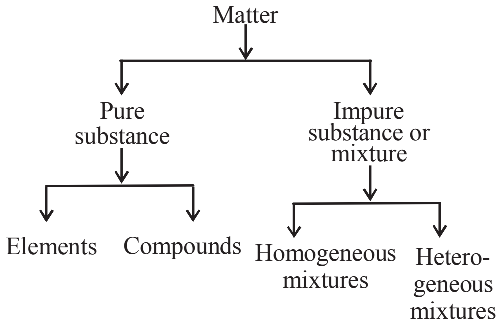
INTRODUCTION

You must have tasted different fruit juices. Some you like and some you don't like. Why do fruit juices taste different? The answer is that they contain different ingredients or we can say, the matter in each juice is different. Scientists sometimes use the term matter instead of composition for description.

To understand the chemical nature of matter, let us understand, what is a **chemical substance**?

In science concept of a substance is a little different from its everyday meaning. In science **substance** refers to a single pure form of matter. Thus we can say that **substance** is a kind of matter that can not be separated into other kinds of matter by any physical process.

Most of the naturally occurring materials on earth are **mixtures** (impure substances). Pure substances are rare.
On the basis of chemical composition matter can be classified as pure substance and impure substance.
This classification can be represented as



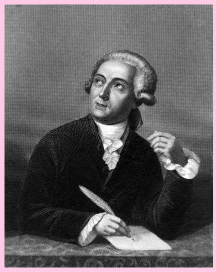
We will now discuss the above classification in detail.

HISTORICAL PREVIEW

The first modern list of chemical elements was given by Antoine Lavoisier's in 1789, known as Elements of Chemistry, which contained thirty-three elements, including light and calorie. By 1818, Jons Jakob Berzelius had determined atomic weights for forty-five of the forty-nine accepted elements. Dmitri Mendeleev had sixty-six elements in his periodic table of 1869.

According to Boyle in the early 20th century, an element was defined as a pure substance that cannot be decomposed into any simpler substance. Put another way, a chemical element cannot be transformed into other chemical elements by chemical processes.

By 1919, there were seventy-two known elements. In 1955, element 101 was discovered and named mendelevium in honour of Mendeleev, the first to arrange the elements in a periodic manner. In October 2006, the synthesis of element 118 was reported; the synthesis of element 117 was reported in April 2010.


Antonine Lavoisier

PURE SUBSTANCE

When a scientists says that some thing is pure, they mean it is a single substance.
A pure substance is one that has a uniform composition and always have the same texture, colour, taste at a given temperature and pressure. A pure substance is homogeneous in nature and it has definite set of properties. It is not possible to change the composition of a pure substance by physical methods. We can further classify the pure substances as elements and compounds.
The purity of a pure substance can be tested by checking its melting point or boiling point. A pure substance has a fixed melting point or boiling point at constant pressure. For example the boiling point of pure water is 373 K at normal pressure (1 atmosphere). The presence of impurities in water increases its boiling point.

idea box

Take pure water in 100 mL beaker and boil it on a Bunsen burner flame. Take a lab thermometer and note down its boiling point. Now weight approximately 10g of table salt (NaCl) and dissolve it uniformly in another beaker filled with pure water. Now boil this aqueous salt solution and measure its boiling point with the help of thermometer and further repeat the above process now by dissolving 10g of copper sulphate. (CuSO4.5H2O). Record your observation.

(a) What is the boiling point of pure water?

(b) Is there any change in boiling point after dissolving common salt?

(c) Do you observe any difference in boiling point later when test is carried out by dissolving copper sulphate in pure water?

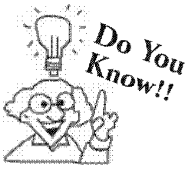
(d) Write conclusion of a above experiment on the basis of your observations.

CHECK Point

Though pure water is a pure substance but sea water is impure. Explain.

SOLUTION

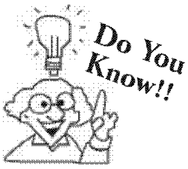
A pure substance has a fixed melting point and boiling point. Sea water freezes at a temperature well below the freezing point of pure water (0°C) and boils at a temperature above the boiling point of pure water (100°C).



Every sample of a pure substance is always the same, no matter what the form. For example water from tap water, sea water, and blood would be exactly same. Water can be mixed with all sorts of other substances, but water itself is always the same substance.

ELEMENTS

Pure substances which are made up of only one kind of atoms are known as elements. *e.g.* Pure gold is an element it contains only gold atoms. Nitrogen gas is an element because it contains only nitrogen atom. Similarly graphite (carbon) in your pencil is an element. Graphite is made up solely of carbon atoms. Elements cannot be broken down into other substances by any chemical means. Individually or in combination, the elements form every object in the world. At present there are around 118 different elements. And of these, you probably use only 30 to 40 in your daily life. All of the elements are inserted in a chart called the periodic table



Atoms of pure elements may bond to each other chemically in more than one way, allowing the pure element to exist in multiple structures (special arrangements of atoms), known as allotropes, which differ in their properties. For example, carbon can be found as diamond, which has a tetrahedral structure around each carbon atom; graphite, which has layers of carbon atoms with a hexagonal structure stacked on top of each other; fullerenes, which have nearly spherical shapes. The ability for an element to exist in one of many structural forms is known as 'allotropy'.

CHECK Point

How many type of atoms you can expect to find in a sample of any element?

SOLUTION

An element is made up of only one type of atom.

Knowledge ENHANCER

In the periodic table, each element is designated by its chemical symbol, which comes from the letters of the elements name. For example, the atomic symbol for carbon is C, and that for cobalt is Co. In many cases, the atomic symbol is derived from the element's latin name. Gold has the atomic symbol Au which comes from its latin name, aurum. Iron has the atomic symbol Fe which comes from its latin name Ferrum.
Every element has a set of distinct characteristics i.e. atomic symbol, atomic numbers and atomic masses.
So the periodic table is listing of all the known elements with their atomic symbols, atomic numbers, and atomic masses.

PERIODIC TABLE

1 H																	2 He
3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
55 Cs	56 Ba	57 La	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
87 Fr	88 Ra	89 Ac	104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Uun	111 Uuu	112 Uub						
				58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu
				90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr

Do You Know!!

At present 118 elements are known, out of which 92 are found in earth crust and rest all are artificially prepared by nuclear reactions. About 90% of the mass of earth crust consists of only five elements-oxygen, silicon, aluminium, iron, and calcium.

CHECK Point

A substance is analysed in a laboratory, and when viewed under an electron microscope, it is revealed that it contains only one kind of atom. Is the substance an element, compound or mixture?

SOLUTION

Element

ATOMICITY OF AN ELEMENT

The number of atoms of an element present in its molecule is called its atomicity. *e.g.*

Element	Atomicity
Hydrogen, H_2	2
Neon, Ne	1
Oxygen, O_2	2
Nitrogen, N_2	2
Helium, He	1
Phosphorus, P_4	4
Sulphur, S_8	8

On the basis of their atomicity elements are classified into following classes :

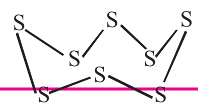
(i) **Monoatomic elements :** These are the elements having atomicity equal to 1. *i.e.* the molecule of such elements consists of only 1 atom *e.g.* He (Helium), Ne (Neon), Cu (Copper), Au (Gold) etc.

(ii) **Diatomic elements :** These are the elements having atomicity equal to 2. *i.e.* the molecule of such elements consists of 2 atoms. *e.g.* Hydrogen (H_2), Oxygen (O_2), Nitrogen (N_2), Chlorine (Cl_2), Bromine (Br_2), Iodine (I_2) etc.

(iii) **Polyatomic elements :** These are the elements having atomicity more than 2. *i.e.* the molecules of such elements consists of more than 2 atoms *i.e.* 3 atoms (*triatomic*), 4 atoms (tetra-atomic) *etc. e.g.* Phosphorus (P_4), Sulphur (S_8) etc.

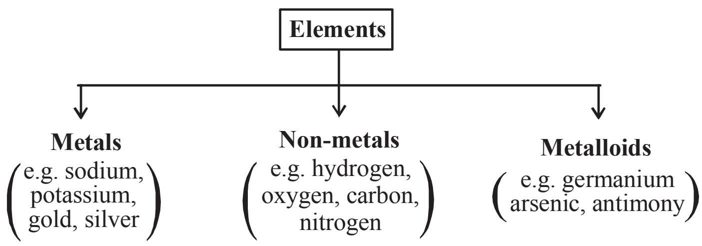
Do You Know!!

In addition to S_8 ring many other modification of sulphur containing 6-20 sulphur atoms per ring have been synthesized in last two decades. The structure of S_8 is



CLASSIFICATION OF ELEMENTS

On the basis of variation in their properties elements can be broadly classified as **metals, non-metals** and **metalloids.**



It is quite difficult to frame an exact definition for metals and non-metals in terms of physical properties, because regardless of what definition is adapted, there are always exceptions.

Following table lists differences in physical properties of metals and non-metals :

TABLE

Differences in Physical properties of metals and non-metals

Property	Metals	Non-metals
1. State	Generally in solid state at room temperature. **Exceptions:** Mercury (Hg), Gallium (Ga). These are liquids at room temperature.	Generally liquid or gas at room temperature. **Exceptions:** Carbon (C), boron (B) and sulphur (S) are solids at room temperature.
2. Hardness	Generally hard. Tungsten and chromium are the hardest metals. **Exceptions:** sodium (Na), Potassium (K), Mercury (Hg), lead (Pb) are not hard. Na can be easily cut with a knife.	Generally soft. **Exception:** Diamond. It consists of carbon (a non-metal). It is the hardest substance known.
3. Lustre	Possess a brilliant lustre. The lustre is because of the fact that metals reflect light from their surface instead of absorbing it.	No lustre. **Exceptions:** Iodine and graphite (*i.e.* a form of carbon) possess lustre.
4. Melting Point	Metals have high melting point. m.p. depends on the strength of metallic bond present in metals. m.p. is high in case of heavy metals. **Exceptions:** Sodium and potassium which are light elements have low melting point.	Non-metals have low melting point. **Exceptions:** carbon, boron and silicon have high melting points.
5. Density	Metals have a high density. **Exceptions:** The density of magnesium, aluminium and titanium is not high because they are light metals.	Non-metals have a low density. **Exceptions:** Diamond (a form of carbon) has a high density.
6. Boiling Point	High boiling point. **Exceptions:** Mercury, gallium.	Low boiling point. **Exceptions:** Carbon, boron and silicon.
7. Ductility	Metals posses ductility *i.e.* they can be drawn into wires. **Exceptions:** Zinc, Mercury, Gallium.	They do not possess the property of ductility. **Exception:** Carbon fibres.
8. Malleability	Metals are malleable. **Exceptions:** Zinc, Mercury and gallium.	Non-metals are generally non-malleable. **Exception:** Carbon fibres.
9. Brittleness	Metals are generally not brittle. **Exceptions:** Zinc	Non- metals are generally brittle.
10. Conductivity	Good conductors of heat and electricity. **Exceptions:** Bismuth and Tungsten which are poor conductors of electricity.	Bad conductors of heat and electricity. **Exceptions:** Graphite and gas carbon are good conductors of electricity.
11. Sonorousness	Metals are sonorous (*i.e.* a sound is produced by them when struck with some hard material)	Non-metals are non-sonorous.

Metalloids : Some elements possess the properties of both metals and non-metals and these elements are called metalloids. *e.g.* Germanium, arsenic, antimony, selenium and tellurium.

COMPOUNDS

When atoms of different elements bond to one another, they make a compound. Sodium atoms and chlorine atoms, for example, bond to make the compound sodium chloride commonly known as table salt. Nitrogen atoms and hydrogen atoms join to make the compound ammonia which is a common household cleaner.

Hence "A chemical compound is a pure substance formed from chemical combination of two or more elements."

Water (H_2O) is considered a pure substance. It consists of two types of atoms (hydrogen atoms and oxygen atom). In water there is a fixed number of hydrogen atoms and oxygen atoms combined together chemically in a definite proportion by weight. In water (H_2O), hydrogen and oxygen combine in a fixed ratio of 1 : 8 (by weight) to form water.

Do You Know!!

Compounds are formed from simple, substances by chemical reaction. Some compounds are formed directly from their constituent elements while other compounds are formed by reaction of of an element with another compound. Compounds are also made by reaction of compounds with each other.

CHECK Point

Are the physical and chemical properties of a compound necessarily similar to those of the elements from which it was composed?

SOLUTION

No, it is nto necessary that physical and chemical properties are similar to those of the elements of which they composed.

A compound possess entirely different physical and chemical properties from those of its constituents.

The sodium chloride NaCl is very different from the elemental sodium and the elemental chlorine used in its formation. Elemental sodium, (Na) consists of nothing but sodium atoms, which form a soft, silvery metal that can cut easily with a knife. Its melting point is 97.5°C and it reacts violently with water. Elemental chlorine, Cl_2 consists of chlorine molecules. This material, a yellow green gas at room temperature is very toxic, and it was used as a chemical warfare agent during world war I. Its boiling point is - 34°C. The compound NaCl is a translucent, brittle, colorless crystal having a melting point of 800°C. It does not react with water as sodium does and not toxic to humans as chlorine is, but it is an essential component of all living organism.

CHECK Point

Hydrogen Sulphide is one of the bad smelling compounds. Rotten eggs get their characteristic bad smell from the hydrogen sulphide they release. Can you confirm from this information that elemental sulphur so is just as smelly ?
How will you prove cuprous oxide is a compound and not an element?

SOLUTION

No, you cannot. Infact the odour of elemental sulphur is negligible as compared with that of hydrogen sulphide. Compounds are truly different from the elements from which they are formed. Hydrogen sulphide is as different from elemental sulphur.

On heating cuprous oxide. It give lustrous Cu metal which has different composition from starting compound

Do You Know!!

Chemical and physical properties of a compound differ from those of its constituents. e.g. water consists of hydrogen (a combustible substance) and oxygen (a supporter of combustion). Water is used to extinguish fire i.e. it is neither combustible nor a supporter of combustion.

MIXTURES

Mixtures are absolutely everywhere you look. Mixtures are the form that are mostly found in nature e.g. rocks, air, or the ocean they are just about anything you find. They are substances held together by physical forces, not chemical.

.Two or more substances (either elements or compounds) can be mixed together in any proportion and the resultant substances so obtained are called **mixtures** (impure substance).

Do You Know!!

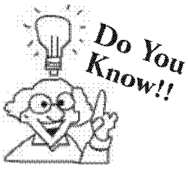
Fizzy drinks we drink are mixture of liquids (water and flavouring) and a gas (carbondioxide) The gas makes the fizzy bubble

Classification of Matter on The Basis of Chemical Composition (Element, Compound, Mixture)

Most of the material around us are a **mixtures.** A mixture contains more than one substance, elements or compounds.

A mixture depicts the properties of its constituting substances. For example, lemonade (nimbu pani) is a mixture of water, lemon juice, sugar and salt. When we drink it we get the sour taste of lemon, salty taste of salt, sweet taste of sugar. Moreover it quenches our thirst as water does.

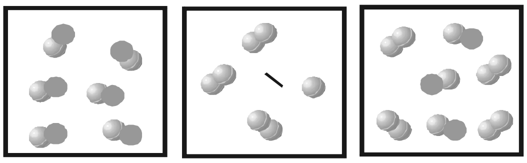
Most materials we considered as mixtures are mixtures of elements, mixtures of compounds, or mixture of element and compounds. For example, steel is a mixture of the elements iron, chromium, nickel, and carbon. Our atmosphere or air is a mixture of the elements nitrogen, oxygen and argon with traces of carbon dioxide and water vapour.



The formation of a mixture is a physical change because each substance in a mixture retains its chemical, identity. Whereas the formation of a compound is a chemical change. Recall that sodium chloride is composed of Na^+ and Cl^- which is entirely different from its elements,

CHECK Point

1. **So far, you have learned about three kinds of substances:**
   elements, compounds, and mixtures. Which box below contains only an element? Which contains only a compound? Which contains a mixture?

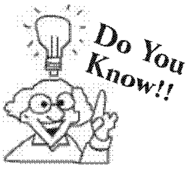


A B C

2. **You have just won a block of pure 24-carat gold. Have you just procured an element, compound or mixture?**

SOLUTION

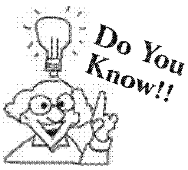
1. The molecules in box A contain two different types of atoms and so are representative of a compound. The molecules in box B each consist of the same atoms and so are representative of an element. Box C is a mixture of the compound and the element.
2. Element



Solutions are mixtures with particle size between 0.1 to 2 nanometers. Light can usually passthrough the solution. If the solute able to absorb visible light then the solution will have a color

Types of mixtures : Depending upon the nature of the components that forms the mixture we can have different types of mixtures.

(i) **Homogeneous mixture :** It is a mixture that has the same composition throughout. e.g. a solution of sugar in water. Such a mixture has two or more components. The composition of such a mixture is variable (e.g. different quantities of sugar are added to the same quantity of water).
Some other examples of homogeneous mixture are alloys, salt in water, copper sulphate solution etc.



Suspensions are mixtures with particles having diameter greater than 1000 nm. They do not transmit light.

(ii) **Heterogeneous mixture :** In such a mixture the particles of each component of the mixture remain separate and can be observed as individual grains under a microscope. e.g. mixture of grains and sand. This type of mixtures contain physically distinct parts and have a non-uniform composition. Some other examples of heterogeneous mixture are a mixture of sodium chloride and iron fillings, a mixture of salt and sulphur, a mixture of oil and water.

Knowledge ENHANCER

There are several types of mixtures. They are:

- *solutions*
- *suspensions*
- *emulsions*
- *colloidal dispersions*

Solutions: *These are mixtures made by mixing a solute and a solvent. The solute is the substance that dissolves. The solvent is the substance that does the dissolving.*

Suspensions: *These are heterogeneous mixtures of a solid and a liquid in which the solid does not dissolve. Suspensions will settle when left standing undisturbed.*

Emulsions: *These are a special type of suspension. This mixture consists of two liquids that do not mix. Since the liquids do not mix, emulsions are heterogeneous. Emulsions will settle into layers when they are left standing undisturbed.*

Colloidal Dispersions: *These are mixtures with characteristics part way between a solution and a suspension. Colloidal dispersions may appear homogeneous but are actually heterogeneous. Colloidal dispersions do not settle when left standing undisturbed for a period of time.*

Examples:			
Solutions	***Suspensions***	***Emulsions***	***Colloidal dispersions***
salt and water	*soil and water*	*oil and vinegar*	*liquid laundry starch and water*
alcohol and water	*sand and water*	*kerosene and water*	*fine pottery clay and water*
copper sulfate and water	*clay and water*	*oil and water*	*mayonnaise*

Do You Know!!

Most homogeneous mixture are also known as solutions and examples of these include pure air, coffees, and even metal alloys.

idea box

You can compare a solution with a mixture that contains non-soluble substances. Stir some sand into a jar of water. Then stir a spoonful of salt into a different jar of water.

Differences between Compounds and Mixtures : These differences between compounds and mixtures are summarized in the table below.

Mixtures	Compounds
(i) The constituents can be separated from one another by physical methods.	(i) The constituent elements can not be separated by physical methods.
(ii) Mixtures vary widely in composition	(ii) They are fixed in their composition by mass of elements present.
(iii) Mixing is not generally accompanied by external effects such as as explosion, evolution of heat or volume changes (for gases)	(iii) Chemical combination is usually accompanied by one or more of these effects (*i.e.* explosion, evolution or absorption of heat etc.)
(iv) The properties of mixtures are the sum of properties of its constituents.	(iv) The properties of a compound are different from those of its constituent elements.

Classification of Homogeneous mixtures : Homogeneous mixtures can be further classified as follows:

(i) **Solid-solid mixture:** Examples of such a mixtures are brass, bronze etc.
A homogeneous mixture of two or more metals (or metal and non-metal) is called an **alloy.** Brass and Bronze are alloys. Alloys find a wide range of applications in industry.

(ii) **Solid-liquid mixture:** Examples of such a mixture are
(a) aqueous solution of salt or sugar
(b) solution of iodine (I_2) in carbon tetra chloride (CCl_4).

(iii) **Liquid-liquid mixture:** Examples of this type of mixture are
(a) rectified spirit (a mixture of gasoline and alcohol)
(b) a mixture of toluene and benzene.

(iv) **Liquid-gas mixture:** This type of mixture is exemplified by liquor ammonia which is a mixture of ammonia gas and water.

(v) **Gas-gas mixture:** Pure air is an example of this type of mixture. Air consists of nitrogen, oxygen, argon *etc*.

Classification of Heterogenous mixtures :

(i) **Solid-gas mixture:** For example, Hydrogen gas adsorbed on palladium.

(ii) **Solid-solid mixture:** For example
(a) gun powder
(b) mixture of sulphur and iron filling

(iii) **Solid-liquid mixture: Example :** Suspension of sulphur in water.

(iv) **Liquid-liquid mixture: Example :** Benzene in water (*i.e.* immiscible liquids).

Points of Differences Between Homogeneous and Heterogeneous mixtures : These are listed in table below :

	Homogeneous mixtures		Heterogeneous mixtures
(i)	Such mixtures have same composition throughout	(i)	Such a mixture has different composition in different parts.
(ii)	The components of such a mixture cannot be seen even under a microscope.	(ii)	The components of such a mixture can be seen even with naked eye.

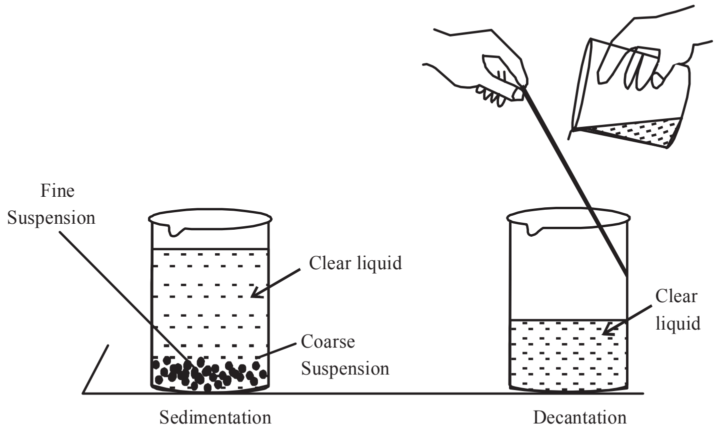
idea box

INSTRUCTION :
Try this experiment only with your teacher as it can be dangerous.
Place the mixture of iron fillings and sulphur in a pyrex glass test tube. Heat the test tube in the flame of Bunsen burner until the mixture glows. Remove the test tube from the flame and allow it to cool. When sufficiently cooled, break the test tube and remove the contents. Examine the contents with a magnet and magnifying glass.
Now find the answer of following questions.
(1) Your magnet will attract the resulting material or not?
(2) You are able to see the yellow Sulphur particles or not?
(3) Are the particles in a mixture in any way different than they were before mixing?

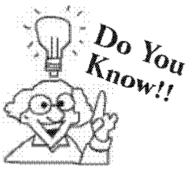
SEPARATING THE COMPONENTS OF A MIXTURE

We have already learnt that most of the natural substances are not chemically pure. Different methods of separation are used to get individual components from the mixtures.
Heterogeneous mixture can be separated into their respective components by simple physical methods like hand picking, sieving, filtration that we use in our daily life.
Sometimes special techniques are used for separation of components of a mixture. Some of these techniques are discussed in details.
The constituents of any mixture can be separated by taking advantage of the difference in their physical properties.

1. **Sedimentation and Decantation :** This method is generally used for separating the constituents of a system in which one component is liquid and the other is a coarse suspension of a solid heavier than the liquid. For example muddy river water.



Separation by sedimentation and decantation.



A sedimentation technique is widely used in the cleaning process of drinking water, while gases cannot be sperated by using sedimentation.

Such a mixture is allowed to stand for some time. The solid particles settle down to the bottom of the vessel under the action of gravity. This settling down of coarse suspensions is called **sedimentation**. The clear supernatant liquid (upper layer of liquid) is carefully transferred to another beaker. This process of mechanically transferring of the clear liquid without disturbing the settled solid particles is called **decantation**.

2. **Filtration:** This method is used for separating insoluble component (suspended impurity *etc.*) of a solution. In principle, a solution containing suspended impurities is made to pass through a porous membrane like filter paper, filter cloth, porous membrane of clay or porous glass filters. Such membranes do not allow the fine particles of suspended impurity to pass through the pores of membrane, thus separating the insoluble/suspended material from the solvent (generally water).

   Sometimes, a small quantity of alum is added to the system before filtration. A typical experimental set up for filtration in the laboratory (using filter paper) is shown in figure.

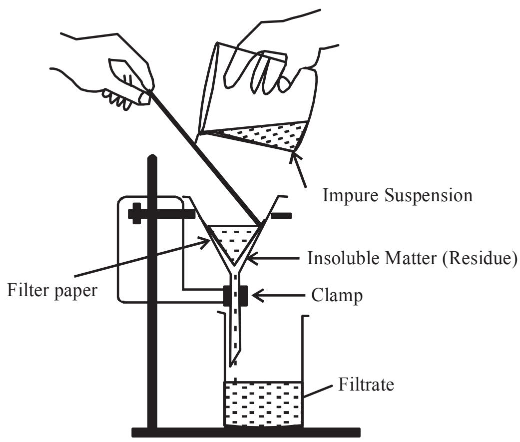
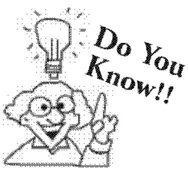


Fig. *Separation by filtration*



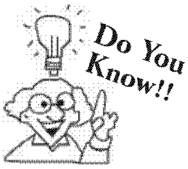
These days very effective and different methods are used for filtration of drinking water like carbon filtration, submicron filtration and Ion exchange

CHECK Point

How might you separate a mixture of sand and salt?

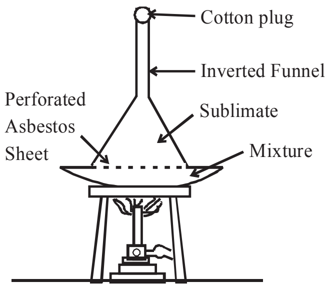
SOLUTION

Dissolve the mixture containing salt and sand in water. Salt dissolves in water whereas sand is insoluble in water. Filter it. Sand will be retained in filter paper while salt solution will be collected as filtrate. Salt is separated from salt solution by distillation the process discussed further in chapter. During distillation water distills and salt remains in the flask.

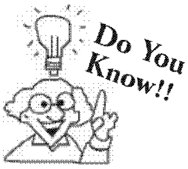


Only those substances whose vapour pressures becomes equal to the atmospheric pressure much before their respective melting points can be purified by sublimation.

3. **Sublimation :** Sublime substances such as camphor, ammonium chloride, naphthalene (moth-balls) and iodine on heating directly pass into vapours, without passing through the liquid phase. The vapours on cooling give back the substance in the crystalline form (known as sublimate). This process, known as **sublimation** is used for separating such substances in which the other constituents are not affected by heating. An experimental set up is shown in figure below :



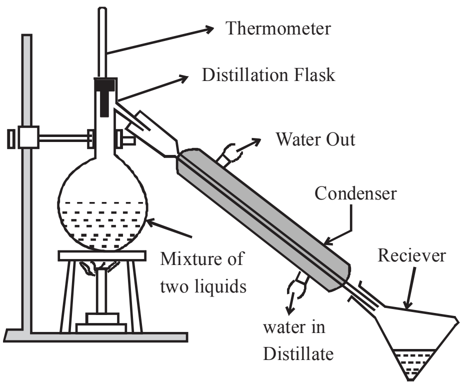
Separation by sublimation



In the universe sublimation occurs on the surface of asteroids and comets, when these are formed by mixtures of solid carbondioxide, water, ammonia and methane.

4. **Distillation :** Two liquids having different boiling points (with a difference of 10-20 K) can be separated by this method. When mixture of two such liquids is heated in a retort (or distillation flask) the low-boiling liquid component first distills over and the other component is left behind in the retort.

A simple experimental set up is shown in figure below



Separation by distillation

Do You Know!!

This method is used for purification of those liquids which do not decompose of their respective boiling points and contain non-volatile impurities.

CHECK Point

How does distillation separate the components of a mixture?

SOLUTION

Distillation is the process of simultaneous vaporization and condensation. The substances having different boiling points are separated by this method. The substance having lower boiling point is distilled first and collected. The liquid having higher boiling point is distilled at higher temperature and is collected in separate container. If the other substance is solid, it remains in distillation flask.

Do You Know!!

Evaporation is a method of separating a soluble solid from the solvent in which it is dissolved. Lemon juice, which is a solution of citric acid in water, separates by evaporation. Water evaporates off from boiling lemon juice. Eventually, only solid crystals of citric acid are left behind.

Do You Know!!

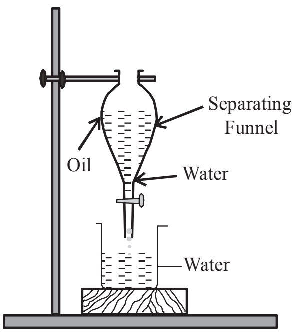
Fractional distillation can not be used to separate a mixture of liquids which form azeotropes i.e. constant boiling mixtures.

5. **Gravity Method :** This method takes advantage of difference in the density of the two components of a mixture. For example two immiscible liquids, *e.g.*, oil and water, can be separated with the help of a separating funnel. The heavier component forms the lower layer, which can be separated by opening the stopcock and collecting it in a clean and dry beaker.

Do You Know!!

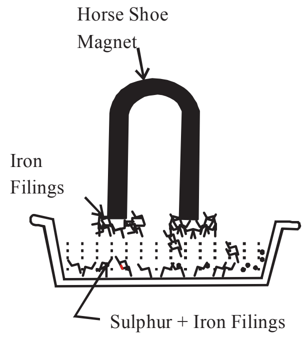
Gravity seperation technique is also known as liquid-liquid extraction or seperating funnel extraction.

A typical set up is shown in figure below :



Separation of two immiscible liquids by a separating funnel (gravity methods)

6. **Magnetic Method :** The magnetic component of a mixture can be separated with the help of an ordinary horseshoe magnet. The mixture is placed in a china or glass plate and the magnet is moved over the mixture. The magnetic material gets attached to the magnet. The process is repeated a number of times till the magnetic material is completely removed.



Separation of a magnetic substance by a magnet.

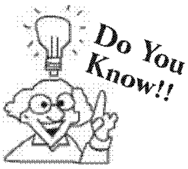
CHECK Point

1. **How can the components of a mixture be separted from one another?**
2. **How might you separate a mixture of iron and sand?**

SOLUTION

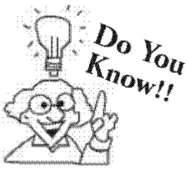
1. Components of a mixture can be separated from one another by simple techniques taking advantage of difference in physical properties of its components. For example a mixture of solid and liquid can be separated by filtration.
   Mixtures can also be separated by taking advantages of a difference in boiling or melting point. For example a mixture of water and some other liquid having different boiling points can be separated by distillation. In this method each liquid distill at a particular temperature. In this way each liquid can be isolated.
2. Mixture of sand and iron is separated by magnet. When magnet is brought over this mixture iron being magnetic in nature sticks to it, leaving behind sand in container.

7. **Crystallisation :** The hot saturated solution is allowed to cool slowly and undisturbed in a beaker or in a crystallising dish. After some time the crystals of the pure compound are formed and impurities are left behind in solution.



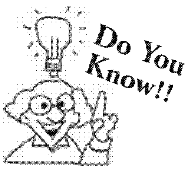
Fractional Crystallization: *This method is used to separate two or more solutes, by making use of their difference in solubility.*

8. **Chromatography :** It is another technique used for the separation of various constituents of a mixture.
   This technique is used to separate the **dissolved constituents** of a mixture. In this technique the dissolved constituents are **adsorbed** over an appropriate adsorbing material (i.e. **adsorbent).**



Chromatography finds application in research laboratories for identification of new substances.

This method makes use of the difference in adsorption of constituents by a surface of an appropriate **adsorbent** (adsorbing material). This method is used to separate the constituents in a mixture of ink. The technique used for this is known as **paper chromatography.**



Scientists can identify the substances in a solution by comparing their chromatograms with those of known substances. This method can be used, for example, to identify the colourings used in foods.

SOME DIFFERENT TYPES OF MIXTURES AND THE PRINCIPLE INVOLVED IN SEPARATION:

1. Solid-solid mixtures

		Components of mixture	Technique used	Principle involved
1.	(a)	Mixture of sulphur (soluble in CS_2) and sand (insoluble).	Solvent extraction	One of the components is soluble in same solvent and the other is insoluble in that solvent.
	(b)	Mixture of ammonium chloride (insoluble) and Iodine (soluble in carbon tetra chloride)		
2.		Mixture of iron filing and sand.	magnetic separation	one component is magnetic and gets attracted towards the magnet.
3.		Mixture of sand (heavier) and saw dust.	Gravity method	One of the component is heavier and the other is lighter than the given liquid.
4.	(a)	Mixture of Iodine (sublime) and sand		One of the component is sublimate and it sublimes on heating.
	(b)	Mixture of ammonium chloride (sublime) and sand.	Sublimation	

2. Separation of Solid-Liquid Mixtures :

		Mixture	Technique
(a)		Mixture of sand and water	Sedimentation (sand is insoluble)
(b)		Mixture of silver chloride (or chalk) and water	Filtration (AgCl or chalk is insoluble)
(c)	(i)	Mixture of sodium chloride (NaCl) and water	Evaporation (NaCl is soluble in water and is not sublime)
	(ii)	Mixture of sulphurand carbon dislphide	sulphur is soluble in carbon disulphide and is not sublime

3. Separation of Liquid-Liquid Mixtures :

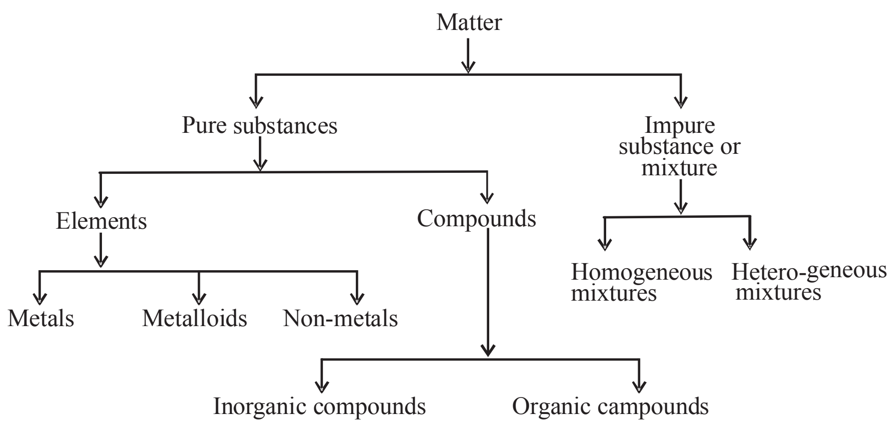
Mixture of petrol and water

Petrol is lighter and water is heavier liquid. They are immiscible liquids. They can be separated by use of **separating funnel**.

SUMMARY

Substance : In science substance refers to any single pure from of matter.

CHEMICAL CLASSIFICATION OF MATTER



Test for purity of a pure substance : By checking its m.p. or b.p. We can test the purity of a pure substance. A pure substance has a fixed melting point or boiling point at constant pressure.

Element : Pure substance in which molecules are made up of only one kind of atoms. .e.g. hydrogen, oxygen, gold, silver, iron etc.

Atomicity of an element : It is the number of atoms of an element present in its molecule. If atomicity is one it is called monoatomic e.g. He, Ne, Ar, Kr etc. If atomicity is two it is called diatomic e.g. H_2, O_2, N_2, Cl_2, Br_2, I_2 etc. If atomicity is more than two it is called polyatomic e.g. P_4, S_8 etc.

Classification of Elements : On the basis of their properties elements can be classified as metals, non-metals or metalloids.

Metals : All the metals are solids except mercury which is a liquid.

Non-metals : All the non-metals are solids or gases except bromine which is a liquid.

Metalloids : The elements which possess the properties of both metals and non-metals are called metalloids. e.g. Germanium, arsenic, antimony, selenium, tellurium etc.

Compound : A chemical compound is a pure substance which contains two or more elements chemically combined in a fixed ratio. e.g. water which consists of hydrogen and oxygen atoms combined together chemically in a definite proportion by weight (oxygen : hydrogen : : 8 : 1)

Mixtures : These are impure substance. Two or more substances (either elements or compounds) can be mixed together in any proportion and the resultant substances obtained are called mixtures.

Types of mixture :

(i) **Homogeneous mixture :** It is a mixture that has the same composition through out. e.g. a solution of sugar in water. The composition of such a mixture is variable.

(ii) **Heterogeneous mixture :** In such a mixture the properties of each component of the mixture remain separate and can be observed in individual grains under a microscope. e.g. a mixture of sand and grains.

This types of mixture contain physically distinct parts and leave non-uniform composition.

Classification of Homogenous mixtures :

(i) **Solid-solid mixture :** Examples are brass, bronze etc.

A homogenous mixture of two or more metals (or non-metals.) is called an alloy.

(ii) **Solid-liquid mixture :** Example are, solution of salt in water, solution of sugar in water, solution of iodine (I_2) in carbon tetrachloride (CCl_4)

(iii) **Liquid-liquid mixture :** Examples are rectified spirit which is a mixture of alcohol and water, any two miscible liquid such as bronze and toluene.

(iv) **Liquid-gas mixture :** Example liquor ammonia which is a mixture of ammonia (gas) in water (liquid).

(v) **Gas-gas mixture :** Example is air which is a mixture of various gases such as oxygen, nitrogen, carbondioxide etc.

Classification of Heterogeneous mixtures :

(i) **Solid-gas mixture :** Example- Hydrogen gas adsorbed on palladium.

(ii) **Solid-solid mixture :** Examples: gun powder, mixture of sulphur and iron filings.

(iii) **Solid-liquid mixture:** Example: suspension of sulphur in water.

(iv) **Liquid- liquid mixture :** Example: Benzene in water or any two immiscible liquids.

Methods for separation of constituents of a mixture

various methods used are :

(i) Sedimentation or decantation.	(ii) Filtration
(iii) Sublimation	(iv) Distillation
(v) Gravity method	(vi) Magnetic method
(vii) Evaporation	(viii) Fractional distillation
(ix) Chromatography	

1 EXERCISE

Multiple Choice Questions :

DIRECTIONS : *This section contains* ***20*** *multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.*

1. Select a pure substance
(a) Brass (b) Bronze
(c) Magnesium (d) All of these

2. Which one of the following is a heterogeneous mixture?
(a) Liquid ammonia (b) Gun powder
(c) Sugar solution in water (d) both (a) and (c)

3. If we add potassium nitrate to ice it will result in
(a) Increase in m.p. of ice.
(b) Decrease in m.p. of ice
(c) Neither (a) nor (b)
(d) Change in colour of ice

4. The atomicity of an element tells about its
(a) chemical properties
(b) physical properties
(c) number of atoms of element present in a molecule
(d) none of these

5. Which of the following is neither an element nor a compound?
(a) Water (b) Air
(c) Glucose (d) Iron

6. The formation of a chemical compound is accompanied with.
(a) Evolution of energy
(b) Absorption of energy
(c) Either absorption or evolution of energy
(d) None of these

7. Which one is the smallest unit of a compound?
(a) An electron (b) An ion
(c) An atom (d) A molecule

8. Which one is suitable method for separation of a mixture of water and sodium chloride?
(a) Decantation (b) Evaporation
(c) Simple distillation (d) Fractional distillation

9. When a solution of common salt in water is distilled, the salt is found in
(a) Distilling flask (b) Receiver
(c) Cooling water (d) Condenser

10. Which one will be most suitable method to separate a mixture of sodium chloride and ammonium chloride?
(a) Fractional distillation
(b) Sublimation.
(c) Sedimentation
(d) Chromatography

11. Separation of a mixture of the solids by fractional crystallisation depends on the difference in their
(a) Densities (b) Solubilities
(c) Shape of crystals (d) Size of crystals

12. Select the mixture that can be easily separated using a separating funnel.
(a) Milk and water
(b) Sugar and water
(c) Oil and water
(d) Petrol and kerosene

13. When a mixture of sand, common salt, glass powder and iodine is heated, a sublimate is formed. The sublimate formed is
(a) Sodium chloride (b) Glass
(c) Iodine (d) Common salt

14. The noble gases or rare gases exist as
(a) Mono-atomic (b) Di-atomic
(c) Tri - atomic (d) Tetra - atomic

15. Which of the following in an element?
(a) Marble (b) Graphite
(c) Washing stone (d) Stone

16. Which of the following is a compound?
(a) Stainless steel (b) Brass
(c) Iron sulphide (d) Diamond

17. Which of the following is not a compound?
(a) Common salt (b) Water
(c) Iron filings (d) Copper sulphate

18. Select the one that has a definite boiling point
(a) True solution (b) Compound
(c) Colloid (d) All of these

19. Which of the following will yield a mixture?
(a) Crushing of marble tile
(b) Breaking of ice-cubes
(c) Addition of sodium metal to water in a china dish
(d) Agitating a detergent with water in a washing machine.

20. Brass is an example of
(a) Compound
(b) Element
(c) Homogeneous mixture
(d) Heterogeneous mixture

More than One Option Correct :

DIRECTIONS : *This section contains* ***9*** *Multiple Choice Questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE OR MORE may be correct.*

1. A pure substance in one.
(a) that has a uniform composition
(b) which always have the same texture
(c) which is either an element or a compound
(d) which is a mixture

2. Hydrogen is
   (a) An element
   (b) Diatomic element
   (c) Gas at room temperature
   (d) A non-metal
3. Select the polyatomic elements.
   (a) O_2 (b) O_3
   (c) P_4 (d) S_8
4. Which of the following metals are solid ?
   (a) Sodium (b) Potassium
   (c) Mercury (d) Manganese
5. Which of the following are metalloids.
   (a) Selenium (b) Silicon
   (c) Arsenic (d) Antimony
6. Which of the following are homogeneous mixtures.
   (a) Brass
   (b) Bronze
   (c) Copper sulphate solution
   (d) Potassium permanganate solution
7. Which of the following is example of gas-gas mixture ?
   (a) Mixture of carbon dioxide and nitrogen
   (b) Air
   (c) Petroleum
   (d) Cold drink
8. Hydrogen gas adsorbed on palladium is
   (a) A solid-gas mixture
   (b) A solid-solid mixture
   (c) Neither (a) nor (b)
   (d) A compound
9. Which of the following is an example of solid-liquid mixture.
   (a) Suspension of sulphur in water
   (b) Muddy water
   (c) Both (a) and (b)
   (d) None of the above

Passage Based Questions :

DIRECTIONS : *Study the given paragraph(s) and answer the following questions.*

We can test the purity of a substance by checking its melting point or boiling point. A pure substance has a fixed melting or boiling point at constant pressure. For pure water the boiling point at normal atmospheric pressure is 373K but water containing some salt boils over a range of temperature above 373K

1. The boiling of sea water at normal atmospheric pressure is.
   (a) 373 K
   (b) more than 373 K
   (c) less than 373 K
   (d) can't be predicted
2. Distilled water obtained from sea water will boil at.
   (a) 373 K (b) 372 K
   (c) 374 K (d) None of these
3. The melting point of ice is
   (a) 273 K (b) 373 K
   (c) 300 K (d) 298 K

Assertion and Reason :

DIRECTIONS : *Each of these questions contains an Assertion followed by reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.*

(a) If both **Assertion** and **Reason** are **correct** and Reason is the **correct explanation** of Assertion.
(b) If both **Assertion** and **Reason** are correct, but Reason is **not the correct explanation** of Assertion.
(c) If **Assertion** is **correct** but **Reason** is **incorrect**.
(d) If **Assertion** is **incorrect** but **Reason** is **correct**.

1. **Assertion :** Pure substances in which molecules are made up of only one kind of atoms are known as elements.
   Reason : Hydrogen, oxygen and nitrogen are elements.
2. **Assertion :** Ozone is a polyatomic element.
   Reason : A molecule of ozone consists of three atoms of oxygen.
3. **Assertion :** Germanium, arsenic and antimony are classified as metalloids.
   Reason : They possess some properties of metals and some properties of metalloids.
4. **Assertion :** With rise of temperature Germanium acts as a semi-conductor
   Reason : It shows metallic property.

Multiple Matching Questions :

DIRECTIONS : *Each question has four statements (A, B, C and D) given in Column I and four statements (p, q, r and s) in Column II. Any given statement in Column I can have correct matching with one or more statement(s) given in Column II. Match the entries in column I with entries in column II.*

1.

	Column 'I'		Column 'II'
A.	Sugar in water	(p)	Homogeneous mixture
B.	Ethanol and Kerosene	(q)	Fractional distillation
C.	Iron fillings and sodium chloride	(r)	Fractional crystallisation
D.	Common salt in water	(s)	Heterogenous mixture

Integer Type Questions :

DIRECTIONS : *Following are integer based questions. Each question, when worked out will result in one integer from 0 to 9 (both inclusive).*

1. Atomicity of Phosphorous
2. How many among the following are solid-solid mixture?
   Gun powder, mixture of sulphur and Iron filling, Benzene in water, suspension of sulphur in water.
3. How many among the followings are compounds?
   Stainless steel, Brass, Iron Sulphide, Diamond, Sodium chloride

SOLUTIONS

Brief Explanations of Selected Questions

1 EXERCISE

Multiple Choice Questions :

1. (c) Magnesium is element where as brass and bronze are mixtures.
2. (b) It is a mixture of charcoal, sulphur and Potassium nitrate.
3. (b) The m.p. of a solid decreases due to presence of impurities.
4. (c)
5. (b) air is a mixture.
6. (c)
7. (d) The smallest unit of a chemical compound is a molecule.
8. (c) It is a mixture of soluble solid in a liquid.
9. (a) It is left behind as residue.
10. (b) Ammonium chloride is sublime where as sodium chloride is not.
11. (b) It depends upon the difference in their solubilities in a solvent.
12. (c) Oil and water are immiscible
13. (c) Iodine is sublime and forms the sublimate.
14. (a) Noble gases exist as mono-atomic (He, Ne, Ar etc.).
15. (b) 16. (c) 17. (c)
18. (b) 19. (d) 20. (c)

More Than One Correct :

1. (a), (b) and (c) 2. (a), (b), (c), (d)
3. (b), (c), (d)
4. (a, b, d)
Mercury is liquid where as others are solid metals.
5. (a), (c) and (d)
6. (a), (b), (c) and (d) 7. (a) and (b)
8. (a) 9. (c)

Passage Based Questions :

1. (b) sea water contains dissolved salt as impurities.
2. (a) Distilled water is pure water.
3. (a) 273 K [m.p. of ice is 0° C or 273K]

Assertion & Reason :

1. (b)
2. (a) [ozone is O_3]
3. (c) They possess some properties of metals and some properties of non- metals.
4. (b)

Multiple Matching Questions :

1. $A \rightarrow (p, r)$; $B \rightarrow (p, q)$; $C \rightarrow (r, s)$; $D \rightarrow (p, q, r)$

Integer Type Questions :

1. 4
2. 2
Gun powder and mixture of sulphur and Iron filling
3. 2
Iron Sulphide and Sodium Chloride

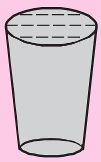
Chapter 7

STATES OF MATTER

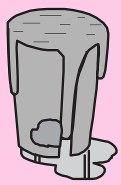
INTRODUCTION

Let us notice everything around us. The things or objects around us, the presence of which can be felt with the help of any of our five senses i.e. sight, touch, smell, hearing and taste are all made up of matter. The food we take, wood, plastic, petrol, water, steam, even the air we breathe are all examples of matter : each of these object has mass and occupies space.

Only one object can occupy the same space at the same time as you can see in figure that first all space in glass is occupied by water but when piece of stone is dropped into it. The water overflows because the stone takes the space of water.



The glass filled with water up to the brim. We can see the water inside the glass.



A stone is dropped into the glass of water. The water overflows because the stone takes the space of the water. The volume of the water that overflows is the volume of the stone.

Matter may be defined as anything that occupies space, has weight and offers resistance to any stress (force). e.g. wood, stone, iron, air, water etc.

Some matter is living. It grows, produce other living things like itself and finally dies. Non living things can do none of these things.

CLASSIFICATION OF MATTER

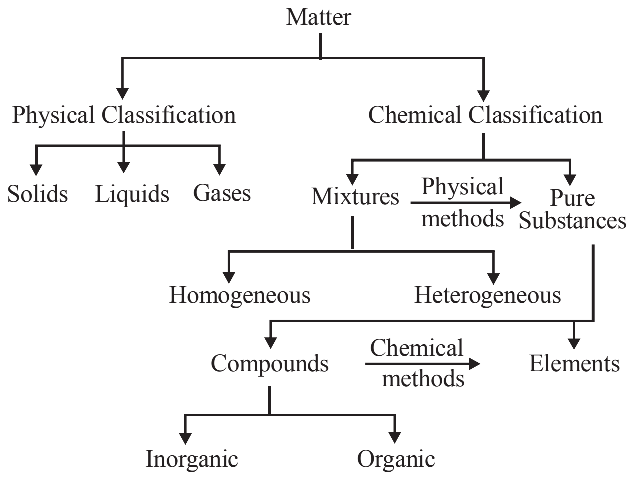
The earlier **Indian Philosophers** classified matter in the form of five basic elements — **"The Panch Tatva"** — Air, Earth, Fire, Sky and Water. It was believed by them that everything living or non-living is made up of these five basic elements.

However the ancient **Greek philosophers** believed that there were only **four basic elements** — Earth, Fire, Air and Water.

At present the scientists have classified the matter in the following two ways :

(i) The **physical classification** based on physical properties of matter, and

(ii) The **chemical classification** based on chemical composition of matter.



In this chapter we will study only about physical classification of matter. According to the physical classification, on the basis of the forces of attraction existing between molecules of a substance, matter can be classified as *solids*, *liquids* and *gases*.

HISTORICAL PREVIEW

Empedocles, a Greek Philosopher and scientist who lived on south coast of Sicily between 492 BCE and 432 BCE, proposed one of the first theories that attempted to describe the things around us. Empedocles argued that all matter was composed of four elements : fire, air, water and earth. The ratio of these four elements effects the properties of matter. Stone was thought to contain a high amount of earth, while a rabbit was thought to have a higher ratio of both water and fire, thus making it soft and giving it life. His theory was popular but it had a number of problems. For example, no matter how many times you break a stone in half, the pieces never resemble any of the core elements of fire, air, water or earth.

Then after few decades, Democritus, another Greek developed a new theory of matter according to which if you continued to cut the stone into smaller and smaller pieces, at some point you would reach a piece so tiny that it could no longer be divided. Democritus called these infinitesimally small piece of matter atoms, meaning "indivisible". He suggested that atoms were specific to the material that they made up of, means atoms of stone were unique to stone and different from the atoms of other materials.

In 1643, Evangelista Torricelli, an Italian mathematician showed that air had weight. Thus it must be made of something physical. Then Daniel Bernoulli, a Swiss mathematician, developed a theory that air and other gases consists of tiny particles which are too small to be seen, and are loosely packed in an empty volume of space.

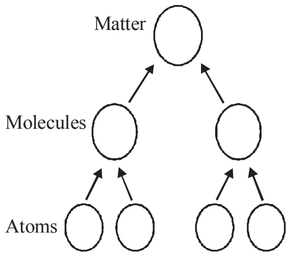
Joseph priestley in 1773 found red mercury clax on heating gives liquid mercury and a strange gas. Later in 1778, Antoine Lavoisier, a French scientist, called this gas priestley's gas oxygen.

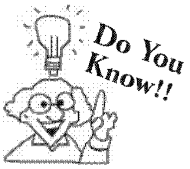
Knowledge ENHANCER

- ***Plasma :*** *This is the state of matter that consists of superenergetic and super excited particles. These particles are in the form of ionized gases. The fluorescent tubes and neon sign bulbs consist of plasma.*
  Fluorescent tubes are filled with helium gas and neon sign bulbs contain neon gas.
- *The Sun and stars glow because of presence of plasma in them. The plasma is created in stars because of very high temperature. The plasma glows with a special colour depending upon the nature of gas.*

Composition of Matter

The smallest particle of matter that is capable of free existence retaining all the characteristics of it is called *molecule* (Sometimes atom and ion) and molecules are made up of atoms. Thus *molecules are the constituent particles of matter* and the characteristics of these molecules and their arrangement in various states of matter was explained by *kinetic molecular theory of matter*.





The light and heat that comes from the sun and stars and electricity that can make lamps light up and motors run, these are not matter these are examples of Energy.

Kinetic Molecular Theory of Matter

This theory visualizes that matter is made up of atoms, molecules or ions that are in random motion and have space between them.

Postulates of Kinetic Theory of Matter

The important postulates of this theory are :

1. Matter is made up of small particles that may be atoms, ions or molecules. The constituent particles of a given substance are identical in all respects.
2. There is an empty space (gap) between the molecules (atoms or ions) which is known as intermolecular space.
3. Molecules exert an attractive force upon each other which is called intermolecular force of attraction. The magnitude of this force depends upon the state of substance (i.e. solid, liquid, gas) and also on the magnitude of intermolecular spaces.
4. The force of attraction that exists between similar molecules is called **"cohesive force"** and that which exists between dissimilar molecules is called **"adhesive force"**.
5. The molecules are always in a state of motion and they possess *kinetic energy* due to their motion.
6. On heating, there occurs an increases in the kinetic energy of the molecules (Kinetic energy $\propto$ Temperature).

CHECK Point

1. **Why matter exist in three different states?**
2. **What is the fourth state of matter?**

SOLUTION

1. Matter exist in three different states because of difference in intermolecular space between molecules of matter. This intermolecular space is lesser in solids higher in gases and intermediate in case of liquids.
2. Fourth state of matter is plasma. It is composed of ionic matrix which exists only at very high temperatures. Such a high temperature may be attained during nuclear fusion reactions.

STATES OF MATTER

We will now study the characteristics of these three states of matter.

The Solid State :

The solid state is characterised by, having a definite shape, distinct boundaries, rigidity and incompressibility.

They are like the children in this picture. In this picture children form a tight circle and cannot move easily. Similarly molecules in solids are tightly packed and cannot move easily.

Do You Know!!

Amorphous solids are, regarded as 'Liquids at all temperatures". Like, if glass is heated gradually, it softens and start to flow without undergoing a definite and abrupt change in to liquid state

Solids are of different types. In some cases differences would be easily recognizable (such as differences in size and colour) like in case of ball and brick, but some differences are not as easy to see. Sugar and salt can look alike, but they are not used in the same way : their functions are different. In case of solids forces are strong and have ionic, covalent, metallic or even hydrogen bonds.

Solids are classified as *crystalline* and *amorphous*. A crystalline solid has a definite and sharp melting point at which it becomes a liquid. Amorphous solids have no precise melting point.

Moreover a crystalline solid has a definite geometry due to definite arrangement of molecules, atoms or ions. In amorphous solid, molecules, atoms or ions does not have any pattern or arrangement and therefore, does not have any definite geometrical shape.

Knowledge ENHANCER

Substances such as pitch and resin are called semisolids; these are actually very viscous liquids, but their flow or change of shape is very slow at ordinary temperatures. Properties in which solids differ from one another include density, hardness, ductility, elasticity etc.

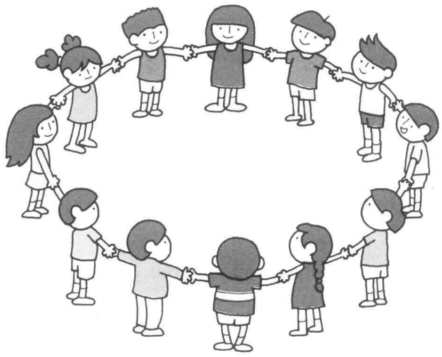
idea box

Cubes or crystals of common salts or sugar can be converted into powder form, but cube of wood can not be. Though both are solids and have almost same characteristics as solids have. Discuss this with your friends and teachers and try to find out answer.

The Liquid State :

The liquid state is characterised as having fluidity, low compressibility and no definite shape. Liquids take the shape of container in which they are stored.

A liquid flows and changes shape because its particles are free to move about. They are not as tightly packed together as in a solid. Compare them with the children in the given picture. The children are farther apart and they do not pull so strongly at one another. They have more space to move about and they can change the shapes of the circle. However, they still have to keep together because they are holding onto one another. This explains why liquids have a definite volume.

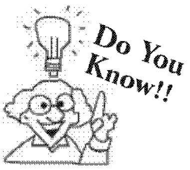


Properties of liquids.

1. Liquids have no definite shape.
2. Liquids have a definite volume.
3. Liquids have much higher density than gases but less than solids.
4. Liquids diffuse like gases but the diffusion is much slower.
5. Liquids are compressible to appropriate extent.
6. Liquids exhibit vapour pressure.

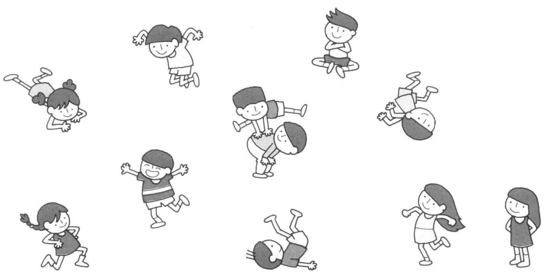
Knowledge ENHANCER

Water is a very important liquid. Roughly 70% of an adult's body is made up of water. Water covers 70 to 75% percent of the earth's surface.



Energy is stored when gases are compressed. Compressed gas exerts a force. Air guns and aerosol cans make use of compressed gases to shoot things out. The tyres of car contain compressed air. It makes the tyres hard. The compressed air in the tyres exerts a force large enough to lift a heavy car.
The balloon floats in air because they are filled with helium, a gas that is less dense than air.

The gaseous state is characterised as having fluidity, high compressibility, no definite boundaries, no definite volume and no definite shape.



A gas has no definite volume or shape because its particles are very far apart. Compare them with the children in the picture above. They are not bound together and so they can move wherever they like. In a big space they move further apart and in a small space they are closer together. In the same way a gas expands to fill a bigger volume. In a smaller space it is compressed.

1. Molecules are sufficiently apart from one another.
2. Force of attraction between gas molecules are almost negligible.

3. Gases possess high compressibility and thermal expansion.
(4) Gases neither have definite shapes nor definite volumes.
(5) Gases generally have low densities.

idea box

Take two metallic bottles filled with water and other filled with air. Fix a rubber balloon at the mouth of both glass bottles. Then both glass bottles are heated at same temperature for the same period of time. Balloon fixed in bottle containing air expands more in comparison to balloon fixed on water containing bottle explain why?

Do You Know!!

Effusion is the motion of gas molecules from high pressure to Low pressure through a small hole.

CHECK Point

1. **What are three properties of a gas that you can measure.**
2. **What will happen when a bicyclist operates the pump to inflate tyre?**

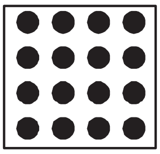
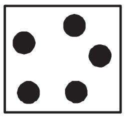
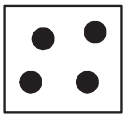
SOLUTION

1. (1) Amount of gas
(2) Volume of gas
(3) Pressure of gas

2. This will result in to decrease in the volume of air in the cylinder and increase its pressure. As a result air will be forced into the bicycle tyre and the tyre will inflate.

COMPARISION OF SOLIDS, LIQUIDS AND GASES

The following table gives a comparative study of solids, liquids and gases based on the kinetic molecular theory of gases.

Property	Solid	Liquid	Gas
1. Arrangement of particles	 Particles are closely packed and have an orderly arrangement	 Particles are loosely packed and there is no orderly arrangement	 Particles are very far apart and are randomly arranged.
2. Forces of attraction between constituent particles.	Very strong force of attraction.	Strong force of attraction.	Weak force of attraction
3. Motion of particles.	Particles can vibrate about a fixed point.	Particles can roll, slide past each other.	Particles are in random motion and are moving about at great speed.
4. Kinetic Energy	Posses low kinetic energy	Possess moderate kinetic energy	Possess high kinetic energy.

Properties of solids, liquids and gases :

The following table lists the properties of three states of matter :

Property	Solids	Liquids	Gases
1. Mass	They have definite mass	They have definite mass	They have definite mass
2. Shape	They have a definite shape.	They take the shape of container in which they are stored.	They do not have a definite shape i.e. take the shape of container.
3. Volume	They have a definite volume	They have a definite volume	They do not have a definite volume and fill the container completely i.e., have the volume of container.
4. Density	In solids the intermolecular space is minimum so they have maximum density.	Intermolecular space is larger than solids so the density of liquids is less as compared to solids.	In gases the intermolecular spaces are maximum so gases have least density.
5. Rigidity	Highly rigid. Particles in solids can vibrate only about their mean position	Less rigid in comparison to solids. In liquids free movement of particles is possible.	Not rigid. In gases particles are free to move in all directions.
6. Diffusion (or miscibility)	No diffusion	Slightly diffusible	Highly diffusible
7. Free surface	Because of their definite shape solids can have any number of free surfaces	Only one upper free surface	No free surface

IDEA BOX

idea box

Go to your kitchen and observe various things. Now make a list of compounds under the category of solids, liquids and gases. Ask your friends to do the same activity at their home. Now compare your list with your friends.

Knowledge ENHANCER

Diffusion : *Diffusion is the gradual mixing of one substance with another.*

Examples of gas diffusion :

1. *The steam from a boiling kettle spreads throughout a room.*
2. *When a gas jar containing clean air is inverted on another gas jar containing bromine (a brown coloured gas) gas, they slowly mixes and the colour becomes light brown in both the jars.*

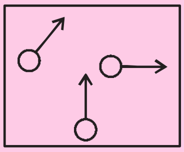
Examples of liquid diffusion :

1. *A dark crystal of potassium manganate (VII) when added to a beaker containing water, the crystal slowly dissolves and yields a purple solution.*

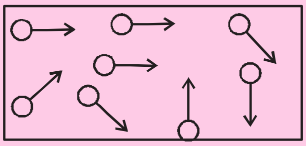
2. The above purple solution [i.e. the solution of potassium manganate (VII)] when diluted with an equal amount of water, it still remains purple but the colour becomes lighter. If we continue adding more water the purple colour still persists. The particles are spreading through out the solution.

Kinetic particle theory : The movement of particles is due to the possession of kinetic energy. More the kinetic energy possessed by particles, faster they move.

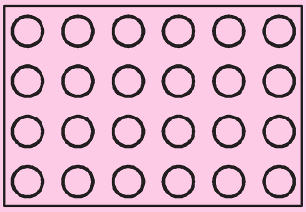
Gas : Gas particles move fastest and move to fill in the whole of the available space.



Liquids : They move slowly and are confined to the limit of the vessel in which they are stored.

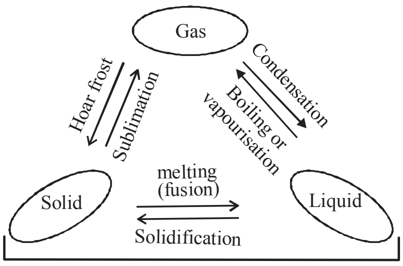


Solids : They possess only sufficient energy to vibrate while held in a regular arrangement.



INTER-CONVERSION BETWEEN STATES OF MATTER

We know from our observation that water can exist in all the three states (i.e. solid, liquid and gas) of matter e.g.
Solid $\rightarrow$ ice
Liquid $\rightarrow$ liquid water
Gas $\rightarrow$ steam or water vapour
Thus we can say that matter can change its state. Moreover the change in state of matter are reversible. The change in state of matter can be brought about by change of pressure or / and by change of temperature.
If we apply high pressure on a gas it changes to a liquid.
If we increase the temperature the solid changes to liquid which is then changes to gas.
The different changes occurring in state of matter by absorption or evolution of heat can be diagrammatically represented as follows :



Interconversion between states of matter

CHECK Point

(1) Solid $\xRightarrow{A}$ Liquid $\xRightarrow{B}$ Gas

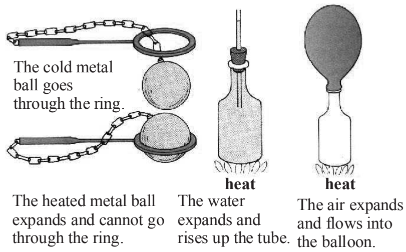
(2) Liquid $\xRightarrow{C}$ Solid

In above equations what is A, B and C?

SOLUTION

A = Melting, B = Boiling and C = Freezing.

Effect of Change of Temperature on States of Matter



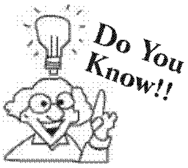
Matter interacts with heat. It expands when it is heated and contracts when it is cooled. On increasing the temperature of solids, the kinetic energy of the particles increases and because of this increase in kinetic energy the particles start vibrating with greater speed. The energy supplied by heat overcomes the intermolecular forces of attraction between solid particles and the particles leave their mean position and start moving more freely.

A stage is reached when the solid melts and gets converted to a liquid. This temperature is called *melting point* of solid.

Melting or **Fusion :** It is the process by which a solid changes to a liquid by absorption of heat.

Melting point : Melting takes place at a constant temperature and this constant temperature is called melting point.

For any pure substance the melting point is fixed. Whenever impurities are present they lower the melting point of a substance. The melting point of ice is 273.16 K.



An increase in pressure increases the melting point but in case of water increasing pressure lowers the melting point.

Boiling : It is the process by which a liquid changes into a gas, this change in state takes place inside the liquid.

In the process of boiling heat is absorbed.

Boiling point : The temperature at which a liquid changes into gas/vapour is known as its boiling point. Boiling point is a bulk phenomenon. Particles from the bulk of liquid gain enough energy to change into vapour state.

Latent heat of fusion : *The quantity of heat (in kilocalorie) required to change its 1 kg mass from solid to liquid state at its melting point is called latent heat of fusion.*

For ice latent heat of fusion = 80 *kilocal/kg*.

Latent heat of vaporization : *It is the quantity of heat required to change its 1 kg mass from liquid to vapour state at its boiling point.*

For water latent heat of vaporisation = 536 kilocal/kg.

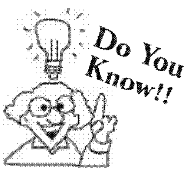
CHECK Point

What do you think about gas and vapour ? Does they mean the same thing ? Explain your answer.

SOLUTION

They mean the same thing. The word vapour is used to describe those gases that usually exist as liquid at room temperature e.g. water exists as liquid at room temperature so when it is in gaseous state it is called water vapour.

Oxygen exists as gas at room temperature so we call it a gas and *not* vapour.



The temperature of boiling liquid remains constant even if heating is continued. On heating a liquid, the average Kinetic energy of liquid molecules increases and therefore temperature of liquid increases till boiling point is reached. At this stage molecule escapes to vapour state resulting in to decrease of energy of system. Thus the energy supplied on heating the liquid after its boiling point is used in compensating the loss of energy due to escaping molecules.

Effect of change of pressure on states of matter

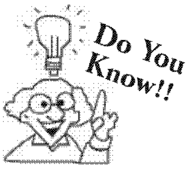
Just as we can change the state of matter by change of temperature, it is possible to bring about a change in state of matter by change of pressure.

For example, with the increase in pressure on gas molecules the volume of gas reduces thus intermolecular distance between gas molecules decreased which results into increase of intermolecular force of attraction between gas molecules. Thus gas gets converted into liquid.

EVAPORATION

The phenomenon of change of a liquid into gas at any temperature below its boiling point is called **evaporation**. Evaporation is a surface phenomenon. The particles of a liquid present on the surface gain energy and leave the surface.

Evaporation is used as a **purification method** to get pure solid from its solution.



Addition of a nonvolatile solute in solvent lowers the vapour pressure thus more amount of heat is required to increase the vapour pressure of solution to atmospheric pressure.

Evaporation is boiling a liquid mixture of a solid dissolved in a solvent so as to evaporate the unwanted solvent and leave the pure solid as residue.

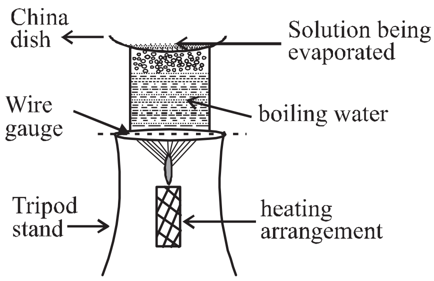
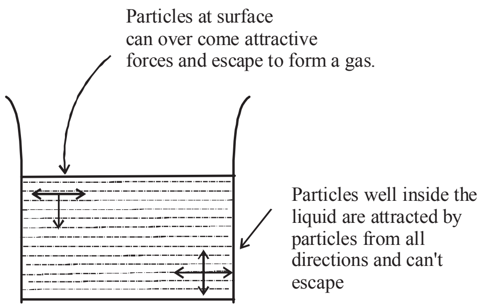
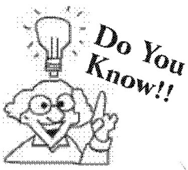
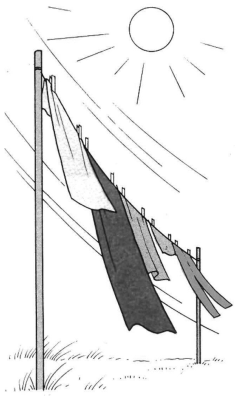


Fig. Evaporation process

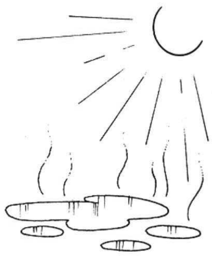




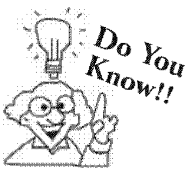
A blow of air or gases through a liquid surface enhances the evaporation. The blown air takes up the liquid molecules from the surface with it.



Evaporation dries our washing clothes. Wet clothes dry faster when hung lengthwise in open sunny places.



Evaporation dries the puddles which form on roads and open spaces after a heavy rain.



Water evaporated quickly when it is heated. As the water boils, it turns into steam.

CHECK Point

1. **Which one out of ethanol and water has more tendency of evaporation?**
2. **Why Evaporation results into cooling?**

SOLUTION

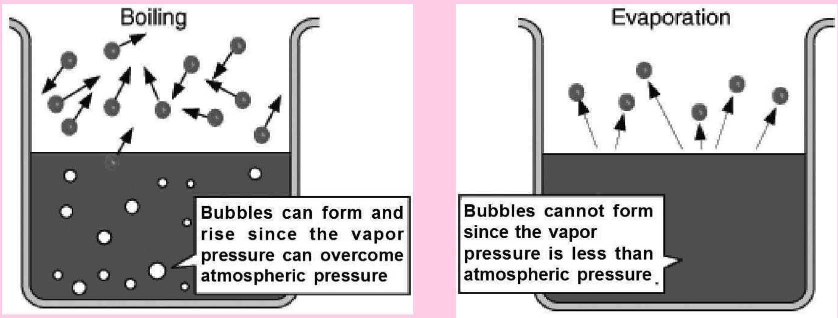
1. Ethanol because intermolecular forces of attraction between molecules is less in comparison to water.
2. Molecules of surface with higher energy escapes out. The remaining molecules are left associated with lower energy, thus temperature falls down.

Knowledge ENHANCER

Difference between Boiling and Evaporation :

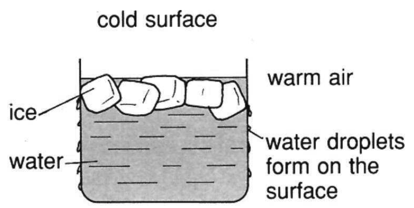
(1) Boiling takes place only at a particular temperature for a liquid. Whereas evaporation occurs at all temperatures.

(2) Boiling is Bulk phenomena i.e. the bubble formation occurs even below the surface. Whereas evaporation is surface phenomena, i.e. bubble formation occurs only on the surface of liquid.

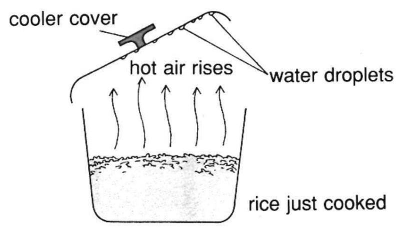


CONDENSATION

It is the process whereby a gas changes into a liquid. This change in state occurs when a gas is cooled. During the process of condensation heat is evolved. On cooling the kinetic energy of particles decreases and hence the velocity. Consequently, the intermolecular distance between the gas molecules decreases and it results in an increase in intermolecular forces of attraction. Because of this the molecules become less free to move about and the gas attains the molecular arrangement of a liquid.



The outer surface of a glass containing cold water becomes wet because of condensation. Water droplets form when the warm, damp air meets the cold glass surface.



Water condenses on the underside of the cover of a cooking pot because the cover is cooler than the rest of the pot.

CHECK Point

Vapour $\xrightarrow{A}$ Liquid

In above equation which phenomena A represents?

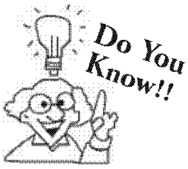
SOLUTION

A = Condensation

FREEZING OR SOLIDIFICATION

It is the process whereby a liquid changes into a solid. Heat is evolved during this process. Freezing takes place at a fixed temperature and this temperature is called freezing point.

Freezing or **Solidification** occurs due to decrease in Kinetic energy of liquid molecules in the form of heat energy. Consequently, the intermolecular distance decreases and intermolecular force of attraction increases. Molecules lose their movement and they vibrate about their fixed mean position. Thus the molecules of a liquid attain the molecular arrangement of a solid.



A pure substance has a fixed freezing point. Impurities if present lower the freezing point. Any increase in pressure lowers the freezing point in case of water but in case of other substances there occurs an increase in freezing point on increasing the pressure.

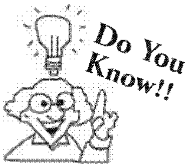
SUBLIMATION

It is the process where by interconversion between solid state and gaseous state occurs without passing through the liquid state

$$Solid \underset{cool}{\overset{heat}{\rightleftharpoons}} gas$$

When solid changes to a gas there occurs an absorption of heat and heat is evolved when gas changes to solid state.

Sublimation takes place at a fixed temperature.



A substance that shows the property of sublimation is called sublime substance. e.g. Iodine, dry ice (solid carbon dioxide), ammonium chloride, camphor etc. The solid obtained on cooling the vapour is called sublimate.

CHECK Point

Solid Camphor $\underset{\textbf{Cool}}{\overset{\textbf{Heat}}{\rightleftharpoons}}$ **gaseous camphor**

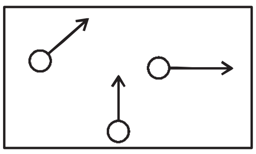
Above equation represents which phenomena?

SOLUTION

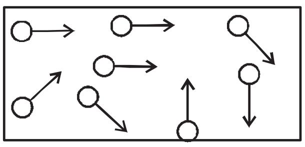
Sublimation.

SUMMARY

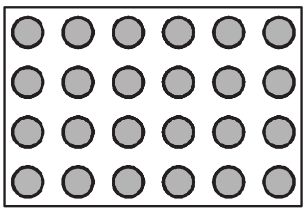
- **Matter :** Any thing which occupies space and has mass is called matter. *Sand, water, oxygen, wood, rock* are all different kinds of matter because all of them occupy space and have mass.
- **States of matter :** Matter exists in three states :
  (a) **Solid :** A substance that is hard and compact is called a solid. A solid had *definite shape* and *definite volume.* e.g. rock, iron etc.
  (b) **Liquid :** A liquid has a *definite volume* but has *no definite shape.* It takes the shape of the vessel in which it is poured. e.g. water, oil etc.
  (c) **Gas :** A gas *neither have definite volume nor a definite shape.* It fills the container completely and takes its shape. e.g. hydrogen, oxygen, air etc.
- **Plasma :** It is the **fourth state of matter.** This is the state of matter that consists of super energetic and super excited particles. These particles are in the form of ionized gases.
- **Atom :** It is the smallest particle of an element that can take part in a chemical reaction. Atoms are building blocks for the whole of matter around us. Atoms may or may not exist in free state.
- **Molecule :** It is defined as the particle of matter that is capable of free existence in nature. Molecule of an element may be made up of two or more similar atoms. e.g. O_2 (made up of two atoms of oxygen), H_2, Cl_2, P_4, S_8 etc. Molecule of a compound is made up of two or more dissimilar atoms e.g. H_2O, NaCl, CS_2 etc.
- *Kinetic molecular theory* visualizes that matter is made up of atoms, molecules or ions in constant motion.
- *Evidence of existence of individual particles* is provided by *diffusion.* Diffusion is the gradual mixing on one substance with another.
- **Examples of Diffusion :**
  (1) **Gas diffusion :**
    (a) the steam from a boiling kettle spreads through out a room.
    (b) A gas jar of clear air on top of a gas jar of brown bromine gas (heavier than air) slowly mixes and the colour becomes light brown in both the jars.
  (2) **Liquid diffusion :**
    (a) A dark crystal of potassium magnate (VII) added to a beaker of water dissolves to give a purple solution.
    (b) The purple solution of potassium manganate (VII), when diluted with equal amount of water, is still purple (but lighter) and the purple colour persists even on further dilution with water i.e. particles are spreading throughout the solution.
- **Movement of particles in solid, liquid and gas :**
  Particle movement is due to the possession of kinetic energy (the more energy they possess, the faster they move)
  Gas : Gas particles move fastest and move to fill whatever space is available.



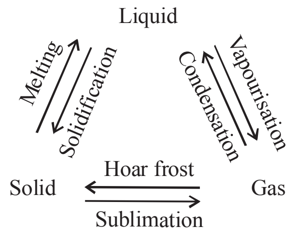
Liquid : Liquid particles move comparatively slowly and are confined to the limit of the vessel in which they are put.



Solid : Solid particles only possess sufficient energy to vibrate while held in a regular arrangement.



- **Inter conversion between states of matter :**



Solid $\xrightleftharpoons[\textit{freezing}]{\textit{melting}}$ Liquid $\xrightleftharpoons[\textit{Condensation}]{\textit{boiling or evaporation}}$ Gas

- **Melting :** It is the process by which a solid changes to liquid by absorption of heat.
- **Melting point :** Melting point (*m.p.*) is the temperature at which solid melts to a liquid. A pure substance melts at a constant temperature.
- **Boiling :** It is the process whereby a liquid changes into gas. This change in state takes place inside the liquid. In this process heat is absorbed.
- **Boiling point :** Boiling point (*b.p.*) is the constant temperature at which a pure liquid boils (changes from liquid to gaseous state).

  [**Note :** Boiling point is dependent on atmospheric pressure.
  lower pressure (*e.g.* up a mountain) $\rightarrow$ lower *b.p.*
  higher pressure (*e.g.* pressure cooker) $\rightarrow$ higher *b.p.*]
- **Melting point of impure solid :** The *m.p.* of impure solid is lower than that of pure substance (e.g. when salt is added to ice on roads).
- **Boiling point of impure liquid :** The boiling point of an impure liquid is higher than that of pure solvent (liquid).
- **Evaporation :** It is the process whereby a liquid changes into a gas. This is a *surface phenomenon.* This process takes place at all temperatures below the boiling point of the liquid. Heat is absorbed during evaporation.
- **Vapour :** Vapour is used to describe those gases that usually exist as liquid at room temperature e.g. water exists as liquid at room temperature so when it is in gaseous state it is called water vapour.
- **Condensation :** It is the process whereby a gas changes into a liquid. This change in state occurs when a gas is cooled. During this process heat is evolved.
- **Freezing or solidification :** It is the process whereby a liquid changes into a solid. Heat is evolved during this process.
- **Freezing point :** The constant temperature at which freezing takes place is called freezing point.

  [**Note :** Numerical value of freezing point and melting point is same].
- **Sublimation :** It is the process whereby inter-conversion between solid and gaseous state occur without passing through the liquid state.

  $$Solid \xrightleftharpoons[cool]{heat} gas$$

  Sublimation takes place at fixed temperature.
- **Sublime substance :** A substance that shows the property of sublimation is called a sublime substance. e.g. Iodine, dry ice, ammonium chloride, benzoic acid. etc.
- **Sublimate :** The solid obtained on cooling the vapours of a sublime substance is called sublimate.

1 EXERCISE

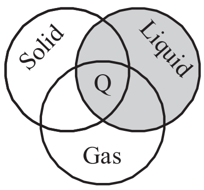
Multiple Choice Questions :

DIRECTIONS : *This section contains **19** multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.*

1. With the increase in pressure the boiling point of a liquid
(a) increase
(b) decreases
(c) remains unchanged
(d) can not be predicted

2. When potassium nitrate is added to ice, it
(a) increases the melting point of ice
(b) decreases the melting point of ice
(c) does not effect the melting point of ice
(d) None of the above is correct

3. Shown below is venn diagram directing different states of matter. Which of the following can be placed at place shown as 'Q' in the diagram?



(a) Dry ice (b) Table salt
(c) Jelly (d) Water

4. The correct order on the basis of intermolecular space is :
(a) solid > liquid > gas
(b) gas > solid > liquid
(c) gas > liquid > solid
(d) liquid > gas > solid

5. The property of diffusion is shown to the maximum extent in
(a) solid (b) liquid
(c) gas (d) Same in all cases

6. When a gas is cooled, the temperature falls and
(a) energy is lost by particles
(b) intermolecular space decreases
(c) intermolecular force of attraction increases
(d) All the above are correct

7. When a solid is allowed to melt, it results in
(a) moving away of particles
(b) coming closer of particles
(c) decrease in speed of particles
(d) decrease in temperature.

8. Select the one that diffuses fastest at room temperature.
(a) Petrol (b) Milk
(c) Perfume (d) Camphor

9. When water vapours condense to form water, the particles of water vapour
(a) gain energy and also gain freedom of motion.
(b) gain kinetic energy and increase their freedom to move about without any significant attraction.
(c) lose energy and come closer, thus their freedom of movement gets restricted.
(d) lose energy and also lose freedom to move about

10. Which of the following is likely to happen when we decrease the inter-particle space in a substance?
(a) an increase in volume occurs.
(b) a decrease in volume occurs
(c) an increase in mass of substance occurs.
(d) a decrease in mass of substance occurs.

11. What happens to the inter-particle attractive forces on increasing distance between particles?
(a) it increases
(b) it decreases
(c) there is no change
(d) None of the above is correct

12. The phenomenon of change of liquid into gas at any temperature below its boiling point is called ______
(a) melting (b) boiling
(c) liquifaction (d) evaporation

13. Evaporation is a
(a) bulk phenomenon
(b) surface phenomenon
(c) Both (a) and (b)
(d) None of these is correct

14. In the process of condensation, a gas changes into liquid and heat is evolved. In the process which form of energy of gaseous molecules is released as heat energy?
(a) kinetic energy (b) potential energy
(c) atomic energy (d) None of these

15. Out of the following. Which is a sublime substance?
(a) Chlorine (b) Fluorine
(c) Iodine (d) Gasoline

16. Which state of matter consists of super energetic particles in the form of ionized gases?
(a) Solid (b) Plasma
(c) Liquid (d) All of these

17. With the increase in temperature of liquid its rate of evaporation.
(a) Increases (b) Decreases
(c) Remains constant (d) Can not be predicted

18. On which of the following rate of evaporation of Liquid depends?
(a) Surface (b) Temperature
(c) Nature of Liquid (d) All of these

19. The force of attraction between similar molecules is called
(a) Cohesive force (b) Adhesive force
(c) Friction force (d) None of these

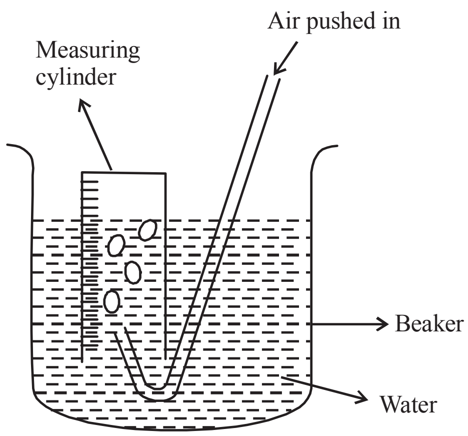
More than One Option Correct :

DIRECTIONS : *This section contains **10** Multiple Choice Questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE OR MORE may be correct.*

1. According to kinetic molecular theory
(a) matter is made up of small particles that may be atoms, molecules or ions.
(b) the empty space (gap) between the particles is known as intermolecular space.
(c) molecules exert an attractive force upon each other.
(d) the molecules always remain fixed.

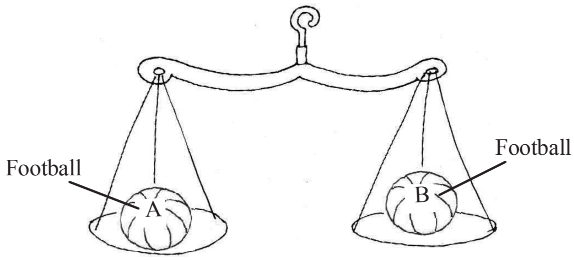
2. Which are/is correct statements?
(a) Matter is continuous in nature.
(b) Of the three state of matter, the one that is most compact is solid state.
(c) In solid state interparticles space (i.e., empty space) is minimum.
(d) The density of solid is generally more than that of a liquid.

3. When air is blown through water placed in a beaker (see figure), we will observe that



(a) air bubbles escape in cylinder
(b) water level increases in beaker
(c) water level decreases in cylinder
(d) water level decreases in beaker

4. A student placed two footballs of the same size and made of the same material in two pans of a balance (see figure).



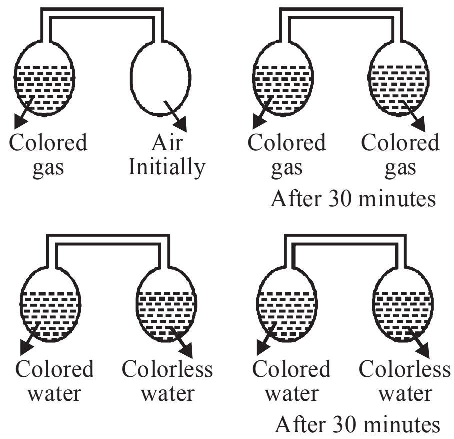
Which of the following is true ?
(a) both balls have same volume of air
(b) both balls have same mass of air
(c) air in ball **A** is more compressed than in ball **B**
(d) all the above are correct

5. Select the correct statement about water
(a) water can exist in all the three forms of matter.
(b) the b.p. of water is 100°C
(c) water can not be compressed
(d) all the above are correct.

6. What is correct on the basis of phenomena of Evaporation ?
(a) It cause cooling effect.
(b) It is favoured by decrease of temperature
(c) Evaporation is a surface phenomena
(d) Evaporation rate increases with increase of the temperature

7. A solution of common salt is prepared and freezed by Raj.
(a) It occurs by evolution of heat
(b) Freezing point of solvent is higher than solution.
(c) No heat change occurs in this process
(d) Kinetic energy of Liquid molecule increases.

8. Sumit placed a solid on table and after 20 minutes he found that volume of solid is reduced and there is no liquid on table as well.
(a) It represents phenomena of sublimation
(b) This phenomena occurs by evolution of heat
(c) Solid substance placed by sumit on table should be sublimate
(d) It represents the phenomena of Boiling

9. 

A student conduct an experiment. Initially he take two connected glass bulbs one containing colored gas other containing air. After few time both bulbs contain colored gas. When he tries to do the same with colored water in one glass bulb and other containing colorless water result obtained was not same.

(a) Rate of diffusion of gas molecules is much faster than Liquid

(b) Rate of diffusion of water molecules is more than gas molecules.

(c) Intermolecular space between gas molecules is very large.

(d) Colorless water will not get colored because it has high density in comparison to gas.

10. Which of the following is gas to room temperature?

(a) Water (b) Carbon dioxide

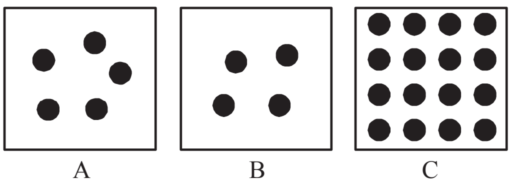
(c) Dry ice (d) Nitrogen

Passage Based Questions :

DIRECTIONS : *Study the given paragraph(s) and answer the following questions.*

Passage - 1

Observe the diagram and answer the questions.



1. The substance A may be

(a) Honey (b) Wax

(c) Hydrogen (d) Plasticide

2. The substance B may be

(a) Hydrogen (b) Liquid

(c) plastics (d) wax

3. The substance C may be

(a) Wax (b) CO_2 gas

(c) Liquid soap (d) Hair oil

Passage - 2

Sheetal once preparing evening tea. In the mean while telephone bell rangs and she rushed upto attend telephone call. She got busy on phone and suddenly heard a sound from kitchen. When she enter back into kitchen. She saw that tea water was boiling with steam over it and cover of tea container is flew a side with water droplets over it.

4. Why cover of tea kettle flew away?

(a) Not properly tight

(b) Tea is explosive in nature

(c) Pressure of steam

(d) None of these

5. Water droplets on cover of tea container represents.

(a) Condensation

(b) Evaporation

(c) Fusion

(d) Sublimation

6. Steam over tea represents phenomena of

(a) Evaporation

(b) Boiling

(c) Fusion

(d) None of these

Assertion and Reason :

DIRECTIONS : *Each of these questions contains an Assertion followed by reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.*

(a) If both **Assertion** and **Reason** are **correct** and Reason is the **correct explanation** of Assertion.

(b) If both **Assertion** and **Reason** are correct, but Reason is **not the correct explanation** of Assertion.

(c) If **Assertion** is **correct** but **Reason** is **incorrect**.

(d) If **Assertion** is **incorrect** but **Reason** is **correct**.

1. **Assertion :** Cooking of rice is difficult at higher altitudes.

Reason : At higher altitudes the boiling point of water increases.

2. **Assertion :** The rate of diffusion is greater in case of gases than in liquids.
Reason: The intermolecular forces of attraction are stronger in gases than in liquids.
3. **Assertion :** During evaporation of liquid there is no change in the temperature of the liquid.
Reason : Kinetic energy associated with particles of matter increases with increase of temperature.
4. **Assertion :** During extreme winter season the roads get blocked in Jammu and Kashmir due to snowfall and to remove ice on roads some salt water is sprinkled on ice.
Reason : The addition of any soluble impurity results in decrease in melting point.

Multiple Matching Questions :

DIRECTIONS : *Each question has four statements (A, B, C and D) given in Column I and four statements (p, q, r and s) in Column II. Any given statement in Column I can have correct matching with one or more statement(s) given in Column II. Match the entries in column I with entries in column II.*

1.	Column 'A'		Column 'B'
A.	Sublimate	(p)	Iodine
B.	Compressibility	(q)	Liquids
C.	Definite Volume	(r)	Solid
D.	Expansion on Heating	(s)	gases

Integer Type Questions :

DIRECTIONS : *Following are integer based questions. Each question, when worked out will result in one integer from 0 to 9 (both inclusive).*

1. How many among the following shows the property of sublimation
Camphor, Napthelene balls, Wax, Iodine, Butter
2. Properties shown by gases among the following Diffusion, definite shape, definite volume, regidity high kinetic energy, Fluidity

SOLUTIONS

Brief Explanations of Selected Questions

1 EXERCISE

Multiple Choice Questions :

1. (a) 2. (b)
3. (d) **Hint :** In this portion different states of water can co-exist
4. (c) 5. (c) 6. (d)
7. (a) **Hint :** Solid changes to liquid so the intermolecular distance increases and molecules move apart, the speed of particles increase and there is no change in temperature because m.p. is a fixed temperature.
8. (c) Diffusion is fastest in gases.
9. (c) When **gas** (water vapour) changes to **liquid** (water) it **loses energy** and its freedom of movement is lesser (i.e. restricted).
10. (b) The volume of gases > liquids > solid where as the inter-particle space follows the order gas > liquid > solid.
11. (b) Force of attraction $\propto \dfrac{1}{\text{interparticle distance}}$
12. (d)
13. (b) Evaporation is a surface phenomenon.
14. (b) potential energy of gas molecules in released in the form of heat energy.
15. (c) 16. (b) 17. (a)
18. (d) 19. (a)

More Than One Option Correct :

1. (a, b, c) The molecules are always in a state of motion.
2. (b, c, d) 3. (a), (b), (c) 4. (a) and (b)
5. (a) and (b) 6. (a), (c), (d) 7. (a), (b), (d)
8. (a), (c) 9. (a), (c) 10. (b), (d)

Passage Based Questions :

1. (a) 2. (a) 3. (a)
4. (c) 5. (a) 6. (b)

Assertion & Reason :

1. (c) Assertion is correct but reason is incorrect. At higher altitudes the pressure is less and due to decrease in pressure the b.p. of water decreases and **not** increases.
2. (a) 3. (d) 4. (a)

Multiple Matching Questions :

1. $A \rightarrow p, r$; $B \rightarrow q, s$; $C \rightarrow q, r$; $D \rightarrow q, r, s, p$

Integer Type Questions :

1. 3
Camphor and Napthelene balls and Iodine
2. 3
Diffusion and high kinetic energy and Fluidity.

Chapter 8

TRANSFORMATION OF MATTER

INTRODUCTION

The science of chemistry mainly deals with the determination of nature and properties of non-living matter which surrounds us and the preparation of new substances, scientifically interesting and generally useful, from the materials that nature has provided us. Chemists are generally interested in the changes which these substances undergo on being subjected to conditions which they normally do not encounter i.e. high temperature, high pressure, extreme cold, contact with materials under varying conditions etc. It is generally from the changes which materials undergo when subjected to such conditions that chemists draw conclusions about their nature.

In the world we live, changes are going on all the time. Water changes from a liquid to vapour on boiling. Plants changes raw materials from air and soil into stems and leaves. Cars and buses burn fuel to move from place to place. If you look around you can find lots of changes. In some changes the material remains the same. For example no matter how finely you crush the salt crystals, the powder that you get is still salty. Nothing has changed except the size. However in some changes a substance changed into a different substance with different properties. For example, on burning wood, it is changed into completely different substances such as carbon dioxide gas and solid ash.

Based on above observations various changes have been classified into physical change and chemical change.

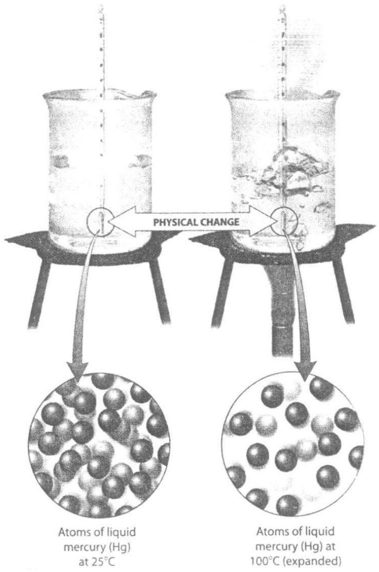
PHYSICAL AND CHEMICAL CHANGES

Physical change

Those properties which describe the look or feel of a substance e.g., colour, hardness, density texture and phase are called physical properties. Every substance has its own characteristic physical properties that we use to identify the substance.

The physical property of a substance can change on changing conditions but the substance remain same or no new substance is created. e.g., if the water is subject to boil, the formation of bubbles show that the water is changing into a gas. The gas disappears into the air as water vapour. If this air touches the lid or plate covering the water, the water vapour will change back into droplet of water. Therefore *a physical change is one in which the substance undergoing the change is not destroyed and no new element or compound is formed. In some physical changes the properties like colour, shape or size of the substance may change e.g. glowing of an electric bulb is a physical change, melting of ice* is a physical change.

Look at the picture of water boiling and changing into steam. Steam is another form of water. Heating water did not create a new material. In changing the water from a liquid to a gas, only the state of the water is changed.



CHECK Point

The melting of gold is a physical change. Why ?

SOLUTION

During a physical change, a substance changes only one or more of its physical properties; its chemical identity does not change. Because melted gold is still gold but in a different form, its melting represents only a physical change.

Chemical change :

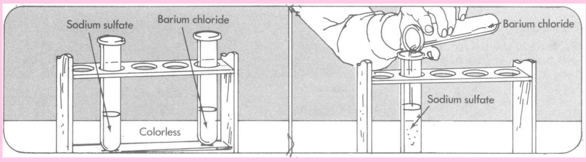
Those properties which allow a chemical substance to react with other substances to transform from one substance to another are known as chemical properties for example copper has the chemical property of reacting with carbon dioxide and water to form a greenish blue solid known as patina.

A chemical change is one in which the identity of the original substance is changed and a new substance or new substances are formed. In a chemical change the properties of the substance before and after the change are entirely different. *e.g. souring of milk, burning of paper, burning of candle etc.*

Look at the picture of the burning candle. The wax of a candle burns into ash and smoke. The original materials are changing into something different.

idea box

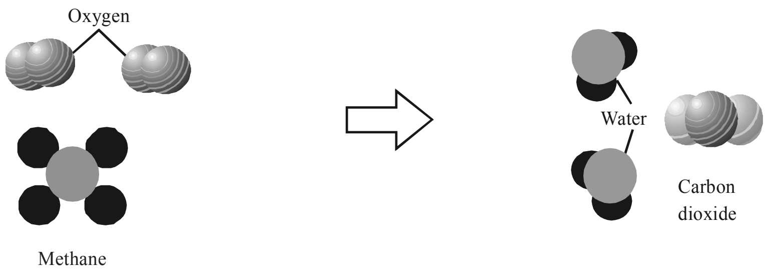
Pour a small amount of sodium sulfate solution in one test tube and a similar amount of barium chloride solution in another test tube. Now pour the contents of one of the test tubes into the other test tube. What do you observe ?



One afternoon after returning home from your school go into your kitchen. Try to find out and list 3 physical and chemical changes occuring in kitchen.

Knowledge ENHANCER

A chemical change involves a change in the way the atoms in the molecules are chemically bonded to one another. A chemical bond is the force of attraction between atoms that holds them together. A methane molecule for example is made of a single carbon atom bonded to four hydrogen atoms, and an oxygen molecule is made of two oxygen atoms bonded each. Figure given below shows the chemical change in which the atoms in a methane molecule and those in two oxygen molecules first pull apart and then form new bonds with different patterns resulting in the formation of carbon dioxide and water molecules.



Therefore, we can say that any change in a substance that involves a rearrangement of the way atoms are bonded is called a chemical change.

HISTORICAL PREVIEW

Henry Cavendish (1731-1810) was exceptionally shy, worked alone, had great wealth, yet rarely spent it. But he made important scientific discoveries. He produced water by exploding oxygen and hydrogen gases together. This caused a chemical change, joining one oxygen atom to two hydrogen atoms. He showed water was a chemical compound. H_2O, not an element as others believed.

CHECK Point

What is the evidence for the following being examples of chemical change ?

(a) Food spoils.

(b) An antacid tablet fizzes in water.

(c) A ring of scum forms around your bathtub.

(d) Iron rusts

(e) A firecracker explodes.

SOLUTION

(a) When food spoils it represents a chemical change because food changes from one chemical substance to another, sometimes this change can be observed by smell (a foul smell is observed).

(b) An antacid tablet fizzes in water, it represents a chemical change because a gaseous substance is formed which can be observed easily.

(c) The formation of scum rings around bath tub represents a chemical change. The scum rings are formed because of deposition of a new substance formed by the action of water in bath tub.

(d) Rusting of iron is a chemical change. Rust is a different chemical substance formed by the action of iron and water in presence of air.

(e) When a fire cracker explodes, new chemical substances are formed so it is a chemical change.

Difference Between Physical and Chemical Change

	Physical change		Chemical change
(i)	It is a temporary change and can be reversed by change of conditions	(a)	It is a permanent change and is an irreversible change.
(ii)	No new substance is formed and the composition of the substance remains unaltered.	(b)	The composition of the substance changes resulting in the formation of one or more new substances.
(iii)	No energy changes occur	(c)	It is generally accompanied by energy changes.

CHECK Point

How can you determine whether an observed change is physical or chemical ?

SOLUTION

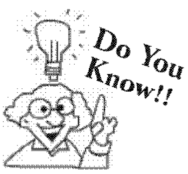
There are two guidelines by which chemical and physical changes can be identified.

1. In a physical change, a change in appearence is the result of a new set of conditions imposed on the same material. Restoring the original conditions restore the original appearence.
2. In a chemical change, a change in appearence is the result of the formation of new material that has its own unique set of properties.

CHEMICAL REACTIONS

So many changes are occuring around us every time. Iron changes to rust. In your body, food changes to blood, bone and tissue. Wood burns and changes to gases which go into the air. Certain rocks can be treated with chemicals to remove metals such as aluminium and iron. Some substances break into simpler substances.

All of these changes are chemical changes. All chemical changes can be written down by means of formulas and equations. What happens when a chemical change takes place ? Atoms or molecules combine with other atoms or molecules. This action is called a chemical reaction.



A chemical reaction involves the making or breaking of chemical bonds. For breaking of bond energy is needed and for making a bond energy is released. Both occur in every chemical reaction. The energy can be evolved or absorbed in the form of heat, light or electrical energy.

In a chemical reaction one group of substances are changed to form another group of substances or technically we can say.

A **chemical reaction** is an *interaction of two or more substances resulting in the formation of one or more new substances.*

During a chemical reaction, atoms rearrange to create one or more new compounds. This activity is nearly summed up in written form as a chemical equation. A chemical equation shows the reacting substances, called reactants, to the left of an arrow that points to the newly formed substances, called products :

$$\text{reactants} \longrightarrow \text{products}$$

Typically, reactants and products are represented by their elemental or chemical formulas. Sometimes molecular models or, simply, names may be used instead. Phases are also often shown : (*s*) for solid, (*l*) for liquid, and (*g*) for gas. Compounds dissolved in water are designated (*aq*) for aqueous solution. Lastly, numbers are placed in front of the reactants or products to show the ratio in which they either combine or form. These numbers are called coefficients, and they represent numbers of individual atoms and molecules. For instance, to represent the chemical reaction in which coal burns in the presence of oxygen to form gaseous carbon dioxide, we write the chemical equation using coefficients of 1.

$$\underset{\text{Reactants}}{1C(s) + 1\,O_2(g)} \longrightarrow \underset{\text{Products}}{1CO_2\,(g)} \quad \text{(Balanced)}$$

For example,

Iron + oxygen ⟶ Iron oxide.

In this example iron and oxygen react to form a new substance iron oxide.
Substances that take part in a chemical reaction are called **reactants**. In the example given above **iron** and **oxygen** are **reactants**.
Substances that are formed as a result of chemical reaction are called **products**. In the above example **iron oxide** is the **product**.
Whenever a chemical change occurs, a chemical reaction is said to have taken place.

Knowledge ENHANCER

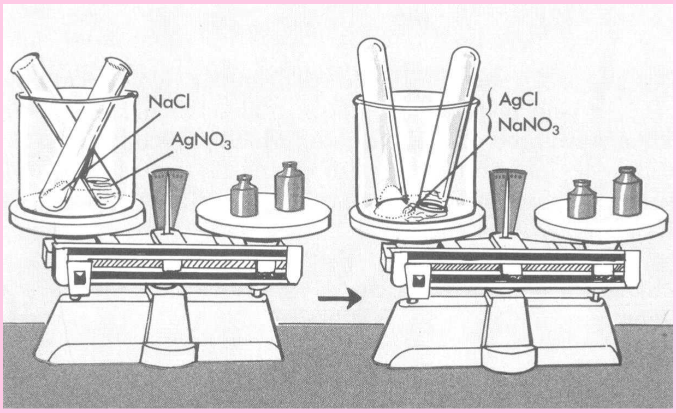
One of the most important principles of chemistry is the law of mass conservation. The law of mass of conservation states that matter is neither created nor destroyed-during a chemical reaction. The atoms present at the beginning of a reaction merely rearrange to form new molecules. This means that no atoms are lost or gained during any reaction. The chemical equation must therefore be balanced. In a balanced equation, each atom must appear on both sides of the arrow the same number of times.

IDEA BOX idea box

Take a small amount of table salt NaCl solution and put it into one test tube. Now take same amount of silver nitrate. $AgNO_3$ solution into another test tube. Put both test tubes into a beaker with open end up. Place tubes on one side of a platform balance and carefully balance with masses on other side.

Now turn both the test tubes upside down in the beaker. The new substances formed are silver chloride AgCl and sodium nitrate $NaNO_3$.

Now check the weight of new substances formed. What do you observe ?



TYPES OF CHEMICAL REACTIONS

(i) **Homogeneous reactions :** In case all the reactants and products of a chemical reaction are in the **same physical state** then such a reaction is known as **homogeneous reactions.**

For representation of a physical state of reactants/products we write (g) for gases, (ℓ) for liquids and (s) for solids state.

Example :

Hydrogen + Chloride $\longrightarrow$ Hydrochloric acid
(gas) (gas) (gas)

$H_2(g) + Cl_2(g) \longrightarrow 2HCl(g)$

(ii) **Heterogeneous reactions :** If in a chemical reaction, all the products and reactants are *not* in the same physical state then such a reaction is known as **heterogeneous reaction**.

Example :

$$\text{Calcium carbonate} \xrightarrow{\text{heat}} \underset{\text{(solid)}}{\text{Calcium oxide}} + \underset{\text{(gas)}}{\text{carbon dioxide}}$$

$$CaCO_3(s) \xrightarrow{\text{heat}} CaO(s) + CO_2(g)$$

(iii) **Exothermic reactions :** A chemical reaction which takes place with evolution of heat is known as an **exothermic reaction.**

Example :

Carbon + Oxygen $\longrightarrow$ Carbon dioxide + Heat

$C(s) + O_2(g) \longrightarrow CO_2(g) + \text{Heat}$

In case of an exothermic reaction the energy required to break the bonds of reactants is less then the energy released during formation of bonds of products and the difference in energy is released as heat energy.

Knowledge ENHANCER

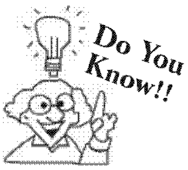
In addition to these reactions we must also know about combustion reaction. Combustion reactions are of great importance. In a combustion reaction a substance reacts with oxygen and release energy. These reactions are exothermic.

The combustion of natural gas is an important source of energy for homes and industry. Natural gas is mainly methane. Its complete combustion produces carbon dioxide and water vapour

Methane + Oxygen $\longrightarrow$ Carbon dioxide + Water

$$CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O$$

Substances such as methane, which undergo combustion readily and give out a large amount of energy, are known as fuels.



Respiration is an exothermic process. We all know that we need energy to stay alive. We get this energy from the food we eat. During digestion, food is broken down into simpler substances. For example, rice, potatoes and bread contain carbohydrates. These carbohydrates are broken down to form glucose. This glucose combines with oxygen in the cells of our body and provides energy. The special name of this reaction is respiration.

$$\underset{\text{(Glucose)}}{C_6H_{12}O_6(aq)} + 6O_2(aq) + 6H_2O(l) \rightarrow 6CO_2(aq) + 12H_2O(l) + \text{energy}$$

(iv) **Endothermic reactions :** A chemical reaction that is accompanied by absorption of heat energy is known as an **endothermic reaction.**

Example :

Nitrogen + Oxygen $\longrightarrow$ Nitric oxide – Heat
(gas) (gas) (gas)

$N_2(g) + O_2(g) \longrightarrow 2NO(g) - \text{Heat.}$

In case of an endothermic reaction the energy absorbed is more than the energy released and the difference in energy is absorbed during the reaction.

Do You Know!!

Rocket ship left off into space and campfire glow red hot as a result of exothermic reactions.

IDEA BOX

idea box

Take about 2 g barium hydroxide in a test tube. Add 1 g of ammonium chloride and mix with the help of a glass rod. Touch the bottom of the test tube with your palm. What do you feel ? Is this an exothermic or endothermic reaction?

(v) **Combination reactions (or Synthesis reactions) :** Those reactions in which two or more substances combine to form a single substance are known as **combination reactions.**

Examples :

(a) When magnesium is burned in air, it combines with the oxygen of air to form magnesium oxide :

Magnesium + Oxygen $\longrightarrow$ Magnesium oxide

$2Mg\,(s) + O_2\,(g) \longrightarrow 2\,MgO\,(s)$

In this reaction two elements, magnesium and oxygen combine to yield a single compound magnesium oxide.

(b) Calcium oxide reacts vigorously with water to produce slaked lime (calcium hydroxide) releasing a large amount of heat.

$$\underset{\text{(Quick lime)}}{CaO(s)} + H_2O(l) \longrightarrow \underset{\text{(Slaked lime)}}{Ca(OH)_2(aq)}$$

(vi) **Decomposition reactions :** Those reactions in which a compound splits up into two or more simpler substances are known as decomposition reactions.

Examples :

(a) When we heat calcium carbonate, it decomposes to form calcium oxide and carbon dioxide.

Calcium carbonate $\xrightarrow{\text{heat}}$ Calcium oxide + Carbon dioxide.

$$CaCO_3(s) \xrightarrow{\text{heat}} CaO(s) + CO_2(g)$$

(b) When water is electrolysed, by passing electric current through acidulated water, it decomposes to give hydrogen and oxygen.

$$2H_2O(\ell) \xrightarrow{\text{Electric current}} 2H_2(g) + O_2(g)$$

These reactions are endothermic. Most of them require heat energy. Decomposition caused by heat energy is called thermal decomposition.

Do You Know!!

Generally Synthesis or Combination reactions are usually exothermic, though they often require an input of heat energy to start them.

However, there is one very important synthesis reaction which is endothermic; namely photosynthesis. This reaction is essential for life on Earth. It takes place in the green leaves of plants and requires energy from sunlight.

CHECK Point

When hydrogen burns in oxygen, water is formed and when water is electrolysed, then hydrogen and oxygen are produced. What type of reaction takes place :

(a) in the first case ?

(b) in the second case ?

SOLUTION

(a) Combination reaction.

(b) Decomposition reaction.

A solution of slaked lime produced by the reaction given is used for white washing walls. Calcium hydroxide reacts slowly with the carbon dioxide in air to form a thin layer of calcium carbonate on the walls. Calcium carbonate is formed after two to three days of white washing and gives a shiny finish to the walls. It is interesting to note that the chemical formula for marble is also $CaCO_3$.

$$\underset{\text{(Calcium hydroxide)}}{Ca(OH)_2(aq)} + CO_2(g) \longrightarrow \underset{\text{(Calcium carbonate)}}{CaCO_3(s)} + H_2O(l)$$

(vii) **Displacement reactions :** Those reactions in which one element takes the place of another element in a compound is known as **displacement reaction.**

Example :

When a piece of zinc metal is placed in copper sulphate solution, zinc sulphate and copper are obtained.

$$Zn(s) + CuSO_4(aq) \longrightarrow ZnSO_4(aq) + Cu(s)$$

In the above reaction zinc has displaced copper from copper sulphate and copper is set free.

Note : Displacement reactions are useful in working out the pattern of reactivity of elements of the same type.

The above displacement reaction occurs because zinc is *more reactive* as compared to copper.

In general, *a more reactive element displaces a less reactive element from its compound*.

CHECK Point

What happens when dilute hydrochloric acid is added to iron filings ?

SOLUTION

Hydrogen gas and iron chloride are produced due to displacement reaction.

The digestion of food in the body is an example of decomposition reaction. When we eat foods like wheat, rice or potatoes, then the starch present in them decomposes to give simple sugars like glucose in the body; and the proteins decompose to form amino acids.

Knowledge ENHANCER

Chemical behaviour of metal –

K
Ca
Na
Mg
Al
Zn
Fe
H
Cu
Ag
Au

*Metals undergo similar kind of reactions. However the "vigour" with which they react is **not** the same. Some metals are more reactive than others. Metals along with hydrogen (a non-metal) are arranged in order of their reactivity in a series, called the **activity series** (see figure)*

This series helps us in understanding the reactivity of metals.

Displacement reaction can be more easily explained by taking example of iron and copper sulphate solution.
When iron nails are dipped in copper sulphate solution the iron nails become brownish in colour and the blue colour of copper sulphate solution fade.

$$\underset{}{Fe(s)} + \underset{\text{(Copper sulphate)}}{CuSO_4(aq)} \longrightarrow \underset{\text{(Iron sulphate)}}{FeSO_4(aq)} + Cu(s)$$

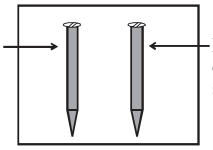
In this reaction, iron has displaced or removed another element, copper from copper sulphate solution.

(viii) **Double displacement reactions :** Those reactions in which two compounds react by an exchange of ions to produce two new compounds are referred to as double displacement reactions.
A double displacement reaction usually occurs in solution and one of the products (being insoluble) precipitates out.
Example:
When silver nitrate solution is added to sodium chloride solution, a white precipitate of silver chloride is formed alongwith sodium nitrate solution.

$$AgNO_3(aq) + NaCl(aq) \longrightarrow AgCl(s)\downarrow + NaNO_3(aq)$$

AgCl (s) formed appears as a white precipitate.
The above double displacement reaction can be represented in ionic form as follows:

$$Ag^+NO_3^- + Na^+Cl^- \longrightarrow Ag^+Cl^- + Na^+NO_3^-$$



(ix) **Isomerisation reactions :** It is a special type of reaction in which the rearrangements of chemical bonds takes place in such a way that a molecule of one compound is changed into the molecule of another compound.
Example:
Ammonium cyanate on heating gets converted into urea.

$$\text{Ammonium cyanate} \xrightarrow{\text{heat}} \text{Urea}$$

$$NH_4CNO \xrightarrow{\text{heat}} NH_2CONH_2$$

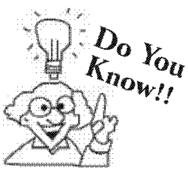
(x) **Neutralization reaction:** In fact neutralization reaction is a **double decomposition type** reaction. In neutralization reaction, an *acid* and a *base* react to form *salt* and *water*.

$$\text{acid + base} \longrightarrow \text{salt + water} \qquad \text{(neutralization reaction)}$$

Examples:

$$\underset{\substack{\text{(base)}\\\text{Sodium}\\\text{hydroxide}}}{NaOH} + \underset{\substack{\text{(acid)}\\\text{Hydro–}\\\text{chloric acid}}}{HCl} \longrightarrow \underset{\substack{\text{(salt)}\\\text{Sodium}\\\text{chloride}}}{NaCl} + \underset{\text{(water)}}{H_2O}$$

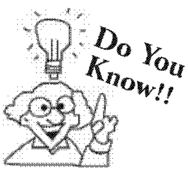
$$\underset{\text{(base)}}{KOH} + \underset{\text{(acid)}}{HCl} \longrightarrow \underset{\text{(salt)}}{KCl} + \underset{\text{(water)}}{H_2O}$$



In a chemical reaction the evolution of gas is represented by an upward arrow (↑) and a precipitate is denoted by a downward arrow (↓).

OXIDATION-REDUCTION REACTIONS

In addition to various categories of chemical reactions discussed so far, some chemical reactions are classified as **Redox** reactions as they involve **Reduction** and **Oxidation** reactions. Which are discussed in detail below.
Oxidation : It is defined as addition of oxygen or removal of hydrogen from a compound.



Both the compounds in such a reaction can be represented by same chemical formula but they have different structures.

Examples :

(i) Carbon + Oxygen $\longrightarrow$ Carbon dioxide

$C + O_2 \longrightarrow CO_2$ (addition of oxygen)

(ii) Iron + Oxygen $\longrightarrow$ Iron oxide

$3Fe + 2O_2 \longrightarrow Fe_3O_4$

(iii) Magnesium + Oxygen $\longrightarrow$ Magnesium oxide

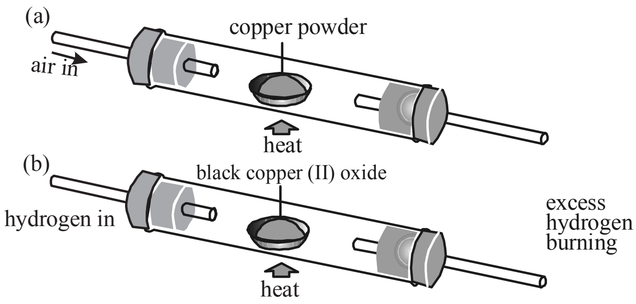
$2Mg + O_2 \longrightarrow 2MgO$

(iv) Hydrogen Sulphide + Chlorine $\longrightarrow$ Hydrogen Chloride

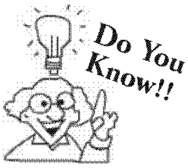
$H_2S + Cl_2 \longrightarrow 2HCl + S$ (Removal of Hydrogen)

(v) Hydrogen chloride + Manganese dioxide $\longrightarrow$ Manganese Chlorine

$4HCl + MnO_2 \longrightarrow MnCl_2 + 2H_2O + Cl_2$

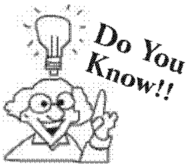


(a) The oxidation of copper to copper oxide.
(b) The reduction of copper oxide back to copper using hydrogen



This type of neutralization is also used to remove carbon dioxide from air-conditioned buildings.

- If a substance gains oxygen during a reaction, it is oxidised.
- If a substance loses oxygen during a reaction, it is reduced.



The most common case of indigestion is the presence of too much acid in the stomach.
The antacids, the medicines used to cure indigestion caused due to presence of too much acid in stomach, contain a suitable miled alkali such as milk of magnesia (i.e. a suspension of magnesium hydroxide in water).

Reduction : It is defined as the addition of hydrogen or removal of oxygen from a compound.

Examples :

Copper oxide + Hydrogen $\longrightarrow$ Copper + Water

$CuO + H_2 \longrightarrow Cu + H_2O$ (Removal of oxygen)

Lead oxide + Carbon $\longrightarrow$ Lead + Carbon dioxide

$2PbO + C \longrightarrow 2Pb + CO_2$

Iodine + Hydrogen Sulphide $\longrightarrow$ Hydrogen Iodide + Sulphur

$I_2 + H_2S \longrightarrow 2HI + S$ (Addition of hydrogen)

A close look at the above oxidation-reduction reaction reveals that if one substance is reduced the other substance of the reactants is oxidized. Thus reduction and oxidation occurs simultaneously. Due to this such reactions are known as **Redox** reactions.

Consider this reaction :

Zinc oxide + Carbon $\rightarrow$ Zinc + Carbon monoxide

oxidation

$$ZnO + C \rightarrow Zn + CO$$

reduction

OXIDISING AND REDUCING AGENTS

Oxidising agent or **oxidant :** Any substance that is capable of giving oxygen to bring about the process of oxidation (or combustion).
Reducing agent : A substance which gives hydrogen or removes oxygen thus causing reduction is known as reducing agent.
Following example helps to understand these.

$$2CuO + H_2 \longrightarrow Cu + H_2O$$

Copper oxide + Hydrogen $\longrightarrow$ Copper + Water

In this hydrogen (H_2) removes oxygen from copper and thus hydrogen (H_2) acts as reducing agent.
Since copper oxide gives oxygen to hydrogen and hydrogen is oxidised to water so in this reaction copper oxide acts as oxidizing agent.

CHECK Point

When a magnesium ribbon is heated, it burns in air to form magnesium oxide. Write a balanced chemical equation for this reaction.
(a) substance oxidised, and
(b) substance reduced.

SOLUTION

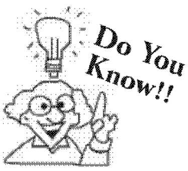
(a) Magnesium (b) Oxygen

idea box

As you have studied there are many types of chemical reactions in chemistry. Study about each of them thoroughly call your friends and also remind them that they should also go through types of chemical reaction. Now play a game with your friends in which you become host and make your all friends contestant. Give them a hint about type of chemical reaction and they have to make a right guess 10 points for right answer and 5 points will be deducted for wrong answer. Let see who wins ?

CHEMICAL EQUATION

A shorthand notation of describing an actual chemical reaction in terms of symbols and formulae alongwith the number of atoms and molecules of the reactants and products is called **chemical equation.** For example, the word equation.



Respiration and photosynthesis, the two life giving reactions, belong to Redox reactions.
In respiration, the food we eat is oxidized and this releases the energy our body needs.
In photosynthesis process, in plants, the carbon dioxide from air get converted to sugar and starch.

Zinc + Sulphuric acid $\longrightarrow$ Zinc sulphate + Hydrogen

Now symbol of zinc is Zn.
Formula of sulphuric acid is H_2SO_4
Formula of zinc sulphate is $ZnSO_4$
Formula of hydrogen is H_2

Putting the symbols and formulae of all substances in the above word equation, we get the following **chemical equation.**

$$\underbrace{Zn + H_2SO_4}_{\text{Reactants}} \longrightarrow \underbrace{ZnSO_4 + H_2}_{\text{Products}}$$

The arrow ($\rightarrow$) sign pointing towards right side is put between the reactants (*i.e.* the substances which react or combine) and products (*i.e.* the new substances produced).

Information Conveyed by the Chemical Equation

To make the point clear, let us consider a chemical equation

$$2H_2 + O_2 \longrightarrow 2H_2O$$

This equation conveys the following information :

(i) Hydrogen combines with oxygen to form water.

(ii) Two molecules of hydrogen combine with one molecule of oxygen to form two molecules of water.

(iii) Two moles of hydrogen combine with one mole of oxygen to form two moles of water.

(iv) 4 ($4 \times 1 = 4$) parts by weight of hydrogen combine with 32 ($2 \times 16 = 32$) parts by weight of oxygen to form 36 ($4 \times 1 + 2 \times 16 = 36$) parts by weight of water.

(v) A molecule of hydrogen consists of two atoms of hydrogen, a molecule of oxygen contains two atoms of oxygen and a molecule of water is made up of two atoms of hydrogen and one atom of oxygen.

(vi) The total weight of reactants is equal to the total weight of products.

Information not conveyed by chemical equation

Or

Limitations of chemical equation

A chemical equation fails to provide the following information.

It does not tell

(i) why and how the reaction occurs ?

(ii) the conditions under which the reaction occurs.

(iii) about the heat changes taking place in the reaction *i.e.* whether heat is absorbed or evolved.

(iv) about the physical state of reactants or products (*i.e.* whether they are solid, liquid or gas).

(v) whether the reaction is reversible or irreversible.

(vi) about the concentration of the reacting substances and the products.

SUMMARY

Physical change : A change in which the state of an object changes but its chemical make up does not change is called physical change. *e.g.* melting of paraffin, evaporation of petrol *etc*.

Chemical change : A change in which new substances with new properties are formed is called a chemical change. *e.g.* changing of milk into curd, rusting of iron objects, digestion of food in our body *etc*.

Characteristics of physical and chemical changes :

	Physical change		Chemical change
1.	Produces no new kind of matter	1.	Always produces a new kind of matter.
2.	Generally reversible	2.	Generally not reversible
3.	not accompanied by great heat changes (except latent heat effects accompanying change of state)	3.	Generally accompanied by considerable heat changes.
4.	Produces no change of mass	4.	Produces individual substances whose masses are different from those of original individual substances.

Chemical Reactions : The chemical changes are called chemical reactions. A chemical reaction is an interaction of two or more substances resulting in the formation of one or more substances. For example

$$\text{Iron} + \text{Oxygen} \longrightarrow \text{Iron oxide}$$

Reactants : Substances that take part in a chemical reaction are called reactants. *e.g.* Iron and oxygen in above example.

Products : Substances that are formed as a result of chemical reaction are called products. For example, iron oxide in the above example.

Types of Chemical Reactions :

(a) **Combination Reaction :** Reactions in which two or more reactants combine to form one product are called combination reaction. For example, hydrogens combines with chlorine to form hydrogen chloride ($H_2 + Cl_2 \longrightarrow 2HCl$).

(b) **Decomposition Reaction :** In such a reaction a chemical compound decomposes into simpler substances. These reactions generally occur under the influence of *heat, light, electricity* or *catalyst*. For example Potassium chlorate on heating decomposes to give Potassium chloride and oxygen.

$$2KClO_3 \xrightarrow[\text{[MNO}_2\text{]}]{\text{Heat}} 2KCl + 3O_2$$

(c) **Displacement Reaction :** In such a reaction, an element reacts with a compound to displace one of the elements in the compound. For example

$$\underset{\text{(Element)}}{Zn} + \underset{\text{(Compound)}}{CuSO_4} \longrightarrow \underset{\text{(Compound)}}{ZnSO_4} + \underset{\text{(Element)}}{Cu}$$

(**Note:** In this type of reaction the element displaces another element from compound because it is more reactive than the element already present]

(d) **Double decomposition reaction :** In such a reaction two compounds react to form two new compounds. The positive ion of one compound combines with the negative ion of the other and *vice-versa*. For example

$$AgNO_3 + NaCl \longrightarrow AgCl + NaNO_3$$

(e) **Isomerization Reaction :** It is a special type of reaction in which rearrangement of chemical compounds takes place in such a way that a molecule of one compound is changed into the molecule of another compound. For example.

$$\underset{\text{Ammonium cyanate}}{NH_4CNO} \xrightarrow{\text{heat}} \underset{\text{Urea}}{NH_2CONH_2}$$

(f) **Neutralization Reaction :** It is actually a **double decomposition type reaction**. In neutralization reaction, an *acid* and a *base* react to form *salt* and *water.*

$$\underset{\text{Sodium Hydroxide}}{NaOH} + HCl \longrightarrow \underset{\text{Sodium chloride}}{NaCl} + H_2O$$

Exothermic and Endothermic reactions :

(a) **Exothermic reactions :** Those reactions in which heat is evolved are known as exothermic reactions. For example

$$N_2 + 3H_2 \longrightarrow \underset{\text{(ammonia)}}{2NH_3} + \text{Heat}$$

(b) **Endothermic reaction :** Those reactions in which heat is absorbed are known as endothermic reactions. For example

$$N_2 + O_2 + \text{Heat} \longrightarrow 2NO$$

or $N_2 + O_2 \longrightarrow 2NO - \text{Heat}$

Homogeneous reaction : When all the reactants and products are in the same physical state, the reaction is known as homogeneous reaction. For example

$$HCl\,(aq) + NaOH\,(aq) \longrightarrow NaCl\,(aq) + H_2O$$

Heterogeneous reactions : When reactants and products are in different physical states, the reaction is known as a heterogeneous reaction. For example

$$CaCO_3(s) + 2HCl\,(aq) \longrightarrow CaCl_2\,(aq) + CO_2\,(g) + H_2O$$

Irreversible reaction : The reactions which proceed only in one direction are called irreversible reactions. For example

$$2KClO_3 \xrightarrow[\text{[MnO}_2\text{]}]{\text{heat}} 2KCl + 3O_2$$

Reversible reactions : Those reactions in which products can react to reform the original reactants are called reversible reaction. For example

$$NH_4Cl\,(s) \rightleftharpoons NH_3\,(g) + HCl\,(g)$$

Oxidaton : It is the process that involves addition of oxygen (or electronegative element) or removal of hydrogen (or electropositive element) from a compound. For example

$$C + O_2 \longrightarrow CO_2$$

$$H_2S + Cl_2 \longrightarrow 2HCl + S$$

[**Notes :** Oxidation reaction are generally exothermic]

Reduction : It is the process that involves addition of hydrogen (or electropositive element) or removal of oxygen (or electro negative element) from a compound. For example.

$$CuO + H_2 \longrightarrow Cu + H_2O$$

$$I_2 + H_2S \longrightarrow 2HI + S$$

Redox reactions : The oxidation and reduction reactions taking place simultaneously are called **redox reactions**.

Oxidising agent or oxidant : Any substance that is capable of giving oxygen to bring about the process of oxidation (or combustion). Thus *oxidising agents* are chemicals that oxidise other chemicals. For example in the reaction.

$$4Fe + 3O_2 \longrightarrow 2Fe_2O_3$$

O_2 Oxidises Fe to Fe_2O_3, so O_2 is oxidising agent. Common oxidising agents are O_2 (g), Cl_2 (g), Cl_2 (aq), conc. H_2SO_4 (sulphuric acid)

Reducing agent: A substance which gives hydrogen or removes oxygen thus causing reduction is known as reducing agent. For example, in the reaction

$$CuO\ (s) + H_2(g) \longrightarrow Cu\ (s) + H_2O\ (\ell)$$

CuO (s) is reduced to Cu (s) and $\mathbf{H_2}$ (g) which brings about this change is called reducing agent common reducing agent are H_2 (g), metals (Na, K *etc.*), carbon, KI (aq) *etc*.

Combustion Reaction: The burning of fuel in presence of oxygen is an example of such a reaction.

In this case *fuel* is oxidised and *oxygen* is the oxidising agent.

Chemical Equation: A chemical equation is a shorthand representation of what happens in a chemical change. It shows which chemicals are involved and in a symbol equation, the ratio of numbers of molecules of reactants and products.

Word equation, e.g.

$$\text{Iron} + \text{Sulphur} \longrightarrow \text{iron (II) sulphide}$$

Symbol equation:

$$Fe(s) + S(s) \longrightarrow FeS(s)$$

Essentials of a chemical equation

(i) It should represent an **actual chemical change.**

(ii) It should be **balanced.**

(iii) It should be **molecular.**

Limitations of a chemical equation

(i) It does not indicate the physical state of the reactants and the products.

(ii) It does not indicate whether heat is evolved or absorbed in the reaction.

(iii) It does not indicate the specific conditions required for the reaction like pressure, catalyst, temperature.

1 EXERCISE

Multiple Choice Questions :

DIRECTIONS : *This section contains **14** multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.*

1. Select the one that is **not** a chemical change ?

(a) Dissolution of ammonia in water.

(b) Dissolution of carbon dioxide in water.

(c) Dissolution of oxygen in water.

(d) None of the above is a chemical change.

2. Select the one that represents a displacement reaction.

(a) $NaCl + AgNO_3 \longrightarrow AgCl + NaNO_3$

(b) $Zn + CuSO_4 \longrightarrow ZnSO_4 + Cu$

(c) $HCl + NaOH \longrightarrow NaCl + H_2O$

(d) $C + O_2 \longrightarrow CO_2$

3. Which one of the following is a combination reaction ?

(a) Formation of a mixture of carbon monoxide and hydrogen. When steam is passed over red hot iron.

(b) Reaction of water with sodium metal to form sodium hydroxide and hydrogen.

(c) $Ca(OH)_2 + Na_2CO_3 \longrightarrow 2NaOH + CaCO_3 \downarrow$

(d) Preparation of stannic chloride (Tin (iv) chloride) by passing chlorine gas into molten tin (Sn).

4. Which of the following involves combination of two elements?

(a) $CaO + CO_2 \longrightarrow CaCO_3$

(b) $4Na + O_2 \longrightarrow 2Na_2O$

(c) $2SO_2 + O_2 \longrightarrow 2SO_3$

(d) $NH_3 + HCl \longrightarrow NH_4Cl$

5. When hydrogen sulphide gas is passed through a blue solution of copper sulphate, a black precipitate of copper sulphide is formed. This is an example of ?

(a) combination reaction

(b) displacement reaction

(c) decomposition reaction

(d) double decomposition reaction

6. A change is said to be a chemical change when

(a) it is accompanied by energy change

(b) it is accompanied by formation of new substances

(c) it is accompanied by change in physical properties

(d) All the above are correct

7. Which one is a decomposition reaction ?

(a) $2HgO \xrightarrow{\text{heat}} 2Hg + O_2$

(b) $CaCO_3 \xrightarrow{\text{heat}} CaO + CO_2$

(c) $2H_2O \xrightarrow{\text{Electrolysis}} 2H_2 + O_2$

(d) All the above reactions are decomposition reactions.

8. The reactions in which two compounds exchange their radicals to form two new compounds are called

(a) displacement reaction

(b) decomposition reaction

(c) double displacement reaction

(d) isomerisation reaction

9. Consider the following reaction and tell which one is the oxidising reagent.

$$SO_2 + 2H_2S \longrightarrow 2H_2O + 3S$$

(a) SO_2 (b) H_2S

(c) H_2O (d) S

10. $CuO + H_2 \longrightarrow Cu + H_2O$

Above reaction is an example of

(a) redox reaction

(b) synthesis reaction

(c) neutralisation reaction

(d) decomposition reaction

11. A substance which gets oxidised, during a redox reaction, is known as

(a) oxidizing agent

(b) reducing agent

(c) either oxidising or reducing agent

(d) None of the above

12. In the reaction :

$$PbO + C \longrightarrow Pb + CO$$

(a) PbO is oxidised (b) PbO is oxidant

(c) C is reductant (d) Both (b) and (c)

13. Which of the following is an isomerization reaction ?

(a) $2Na + 2H_2O \longrightarrow 2NaOH + H_2$

(b) $2KBr + Cl_2 \longrightarrow 2KCl + Br_2$

(c) $ZnCO_3 \xrightarrow{heat} ZnO + CO_2$

(d) $NH_4CNO \longrightarrow NH_2CONH_2$

14. Chemical equations are balanced in accordance with the

(a) Dalton's law

(b) Law of conservation of mass

(c) Law of definite composition

(d) None of the above

More than One Option Correct :

DIRECTIONS : *This section contains 9 Multiple Choice Questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE OR MORE may be correct.*

1. The reaction

$$NaOH + HCl \longrightarrow NaCl + H_2O$$

is an example of

(a) Displacement reaction

(b) Double decomposition reaction

(c) Neutralization reaction

(d) Isomerisation reaction

2. Most of the chemical changes are found to undergo

(a) a change in the weight of the reacting substances

(b) a change in temperature

(c) a change in energy

(d) all of the above are incorrect

3. Select the incorrect statements.

(a) A physical change can be easily reversed by a slight change of conditions.

(b) A physical change is accompanied by a change in color of the reacting substances.

(c) It is possible to characterise a chemical reaction by only a change in state

(d) All the above are incorrect.

4. The reaction :

$$C\,(s) + O_2\,(g) \longrightarrow CO_2\,(g) + \text{heat is}$$

(a) a homogeneous reaction

(b) a heterogeneous reaction

(c) an exothermic reaction

(d) an endothermic reaction

5. The reaction :

$$N_2\,(g) + O_2\,(g) + \text{heat} \longrightarrow 2NO\,(g)$$

(a) a homogeneous reaction

(b) a heterogeneous reaction

(c) a combination reaction

(d) an endothermic reaction

6. The reaction;

$$2Mg\,(s) + O_2\,(g) \longrightarrow 2\,MgO\,(s) \text{ is}$$

(a) a combination reaction

(b) a synthesis reaction

(c) a reaction in which an element and a compound react to produce a compound

(d) a reaction in which two elements react to produce a compound

7. In the respiration

(a) the food that we eat is oxidised

(b) the food that we eat is reduced

(c) energy is released

(d) energy is absorbed

8. A burning of candle represents

(a) physical change (b) volume change

(c) chemical change (d) none of the above

9. A balanced chemical equation should represent

(a) Temperature of reaction

(b) Catalyst involved

(c) Atomicity of each element

(d) Physical states of reactant and products

Passage Based Questions :

DIRECTIONS : *Study the given paragraph(s) and answer the following questions.*

Passage

Take appropriate quantity of $CuSO_4$ from your school laboratory and dissolve it in small quantity of water to obtain blue coloured aqueous solution. Now put 3-5 pieces of medium size iron nail in it. After sometime colour of solution starts changing from blue to green.

1. Which type of reaction is responsible for this change ?

(a) Displacement (b) Combination

(c) Decomposition (d) Double decomposition

2. Correct chemical equation for process.

(a) $2Fe + CuSO_4 \longrightarrow Fe_2SO_4 + Cu$

(b) $Fe + CuSO_4 \longrightarrow FeSO_4 + Cu$

(c) $FeSO_4 + Cu \longrightarrow CuSO_4 + Fe$

(d) None of the above

3. Above phenomena represents ?

(a) Physical change

(b) Chemical change

(c) Both physical and chemical

(d) None of the above

Assertion & Reason :

DIRECTIONS : *Each of these questions contains an Assertion followed by reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.*

(a) If both **Assertion** and **Reason** are **correct** and Reason is the **correct explanation** of Assertion.
(b) If both **Assertion** and **Reason** are correct, but Reason is **not the correct explanation** of Assertion.
(c) If **Assertion** is **correct** but **Reason** is **incorrect**.
(d) If **Assertion** is **incorrect** but **Reason** is **correct**.

1. **Assertion :** Burning of paper is a physical change.
Reason : The products formed on burning of paper can not be easily converted back to paper.

2. **Assertion :** The change of water from liquid to steam on heating is a physical change.
Reason : The volume remains same and the change involves a change in composition.

3. **Assertion :** By passing electric current through water, we can decompose it into hydrogen and oxygen.
Reason : Water is a compound whereas hydrogen and oxygen are elements.

4. **Assertion :** The formation of rust is a chemical change.
Reason : For the formation of rust iron must be exposed to air and water.

5. **Assertion :** A redox reaction involves both oxidation and reduction.
Reason : $SO_2 + 2H_2S \longrightarrow 2H_2O + 3S$ is an example of redox reaction.

Multiple Matching Questions :

DIRECTIONS : *Each question has four statements (A, B, C and D) given in Column I and four statements (p, q, r and s) in Column II. Any given statement in Column I can have correct matching with one or more statement(s) given in Column II. Match the entries in column I with entries in column II.*

1.

	Column I		Column II
A.	$C + O_2 \rightarrow CO_2 + \text{heat}$	(p)	Exothermic
B.	$CaCO_3 \xrightarrow{\Delta} CaO + CO_2$	(q)	Combination
C.	$N_2 + O_2 \rightarrow 2NO$	(r)	Endothermic
D.	$HCl + NH_4OH \rightarrow NH_4Cl + H_2O + 12.3kJ$ heat	(s)	Decomposition

2.

	Column I		Column II
A.	$C + O_2 \rightarrow CO_2$	(p)	Displacement
B.	$H_2S + Cl_2 \rightarrow 2HCl + S$	(q)	Oxidation
C.	$I_2 + H_2S \rightarrow 2HI + S$	(r)	Combination
D.	$H_2 + Cl_2 \rightarrow 2HCl$	(s)	Reduction

Integer Type Questions :

DIRECTIONS : *Following are integer based questions. Each question, when worked out will result in one integer from 0 to 9 (both inclusive).*

1. In the following reaction

$$2KClO_3 \xrightarrow[\text{[MnO}_2\text{]}]{\text{heat}} 2KCl + xO_2$$

The value of x is

2. How many of them are chemical changes.
melting of ice, glowing of bulb, souring of milk, burning of paper, burning of candle

SOLUTIONS

Brief Explanations of Selected Questions

1 EXERCISE

Multiple Choice Questions :

1. (c)
2. (b)
3. (d) [**Hint :** $Sn + 2Cl_2 \longrightarrow SnCl_4$]
4. (b) [**Hint :** Both Na and O_2 are elements]
5. (d) [**Hint** : $\underset{\text{(Blue)}}{CuSO_4} + H_2S \longrightarrow \underset{\text{(Black)}}{CuS\downarrow} + H_2SO_4$]
6. (d)
7. (d) [**Hint :** In all these the reactants decompose to form simpler products.]
8. (c)
9. (a) In this reaction SO_2 acts as the oxidising reagent.
10. (a) [**Hint :** It involves oxidation and reduction.]
11. (b)
12. (d) [**Hint :** In this carbon reduces PbO to Pb so carbon (C) is reducing agent (reductant) PbO acts as oxidising agent (oxidant) as it oxidises C to CO.
13. (d)
14. (b)

More Than One Option Correct :

1. (b, c)
2. (a, b, c)
3. (b,c)
4. (b, c)
5. (a, d,)
6. (a, b, d)
7. (a, c)
8. (a, b, c)
9. (a, b, d)

Passage Based Questions :

1. (a) This is because of the displacement reaction.
2. (b) $\underset{\text{(Blue)}}{CuSO_4(aq)} + Fe \longrightarrow \underset{\text{(Green)}}{FeSO_4} + Cu(s)$
3. (a) As the proportion of salt dissoved in water is changing because of formation of new salt ($FeSO_4$). Thus as the proportion of solution is changing and new salt is formed which have different physical properties. Thus phenomina also represents physical change.

Assertion & Reason :

1. (d)
2. (c) Both incorrect.
3. (b)
4. (b)
5. (b)

Multiple Matching Questions :

1. $A \to (p, q)$; $B \to (r, s)$; $C \to (q, r)$; $D \to (p)$
2. $A \to (q, r)$; $B \to (p, q, s)$; $C \to (p, s, q)$; $D \to (r, s)$

Integer Type Questions :

1. 3

$$2KClO_3 \xrightarrow[{[MnO_2]}]{\text{heat}} 2KCl + 3O_2$$

2. 3

souring of milk, burning of paper and burning of candle.